

User Manual

MM32F0020

32-bit Microcontroller Based on Arm®Cortex®-M0 Core

Version: 1.01

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1 System and Memory Architecture

1.1 System architecture overview

MM32F0020 product is a 32-bit microcontroller product based on Arm®Cortex®-M0 processor. It features high performance and low power consumption. It adopts the matrix bus structure, which includes one AHB host: CPU as well as three slaves: SRAM, Flash memory, AHB bus (including bus bridge from AHB to APB), and a variety of devices connected to APB bus.

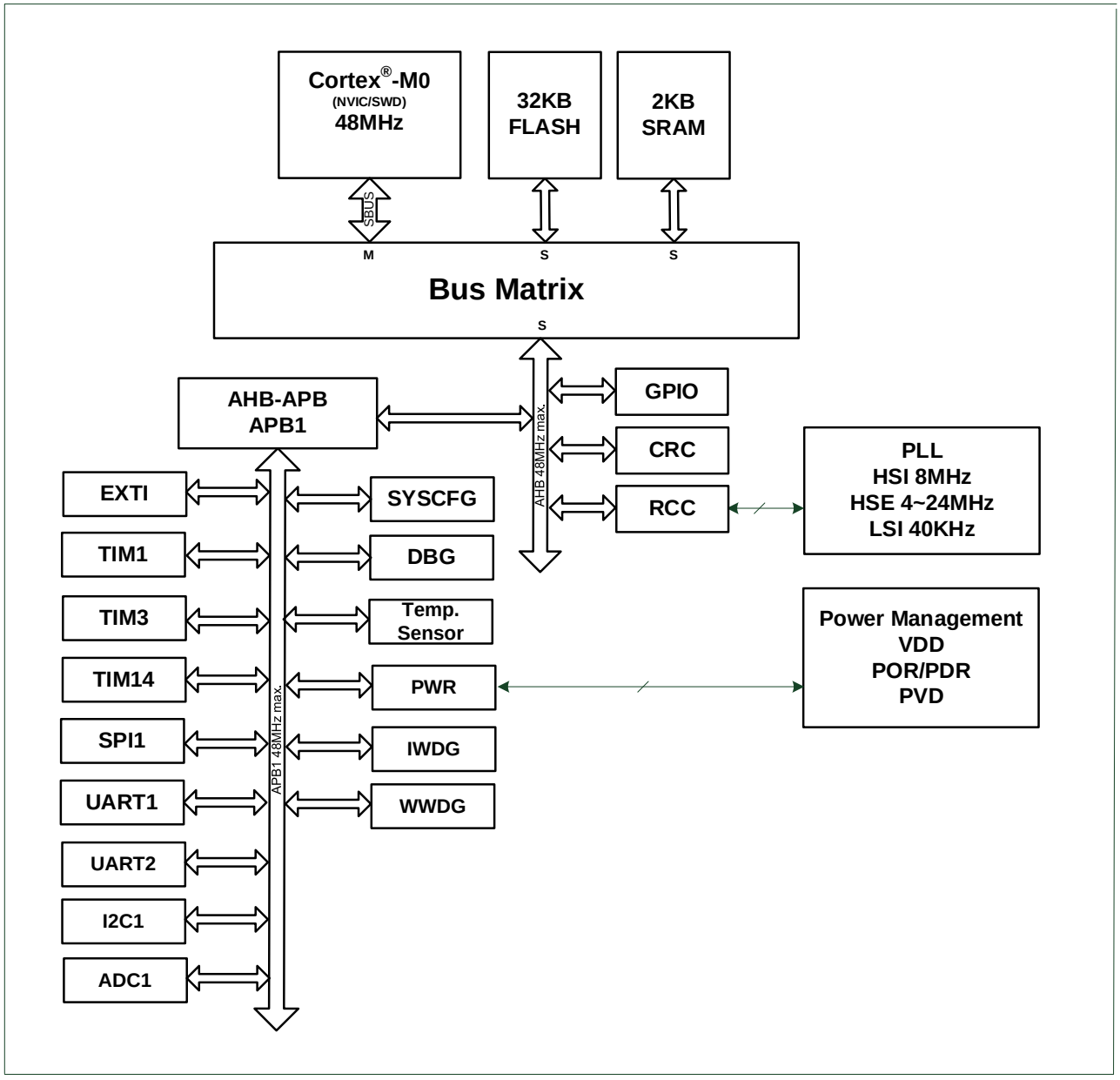


Figure 1-1 System architecture

1.1.1 System bus (SBUS)

The system bus connects the CPU core and the bus matrix to construct a passage between the CPU and the bus matrix for data transfer, and the CPU can act as a host to drive the bus, eventually achieving read and write operations to slaves such as APB peripherals, SRAM, flash memory, etc.

1.1.2 BUS Matrix

The bus matrix consists of an AHB interconnected matrix, an AHB bus and a bridged APB bus. Peripherals (RCC, GPIO and CRC) of the AHB bus are connected to the system bus via the AHB interconnected matrix. The connections between the APB and AHB buses exchange data via the AHB1APB bridge. When the APB registers are accessed by 8-bit 16-bit, the APB is automatically widened to 32-bit, and similarly, the AHB1APB bridge has an automatic widening function.

1.2 Memory introduction

The system contains a linear address space with an addressing range of 4GB, where program and data memory, peripheral registers and I/O interface registers are distributed in different address ranges within the 4GB space. Specific information on the address mapping can be found in the memory map table.

1.2.1 Memory map and register addressing

For the memory map, please refer to the memory map diagram in the corresponding chapter of each peripheral.

Table 1-1 Memory mapping

Bus	Address range	Size	Peripheral
Flash	0x0000 0000 - 0x0000 7FFF	32 KB	Main flash memory, system memory or SRAM depends on the BOOT configuration
	0x0000 8000 - 0x07FF FFFF	~127 MB	Reserved
	0x0800 0000 - 0x0800 7FFF	32 KB	Main Flash memory
	0x0801 0000 - 0x1FF0 FFFF	~383 MB	Reserved
	0x1FFE 0000 - 0x1FFE 01FF	0.5 KB	Reserved
	0x1FFE 0200 - 0x1FFE 0FFF	3 KB	Reserved
	0x1FFE 1000 - 0x1FFE 11FF	0.5 KB	Reserved
	0x1FFE 1200 - 0x1FFE 1BFF	2.5 KB	Reserved
	0x1FFE 1C00 - 0x1FFF F3FF	~256 MB	Reserved

Bus	Address range	Size	Peripheral
	0x1FFF F400 - 0x1FFF F7FF	1 KB	System memory
	0x1FFF F800 - 0x1FFF F80F	16 B	Option bytes
	0x1FFF F810 - 0x1FFF FFFF	2 KB	Reserved
SRAM	0x2000 0000 - 0x2000 07FF	2KB	SRAM
	0x2000 4000 - 0x2FFF FFFF	~255 MB	Reserved
APB1	0x4000 0000 - 0x4000 03FF	1 KB	Reserved
	0x4000 0400 - 0x4000 07FF	1 KB	TIM3
	0x4000 0800 - 0x4000 0BFF	8 KB	Reserved
	0x4000 2800 - 0x4000 2BFF	1 KB	Reserved
	0x4000 2C00 - 0x4000 2FFF	1 KB	WWDG
	0x4000 3000 - 0x4000 33FF	1 KB	IWDG
	0x4000 3400 - 0x4000 37FF	1 KB	Reserved
	0x4000 3800 - 0x4000 3BFF	1 KB	Reserved
	0x4000 4000 - 0x4000 43FF	1 KB	Reserved
	0x4000 4400 - 0x4000 47FF	1 KB	UART2
	0x4000 4800 - 0x4000 4BFF	3 KB	Reserved
	0x4000 5400 - 0x4000 57FF	1 KB	I2C1
	0x4000 5800 - 0x4000 5BFF	1 KB	Reserved
	0x4000 5C00 - 0x4000 5FFF	1 KB	Reserved
	0x4000 6000 - 0x4000 63FF	1 KB	Reserved
	0x4000 6400 - 0x4000 67FF	1 KB	Reserved
	0x4000 6800 - 0x4000 6BFF	1 KB	Reserved
	0x4000 6C00 - 0x4000 6FFF	1 KB	Reserved
	0x4000 7000 - 0x4000 73FF	1 KB	PWR
	0x4000 7400 - 0x4000 FFFF	35 KB	Reserved
	0x4001 0000 - 0x4001 03FF	1 KB	SYSCFG
	0x4001 0400 - 0x4001 07FF	1 KB	EXTI
	0x4001 0800 - 0x4001 23FF	7 KB	Reserved
	0x4001 2400 - 0x4001 27FF	1 KB	ADC1
	0x4001 2800 - 0x4001 2BFF	1 KB	Reserved
	0x4001 2C00 - 0x4001 2FFF	1 KB	TIM1
	0x4001 3000 - 0x4001 33FF	1 KB	SPI1
	0x4001 3400 - 0x4001 37FF	1 KB	DBG
	0x4001 3800 - 0x4001 3BFF	1 KB	UART1
	0x4001 3C00 - 0x4001 3FFF	1 KB	Reserved
	0x4001 4000 - 0x4001 43FF	1 KB	TIM14
	0x4001 4400 - 0x4001 47FF	1 KB	Reserved
	0x4001 4800 - 0x4001 4BFF	1 KB	Reserved
	0x4001 4C00 - 0x4001 7FFF	13 KB	Reserved
AHB	0x4002 0000 - 0x4002 03FF	1 KB	Reserved
	0x4002 0400 - 0x4002 0FFF	3 KB	Reserved
	0x4002 1000 - 0x4002 13FF	1 KB	RCC

Bus	Address range	Size	Peripheral
	0x4002 1400 - 0x4002 1FFF	3 KB	Reserved
	0x4002 2000 - 0x4002 23FF	1 KB	Flash Interface
	0x4002 2400 - 0x4002 2FFF	3 KB	Reserved
	0x4002 3000 - 0x4002 33FF	1 KB	CRC
	0x4002 3400 - 0x4002 FFFF	47 KB	Reserved
	0x4003 0000 - 0x4003 03FF	1 KB	Reserved
	0x4003 0400 - 0x47FF FFFF	~127 MB	Reserved
	0x4800 0000 - 0x4800 03FF	1 KB	GPIOA
	0x4800 0400 - 0x4800 07FF	1 KB	GPIOB
	0x4800 0800 - 0x4800 0BFF	1 KB	Reserved
	0x4800 0C00 - 0x4800 0FFF	1 KB	Reserved
	0x4800 1000 - 0x5FFF FFFF	~384 MB	Reserved

1.2.2 Embedded SRAM

It features up to 2K bytes of a built-in static SRAM. It can be accessed in byte (8 bits), half-word (16 bits) or word (32 bits). The starting address of SRAM is 0x2000 0000.

SRAM can be accessed by CPU with the fastest system clock and without any waiting.

1.2.3 FLASH memory overview

The Flash memory is composed of two storage areas:

- The main Flash memory block includes the application data and user data area.
- The information block includes the option bytes and system memory:
  - Option bytes - Containing hardware and user storage protection configuration options.
  - System memory - Containing boot loader code.

The Flash memory interface based on the AHB protocol, executes commands and data access. The prefetch buffer function of the Flash interface can speed up the code execution speed of CPU.

1.2.4 Boot configuration

Three different boot modes can be selected through BOOT0 pin level state and nBOOT1 bit, as shown in the table below:

Table 1-2 Boot mode

BOOT Mode	Selection of BOOT mode	
	nBOOT1 bit	BOOT0 pin
Main Flash memory	x	0

BOOT Mode	Selection of BOOT mode	
	nBOOT1 bit	BOOT0 pin
System memory	1	1
Built-in SRAM	0	1

The user selects the three boot modes by setting the nBOOT1 bit and BOOT0 pin level logic values. After the device is reset, the chip determines the boot mode based on the nBOOT1 bit and BOOT0 pin values. Among them, the set value of the boot mode pin needs to be maintained in standby mode because, every time when waking up from standby mode, the CPU will resample the values of nBOOT1 bit and BOOT0 to redetermine the boot mode.

After the device is reset, the CPU will begin to obtain the stack top value from the address 0x0000 0000, then obtain the base address of the boot code from address 0x0000 0004, and execute the program from the base address.

There are three main BOOT modes: main Flash memory, system memory and built-in SRAM.

BOOT from the main Flash memory (BOOT0 = 0): The starting address of the main Flash memory is 0x0800 0000. When it is chosen as the BOOT mode, it is mapped to the BOOT storage space (0x0000 0000) , but the Flash memory content can still be accessed from the starting address (0x0800 0000). When the main Flash memory is chosen as the BOOT mode, both the BOOT address and starting address can access the Flash memory.

BOOT from the system memory (nBOOT1= 1 & BOOT0 = 1): The starting address of the system memory is 0x1FFF F400. When it is chosen as the BOOT mode, it is mapped to the BOOT storage space (0x0000 0000), but the system memory content can still be accessed from the starting address (0x1FFF F400). When the system memory is chosen as the BOOT mode, both the BOOT address and starting address can access the system memory.

BOOT from the built-in SRAM (nBOOT1= 0 & BOOT0 = 1): The starting address of the built-in SRAM memory is 0x2000 0000. When it is chosen as the BOOT mode, it is mapped to the BOOT storage space (0x0000 0000), but the built-in SRAM memory content can still be accessed from the starting address (0x2000 0000). When the built-in SRAM is chosen as the BOOT mode, both the BOOT address and starting address can access the built-in SRAM.

1.2.5 Boot loader

The boot loader after leaving the factory is stored in the system memory, and ISP programming can be performed by the serial port.

2 Embedded FLASH

2.1 Overview

The embedded Flash supports the on-chip Main Flash up to 32KBytes. It also provides an option byte block and a system BOOT block (support chip Boot) as well as reserved encryption space for special application scenarios. The Flash memory controller supports read operations, page erase and whole-chip erase. These can be write into the Flash memory by 16-bit (half word) programming, with the erasing life span up to 100,000 times. When reading data, the Flash memory controller supports the data interface with the prefetch buffer to support the MCU running at a higher main frequency.

2.2 Flash composition and description

2.2.1 Flash composition

- The Flash memory is organized as 32-bit wide memory cells that can be used for storing both code and data.
- The main Flash memory block is divided into 32 pages (1Kbyte per page) or 8 write protection blocks (4Kbytes per block).
- The main Flash memory block can be erased by page (per 1Kbyte) and set with write protection by page.
- The write protection is set by taking the unit of 4 pages (4Kbytes) as one write protection block.
- The entire on-chip Flash is composed of two parts: Main storage block and information storage block.
- The main storage block is used to store user code and data, the user code can erase, program and read the main memory. Each 1Kbyte is called one page in the main storage block, which can perform the smallest unit of erasing. In addition, write protection is allocated in the unit of 1 write protection area (4Kbytes, 4 pages=1 write protection block), as shown in Table 2-1.

Table 1- 1 Flash storage space

Module	Block name	Page name	Address	Size (byte)
Main storage block	Write protection area 0	Page 0	0x0800 0000 - 0x0800 03FF	1K
		Page 1	0x0800 0400 - 0x0800 07FF	1K
		Page 2	0x0800 0800 - 0x0800 0BFF	1K
		Page 3	0x0800 0C00 - 0x0800 0FFF	1K
	…	…	…	…

Module	Block name	Page name	Address	Size (byte)
	Write protection area 7	Page 28	0x0800 7000 - 0x0800 73FF	1K
		Page 29	0x0800 7400 - 0x0800 77FF	1K
		Page 30	0x0800 7800 - 0x0800 7BFF	1K
		Page 31	0x0800 7C00 - 0x0800 7FFF	1K

In the information storage block, except for the“System Memory ISP” area that is factory locked and cannot be written by the user, other areas can be read and written by the user under certain conditions. The information memory can be divided into encryption space, system memory ISP and option byte.The encryption space is used to store datas and codes that need special protection and the system memory ISP is used to store the default ISP Bootloader. The first 12 bytes of the option byte is the write and read protection information of the main memory and the remaining bytes can be used to store user-specific data. The user can erase, program and read the information storage block through a specified procedure. As ISP is used to cure ISP upgrade code, it doesn't support the user to erase and program.

Table 1-2 Information block

Module	Name	Address	Size (byte)
Information block	Encryption space	0x1FFE 1000 - 0x1FFE 1BFF	3K
	System storage ISP	0x1FFF F400 - 0x1FFF F7FF	1K
	Option byte	0x1FFF F800 - 0x1FFF F9FF	0.5K

2.2.2 Option byte description

In the option byte page, the content mainly includes write protection enable, watchdog enable, etc. The Flash controller can disable the write function of the main memory by setting the value in the option byte to prevent illegal write and it can also enable the hardware watchdog. Relevant information is stored in the option byte. The content in the option bytes, after modification, will only be valid after reset or power-on again. It will be written in the complement mode, such as nUser, nData, etc. Each time after system reset, the option byte will reload the data of optional byte information block and make corresponding judgments and status changes. These sates are stored in the option byte registers (FLASH_OBR and FLASH_WRPR). Each selection bit in the information block has the corresponding inverse code bit, which is used to verify whether the selected bit is correct when loading the selected bit. If difference is detected during the loading process, an option byte error flag (OPTERR) will be produced. The interrupt, if turned on, will be triggered.

The organizational structure of the option byte in the option byte block is given in the table below (the value in bit 15~8 is the inverse code of option byte in bit 7~0):

Table 1-3 Organizational structure of option byte

Address	[15: 8]	[7: 0]	Default value
0x1FFF F800	nRDP	RDP	0xFFFF
0x1FFF F802	nUSER	USER	0xFFFF

Address	[15: 8]	[7: 0]	Default value
0x1FFF F804	nData0	Data0	0xFFFF
0x1FFF F806	nData1	Data1	0xFFFF
0x1FFF F808	nWRP0	WRP0	0xFFFF

Table 2-1 USER bit meaning

	Bit	Field	Type	Default	Description	FLASH_OBR
RDP	7: 0	RDP	rw	0xFF	0xA5	FLASH_OBR. Bit1
nRDP	15: 8	nRDP	rw	0xFF	0x5A	
User Byte	0	WDG_SW	rw	0x01	0: Hardware watchdog 1: Software watchdog	FLASH_OBR. Bit2
	1	nRST_STOP	rw	0x01	0: In Stop mode, the reset is produced 1: In Stop mode, the reset is not produced	FLASH_OBR. Bit3
	2	nRST_STDBY	rw	0x01	0: In Standby mode, the reset is produced 1: In Standby mode, the reset is not produced	FLASH_OBR. Bit4
	3	Reserved	rw	0x01	Reserved as 0x01	Reserved
	4	nBOOT1	rw	0x01	0: nBOOT1=0 1: nBOOT1=1	FLASH_OBR. Bit6
	5	OBR_nRST	rw	0x01	NRST MUX PA10 1: NRST function 0: GPIO function	FLASH_OBR. Bit7
	6	Reserved	rw	0x01	Reserved as 0x01	Reserved
	7	Reserved	rw	0x01	Reserved as 0x01	Reserved
DATA0 Byte	0	DATA0.Bit0	rw	0x01	User defined	FLASH_OBR. Bit10
	1	DATA0.Bit1	rw	0x01	User defined	FLASH_OBR. Bit11
	2	DATA0.Bit2	rw	0x01	User defined	FLASH_OBR. Bit12
	3	DATA0.Bit3	rw	0x01	User defined	FLASH_OBR. Bit13
	4	DATA0.Bit4	rw	0x01	User defined	FLASH_OBR. Bit14
	5	DATA0.Bit5	rw	0x01	User defined	FLASH_OBR. Bit15
	6	DATA0.Bit6	rw	0x01	User defined	FLASH_OBR. Bit16
	7	DATA0.Bit7	rw	0x01	User defined	FLASH_OBR. Bit17

	Bit	Field	Type	Default	Description	FLASH_OBR
DATA1 Byte	0	DATA1.Bit0	rw	0x01	User defined	FLASH_OBR. Bit18
	1	DATA1.Bit1	rw	0x01	User defined	FLASH_OBR. Bit19
	2	DATA1.Bit2	rw	0x01	User defined	FLASH_OBR. Bit20
	3	DATA1.Bit3	rw	0x01	User defined	FLASH_OBR. Bit21
	4	DATA1.Bit4	rw	0x01	User defined	FLASH_OBR. Bit22
	5	DATA1.Bit5	rw	0x01	User defined	FLASH_OBR. Bit23
	6	DATA1.Bit6	rw	0x01	User defined	FLASH_OBR. Bit24
	7	DATA1.Bit7	rw	0x01	User defined	FLASH_OBR. Bit25

Note: In the write protection values, one bit corresponds to four pages, namely 4096 Bytes.

2.2.3 Encryption space description

In the encryption space, the Flash controller can, via the specified key pairing, achieve the read and write-protect of the content to prevent illegal access.

2.3 Flash option and procedure

2.3.1 Flash read operation

User code and data are stored in the main memory block and the Flash memory controller can read data or instructions in the 8bit/16bit/32bit. The main Flash module is uniformly addressed like ordinary peripherals. Based on the read and write protection requirements, any read or write operations on the content of the main memory block shall undergo a specific judgment process to prevent illegal read and write.

The Flash memory executes instruction fetching and date fetching via the AHB bus according to the method set in the Flash access control register (FLASH_ACR). Based on AHB clock, set the corresponding access latency and enable the prefetch value buffer. By doing so, the instruction fetch speed of CPU can be increased, thereby raising the running speed of CPU. The access latency if less than or equal to 24MHz in SYSCLK, can be set as 0. For an increment of 24MHz, an additional latency is required.

After power-on reset, the Flash memory controller sets the the prefetch buffer to be open by default. If the prefetch instruction buffer function needs to be closed or reopened, SYSCLK must be set below 24MHz, and AHB clock can close or reopen the prefetch instruction buffer function provided that no frequency division has been exerted (SYSCLK must be equal to HCLK).

In order to protect the correct reading of Flash, the speed ratio of the prefetch instruction controller must be specified in the LATENCY[2:0] in the Flash access control register. The value is equal to the number of waiting cycles inserted from each Flash access to the next access. After reset, this value is zero by default, that is, no waiting cycle is inserted and the corresponding system clock is also reset as the built-in clock HSI=8MHz. If the system clock is to be modified after reset, the safe value of LATENCY[2:0] shall be configured first. When the AHB clock prescaler is greater than 1, the corresponding access latency of the prefetch instruction buffer should also be set.

Table 2-2 Latency setting relation

SYSCLK	AHB DIV	Latency
0MHz < SYSCLK <=24MHz	1	0
24MHz < SYSCLK <= 48MHz	1	1
48MHz < SYSCLK <= 72MHz	1	2
72MHz < SYSCLK <= 96MHz	1	3
96MHz < SYSCLK <= 120MHz	1	4

2.3.2 Flash programming mode and operation procedure

The embedded Flash supports three programming modes below.

Table 1-6 Programming Mode

Programming mode	Programming description
In circuit programming (ICP)	ICP means that the user code is burned into MCU by a specific programmer using SWD interface and changing Flash content.
In system programming (ISP)	ISP means that the user code is burned into MCU by ISP Firmware based on a specific UART interface and changing Flash content.
In application programming (IAP)	Different from ICP and ISP mode, IAP (in application programming) can use any MCU supported communication interface (UART, I2C, SPI, CAN, USB, etc.) to download program or data. IAP allows the user to rewrite the application during program operation, under the precondition that some programs must be preburned by ICP or ISP mode.

The burning and erasing can be completed within the working voltage range of the whole product. When writing or erasing the Flash space, the internal oscillator (HSI) must be turned on and the AHB clock must be greater than or equal to 8MHz.

As long as CPU does not access Flash space, the ongoing Flash write operation will not hamper the running of CPU (running from RAM or ISP). When writing or erasing the Flash, the read access to the Flash will always encounter a bus pause, and it will not continue until the writing or erasing operation is completed. Therefore, it is not allowed to fetch instruction and access data when the Flash is written or

erased.

The programming of Flash is composed of a series of actions, mainly including:

- Unlock and protect Flash operation
- Erase Flash (page erasing and whole-chip erase)
- Programme Flash (half-word programming)
- Unlock and protect the operations of each sector in the information block (such as option byte)
- Erase each sector in the information block (such as option byte)
- Programme (half-word programming) each sector in the information block (such as option byte)

ISP, IAP programming procedure

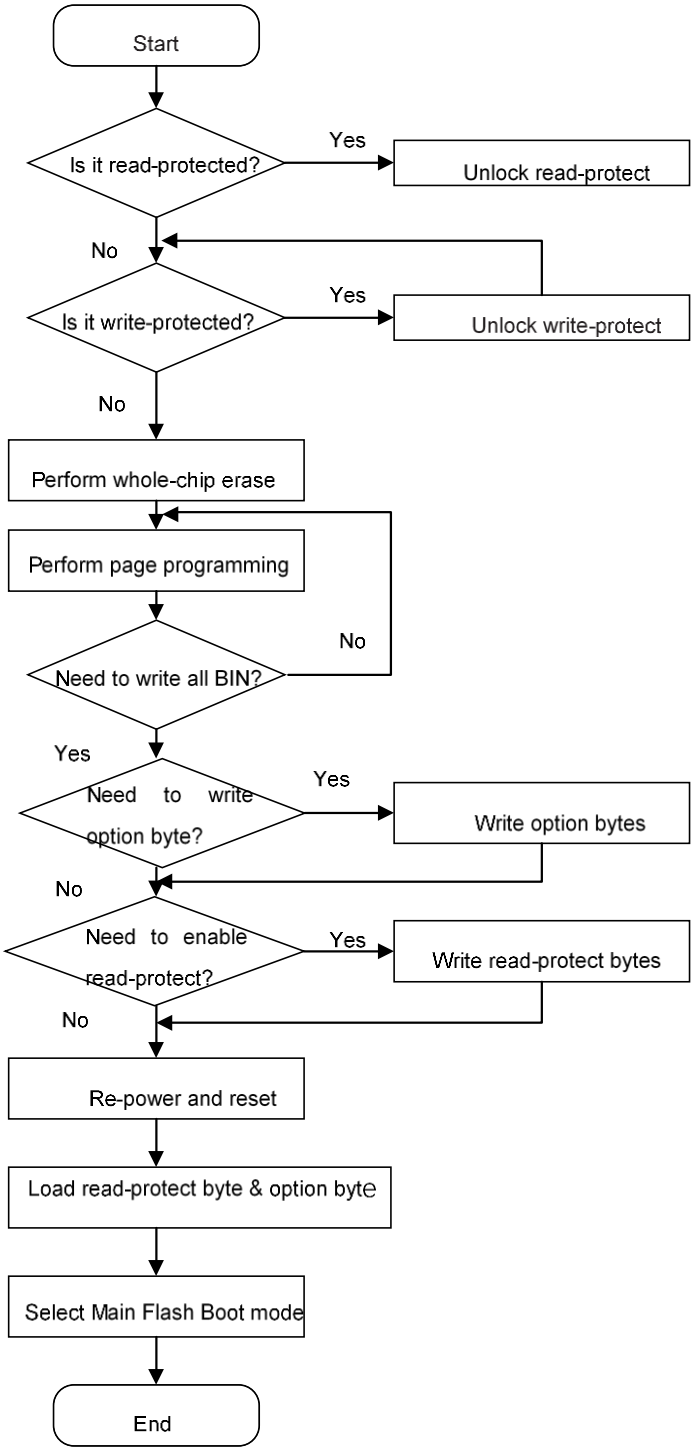


Figure 2-1 ISP programming procedure

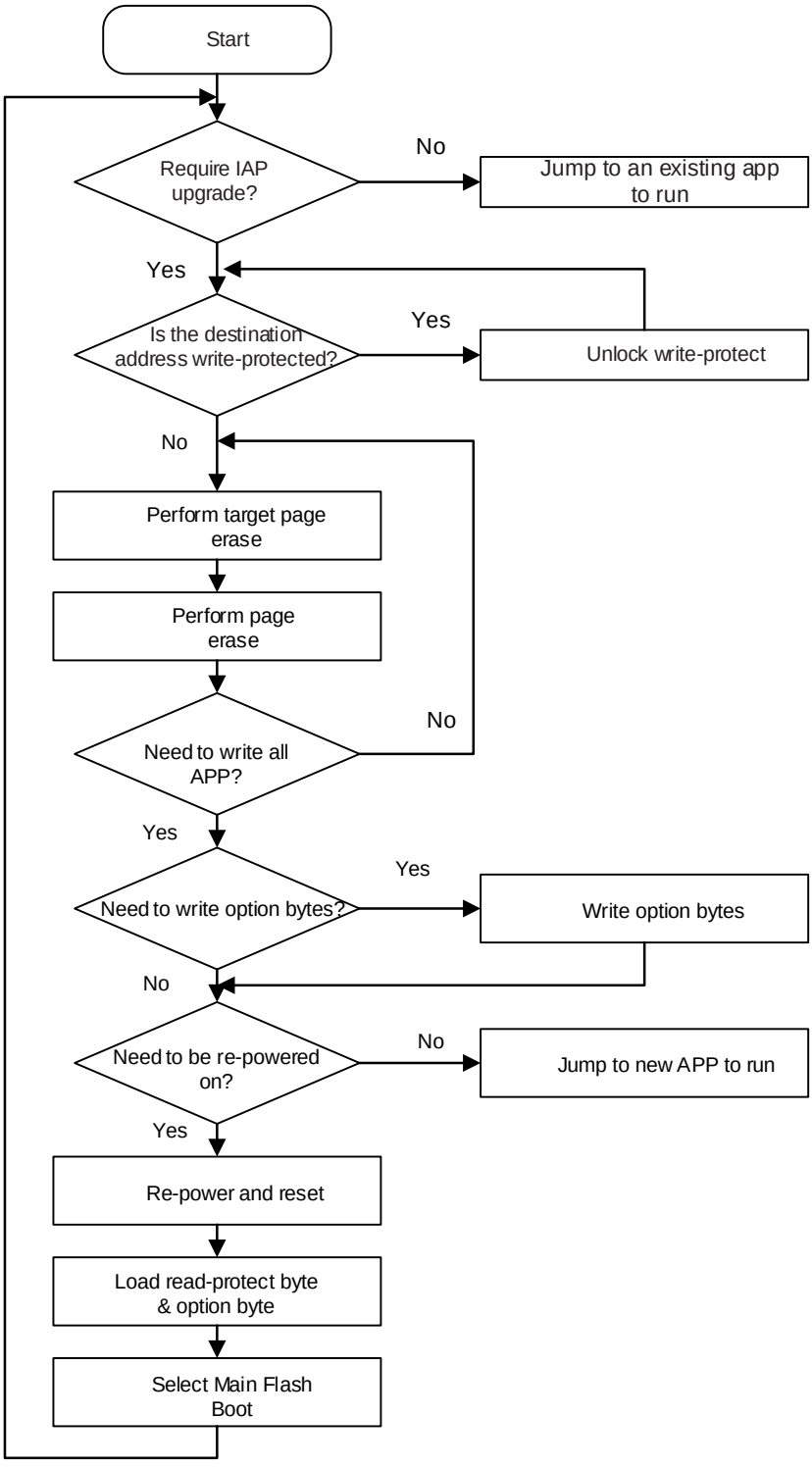


Figure 2-2 IAP programming procedure

2.3.3 Disable and enable the operation restriction on Flash block

The embedded Flash memory is in a protected status after reset, which can avoid accidental operations that damage the Flash storage space, such as page erase, whole chip erase and value writing. FLASH_CR register is locked after reset, and the LCOK bit of FLASH_CR is set as 1 by the controller

module. Only by writing 0x45670123 and 0xCDEF89AB respectively to the FLASH_KEYR register for unlocking and setting the LCOK bit of FLASH_CR as 0, can FLASH_CR be accessed. Otherwise, the FLASH_CR register is not allowed to be rewritten.

Set the LOCK bit of FLASH_CR LOCK as 1 by software to lock again, making the Flash memory in the protected status.

Disable protection operation code:

```
#define FLASH_KEY1              ((unsigned int)0x45670123)
#define FLASH_KEY2       ((unsigned int)0xCDEF89AB)
void FLASH_Unlock()
{
    FLASH->KEYR = ((unsigned int)0x45670123);
    FLASH->KEYR = ((unsigned int)0xCDEF89AB);
}
```

Operations that do not conform to the above sequence and writing the wrong value will lock FLASH_CR and trigger a bus error till the next reset.

Enable protection operation code:

```
#define FLASH_CR_LOCK_Pos           (7)
#define FLASH_CR_LOCK               (0x01U << FLASH_CR_LOCK_Pos)
void FLASH_Lock(void)
{
    FLASH->CR |= FLASH_CR_LOCK;
}
```

2.3.4 Disable and enable the operation restriction on option byte block

After the Flash memory controller is reset, its option byte block is write-protected by default, and readable at any time. In order to avoid destructive operations such as block erasing and writing value on the option byte area and read protection setting area, after reset, FLASH_CR register enters the locked status and the LOCK bit of FLASH_CR is set as 1 by the controller module, while OPTWRE bit is cleared as 0 by the controller module. Therefore, it is necessary to write 0x45670123 and 0xCDEF89AB to the FLASH_KEYR register to unlock the FLASH. After the LOCK bit of FLASH_CR is 0, the option byte block are unlocked. Writing 0x45670123 and 0xCDEF89AB to FLASH_OPT_KEYR register, the hardware will set the OPTWRE bit of FLASH_CR register as 1. By doing so, block erasing and and half-word programming can be performed on the option byte area. The OPTWRE bit of the FLASH_CR register can be set as 0, thereby disabling block erasing and half-word programming on the option byte area.

Table 1-7 Changes of status in the protection setting

Set and status	Main Flash block	Information block	Description
Power on and reset The status of Flash controller is FLASH_CR.LOCK=1 FLASH_CR.OPTWRE=0	Protection	Protection	Enable operation protection of main Flash block Enable operation protection of option byte block
Set FLASH_KEYR=0x45670123 FLASH_KEYR=0xCDEF89AB The status of Flash controller is FLASH_CR.LOCK=0 FLASH_CR.OPTWRE=0	Disable protection	Protection	Disable operation protection of main Flash block, execute whole chip erasing, page erasing, half-word programming of main Flash block. Keep enabling operation protection of option byte block. It disables block erasing and half-word programming of option byte area.
FLASH_KEYR=0x45670123 FLASH_KEYR=0xCDEF89AB FLASH_OTPKEYR=0x45670123 FLASH_OTPKEYR=0xCDEF89AB The status of Flash controller turns FLASH_CR.LOCK=0 FLASH_CR.OPTWRE=1	Disable protection	Disable protection	Disable operation protection of main Flash block, execute whole chip erasing, page erasing, half-word programming of main Flash block. Disable operation protection of option byte block. It enables block erasing and half-word programming of option byte area.
Set FLASH_CR.OPTWRE=0 Keep FLASH_CR.LOCK=0	Disable protection	Enable protection	It disables operation protection of main Flash block, and enables operation protection on option byte block
Set FLASH_CR.OPTWRE=0 Set FLASH_CR.LOCK=1	Enable protection	Enable protection	Enable operation protection of main Flash block as well as option byte block

Disable protection operation code:

```
#define FLASH_KEY1              ((unsigned int)0x45670123)
#define FLASH_KEY2       ((unsigned int)0xCDEF89AB)
void FLASH_Unlock(void)
{
    FLASH->KEYR = ((unsigned int)0x45670123);
    FLASH->KEYR = ((unsigned int)0xCDEF89AB);
}
```

Enable protection operation code:

```
#define FLASH_CR_LOCK_Pos           (7)
#define FLASH_CR_LOCK               (0x01U << FLASH_CR_LOCK_Pos)
void FLASH_Lock(void)
```

```
{
    FLASH->CR |= FLASH_CR_LOCK;
}
```

Disable operation code of option byte as well as read protection setting area:

```
#define FLASH_KEY1              ((unsigned int)0x45670123)
#define FLASH_KEY2       ((unsigned int)0xCDEF89AB)
void FLASH_OPT_Unlock (void)
{
    FLASH->OPTKEYR = FLASH_KEY1;
    FLASH->OPTKEYR = FLASH_KEY2;
}
```

Enable operation code of option byte as well as read protection setting area:

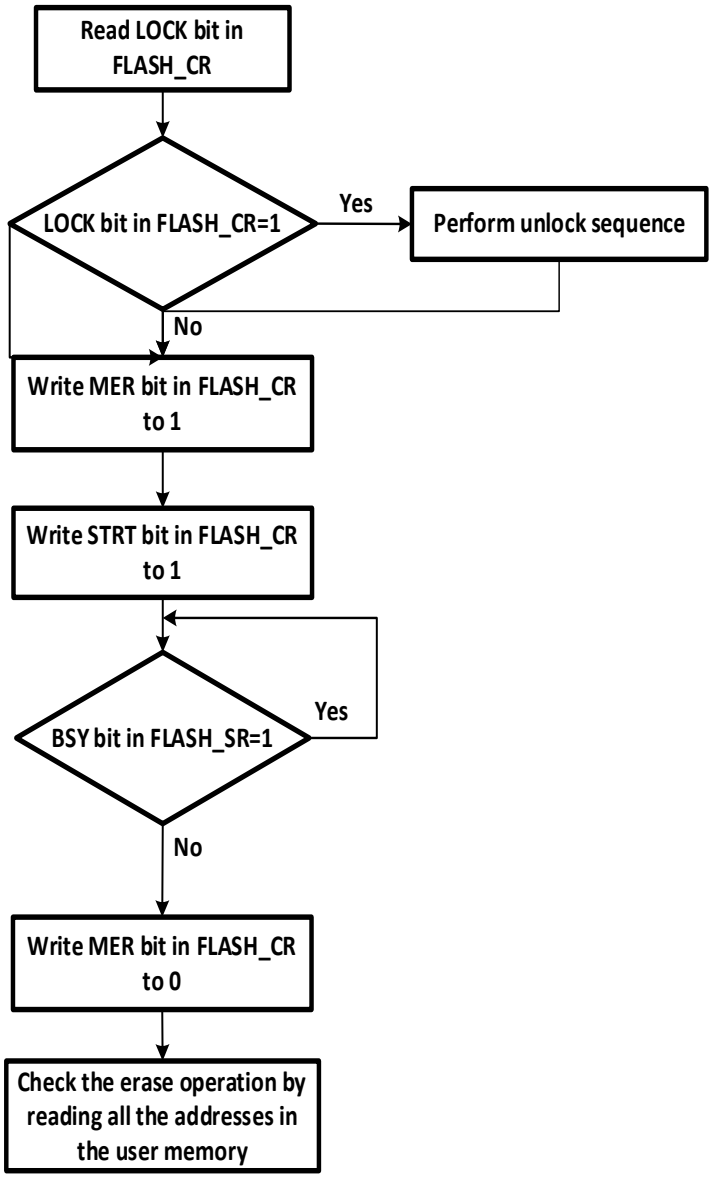
```
#define FLASH_CR_OPTWRE_Pos        (9)
#define FLASH_CR_OPTWRE              (0x01U << FLASH_CR_OPTWRE_Pos)
void FLASH_OPT_Lock(void)
{
    FLASH->CR &= ~FLASH_CR_OPTWRE;
}
```

2.3.5 Main Flash block erasing

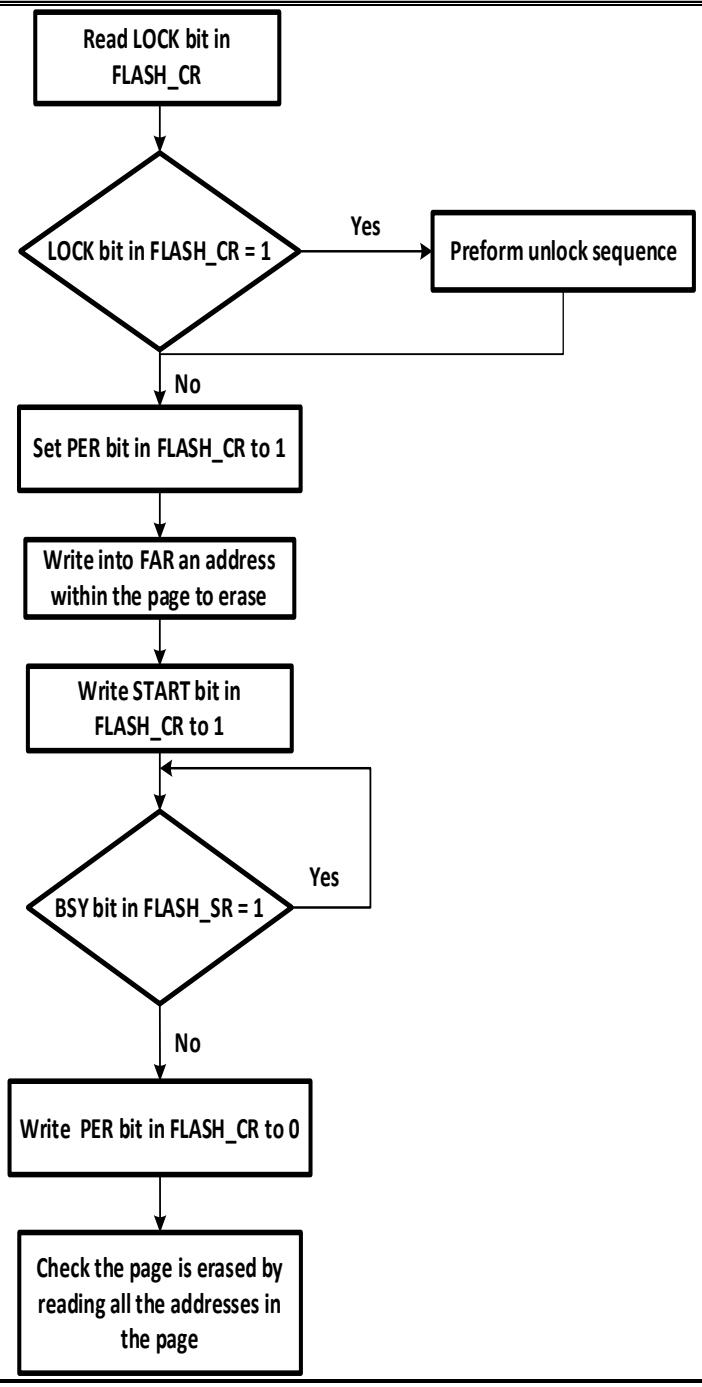
The Flash memory controller supports the whole chip erasing of main Flash block and the erasing of pages inside the main Flash memory in the unit of page.

The whole chip erasing function will initialize all the contents of the main Flash memory block and set all values as 0xFFFF. However, the information block will not be affected by this instruction.

Main Flash memory Erase procedure:

Steps	Simple flowchart
(1)Read the LOCK bit of FLASH_CR to ensure that Flash protection is disabled, (2) Otherwise execute Flash Unlock (3) Set MER bit in the FLASH_CR register as 1 and enable the whole chip erasing of main Flash (4) Set STRT bit in the FLASH_CR register as 1 (5) Start the whole chip erase of the main Flash (6) Check BUSY bit in FLASH_SR (7) Wait for completion of whole erase of main Flash (8) Clear the MER bit of FLASH_CR register, and make this bit as 0 (9) Optionally conduct Blank check to ensure erasing success	

Flash memory Page Erase procedure:

Steps	Simple flowchart
(1) Read the LOCK bit of FLASH_CR to ensure that Flash protection is disabled, otherwise, execute Flash Unlock (2) Set PER bit in FLASH_CR register as 1 and enable page erasing (3) Write the base address of the to-be-erased page to FLASH_AR register (4) Set STRT bit in FLASH_CR register as 1 and start page erasing (5) Check BUSY bit in FLASH_SR, (6) Wait for page erasing to finish (7) Clear the PER bit of FLASH_CR register, and make this bit as 0 (8) Select content of this page to check and ensure the success of page erasing (9) For continuous erasing of multiple pages, repeat 2~7 operations till the completion of erasing all pages	

2.3.6 Main Flash block programming

The main Flash memory only supports 16-bit half-word programming to change the content of main storage Flash memory block. If it is programmed with a length of 32-bit whole word or 8-bit byte, a hardware error interrupt is caused. When the PG bit of FLASH_CR is 1, a half-word (16 bits) can be directly written to the corresponding address, namely one programming operation.

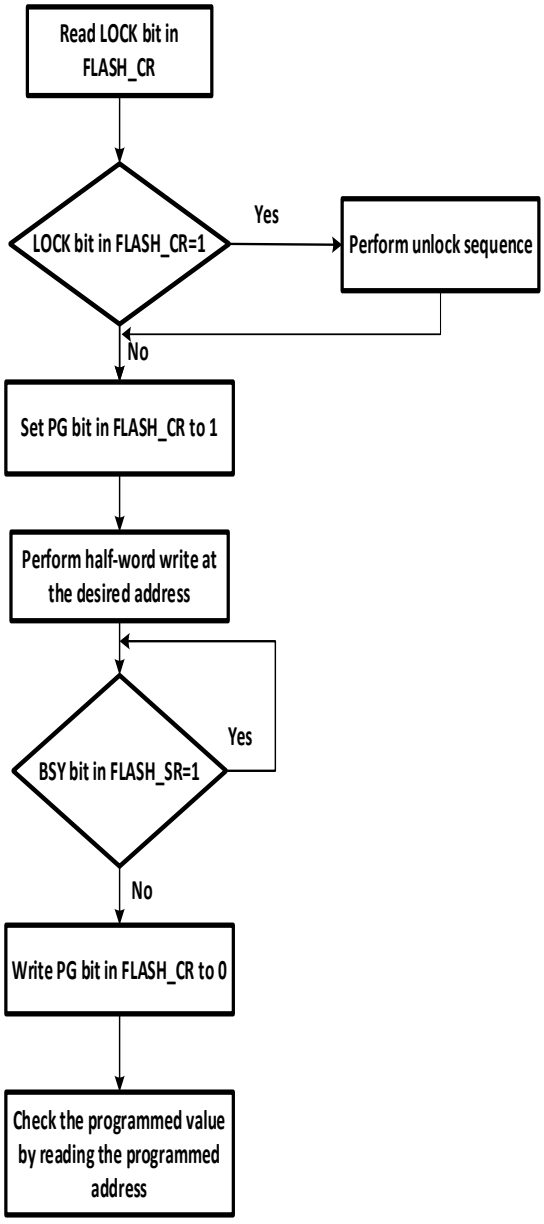
The main Flash memory controller will pre-read whether the half word to be programmed is whole 1

(whether it is 0xFFFF); if not, the programming will be automatically canceled and a programming error warning will be displayed on the PGERR bit of FLASH_SR register.

If the write protection bit in FLASH_WRRP of the corresponding write protection block of the address to be programmed is valid, there will be no programming action; but a programming error warning is generated. It will also produce programming error alarm. After programming, EOP bit in FLASH_SR register will produce the prompt.

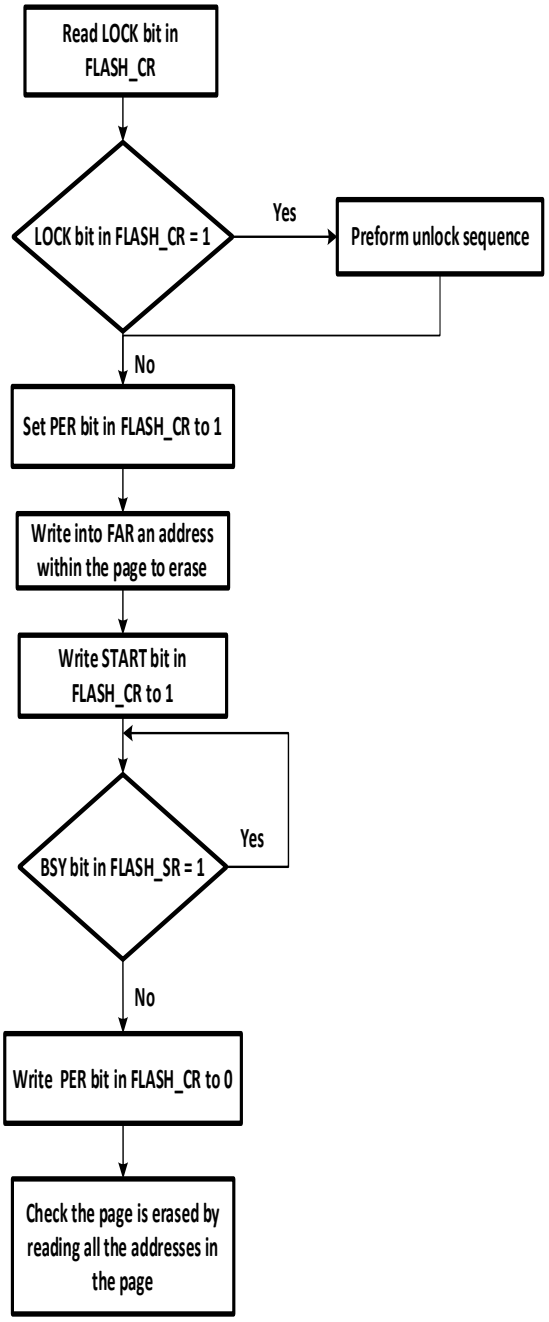
Note: When the CPU enters the power saving mode, an error will be produced to the Flash memory operation via SWD interface. It is necessary to avoid running interrupt programs while perform erasing or programming operations.

Specific steps for the register setting of main Flash block programming:

Steps	Simple flowchart
(1)Read the LOCK bit of FLASH_CR to ensure that Flash protection is disabled, otherwise, execute Flash Unlock (2) Set PG bit in FLASH_CR register as 1 and enable half-word programming (3) Write data to the target address in a unit of half word, and align the target address with half word (4) Check BUSY bit in FLASH_SR (5) Wait for half-word programming to finish (6) Clear the PG bit of FLASH_CR register, and make this bit as 0 (7) Select to read target address data to ensure the success of half-word programming (8) For more continuous half-word programming, repeat 2~6 operations, till the completion of half-word programming of all target addresses	

2.3.7 Option byte block erasing

Specific steps for the register setting of option byte block erasing:

Steps	Simple flowchart
(1) Read the LOCK bit of FLASH_CR to ensure that Flash protection is disabled, otherwise, execute Flash Unlock (2) Read the OPTWRE bit of FLASH_CR to ensure that the option block protection is disabled. Otherwise, execute option block Unlock (3) Write block base address to be erased to FLASH_AR register (4) Set OPTER bit in FLASH_CR register as 1 and enable option byte block erasing (5) Set STRT bit in FLASH_CR register as 1 (6) Check BUSY bit in FLASH_SR (7) Wait for block erasing to finish (8) Clear the OPTER bit of FLASH_CR register, and set the value as 0 (9) Select to check the block content to ensure the success of the block erasing (10) For continuous erasing of multiple blocks, repeat 3~8 operations, till the completion of erasing all option blocks	

2.3.8 Option byte block programming

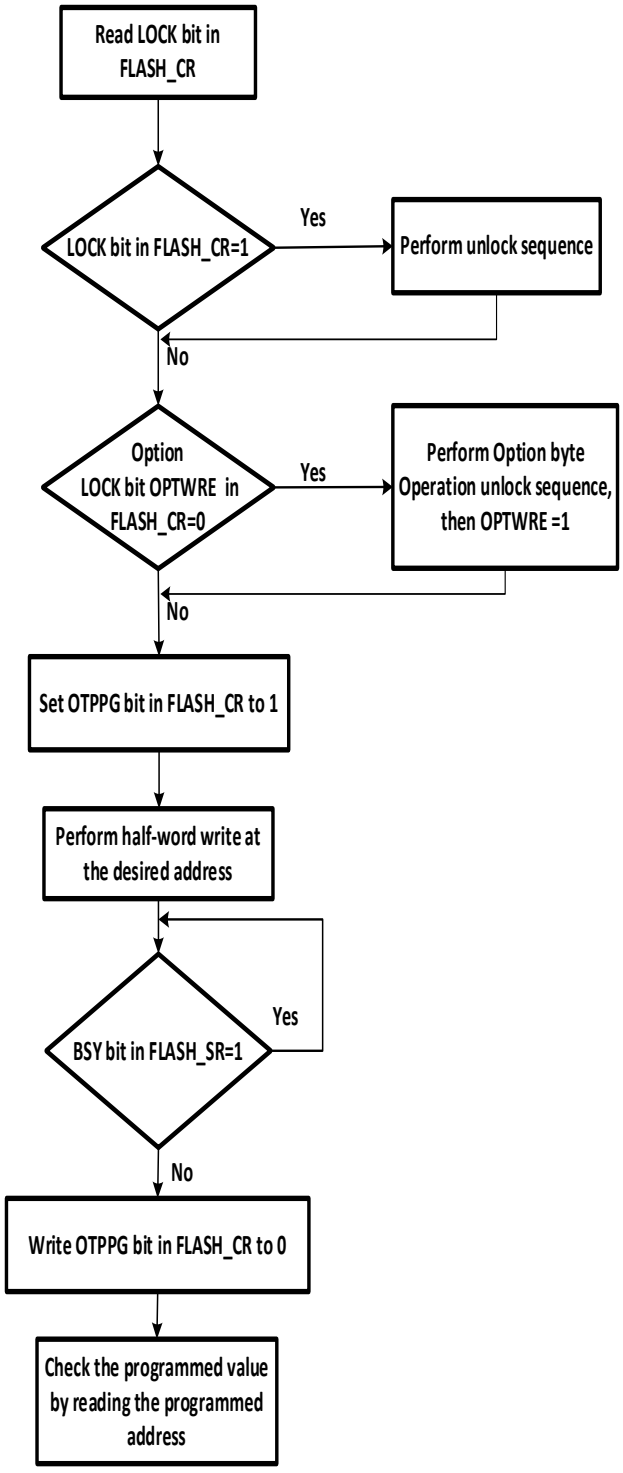
The programming of the option byte block is different from that of the main Flash memory block address, because the written value after reset is loaded to the configuration option, which requires more stringent protection. After the access restriction on the Flash memory controller is removed, the access

restriction on the option byte block and read protection area should also be disabled. After this operation is completed, the OPTWRE bit in the FLASH_CR register will be set as 1 before the follow-up programming is allowed.

The valid data of option byte is the low 8 bits, and the high 8 bits are the inverse code of low 8 bits, thus forming the 16-bit data. During programming, the software sets the high 8 bits as the inverse code of low 8 bits to ensure that the written value of the option byte is always correct, and then write 16-bit data in sequence.

The change of optional byte is only valid after the system is powered-on and reset.

Specific steps for the register setting of option byte block half-word programming:

Steps	Simple flowchart
(1) Read the LOCK bit of FLASH_CR to ensure that the Flash protection is disabled. Otherwise, execute Flash Unlock (2) Unlock Read the OPTWRE bit of FLASH_CR to ensure that the option block protection is disabled. Otherwise, execute option block Unlock (3) Set OPTPG bit of FLASH_CR register as 1 and enable option byte block half-word programming (4) Write data (half word) to target address and align target address by half word (5) Check BUSY bit of FLASH_SR (6) Wait for the completion of half word programming (7) Clear the OPTPG bit of FLASH_CR register, and set the value as 0 (8) Select to read target address data and ensure the success of half-word programming (10) For continuous programming of multiple half words, repeat 3~7 operations, till the completion of half word programming of all target addresses	

2.3.9 Flash memory protection

The main Flash block enable read protection can prevent the codes of the main Flash block from being read by unreliable codes and the accidental erasing and programming of main Flash memory block during program runaway. The read protection enable operation works for the whole main memory block

while the minimum unit of enabling write protection is a write protection block (namely 4 pages).

2.3.9.1 Main Flash block read protection

The main Flash block enables or disables read protection is to set RDP half word by built-in SRAM or ICP, ISP, and works when loading new RDPs after the system is re-powered on and reset. It works by executing one power-on and reset rather than system reset, if read protection is set.

Enable read protection

According to the operation mode of half word programming in the option byte area, write in sequence the half word of RDP to the corresponding address

- Set the address value of FLASHAR as 0x1FFFF800, and execute erasing of the option block.
- According to the operation mode of half word programming in the option byte area, write in sequence half words of 0x807F to the corresponding address.
- Conduct a power-on reset to reload the option byte and the read protection is then disabled.

When RDP word includes the following values, the main Flash block is placed in the protected status after being re-powered on and reset.

Table 1-8 Flash read protection status

Enable read protection	Read protection status
Erase 0x1FFFF800 option block Write 0x807F half word to the corresponding address 0x1FFFF800 Disable read protection after being re-powered on and reset	Protection

When read protection half word is written within the corresponding value:

1. Only read operations to the main Flash memory from user code (start from main Flash memory in non-debug mode) are allowed.

2. After read protection, operations to the Flash are disabled in the debug mode (SRAM boot and debug mode).

3. MCU can be programmed by the code executed in the main Flash memory (enable functions such as IAP or data storage), but write or erasing operations in debug mode or booting from the internal SRAM (except for whole chip erasing).

4. All functions of loading and executing code to the built-in SRAM through SWD remain valid or it can also boot from the built-in SRAM. This function can be used to disable read protection.

5. Access to the main Flash memory by executing code from the built-in SRAM or access to the Flash via DMA, SWV (Serial Wire Viewer) and SWD (Serial Wire Debug) will all be disabled.

Disable read protection

Disable read protection from built-in SRAM or ICP mode:

1. Set FLASH_AR address value as 0x1FFFF800 and execute the operation block erase.

2. According to the operation mode of half word programming in the option byte area, write in sequence half words of 0x5AA5 to the corresponding address.

3. Set FLASHAR address value as 0x08000000 and execute main FLASH whole chip erasing.

4. Conduct a power-on reset to reload the option byte, and disable read protection.

Table 1- 9 Flash disable read protection status

Disable read protection status	Read protection status
Erase 0x1FFFF800 option block Write 0x5AA5 half word to the corresponding address 0x1FFFF800 Erase 0x08000000 main Flash whole chip Power on and reset, disable read protection	Disable read protection

Note: 1. If the corresponding address value of the option byte block is not 0xFFFF, execute the erasing of option byte block first, which will not lead to an automatic erasing of the whole chip or change the read protection status. 2.Whole chip erasing must be performed on the main Flash of 0x08000000.

2.3.9.2 Main Flash block write protection

Enable write protection

The write protection is enabled by setting WRP bit of WRP0~WRP15 in the option byte block as 0. After system reset, load the new option byte to enable write protection. An attempt to write or erase a write protection area page will assert WRPRTERR flag bit in the FLASH_SR.

Table 2-3 Write protection area

Address	[15: 8]	[7: 0]	Default	Note
0x1FFF F808	nWRP0	WRP0	0xFFFF	

Disable write protection

There are two scenarios for disabling the write protection:

1. Scenario 1: Disable write and read protection:

a. Use the OPTER bit in the Flash control register (FLASH_CR) to erase the whole option byte block; write a half word of 0x5AA5 to the corresponding address 0x1FFFF800;

b. Perform a whole-chip erasing on the main Flash block of 0x08000000;

c. Reset the system,and reload the option byte (including the new WRP byte); write protection is disabled. Use of this method will disable the write protection of the whole chip main Flash module and erase the whole chip of main Flash block.

2. Scenario 2: Disable write protection and keep the read protection valid. This scenario is

commonly seen in the user's startup program:

a. Use OPTER bit of Flash control register (FLASH_CR) to only erase the whole option byte block;

b. Reset the system and reload option byte (including the new WRP byte); write protection is disabled.

Use of this method will disable the write protection of the whole main Flash module and keep the read protection valid.

2.4 Register

2.4.1 Register overview

Table 111 FLASH register overview

Offset	Acronym	Register Name	Reset
0x00	FLASH_ACR	Flash access control register	0x00000038
0x04	FLASH_KEYR	FPEC key register	0x00000000
0x08	FLASH_OPTKEYR	Flash OPTKEY register	0x00000000
0x0C	FLASH_SR	Flash status register	0x00000000
0x10	FLASH_CR	Flash control register	0x00000080
0x14	FLASH_AR	Flash address register	0x00000000
0x1C	FLASH_OBR	Option byte register	0x03FFFC1C
0x20	FLASH_WRPR	Write protection register	0xFFFFFFFF

Note: Flash register only supports 32-bit access

2.4.2 FLASH_ACR Flash Access Control Register

Offset address:0x00

Reset value:0x0000 0038

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.										PRFTBS	PRFTBE	Res.	LATENCY		
										r	rw		rw		

Bit	Field	Description
31: 6	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
5	PRFTBS	Prefetch buffer status 0: Prefetch buffer close 1: Prefetch buffer open
4	PRFTBE	Prefetch buffer enable 0: Close prefetch buffer. 1: Open prefetch buffer. Note 1: Only when LATENCY is 0, the prefetch buffer switch can be controlled by this bit; Note 2: When the HPRE[3] bit in RCC_CFGR is set to a value (not 0), the hardware of the prefetch buffer is automatically enabled, and the PRFTBS bit is set to 1.
3	Reserved	Reserved, must be kept at reset value.
2: 0	LATENCY	Latency These bits indicate the ratio of SYSCLK (system clock) period to Flash memory access time. 000: zero wait state, 0 < SYSCLK ≤ 24MHz 001: one wait state, 24MHz < SYSCLK ≤ 48MHz 010: two wait states, 48MHz < SYSCLK ≤ 72MHz 011: three wait states,72MHz < SYSCLK ≤ 96MHz 100: four wait states,96MHz < SYSCLK ≤ 120MHz

2.4.3 FLASH_KEYR FPEC Key Register

Offset address:0x04

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	19	18	17	16	15
FKEYR															
w															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FKEYR															
w															

Bit	Field	Description
31: 0	FKEYR	FPEC Flash key These bits are used to input FPEC unlock key.

Note: All these bits are write-only and return to 0 when read.

2.4.4 FLASH_OPTKEYR Flash OPTKEY Register

Offset address:0x08

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OPTKEYR															

w															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OPTKEYR															
w															

Bit	Field	Description
31: 0	OPTKEYR	Option byte key These bits are used as the input option byte key to disable OPTWRE.

Note: All these bits are write-only and return to 0 when read.

2.4.5 FLASH_SR Flash Status Register

Offset address:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.										EOP	WRPRT ERR	Res.	PGERR	Res.	BSY
										rw	rw		rw		r

Bit	Field	Description
31:6	Reserved	Reserved, must be kept at reset value.
5	EOP	End of operation When the Flash memory operation (programming I erasing) is completed, the hardware sets this bit as “1” and write “1” to clear the bit status.
4	WRPRTERR	Write protection error When an attempt is made to programme the write-protected Flash memory address, the hardware sets this bit as “1” and write “1” to clear the bit status.
3	Reserved	Reserved, must be kept at reset value.
2	PGERR	Programming error When an attempt is made to programme an address whose content is not “0x FFFF”, the hardware sets this bit as as “1” and write “1” to clear the bit status Note: Prior to programming, clear the STRT bit of FLASH_CR register.
1	Reserved	Reserved, must be kept at reset value.
0	BSY	Busy This bit indicates that the operation of Flash memory is in progress. Before Flash operation starts, this bit is set as “1”. After the operation or in case of an error, this bit is cleared as “0”.

2.4.6 FLASH_CR Flash Control Register

Offset address:0x10

Reset value:0x0000 0080

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						OPTWRE	Res.	LOCK	STRT	OPTER	OPTPG	Res.	MER	PER	PG
						rw		rw	rw	rw	rw		rw	rw	rw

Bit	Field	Description
31:10	Reserved	Reserved, must be kept at reset value.
9	OPTWRE	Option byte write enable When this bit is “1”, the option byte is allowed to be programmed. When the correct key sequence is written in the FLASH_OPTKEYR register, the bit is set as “1”. This bit can be cleared by writing 0.
8	Reserved	Reserved, must be kept at reset value.
7	LOCK	Lock Only write “1”. When the bit is “1”, it indicates that FPEC and FLASH_CR are locked. When the correct unlocking sequence is detected, the hardware automatically clears this bit as “0”. After an unsuccessful unlocking operation, this bit cannot be changed before the next system reset.
6	STRT	Start An erasing is triggered when this bit is “1”. The bit can only be set to “1” by software and automatically be cleared to “0” when BSY turns into “1”.
5	OPTER	Option byte erase Erase option byte.
4	OPTPG	Option byte programming Programme option byte.
3	Reserved	Reserved, must be kept at reset value.
2	MER	Mass erase Choose to erase all user pages.
1	PER	Page erase Choose to erase page.
0	PG	Programming Choose to conduct programming.

2.4.7 FLASH_AR Flash Address Register

Offset address:0x14

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FAR															
w															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FAR															
w															

Bit	Field	Description
31: 0	FAR	Flash Address Select the page to be erased when performing page erasing. Note: When the BSY bit in FLASH_SR is “1”, the register cannot be written.

The address is modified by the hardware to the currently used one. During the operation of page erasing, the register is modified to specify the page to be erased.

2.4.8 FLASH_OBR Option Byte Register

Offset address:0x1C

Reset value:0x03FF FC1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.						Data1								Data0	
						r								r	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Data0						Res.			nBOOT1	Res.	nRST_STDBY	nRST_STOP	WDG_SW	Res.	OPTERR
r									r		r	r	r		r

Bit	Field	Description
31: 26	Reserved	Reserved, must be kept at reset value.
25: 18	Data1	Data1
17: 10	Data0	Data0
9: 7	Reserved	Reserved, must be kept at reset value.
6	nBOOT1	nBOOT1
5	Reserved	Reserved, must be kept at reset value.
4	nRST_STDBY	Reset event in standby mode 0: Reset in standby mode 1: No reset in standby mode

Bit	Field	Description
3	nRST_STOP	Reset event in stop mode 0: Reset in stop mode 1: No reset in stop mode
2	WDG_SW	Select watchdog event 0: Hardware watchdog 1: Software watchdog
1	RDPRT	Read protection level status When setting to “1” , it indicates that the Flash memory is read protected. Note: The bit is read-only.
0	OPTERR	Option byte error When this bit is “1”, it indicates that the option byte does not match its inverse code. Note: The bit is read-only.

2.4.9 FLASH_WRPR Write Protection Register

Offset address:0x20

Reset value:0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								WRP							
								r							

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value.
7:0	WRP	Write protection The register includes the write protection option byte loaded by OBL. 0:Write protection enabled 1:Write protection disabled

3 PWR Power Controller

Power controller (PWR) mainly involves features such as chip's power supply, power manager and low-power mode.

3.1 Power supply system

The chip is powered by four types of power supply below:

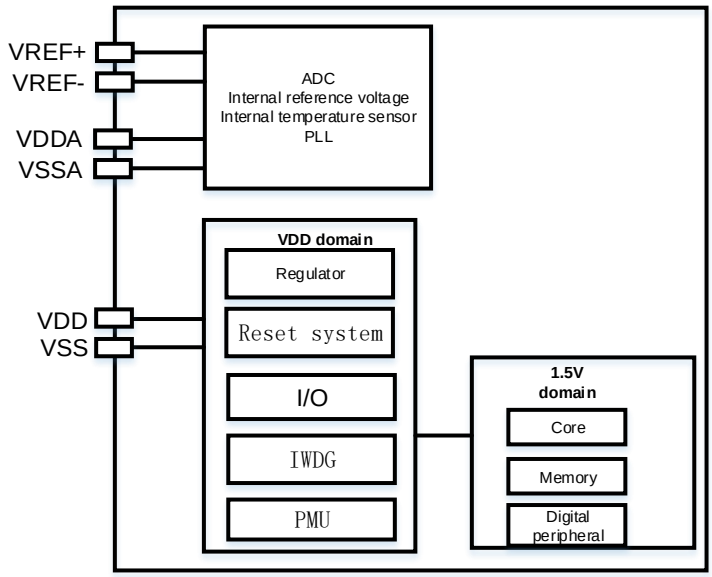


Figure 3-1 Power control block diagram

- The analog power supply provided by VDDA and VSSA provides voltage for the analog module of the chip and is used for ADC module, internal reference voltage sensor and PLL.
- The digital power is supplied by VDD and VSS provides power for digital part and I/O pins.

Note:

- In the power supply system, it's required to connect the 10uF and 100nF capacitor in the corresponding power pin, and try to get close to the chip power pin.
- Among packages with VDDA and VSSA, VDDA and VSSA cannot be pending, and the voltage difference of VDD and VDDA needs to be lower than 50mV.
- Among packages without VDDA and VSSA,VDDA and VSSA must be connected to VDD and VSS within the package.

3.1.1 Analog module power supply

The VDDA mainly supplies power for the internal analog circuit of the chip. It is mainly composed of

ADC module, internal reference voltage sensor, PLL, etc. Therefore, the stability of the power supply affects the performance of the analog module.

ADC accuracy partially depends on the power stability of ADC module. For ADC application of high precision, in order to filter and mask ADC sampling interference by the burr on printed circuit board and improve ADC conversion accuracy, ADC uses an independent external power for stable external power supply.

- ADC power pin is VDDA
- Independent power ground of ADC is VSSA

3.1.2 Digital module power supply

VDD and VSS are the digital module power ports of the chip. They mainly provide power for IO, as well as core, built-in digital peripherals and memory via the voltage regulator.

3.1.3 VDD domain

VDD domain mainly provides power for PMU and IWDG and keeps working state after power-on.

3.1.4 1.5V domain

1.5V domain mainly provides power for the core, memories and peripherals of the chip, and keeps enable after power-on by default. When entering the low-power mode, the chip hardware will choose to turn off the 1.5V domain and it automatically turns on after wakeup. It has the following several working states:

Run Mode: 1.5V domain operates in normal power mode, memories and peripherals are working normally.

Sleep mode: 1.5V domain operates in normal power mode, CPU enters the sleep mode, while memories and peripherals work in a normal power consumption mode.

Stop mode: 1.5V domain works in a low power consumption mode, and only keeps the content of registers and RAM.

Deep stop mode: 1.5V domain works in a lower power consumption mode, and only keeps the content of registers and RAM.

Standby mode: 1.5V domain stops power supply. the contents of registers and SRAM are completely lost.

3.2 Power manager

3.2.1 Power on reset and power down reset

The chip has an integrated POR/PDR circuitry. When the power voltage reaches the minimum working voltage, the system works normally. When the power voltage is lower than the minimum working voltage, the system doesn't work. When the chip is powered on or powered down, the power voltage reaches the minimum working voltage of the chip during the power-on operation and the chip will produce a power-on reset. Similarly, during the chip power-down operation, the chip will produce a power-down reset when the voltage is lower than the minimum working voltage.

When VDD is lower than the specified limit voltage POR/PDR, the system remains reset, and NRST reset pin is in low level. Please refer to the electric characteristics section of the Data Manual for details on the reset duration ($t_{RSTTEMPO}$) power-on reset and power-down reset.

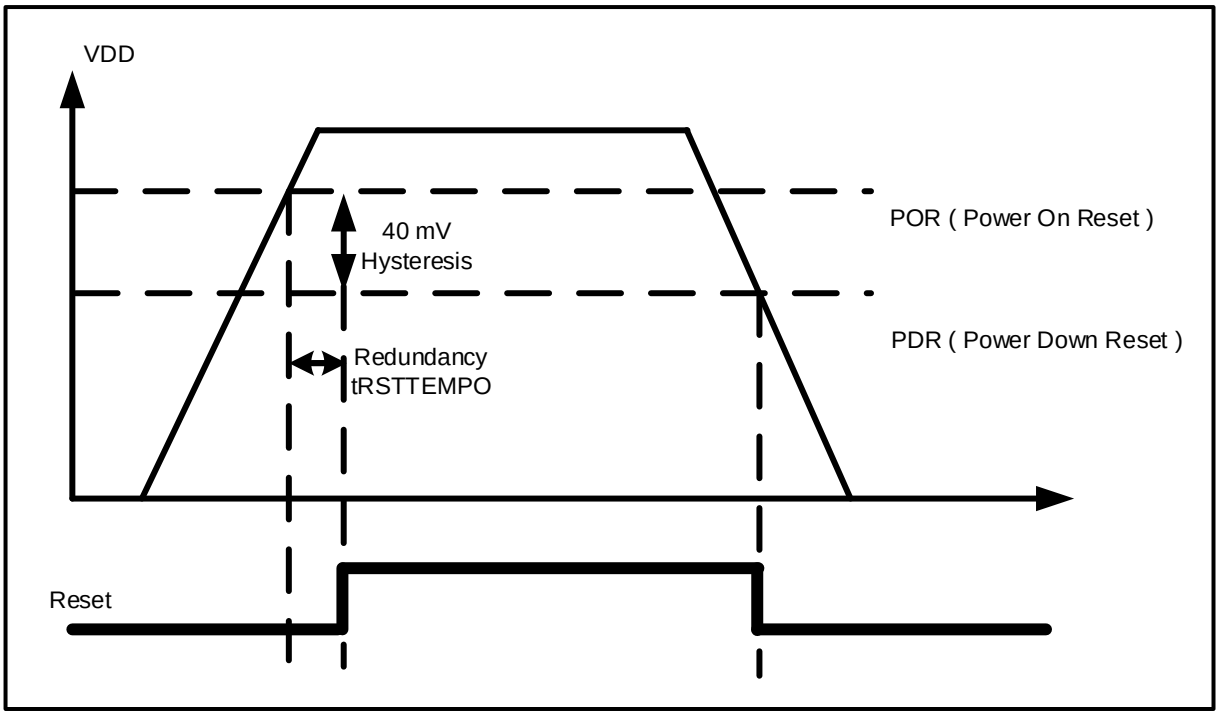


Figure 3-2 Power-on reset and power-down reset waveform

3.2.2 Programmable voltage detector

The Programmable Voltage Detector (PVD) can be used to monitor the power voltage of the chip. When the power voltage drops below a specified threshold, it also produces an interrupt if enabled through the EXTI registers, and the software can conduct an emergency treatment. When the power voltage returns to a specified threshold higher than the original one, it also produces an interrupt if enabled through the EXTI registers. The power supply is resumed after the software processing. There

is a fixed difference between the power-down threshold and power-rise threshold, and this is PVD lag voltage. This difference can be detected through the listed PVD threshold data. The purpose of introducing the difference is to prevent the frequently interrupt caused by voltage fluctuation around the threshold.

The user can set the threshold voltage of the PLS bit in the power control register PWR_CR.PLS through the software and compare this value with the power supply voltage of the chip for power monitoring.PVD can be enabled by setting the PWR_CR.PVDE bit , PWR_CSR. PVDO flag in the power control/status register is to indicate whether VDD is higher or lower than PLC bit threshold voltage in the PWR_CR.PLS bit.

PVD interrupt corresponds to the external interrupt EXTI16. If the user has the EXTI 16 of external interrupt, the event will produce interrupt and enters interrupt service function. When VDD drops below the PVD threshold or VDD rises above the PVD threshold, PVD interrupt will produce (PVD reset also produces via software configuration) according to rising/falling edge of the EXTI 16 of external interrupt. The user may perform some corresponding operations during interrupt. For example, when the condition triggers and power down speed is slower than the execution time of interrupt processing procedure, the emergency stop may be executed if the system needs to enter the special protection status. Save some important system data, and conduct corresponding protection operation on the peripherals.

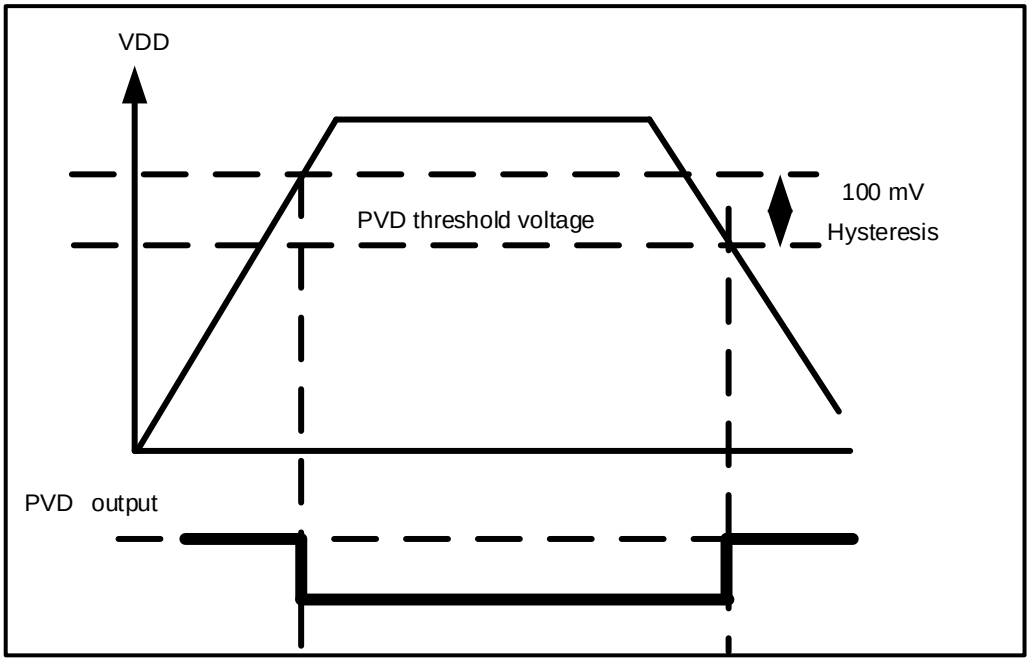


Figure 3-3 PVD threshold waveform

3.3 Power control

3.3.1 Power control overview

MCU keep run after system reset or power reset, when MCU doesn't work, MCU can save power by low power mode. When the MCU needs to start working again, it can wakeup the MCU through various wakeup sources, so as to save the current consumption of the chip.

The chip has four low power modes with different power consumption, wakeup time and wakeup sources. Upon need, the user may choose the optimal low power mode.

Four low power modes of the chip:

Sleep mode: CPU stops, All chip peripherals including CPU peripherals like NVIC, SysTick still run.

Stop mode: 1.5V domain only works at lower power mode. It only keeps content of the registers and RAM and CPU and all peripheral clocks stop.

Deep stop mode: 1.5V power domain works at lower power mode. It only keeps content of the registers and RAM and CPU and all peripheral clocks stop.

Standby mode: 1.5V domain stops power supply. the contents of this domain register and SRAM are lost.

Besides, under run mode, reduce power consumption by any mode below:

- Reduce system clock frequency: While meeting system demand, select lower speed time frequency or adopt the cyclic changeover of high speed clock and low speed clock to save power.
- Disable unused peripheral clock of APB and AHB bus: The user only enables the required application clock signal. Other excessive clocks are disabled.
- Select lower voltage power: The higher the power voltage, the higher the power consumption of the chip. Select the proper power voltage within the chip’s safety voltage range.

Table 4- 1 Low power modes

Mode	Entry mode	Wakeup mode	Influence on 1.5V domain clock	Influence on VDD domain clock	Voltage regulator	Influence on data and register	Precaution
Sleep Mode	WFI (Wait for Interrupt)	Any interrupt	CPU clock off, no influence on	N/A	On		External clock

Mode	Entry mode	Wakeup mode	Influence on 1.5V domain clock	Influence on VDD domain clock	Voltage regulator	Influence on data and register	Precaution
	WFE(Wait for Event)	Wakeup event	other clock include ADC clock				remains, and the content of register and SRAM remains
Stop Mode	Clear LPDS bit; set SLEEPDEEP bit; WFI or WFE;	Any external interrupt (setup in external interrupt register) or event, IWDG interrupt wakeup without reset	All 1.5V domain clocks close	PLL, HSI and HSE oscillators close	On	Content of register and SRAM remains	Unused GPIO before entering low power should be equipped with analog input status
Deep Stop Mode	Clear PDDS bit; set LPDS bit; SLEEPDEEP bit; WFI or WFE;	Any external interrupt (setup in external interrupt register), IWDG interrupt wakeup without reset			On	Content of register and SRAM remains	Unused GPIO before entering low power should be equipped with analog input status
Standby Mode	Set PDDS bit; SLEEPDEEP bit; WFI or WFE;	WKUP pin, NRST pin peripheral reset, IWDG reset			Off	The contents of register and SRAM are lost	Wakeup equals to chip reset

3.3.2 Reducing system clock under run mode

While meeting use demand, select low frequency or adopt the cyclic changeover of high speed clock and low speed clock to save power.

The system clock of the chip may be flexibly configured. The user may select the system clock of different sources or reduce the frequency of any system clock (SYSCLK, HCLK, PCLK1) by configuring different clock dividers.

Before entering sleep mode, reducing the clock frequency of peripherals can effectively save power consumption in sleep mode.

3.3.3 External clock control

While the chip executes the program, close the peripheral clock to reduce power consumption.

In sleep mode, close peripheral clock before executing WFI or WFE instruction and effectively reduce current consumption of peripheral under sleep mode.

The peripheral is mainly on the AHB and APB bus. Enable the register (RCC_AHBENR) by setting AHB peripheral clock, and enable the register (RCC_APB1ENR) by setting APB peripheral clock.

3.3.4 Sleep mode

3.3.4.1 Enter sleep mode

Execute WFI (Wait For Interrupt)/WFE (Wait for Event) instruction, and request MCU to enter the Sleep mode. According to the value of SLEEPONEXIT bit inside the CPU system control register, there are two options available for selecting the sleep mode entry mechanism:

If SLEEPONEXIT bit is cleared, when WFI or WFE is executed, the MCU immediately enters the sleep mode.

If SLEEPONEXIT bit is asserted and the system exits from the interrupt processing program with the lowest priority, the MCU immediately enters the sleep mode.

Under the sleep mode, all I/O pins keep the status under the run mode.

3.3.4.2 Exit sleep mode

The sleep mode is waken up immediately after interrupt or event.

Table 3-1 SLEEPNOW mode

SLEEPNOW mode	Description
Entry	Execute WFI (Wait for Interrupt) or WFE(Wait for Event) instructions under conditions below: SLEEPDEEP = 0 SLEEPONEXIT = 0
Exit	If executing WFI, enter sleep mode: interrupt wakeup (refer to interrupt vector table) If executing WFE, enter sleep mode: event wakeup(refer to wakeup event management)
Wakeup latency	Immediate wakeup

Table 3-2 SLEEPONEXIT mode

SLEEPONEXIT mode	Description
Entry	Execute WFI (Wait for Interrupt) instructions under conditions below: SLEEPDEEP = 0 SLEEPONEXIT = 1
Exit	Interrupt (refer to interrupt vector table)
Wakeup latency	Immediate wakeup

3.3.5 Stop Mode

Under stop mode, CPU enters the deep sleep mode, and all clocks in the 1.5V domain are stopped. Functions of PLL, HSI and HSE oscillators are disabled. Contents of SRAM and registers are kept.

3.3.5.1 Enter stop mode

The independent control bit can be programmed. The stop mode has two control modes according to different clock sources that wakeup:

- Stop mode to wait for external interrupt line WFI mode: Configure PDDS =0, LPDS =0 of power control register (PWR_CR); SLEEPDEEP=1 of CPU system control register (SCR).When WFI is executed, the MCU immediately enters the stop mode.
- Stop mode to wait for external event WFE : Configure PDDS =0, LPDS =0 of power control register (PWR_CR); SLEEPDEEP=1 of CPU system control register (SCR). When WFE is executed, the MCU immediately enters the stop mode.

Functions below can be selected when entering the stop mode:

- Independent watchdog (IWDG): Initiate the independent watchdog by the key register or hardware written into the independent watchdog. The independent watchdog may select interrupt or reset to wakeup chip. When the interrupt wakes up the chip, MCU continues executing the procedure before entering low power. MCU executes reset when the reset is waken. The user may select to close LSI clock source, and thus close independent watchdog.
- Internal low speed oscillator (LSI oscillator): Set by the LSION bit of RCC control/status register

(RCC_CSR).

If ADC is not closed before entering the stop mode, ADC continues to consume current. Close the peripheral by setting ADEN bit of the register ADC_ADCFG as 0. Other unused GPIO need to be set to analog input mode, otherwise there will be current consumption.

3.3.5.2 Exit stop mode

When one interrupt or wakeup event leads to exiting stop mode, the system clock source hardware is auto selected as the HSE oscillator. The user needs to make reconfiguration when other clock source is selected as the clock source.

When the voltage regulator is in run mode and the system exits from the stop mode, there will be an extra startup delay.

Table 4-7 Stop mode

Stop mode	Description
Entry	Execute WFI(Wait for Interrupt) or WFE(Wait for Event) instructions under conditions below: Set the SLEEPDEEP bit of CPU system control register; Reset the PDDS bit and LPDS of the power control register (PWR_CR); Note: In order to enter the stop mode, all external interrupt request bit (interrupt event pending register EXTI_PR)flags must be cleared. Or else, the entry procedure of stop mode will be jumped, and the procedure continues to run.
Exit	Execute WFI(Wait for Interrupt) instructions under conditions below: Any external interrupt line is set as interrupt mode (corresponding external interrupt vector must be enabled in NVIC), refer to interrupt vector table Wait for Event; Execute WFE (Wait for Event) instructions under conditions below: Any external interrupt line is set as event mode, watchdog interrupt;
Wakeup latency	HSI wakeup time and extra time produced by voltage regulator wakeup
Precaution	When entering the stop mode, the unused GPIO is set as analog input mode

3.3.6 Deep Stop Mode

The deep stop mode refers to a kind of low power mode that has taken into consideration the peripheral clock control and control mechanism of voltage regulator based on CPU deep sleep mode. Under the deep stop mode, all clocks are stopped under 1.5V domains. Functions of PLL, HSI and HSE oscillator are disabled. Contents of SRAM and registers are reserved.

3.3.6.1 Enter deep stop mode

Programme the independent control bit. The deep stop mode has two control methods according to different clock sources that have been waken:

- Deep stop mode to wait for external interrupt line WFI: Configure PDDS =0, LPDS =0 of power control register (PWR_CR); SLEEPDEEP=1 of CPU system control register (SCR). When WFI is executed, the MCU immediately enters the deep stop mode.
- Deep stop mode to wait for external event WFE: Configure PDDS =0, LPDS =1 of power control register (PWR_CR); SLEEPDEEP=1 of CPU system control register (SCR). When WFE is executed, the MCU immediately enters the deep stop mode.

Functions are selected when entering the stop mode:

- Independent watchdog (IWDG): Initiate the independent watchdog by the key register or hardware written into the independent watchdog. The independent watchdog may select interrupt or reset to wakeup chip. When the interrupt wakes up the chip, MCU continues executing the procedure before entering low power. MCU executes reset when the reset is waken. The user may select to close LSI clock source, and thus close independent watchdog.
- Internal low speed oscillator (LSI oscillator): Set by the LSION bit of RCC control/status register (RCC_CSR).

If ADC is not closed before entering the deep stop mode, ADC continues to consume current. Close the peripheral by setting ADEN bit of the register ADC_ADCFG as 0. The analog input mode should be set for the unused GPIO. Or else, there is still current consumption.

3.3.6.2 Exit deep stop mode

When one interrupt or wakeup event leads to exiting deepstop mode, the system clock serves as HSI oscillator. The user needs to make reconfiguration when other clock source is selected as the clock source.

When the voltage regulator is in run mode and the system exits from the deep stop mode, there will be an extra startup delay.

Table 3-3Deep stop mode

Deep stop mode	Description
Entry	Execute WFI (Wait for Interrupt) or WFE(Wait for Event) instructions under conditions below: Assert SLEEPDEEP bit in CPU system control register; Reset PDDS and asset LPDS in the reset power control register (PWR_CR); Note: In order to enter the deep stop mode, all external interrupt request bit (interrupt event pending register EXTI_PR) Flag must be cleared. Or else, the entry procedure of deep stop mode will be jumped, and the procedure continues to run.

Deep stop mode	Description
Exit	Execute WFI(Wait for Interrupt) instructions under conditions below: Any external interrupt line is set as interrupt mode (corresponding external interrupt vector must be enabled in NVIC), refer to interrupt vector table Wait for Event; Execute WFE (Wait for Event) instructions under conditions below: Any external interrupt line is set as event mode, such as watchdog interrupt;
Wakeup latency	HSI wakeup time and extra time produced by voltage regulator wakeup
Precaution	When entering the stop mode, the unused GPIO is set as analog input mode

3.3.7 Standby Mode

Under the standby mode, close the voltage regulator when CPU is in deep sleep mode. The entire 1.5V domain is cut off, and PLL, HSI and HSE oscillator are powered off. Contents of SRAM and registers are lost and only standby circuit sustain power supply. The standby mode is the lowest power mode of the chip, and the program begins to be executed after wakeup.

3.3.7.1 Enter standby mode

Program the independent control bit. The mode to enter the standby mode:

Configure PDDS = 1 of the power control register (PWR_CR); SLEEPDEEP=1 of CPU system control register (SCR). When WFI/WFE is executed, MCU immediately enters the standby mode.

Functions below can be selected when entering the standby mode:

- Independent watchdog (IWDG): Initiate the independent watchdog by the key register or hardware written into the independent watchdog. The independent watchdog may select interrupt or reset to wakeup chip. MCU executes reset when the reset is waken. The user may select to close LSI clock source, and thus close independent watchdog.
- Internal low speed oscillator (LSI oscillator): Set by the LSION bit of RCC control/status register (RCC_CSR).

3.3.7.2 Exit standby mode

When the standby mode is waken up by the wakeup event and exits, produce the standby reset. The contents of SRAM and registers are mission. the system exits from the standby mode, and there will be an extra startup delay.

Table 3-4 Standby mode

Standby mode	Description
Entry	Execute WFI (Wait for Interrupt) or WFE (Wait for Event) under conditions below: Assert SLEEPDEEP bit of CPU system control register; Assert PDDS bit of power control register (PWR_CR); Reset WUF bit of power control/status register (PWR_CSR); Note: In order to ensure that the chip can normally wakeup, close interrupt enable prior to entering Standby mode.
Exit	WKUP pin rising edge or falling edge
Wakeup latency	Startup of voltage regulator during reset
Precaution	Configure the correct wakeup source when entering the standby mode. Or else, when entering the standby mode, secondary download is impossible.

Input/output port status under standby mode

Under standby mode, all I/O pins are in high resistance status, except for pins below:

- Enabled wakeup pin, the user needs to correctly configure the corresponding wakeup edge signal

3.3.7.3 Rapid wakeup function

Configure the STDBY_FS_WK bit of PWR_CR register, and adjust the valid time from the occurrence of wakeup event to its effectiveness.

3.4 Debug mode

By default, the debug connection will be lost when the microprocessor enters the stop or standby mode during the microprocessor debug. It's because CPU core missing the clock. Therefore, it's required to set the correct wakeup source.

3.5 Power control register

Access: Compared with the standard APB read and write, the extra APB period is needed when reading/writing PWR registers.

Table 4-10 Power control register overview

Offset	Acronym	Register Name	Reset
0x00	PWR_CR	Power control register	0x00000600
0x04	PWR_CSR	Power control status register	0x00000000
0x08	PWR_CR1	Power control register 1	0x00000000
0x24	PWR_CFGR	Power configuration register	0x00000160

3.5.1 PWR_CR Power control register

Address offset:0x00

Reset value:0x00000600 (clear when wakeup from standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STDBY_FS_WK		Res	PLS				Res				PVDE	CSBF	CWUP	PDDS	LPDS
rw			rw								rw	w	w	rw	rw

Bit	Field	Description
31:16	Reserved	Reserved, always read as 0
15:14	STDBY_FS_WK	Rapidly wakeup standby mode selection bit 00: 9 LSI period wakeup 01: 7 LSI period wakeup 10: 5 LSI period wakeup 11: 2 LSI period wakeup Configure the register as the non “0” value before using the standby mode
13	Reserved	Reserved, always read as 0
12:9	PLS	PVD threshold value selection 0001：2.0V 0010：2.3V 0011：2.6V 0100：2.9V 0101：3.2V 0110：3.5V 0111：3.8V 1000：4.1V 1001：4.4V 1010：4.7V Others:Reserved
8:5	Reserved	Reserved, always read as 0
4	PVDE	PVDE: PVD enable 1: PVD enable 0: PVD disable
3	CSBF	Clear standby flag Always read as 0 1: Clear SBF standby flag(write) 0: Invalid

Bit	Field	Description
2	CWUP	Wakeup flag of clearing wakeup pin 0 1: Clear the corresponding wakeup flag of PWR_CSR register 0: N/A
1	PDDS	Power down deepsleep 1: Enter standby mode when the CPU enters deepsleep 0: Enter stop mode when the CPU enters deepsleep
0	LPDS	Low power deepsleep PDDS = 0, it works together with PDDS bit 1: Voltage regulator in low-power mode during stop mode 0: Voltage regulator on during stop mode When entering stop mode, the current when LPDS = 1 is lower than current when LPDS =0. Details given in the corresponding chip data manual

3.5.2 PWR_CSR Power control/status register

Address offset:0x04

Reset value:0x00000000 (not cleared when wakeup from the standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res							EWUP	Res					PVDO	SBF	WUF
							rw						r	r	r

Bit	Field	Description
31:9	Reserved	Reserved, always read as 0
8	EWUP	Enable wakeup pin 0 1: WKUP pin is used to wakeup CPU from the standby mode, WKUP pin is enforced as input pull down (WKUP pin rising edge wakes up the system from the standby mode) 0: WKUP pin is the universal I/O. WKUP pin event won't wakeup CPU from standby mode
7:3	Reserved	Reserved, always read as 0
2	PVDO	PVD output status 1: PVD output high When VDD high than a threshold is selected by PLS 0: PVD output low When VDD drops below a threshold is selected by PLS Note: PVDE must be enabled.
1	SBF	Standby flag The bit is set by hardware, and cleared only by POR/PDR (power on/power down bit) or setting the CSBF bit of power control register 1 (PWR_CR). 1: System enters standby mode 0: System not in standby mode

Bit	Field	Description
0	WUF	Wakeup flag of wakeup pin 0 1: It has detected that the wakeup pin has the wakeup event 0: It has not detected that the wakeup pin has the wakeup event

3.5.3 PWR_CR1 Power control register 1

Address offset:0x0C

Reset value:0x00000000

The register won't be reset when exiting the standby mode, or reset when RCC_APBRSTR1.PWRRST is effective

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res															WP
															rw

Bit	Field	Description
31:1	Reserved	Reserved, always read as 0
0	WP	Polarity detection of wakeup pin 0 1: low level detection (falling edge), wakeup pin must be configured as high level first 0: high level detection (rising edge), wakeup pin must be configured as low level first

3.5.4 PWR_CFGR Power configuration register

Address offset:0x24

Reset value:0x00000160

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res						LSICAL					LSICALSEL				
						rw					w				

Bit	Field	Description
31:6	Reserved	Reserved, always read as 0

Bit	Field	Description
9:5	LSICAL	Internal low-speed clock calibration bits When the system starts, these bits are automatically initialized to the factory calibration value, and the user can write other calibration values, but the readout is always the factory calibration value. If LSICALSEL = 0x1F, the written value can recalibrate the LSI frequency, otherwise the written value has no effect.
4:0	LSICALSEL	Internal low speed clock calibration select mode The initial value is 0. When writing a different value: Write 1F: Select the value of register LSICAL. It's still read as 0 after writing 1F. Write others: Select the default calibration value

4 SYSCFG System Controller

4.1 Overview

The chip is composed of a set of system configuration registers. Main functions of these registers:

- Manage external interrupt connected to GPIO port (pin configuration)
- Remap memory to code initial area
- System level configuration of some peripherals

4.2 Register

4.2.1 Register overview

Table 4-1 SYSCFG register overview

Offset	Acronym	Register Name	Reset
0x00	SYSCFG_CFGR	SYSCFG configuration register	0x0000000X
0x08	SYSCFG_EXTICR1	SYSCFG external interrupt configuration register 1	0x00000000
0x0C	SYSCFG_EXTICR2	SYSCFG external interrupt configuration register 2	0x00000000
0x10	SYSCFG_EXTICR3	SYSCFG external interrupt configuration register 3	0x00000000
0x14	SYSCFG_EXTICR4	SYSCFG external interrupt configuration register 4	0x00000000
0x18	SYSCFG_PADHYS	SYSCFG PAD configuration register	0x00000000

4.2.2 SYSCFG_CFGR Configuration Register

This register has two control bits MEM_MODE, and is used to configure the mapping of different memories to the initial address 0x00000000. The software configures these two control bits to mask BOOT selection. After reset, this control bit is the actual BOOT mode configuration.

Address offset:0x00

Reset value:0x0000 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														MEM_MODE	
														rw	

Bit	Field	Description
31:2	Reserved	Reserved, must be kept at reset value.
1:0	MEM_MODE	Memory selection Bit Control memory internal mapping to address 0x0000 0000. Set and clear these bits by software. After reset, these bit values are determined by BOOT0 pin configuration value and nBOOT1 bit value. x0: Main Flash memory mapping to 0x0000 0000 01: System Flash mapping to 0x0000 0000 11: Built in RAM mapping to 0x0000 0000

4.2.3 SYSCFG_EXTICR1 External Interrupt Configuration Register 1

Address offset:0x08

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3				EXTI2				EXTI1				EXTI0			
rw				rw				rw				rw			

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value.
15:0	EXTIx	EXTIx configuration (x=0 …3) Select the input source of EXTIx external interrupt. 0000: PA[x] pin 0001: PB[x] pin

4.2.4 SYSCFG_EXTICR2 External Interrupt Configuration Register 2

Address offset:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7				EXTI6				EXTI5				EXTI4			
rw				rw				rw				rw			

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value.
15:0	EXTIx	EXTIx configuration (x=4…7) Select the input source of EXTIx external interrupt. 0000: PA[x] pin 0001: PB[x] pin

4.2.5 SYSCFG_EXTICR3 External Interrupt Configuration Register 3

Address offset:0x10

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11				EXTI10				EXTI9				EXTI8			
rw				rw				rw				rw			

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value.
15:0	EXTIx	EXTIx configuration (x=8…11) Select the input source of EXTIx external interrupt. 0000: PA[x] pin 0001: PB[x] pin

4.2.6 SYSCFG_EXTICR4 External Interrupt Configuration Register 4

Address offset:0x014

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15				EXTI14				EXTI13				EXTI12			
rw				rw				rw				rw			

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value.
15:0	EXTIx	EXTIx configuration (x=12…15) Select the input source of EXTIx external interrupt. 0000: PA[x] pin 0001: PB[x] pin

4.2.7 SYSCFG_PADHYS PAD Configuration Register

Address offset:0x18

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															I2C1_MODE_SEL
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															

Bit	Field	Description
31:17	Reserved	Reserved, must be kept at reset value.
16	I2C1_MODE _SEL	I2C1 port mode selection bit 0: Open drain mode 1: Push pull mode
15:0	Reserved	Reserved, must be kept at reset value.

5 RCC Clock and Reset

5.1 Reset unit

5.1.1 Overview

There are two types of resets: power reset and system reset.

5.1.2 Function block diagram

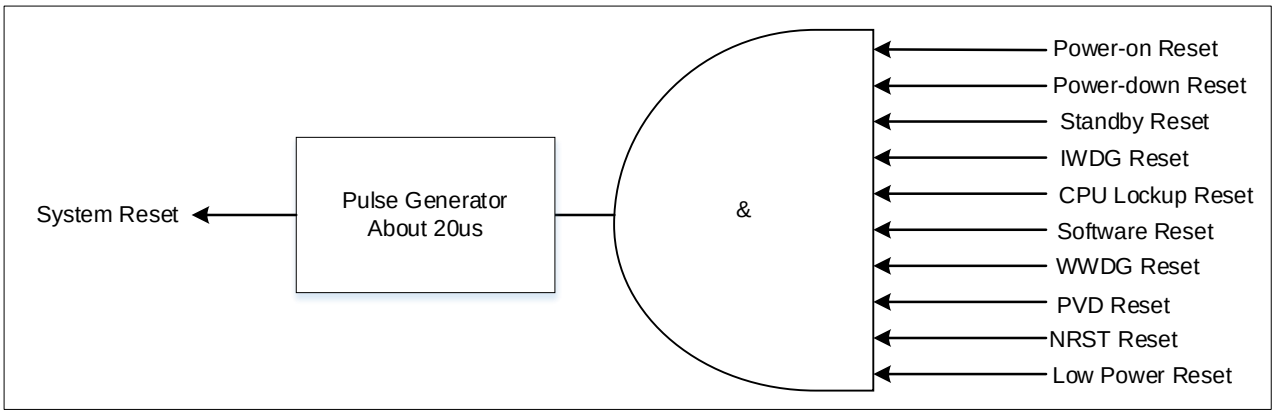


Figure 5-1 Reset function block diagram

5.1.3 Main characteristics

- Reset event judgment: Judge through the reset flag bit of the control status register (RCC_CSR).
- Power on reset: reset all registers.
- System reset: Except that the reset flag in RCC_CSR, internal low-speed oscillator enable flag, standby and wake-up flag in PWR_CSR and DBG_CR are not affected by system reset, other registers are reset by the system.

5.1.4 Functional description

5.1.4.1 Power reset

Power reset includes:

- Power-on reset
- Power-down reset

- Standby reset

5.1.4.2 System reset

System reset includes:

- NRST reset
- WWDG reset
- IWDG reset
- Software reset
- CPU lockup reset
- PVD reset
- Low power reset

NRST Reset:

- When importing the low level through NRST Pin, the external reset will produce.

WWDG reset:

- Feed when the counter value is bigger than the setting value of configuration register (WWDG_CFGR), the window watchdog reset will produce.
- When the counter value is less than 0x40, the window watchdog reset will produce.
- Refer to the chapter of window watchdog for details.

IWDG reset:

- The counter begins ramp down from its reset value 0xFFF. When it ramps down to 0x0000, the independent watchdog reset will produce.
- If the program is anomaly, the feed will not be normal, and the independent watchdog reset will produce.
- Refer to the chapter of window watchdog for details.

Software reset:

- Set SCB_AIRCR[SYSRESETREQ] as 1, and software reset will produce.

CPU lockup reset:

- Configure the LOCKUPEN bit of control status register (RCC_CSR) as 1, and enable CPU

lockup reset.

- CPU lockup reset will produce when CPU enters the lockup status.

PVD reset:

- Configure the PVDRSTEN bit of control status register (RCC_CSR) as 1, enable PVD reset.
- Configure the PVDE bit of power control register 1 (PWR_CR1) as 1, enable PVD.
- Configure the PLS bit of power control register 1 (PWR_CR1) and select the PVD threshold.
- Detect VDD power. When the VDD power is lower than the selected threshold voltage, PVD reset will produce.

Low power reset:

- To prevent the application from entering the low power mode by error, configure the nRST_STDBY bit or nRST_STOP bit in option byte space, and conduct the system reset before entering the low power mode by error.
  - Configure the nRST_STDBY bit of the option byte space as 0, the system will be reset rather than enter the standby mode.
  - Configure the nRST_STOP bit of the option byte space as 0, the system will be reset rather than enter the stop mode or deep stop mode.
  - Refer to the Embedded Flash section for more details.

5.2 Clock unit

5.2.1 Overview

Four configurable independent clocks:

- High-speed external clock (HSE)
- High-speed internal clock (HSI)
- Phase locked loop (PLL)
- Low-speed internal clock (LSI)

5.2.2 Function block diagram

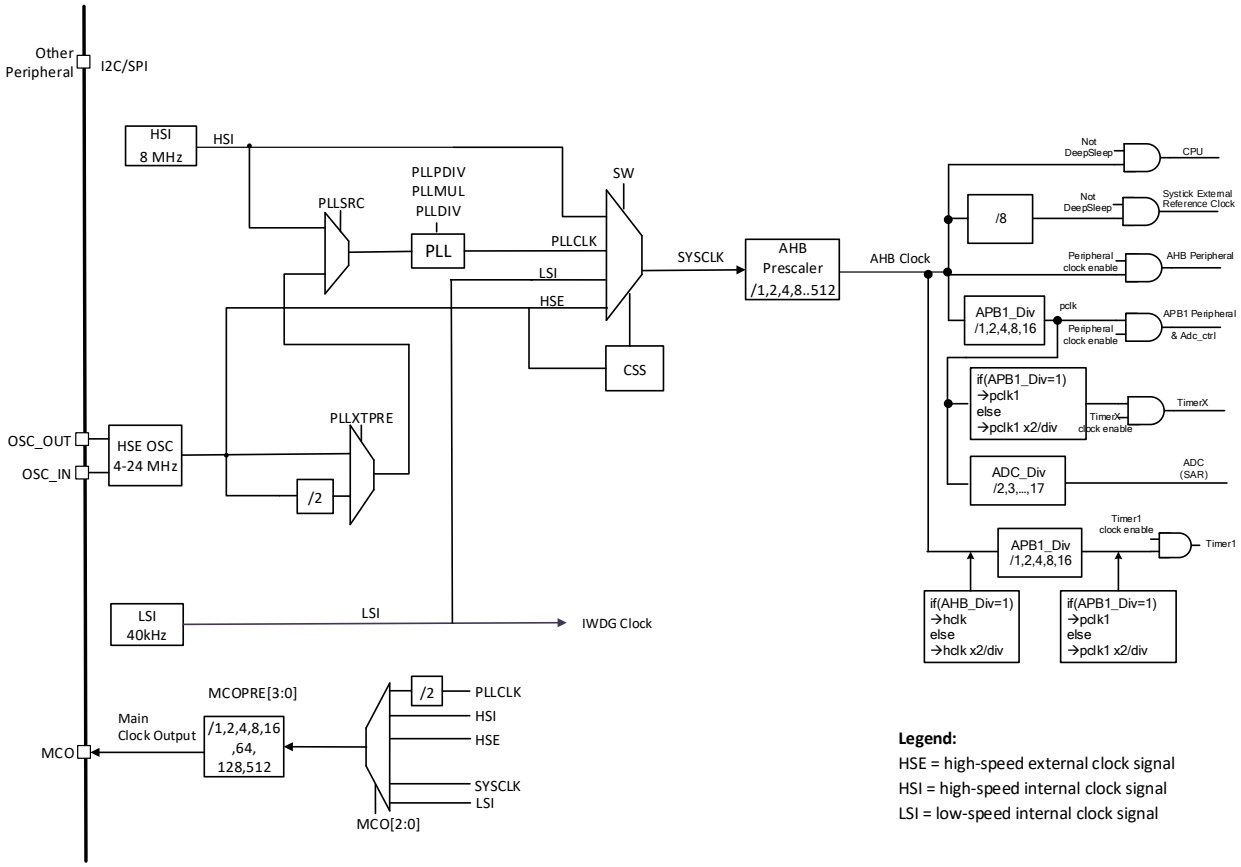


Figure 5-2 Clock tree

5.2.3 Main characteristics

Respectively configure the clock frequency of AHB and APB1 bus via the prescaler control bit of the clock register (RCC_CFGR). The maximum frequency of AHB and APB1 bus clock is 48MHz.

5.2.4 Functional description

5.2.4.1 External high-speed clock (HSE)

Two varieties of high-speed external clock source:

- External high-speed crystal oscillator/ceramic resonator input clock

It's very accurate.

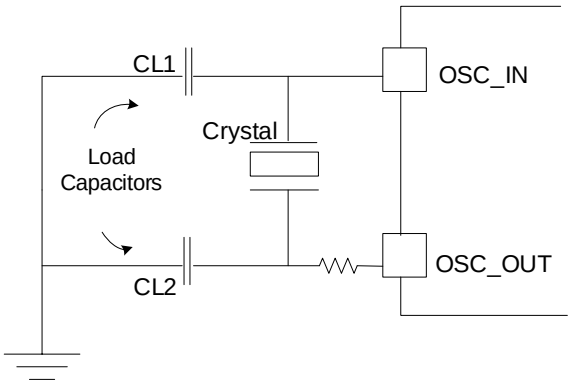


Figure 5-3 high-speed crystal oscillator/ceramic resonator

The figure above is the input module block diagram of high-speed external crystal oscillator/ceramic resonator. Please refer to the datasheet for details.

- External high-speed input speed

This clock input mode is selected by configuring the HSEBYP and HSEON bit of the clock control register (RCC_CR).

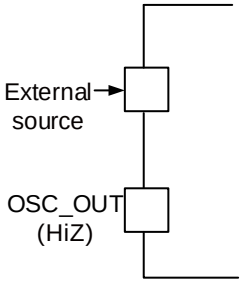


Figure 5-4 External high-speed input speed

The figure above is the module block diagram of high-speed external input clock

The high-speed external clock has characteristics below:

- External high-speed clock source or crystal oscillator must be provided
- Input frequency range 4~24MHz
- External high-speed clock signal (square wave, sine wave) with 50% duty cycle. Please refer to the chapter of electric characteristics of the data manual for details
- When HSE is enabled, OSC_IN pin must be driven while OSC_OUT pin should remain floating when not being used.

Precaution:

- When the high-speed external input clock is used, it's mandatory to set the HSEBYP bit of clock control register (RCC_CR) as '1'.
- Once HSE is enabled, relevant configuration of HSE cannot be changed. When the configuration

needs to be changed, it's mandatory to disable HSE first.

The process to open the HSE configuration:

- Configure the HSEON bit of clock control register (RCC_CR) as 1, and enable HSE.
- Set the HSERDY bit of clock control register (RCC_CR) as 1. It means that HSE is ready, and valid clock signal will be output. At this time, it may be selected as the system clock or peripheral clock source.

5.2.4.2 High-speed internal clock (HSI)

HSI clock signal is produced by the internal 8MHz oscillator. HSI clock source is opened by default when the chip is power on.

Advantage: HSI oscillator can provide the system clock without any external device, and the startup time is faster than HSE crystal oscillator.

Disadvantage: Even after calibration, the frequency may be as precise as the external crystal oscillator or ceramic resonator.

HSI features below:

- It may be directed used as the system clock
- It may be used as PLL input clock source
- In case of fault of HSE crystal oscillator, HSI signal may be used as the backup clock source. Refer to the chapter of clock security system (CSS)

Steps to enable HSI configuration:

- Configure the HSION bit of clock control register (RCC_CR) as 1, and enable HSI;
- Set the HSIRDY bit of clock control register (RCC_CR) as 1. It means that HSI is ready, and valid clock signal will be output. At this time, it may be selected as the system clock or peripheral clock source.

Precaution:

- Once HSI is enabled, relevant configuration of HSI cannot be changed. When the configuration needs to be changed, it's mandatory to disable HSI first.

5.2.4.3 Phase locked loop (PLL)

There are three varieties of PLL input reference clock. Selection can be made according to the PLLXTPRE and PLLSRC of PLL control register (RCC_PLLCFGR):

- HSI oscillator output clock
- HSE output clock
- Two dividing frequencies of HSE output clock

Correlation of PLL input clock f_{PLL_IN} and f_{PLL_OUT}:

$$\frac{f_{PLL_IN}}{(PLLDIV[2:0]+1)*(PLLPDIV[2:0]+1)} = \frac{f_{PLL_OUT}}{PLLMUL[7:0]+1}$$

PLLMUL, PLLPDIV and PLLDIV are the frequency dividing ratio set control bits of PLL multiplier and divider.

Precaution:

- Before enabling PLL, it's mandatory to set PLL (select PLL input clock source and select multiplier frequency factor and frequency divider factor). Once enabling PLL, PLL configuration cannot be changed. In case of change of configuration, first disable PLL.
- PLLDIV of PLL configuration register (RCC_PLLCFGR) is recommended to be set as the odd number (1 or 3 or 5 or 7).

Enable PLL configuration steps: (select HSI as PLL clock source)

- Configure HSION bit of clock control register (RCC_CR) as 1, enable HSI.
- Set the HSIRDY bit of clock control register (RCC_CR) as 1. It means that HSI is ready, and valid clock signal will be output. At this time, it may be selected as the system clock or peripheral clock source.
- Configure PLLMUL (multiplier factor), PLLPDIV (PLL_IN divider factor) and PLLDIV (divider factor) control bit of PLL configuration register (RCC_PLLCFGR).
- Configure PLLON bit of clock control register (RCC_CR) as 1, enable PLL.
- Set the PLLRDY bit of clock control register (RCC_CR) as 1. It means that PLL is ready, and valid clock signal will be output. At this time, it may be selected as the system clock or peripheral clock source.

Enable PLL configuration steps: (select HSI as PLL clock source)

- Configure HSEON bit of clock control register (RCC_CR) as 1 , enable HSE.
- Set the HSERDY bit of clock control register (RCC_CR) as 1. It means that HSE is ready, and

valid clock signal will be output. At this time, it may be selected as the system clock or peripheral clock source.

- Configure the PLLSRC bit of PLL configuration register (RCC_PLLCFGR) as 1, and select HSE clock as the PLL input clock source.
- Configure the PLLXTPRE bit of PLL configuration register (RCC_PLLCFGR), and select HSE or HSE2 division clock as the PLL input clock source.
- Configure the PLLMUL (multiplier factor), PLLPDIV (PLL_IN divider factor) and PLLDIV (divider factor) of PLL configuration register (RCC_PLLCFGR).
- Configure PLLON bit of clock control register (RCC_CR) as 1, enable PLL.
- Set the PLLRDY bit of clock control register (RCC_CSR) as 1. It means that PLL is ready, and valid clock can be exported. At this time, it can be selected as the system clock or peripheral clock source.

5.2.4.4 Low-speed internal clock (LSI)

LSI acts as the low power clock source, and it provides the clock source for the independent watchdog. The clock center frequency is around 40kHz. Refer to the chapter of electric characteristics of data manual.

Steps to enable LSI configuration:

- Set the LSION bit of control status register (RCC_CSR) as 1, enable LSI.
- Set the LSIRDY bit of control status register (RCC_CSR) as 1. It means that LSI is ready, and valid clock can be exported.

Precaution:

- Once LSI is enabled, relevant LSI configuration cannot be changed. If it's required to change configuration, disable LSI first.

5.2.4.5 Interrupt

Table 5-1 RCC global interrupt

Interrupt event	Event flag bit	Enable control bit	Flag clear bit
RCC_PLLRDY	PLLRDYF	PLLRDYIE	PLLRDYC
RCC_HSERDY	HSERDYF	HSERDYIE	HSERDYC
RCC_HSIRDY	HSIRDYF	HSIRDYIE	HSIRDYC
RCC_LSIRDY	LSIRDYF	LSIRDYIE	LSIRDYC

Note: The above flag bit, control bit and clear bit can be configured by the clock interrupt register

(RCC_CIR).

5.2.4.6 System clock selection (SWS)

Four system clock sources:

- High-speed internal clock (default after HSI power on)
- High-speed external clock (HSE)
- Phase locked loop (PLL): When HSI (or HSE) as PLL clock source, HSI (or HSE) cannot be disabled
- Low-speed internal clock (LSI)

System clock configuration steps:

- Enable the required system clock source (HSI, HSE, PLL, LSI). Each clock is enabled differently. For specific method, please view (chapter of HSI, HSE, PLL, LSI).
- Set the bit of the selected clock source RDY signal as 1. It means that the system clock source is ready (when the target clock source is ready, the system clock may change over).
- Select the system clock by configuring the SW bit of clock configuration register (RCC_CFGR).
- Read the SWS bit of clock configuration register (RCC_CFGR), and judge the current system clock source.

5.2.4.7 System clock frequency switch

The system clock frequency switches from the low-speed to high-speed or from high-speed to low-speed. It's recommended to switch the medium speed frequency for transition. The interval switch time from high-speed to low-speed is 1us at least.

5.2.4.8 Peripheral reset

Achieve the corresponding peripheral software reset by APB1 peripheral reset register (RCC_APB1RSTR) and AHB peripheral reset register (RCC_AHBRSTR).

5.2.4.9 Clock security system (CSS)

The specific operation method of clock security system includes the software enable and interrupt processing.

Software enable: Enable clock security system by setting the CSSON bit of the clock control register (RCC_CR) as 1. The clock detector enable is delayed after HSE oscillator enables and becomes ready,

and is disabled when the HSE oscillator stops.

Trigger interrupt: When detecting HSE clock fault, CSS will automatically stop HSE oscillator, send the clock fault event to the brake input terminal of advanced timer, trigger the CSS interrupt and request the software to execute rescue operation. CSS interrupt is connected to NMI interrupt of CPU.

Interrupt judgment flag and clearing mode: When CSSF bit of the clock interrupt register (RCC_CIR) is set as 1 by hardware, it means that HSE clock failure leads to clock security system interrupt. Write the CSSC bit of clock interrupt register (RCC_CIR) as1 by software to clear the security system interrupt flag bit CSSF.

Precaution:

- Once enabling CSS, in case of HSE clock fault, CSS interrupt produces, and automatically generates NMI interrupt. Unless CSS interrupt pending bit is cleared, NMI will be executed continuously. Therefore, during NMI processing, the user must clear CSS interrupt by setting CSSC bit in the clock interrupt register (RCC_CIR) as 1.
- In case of fault, if HSE oscillator is directly used as system clock, the system clock will automatically change over to the high-speed internal oscillator and disable HSE oscillator; if HSE oscillator is used as PLL input clock and PLL clock is used as system clock, PLL will also be disabled.
- Before enabling the clock security system, enable LSI. For details, please refer to the chapter of low-speed internal clock.

Steps when HSE serves as the system clock:

- Configure HSEON bit of clock control register (RCC_CR) as 1, enable HSE.
- Set the HSERDY bit of clock control register (RCC_CSR) as 1. It means that HSE is ready, and valid clock can be exported.
- Configure the SW bit of clock configuration register (RCC_CFGR) as 2'b10 and select HSE as system clock.
- Read the SWS bit of clock configuration register (RCC_CFGR) and judge whether the current system clock source is HSE.
- Configure LSION bit of control status register (RCC_CSR) as 1, enable LSI.
- Set the LSIRDY bit of clock control register (RCC_CSR) as 1. It means that LSI is ready, and valid clock can be exported.
- Set the CSSON bit of clock control register (RCC_CR) as 1, and enable clock safety system.

Steps when HSE is directly as system clock configuration:

- Configure HSEON bit of clock control register (RCC_CR) as 1, enable HSE.
- Set HSERDY bit of clock control register (RCC_CR) as 1. It means HSE is ready, and may export valid clock.
- Configure PLLSRC bit of PLL configuration register (RCC_PLLCFGR) as 1, and select HSE clock as PLL input clock.
- Configure PLLXTPRE bit of PLL configuration register (RCC_PLLCFGR) as 1, and select HSE or HSE2 division as PLL input clock.
- Configure the PLLMUL (multiplier factor), PLLPDIV (PLL_IN divider factor) and PLLDIV (divider factor) control bit of PLL configuration register (RCC_PLLCFGR).
- Configure PLLON bit of clock control register (RCC_CR) as 1, enable PLL.
- Set the PLLRDY bit of clock control register (RCC_CR) as 1. It means that PLL is ready, and valid clock can be exported. At this time, it may be selected as the system clock or peripheral clock source.
- Configure the SW bit of clock configuration register (RCC_CFGR) as 2'b10 and select PLL as system clock.
- Read the SWS bit of clock configuration register (RCC_CFGR) and judge whether the current system clock source is PLL.
- Configure LSION bit of control status register (RCC_CSR) as 1, enable LSI.
- Set the LSIRDY bit of clock control register (RCC_CSR) as 1. It means that LSI is ready, and valid clock can be exported.
- Set the CSSON bit of clock control register (RCC_CR) as 1, and enable clock safety system.

5.2.4.10 Micro-controller clock output (MCO)

The micro-controller clock output (MCO) allows the clock to export to the external MCO pin. The configuration register of the corresponding GPIO port must be configured as the multiplex output function. Select any one of the 6 clock signals below as MCO output clock:

Table 5-2 Correlation of MCO and clock source

MCO bit of clock configuration register (RCC_CFGR)	Clock source
001	APB1_CLK
010	LSI
100	SYSCLK
101	HSI
110	HSE
111	PLL/2

5.2.4.11 Timer clock

When PPRE1 bit [10: 8] of clock configuration register (RCC_CFGR) is 3'b0xx (no frequency division), the clock frequency of the timer hung on APB1 is the same as APB1 bus frequency; otherwise the clock frequency of the timer hung on APB1 is twice of APB1 bus frequency.

When the HPRE bit [7: 4] of the clock configuration register (RCC_CFGR) is 4'b0xxx (no frequency division), the clock frequency of the advanced timer (TIM1) is the same as the clock frequency of the timer hung on APB1; otherwise the clock frequency of the advanced timer (TIM1) is twice the clock frequency of the timer hung on APB1.

5.2.4.12 Independent watchdog clock

When the independent watchdog is enabled by hardware, LSI oscillator will be automatically opened, and cannot be closed;

When the independent watchdog is enabled by software, LSI oscillator needs to be enabled and opened by software. When LSI oscillator is ready, the clock is supplied to IWDG, and LSI can be closed by software.

5.3 Register description

5.3.1 Register overview

Table 5-3 RCC register overview

Offset	Acronym	Register Name	Reset
0x00	RCC_CR	Clock Control Register	0x00000021
0x04	RCC_CFGR	Clock Configuration Register	0x00000000
0x08	RCC_CIR	Clock Interrupt Register	0x00000000
0x10	RCC_APB1RSTR	APB1 Peripheral Reset Register	0x00000000
0x14	RCC_AHBENR	AHB Peripheral Enable Register	0x00000014
0x1C	RCC_APB1ENR	APB1 Peripheral Enable Register	0x000000C0
0x24	RCC_CSR	Control Status Register	0x08000000
0x28	RCC_AHBRSTR	AHB Peripheral Reset Register	0x00000000
0x2C	RCC_CFGR2	Clock Configuration Register 2	0x00000000
0x40	RCC_SYSCFG	System Configuration Register	0x00000103
0x50	RCC_PLLCFGR	PLL Configuration Register	0x00000000

5.3.2 RCC_CR Clock Control Register

Address offset : 0x00

Reset value: 0x0000 0021

Access: No wait state, word, half word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.						PLLRDY	PLLON	Res.				CSSON	HSEBYP	HSERDY	HSEON
						r	rw					rw	rw	r	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.							HSEOUTPUTSEL	HSEDEGLITCHSEL	HSEDEGLITCHBYP	HSELPFSEL	HSELPFBYP	Res.		HSIRDY	HSION
							rw	rw	rw	rw	rw			r	rw

Bit	Field	Description
31: 26	Reserved	Reserved, must be kept at reset value.
25	PLLRDY	PLL clock ready flag is set by hardware. 0: PLL not ready 1: PLL ready
24	PLLON	PLL enable Set '1'or clear '0' via software. When entering stop or standby mode, clear by hardware. If PLL clock is used as system clock or selected as system clock, the bit cannot reset. 0: Disable PLL 1: Enable PLL
23: 20	Reserved	Reserved, must be kept at reset value.
19	CSSON	Clock security system enable Set '1'or clear '0' via software. 0: Disable clock security monitor 1: If the high-speed external oscillator is ready, enable clock security monitor
18	HSEBYP	External high-speed clock bypass Set ‘1' by software to bypass external crystal oscillator. Only when HSE oscillator is disabled, write HSEBYP bit. 0: Disable high-speed external crystal oscillator bypass mode 1: Enable high-speed external crystal oscillator bypass mode
17	HSERDY	External high-speed clock ready flag Set by hardware. 0: high-speed external crystal oscillator is not ready 1: high-speed external crystal oscillator is ready

Bit	Field	Description
16	HSEON	External high-speed clock enable Set '1'or clear '0' by software. When entering the standby or stop mode, the bit is cleared '0' by hardware. When HSE has been or will be selected as the clock source by system clock, this bit is forbidden to be reset. 0: Disable high-speed external crystal oscillator 1: Enable high-speed external crystal oscillator
15: 9	Reserved	Reserved, must be kept at reset value.
8	HSEOUTPUTSEL	Output mode selection 0: direct output 1: filter output
7	HSEDEGLITCHSEL	Deglitching width control bit 0: Typical value 8ns 1: Typical value 5ns
6	HSEDEGLITCHBYP	Bypass control bit of deglitching circuit 0: Normally adopt the deglitching function 1: Bypass the deglitching function
5	HSELPFSEL	LPF filter mode control 0: Signal amplification output after LPF 1: Output after LPF
4	HSELPFBYP	Enable signal that LPF input signal directly output 0: Output after LPF 1: LPFIN direct output
3: 2	Reserved	Reserved, must be kept at reset value.
1	HSIRDY	Internal high-speed clock ready flag Set ‘1' by hardware. It means that the internal clock is ready. When HSION bit is cleared, HSIRDY turns '0' after 3 AHB clock periods. 0: Internal high-speed clock is not ready 1: Internal high-speed clock is ready
0	HSION	Internal high-speed clock enable Set '1'or clear '0' by software. This bit is set to '1' by hardware to force the internal oscillator to enable when exiting Standby or Stop mode or when an external oscillator is used as the system clock and a fault occurs. When HSI has been or will be selected as the clock source by system clock, this bit is forbidden to be reset. 0: Disable internal high-speed clock 1: Enable internal high-speed clock

5.3.3 RCC_CFGR Clock Configuration Register

Address offset : 0x04

Reset value: 0x0000 0000

Access: No wait state, word, half word and byte access

Only when access occurs during clock change over, insert 1 or 2 wait period

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.					MCO			Res.							
					rw										
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.					PPRE1			HPRE				SWS		SW	
					rw			rw				r		rw	

Bit	Field	Description
31: 27	Reserved	Reserved, must be kept at reset value.
26: 24	MCO	Micro controller clock output Set '1' or clear '0' by software 000: No clock output 001: APB1_CLK output 010: LSI clock output 011: No clock output 100: System clock (SYSCLK) output 101: HSI clock output 110: HSE clock output 111: PLL clock 2 output after division Note: This clock output may be stopped when starting or switching MCO clock source. When system clock is exported to MCO pin, please guarantee that the output clock frequency is no more than 50MHz
23: 11	Reserved	Reserved, must be kept at reset value.
10: 8	PPRE1	APB1 prescaler coefficient Set the software to control the APB1 clock (PCLK2) prescaler factor. 0xx: HCLK no division 100: HCLK 2 division 101: HCLK 4 division 110: HCLK 8 division 111: HCLK 16 division

Bit	Field	Description
7: 4	HPRE	AHB prescaler coefficient Set the software to control the APB clock prescaler factor. 0xxx: SYSCLK no division 1000: SYSCLK 2 division 1001: SYSCLK 4 division 1010: SYSCLK 8 division 1011: SYSCLK 16 division 1100: SYSCLK 64 division 1101: SYSCLK 128 division 1110: SYSCLK 256 division 1111: SYSCLK 512 division Note: When the prescaler coefficient of AHB clock is greater than 1, it's mandatory to enable prefetch buffer. See the chapter of flash memory for details.
3: 2	SWS	System clock selection status 00: Select HSI as system clock 01: Select HSE as system clock 10: Select PLL output as system clock 11: Select LSI output as system clock
1: 0	SW	System clock switch Select system clock source by software configuration When returning from stop or standby mode or directly or indirectly as system clock HSE produces fault, the hardware will enforce selection of HSI as system clock. 00: Select HSI as system clock 01: Select HSE as system clock 10: Select PLL output as system clock 11: Select LSI output as system clock

5.3.4 RCC_CIR Clock Interrupt Register

Address offset : 0x08

Reset value: 0x0000 0000

Access: No wait period, word, half word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.								CSSC	Res.		PLLRDYC	HSERDYC	HSIRDYC	Res.	LSIRDYC
								w1c			w1c	w1c	w1c		w1c
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			PLLRDYIE	HSERDYIE	HSIRDYIE	Res.	LSIRDYIE	CSSF	Res.		PLLRDYF	HSERDYF	HSIRDYF	Res.	LSIRDYF

	rw	rw	rw		rw	r		r	r	r		r

Bit	Field	Description
31: 24	Reserved	Reserved, must be kept at reset value.
23	CSSC	Clock security system interrupt clear Set ‘1' by software to clear security system interrupt flag bit CSSF. 0: Invalid 1: Clear security system interrupt flag bit CSSF
22: 21	Reserved	Reserved, must be kept at reset value.
20	PLLRDYC	PLL ready interrupt clear Set ‘1' by software to clear PLL ready interrupt flag bit PLLRDYF. 0: Invalid 1: Clear PLL ready interrupt flag bit PLLRDYF
19	HSERDYC	HSE ready interrupt clear Set ‘1' by software to clear HSE ready interrupt flag bit HSERDYF. 0: Invalid 1: Clear HSE ready interrupt flag bit HSERDYF
18	HSIRDYC	HSI ready interrupt clear Set ‘1' by software to clear HSI ready interrupt flag bit HSIRDYF. 0: Invalid 1: Clear HSI ready interrupt flag bit LSERDYF
17	LSERDYC	LSE ready interrupt clear Set ‘1' by software to clear LSE ready interrupt flag bit LSERDYF 0: Invalid 1: Clear LSE ready interrupt flag bit LSERDYF
16	LSIRDYC	LSI ready interrupt clear Set ‘1' by software to clear LSI ready interrupt flag bit LSIRDYF 0: Invalid 1: Clear LSI ready interrupt flag bit LSIRDYF
15: 13	Reserved	Reserved, must be kept at reset value.
12	PLLRDYIE	PLL ready interrupt enable Set ‘1' by software to enable or clear '0' to disable PLL ready interrupt. 0: Disable PLL ready interrupt 1: Enable PLL ready interrupt
11	HSERDYIE	HSE ready interrupt enable Set ‘1' by software to enable or clear '0' to disable external oscillator ready interrupt. 0: Disable HSE ready interrupt 1: Enable HSE ready interrupt

Bit	Field	Description
10	HSIRDYIE	HSI ready interrupt enable Set ‘1' by software to enable or clear '0' to disable external oscillator ready interrupt. 0: Disable HSI ready interrupt 1: Enable HSI ready interrupt
9	Reserved	Reserved, must be kept at reset value.
8	LSIRDYIE	LSI ready interrupt enable Set ‘1' by software to enable or clear '0' to disable internal 40KHz oscillator ready interrupt. 0: Disable LSI ready interrupt 1: Enable LSI ready interrupt
7	CSSF	Clock security system interrupt flag When the external oscillator produces fault, set ‘1' by hardware. Clear by setting CSSC bit as '1'by software. setting CSSC position ‘1' by software. 0: Produce security system interrupt without HSE clock failure 1: Clock security system interrupt caused by HSE clock failure
6: 5	Reserved	Reserved, must be kept at reset value.
4	PLLRDYF	PLL ready interrupt flag When PLL is ready, set ‘1' by hardware. Clear by setting PLLRDYC bit as '1'by software. 0: Produce clock ready interrupt without PLL lock 1: Clock ready interrupt caused by PLL lock
3	HSERDYF	HSE ready interrupt flag When the high-speed external clock is ready, set ‘1' by hardware. Clear by setting HSERDYC bit as '1'by software. 0: Produce clock ready interrupt without external oscillator 1: Clock ready interrupt caused by external oscillator
2	HSIRDYF	HSI ready interrupt flag When the high-speed internal clock is ready, set ‘1' by hardware. Clear by setting HSIRDYC bit as '1'by software. 0: Produce clock ready interrupt without internal HSI oscillator 1: Clock ready interrupt caused by internal HSI oscillator
1	LSERDYF	LSE ready interrupt flag When the low-speed external clock is ready, set ‘1' by hardware. Clear by setting LSERDYC bit as '1'by software. 0: Produce clock ready interrupt without external 32KHz oscillator 1: Clock ready interrupt caused by external 32KHz oscillator
0	LSIRDYF	LSI ready interrupt flag When the low-speed internal clock is ready, set ‘1' by hardware. Clear by setting LSIRDYC bit as '1'by software. 0: Produce clock ready interrupt without internal 40KHz oscillator 1: Clock ready interrupt caused by internal 40KHz oscillator

5.3.5 RCC_APB1RSTR APB1 Peripheral Reset Register

Address offset : 0x10

Reset value: 0x0000 0000

Access: No wait period, word, half word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	SYSCFG	DBG	PWR	Res.						I2C	Res.			UART2	UART1
	rw	rw	rw							rw				rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			SPI	WWDG	Res.	ADC	Res.					TIM14	TIM1	TIM3	Res.
			rw	rw		rw						rw	rw	rw	

Bit	Field	Description
31	Reserved	Reserved, must be kept at reset value.
30	SYSCFG	SYSCFG Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
29	DBG	DBG Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
28	PWR	PWR reset (Power Interface Reset) Set to '1' or clear '0' by software 0: Invalid 1: Reset
27: 22	Reserved	Reserved, must be kept at reset value.
21	I2C	I2C Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
20: 18	Reserved	Reserved, must be kept at reset value.
17	UART2	UART2 Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
16	UART1	UART1 Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset

Bit	Field	Description
15: 13	Reserved	Reserved, must be kept at reset value.
12	SPI	SPI Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
11	WWDG	Window Watchdog Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
10	Reserved	Reserved, must be kept at reset value.
9	ADC	ADC Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
8: 4	Reserved	Reserved, must be kept at reset value.
3	TIM14	TIM14 Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
2	TIM1	TIM1 Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
1	TIM3	TIM3 Reset Set to '1' or clear '0' by software 0: Invalid 1: Reset
0	Reserved	Reserved, must be kept at reset value.

5.3.6 RCC_AHBENR AHB Peripheral Clock Enable Register

Address offset : 0x14

Reset value: 0x0000 0014

Access: No wait period, word, half word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.													GPIOB	GPIOA	Res.
													rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									CRC	Res.	FLASH	Res.	SRAM	Res.	
									rw		rw		rw		

Bit	Field	Description
31: 19	Reserved	Reserved, must be kept at reset value.
18	GPIOB	GPIOB Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
17	GPIOA	GPIOA Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
16: 7	Reserved	Reserved, must be kept at reset value.
6	CRC	CRC Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
5	Reserved	Reserved, must be kept at reset value.
4	Flash	FLASH Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
3	Reserved	Reserved, must be kept at reset value.
2	SRAM	SRAM Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
1: 0	Reserved	Reserved, must be kept at reset value.

5.3.7 RCC_APB1ENR APB1 Peripheral Clock Enable Register

Address offset : 0x1C

Reset value: 0x0000 00C0

Access: No wait period, word, half word and byte access

Note: When peripheral clock is not enabled, the software cannot read the value of peripheral register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	SYSCFG	DBG	PWR	Res.						I2C	Res.			UART2	UART1
	rw	rw	rw							rw				rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			SPI	WWDG	Res.	ADC	Res.	EXTI	IWDG	Res.		TIM14	TIM1	TIM3	Res.
			rw	rw		rw		rw	rw			rw	rw	rw	

Bit	Field	Description
31	Reserved	Reserved, must be kept at reset value.
30	SYSCFG	SYSCFG Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
29	DBG	DBG Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
28	PWR	Power Interface Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
27: 22	Reserved	Reserved, must be kept at reset value.
21	I2C	I2C Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
20: 18	Reserved	Reserved, must be kept at reset value.
17	UART2	UART2 Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
16	UART1	UART1 Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
15: 13	Reserved	Reserved, must be kept at reset value.
12	SPI	SPI Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
11	WWDG	Window Watchdog Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
10	Reserved	Reserved, must be kept at reset value.
9	ADC	ADC Interface Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
8	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
7	EXTI	EXTI Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
6	IWDG	IWDG Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
5: 4	Reserved	Reserved, must be kept at reset value.
3	TIM14	Timer14 Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
2	TIM1	Timer1 Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
1	TIM3	Timer3 Clock Enable Set to '1' or clear to '0' by software 0: Clock disable 1: Clock Enable
0	Reserved	Reserved, must be kept at reset value.

5.3.8 RCC_CSR Control Status Register

Address offset : 0x24

Reset value: 0x0800 0000

Access: 0-3 wait period, word, half word and byte access

In case of continuous access to the register, insert the wait state.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LPWRRSTF	WWDGRSTF	IWDGRSTF	SFTRSTF	PORRSTF	PINRSTF	Res.	RMVF	LOCKUPF	PVDRSTF	Res.					
r	r	r	r	r	r		w1c	r	r						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								LOCKUPEN	PVDRSTEN	Res.				LSIRDY	LSION
								rw	rw					r	rw

Bit	Field	Description
31	LPWRRSTF	Low power reset flag In case of low power reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software. 0: No low power reset 1: Low power reset
30	WWDGRSTF	Window watchdog reset flag In case of window watchdog reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software. 0: No window watchdog reset 1: Window watchdog reset
29	IWDGRSTF	Independent watchdog reset flag In case of independent watchdog reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software. 0: No independent watchdog reset 1: Independent watchdog reset
28	SFTRSTF	Software reset flag In case of software reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software. 0: No software reset 1: Software reset
27	PORRSTF	POR/PDR reset flag In case of POR/PDR reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software. 0: No POR/PDR reset 1: POR/PDR reset
26	PINRSTF	NRST PIN reset flag In case of NRST PIN reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software 0: No NRST PIN reset 1: NRST PIN reset
25	Reserved	Reserved, must be kept at reset value.
24	RMVF	Remove reset flag Clear the reset flag by setting '1'by software. 0: Invalid 1: Clear reset flag
23	LOCKUPF	CPU lockup reset flag In case of CPU lockup reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software 0: No CPU lockup reset 1: CPU lockup reset

Bit	Field	Description
22	PVDRSTF	PVD reset flag In case of PVD reset, set ‘1' by hardware, and enable clear by power reset or write RMVF bit to clear by software 0: No PVD reset 1: PVD reset
21: 8	Reserved	Reserved, must be kept at reset value.
7	LOCKUPEN	CPU lockup reset enable 0: Disable CPU lockup reset 1: Enable CPU lockup reset
6	PVDRSTEN	PVD reset enable 0: Disable PVD reset 1: Enable PVD reset
5: 2	Reserved	Reserved, must be kept at reset value.
1	LSIRDY	Internal low-speed oscillator ready Setting '1'or clearing '0' by hardware indicates whether the internal 40KHz oscillator is ready. When LSION is cleared, LSIRDY is cleared after 3 AHB clocks. 0: Internal 40KHz oscillator clock not ready 1: Internal 40KHz oscillator clock ready
0	LSION	Internal low-speed oscillator enable Setting '1'or clearing '0' by software, or clear by power reset 0: Disable internal 40KHz oscillator 1: Enable internal 40KHz oscillator

5.3.9 RCC_AHBRSTR AHB Peripheral Reset Register

Address offset : 0x28

Reset value: 0x0000 0000

Access: No wait period, word, half word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.													GPIOB	GPIOA	Res.
													rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									CRC	Res.					
									rw						

Bit	Field	Description
31: 19	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
18	GPIOB	GPIOB Reset Set '1' or clear '0' by software 0: Invalid 1: Reset
17	GPIOA	GPIOA Reset Set '1' or clear '0' by software 0: Invalid 1: Reset
16: 7	Reserved	Reserved, must be kept at reset value.
6	CRC	CRC Reset Set '1' or clear '0' by software 0: Invalid 1: Reset
5: 0	Reserved	Reserved, must be kept at reset value.

5.3.10 RCC_CFGR2 Clock Configuration Register 2

Address offset : 0x2C

Reset value: 0x0000 0000

Access: No wait state, word, half word and byte access

Only when the access occurs during clock changeover, insert 1 or 2 wait periods.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								MCOPRE							
								rw							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															

Bit	Field	Description
31: 24	Reserved	Reserved, must be kept at reset value.
23: 20	MCOPRE	MCO clock prescaler factor 0xxx: No division 1000: MCO 2 division 1001: MCO 4 division 1010: MCO 8 division 1011: MCO 16 division 1100: MCO 64 division 1101: MCO 128 division 1110: MCO 256 division 1111: MCO 512 division

Bit	Field	Description
15: 0	Reserved	Reserved, must be kept at reset value.

5.3.11 RCC_SYSCFG System Configuration Register

Address offset : 0x40

Reset value: 0x0000 0103

Access: 0-3 wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	HSELPFEN	Res.				HSE_RFB_SEL		Res.							PROG_CHECK_EN
	rw					rw									r

Bit	Field	Description
31: 15	Reserved	Reserved, must be kept at reset value.
14	HSELPFEN	External crystal low-pass filter enable 0: Disable 1: Enable
13: 10	Reserved	Reserved, must be kept at reset value.
9: 8	HSE_RFB_SEL	Feedback resistor selection. 00: 2M 01: 1M (applicable to 8M~12M crystal by default) 10: 500K 11: 200K
7: 1	Reserved	Reserved, must be kept at reset value.
0	PROG_CHECK_EN	Check whether the data in Flash is 0xFF when writing Flash 1: Check (hardware is fixed to 1) 0: Not check

5.3.12 RCC_PLLCFGR PLL Configuration Register

Address offset : 0x50

Reset value: 0x0000 0000

Access: No wait state, word, half word and byte access

Only when the access occurs during clock changeover, insert 1 or 2 wait periods.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PLLPDIV			Res.					PLLMUL							
rw								rw							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.					PLLDIV			Res.		PLL_LDS		PLL_ICTRL		PLLXTPRE	PLLSRC
					rw					rw		rw		rw	rw

Bit	Field	Description
31: 29	PLLPDIV	PLL Pre-divide Factor
28: 24	Reserved	Reserved, must be kept at reset value.
23: 16	PLLMUL	PLL Multiplication Factor
15: 11	Reserved	Reserved, must be kept at reset value.
10: 8	PLLDIV	PLL divide factor
7: 6	Reserved	Reserved, must be kept at reset value.
5: 4	PLL_LDS	PLL lock detector accuracy selection
3: 2	PLL_ICTRL	PLL CP current control signal The default value is 2'b11. When PLL input clock source is lower than 8MHz, it's recommended to set this bit as 2'b01.
1	PLLXTPRE	Division selection when HSE clock is used as PLL input Used as PLL input clock when setting '1' or clearing by software to divide HSE clock Only when PLL is disabled, write this bit. 0: HSE clock no division 1: HSE clock 2 division
0	PLLSRC	PLL input clock source selection Set '1'or clear by software to select PLL as input clock source. Only when PLL is disabled, write this bit. 0: HSI clock used as PLL input clock 1: HSE clock used as PLL input clock

Note: Refer to the PLL section of the functional description to configure the RCC_PLLCFGR register.

6 CRC Cyclic Redundancy Check Calculation Unit

6.1 Overview

The CRC (cyclic redundancy check) calculation unit is used to get a 8-, 16- or 32-bit CRC check code based on a fixed generator polynomial. This technique is used to verify data transmission or storage integrity.

6.2 Main characteristics

- Uniform polynomial for calculation 0x4C11DB7:

$X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$

- This polynomial is applied to the Ethernet polynomial.
- Supports input 8-, 16- or 32-bit data register, output 32-bit data register
- Hardware computation done in 3 HCLK cycles
- Provides 32-bit data register (can be used for temporary calculation storage)
- Supports CRC-32 and CRC-32/MPEG-2 algorithms
- Supports to select the input/output to be big-endian or little-endian

6.3 Functional description

6.3.1 Functional block diagram

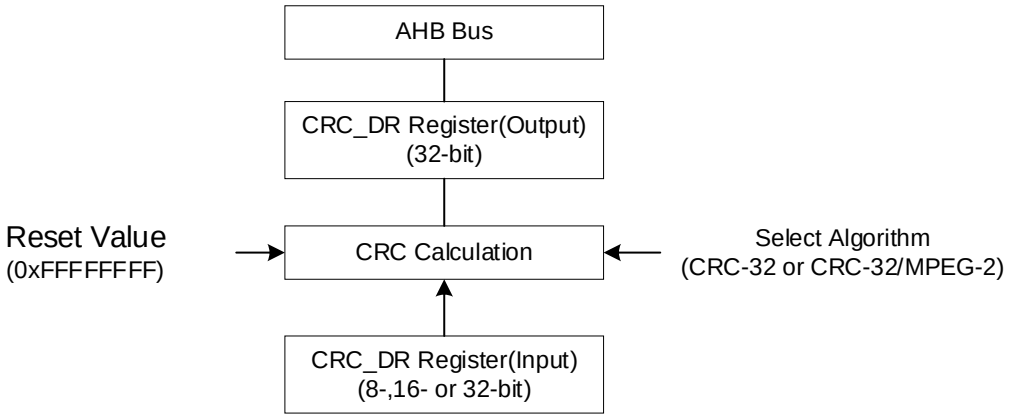


Figure 6-1 CRC functional block diagram

6.3.2 Functional description

- CRC calculator is conducted by the write operation to the CRC data register (CRC_DR). A new CRC calculation result is created from the CRC calculator of the previous CRC result and the

new write. Such result will be stored in the CRC data register (CRC_DR).

- CRC calculator allows consecutive write accesses against the CRC data register (CRC_DR) for calculation. Finally, the CRC calculation result is gained by reading the CRC data register (CRC_DR).
- The CRC data register (CRC_DR) and CRC middle data register (CRC_MIR) can be reset to the default value of 0xFFFFFFFF by setting the RST bit in the CRC control register (CRC_CR). This operation does not affect the contents of the CRC independent data register (CRC_IDR).
- An 8-bit independent data register (CRC_IDR) is provided as a buffered register. Write and read accesses to it can be conducted at any time.
- Different CRC algorithms can be set by using the AS bit in the CRC control register (CRC_CR).
- The bit width for input can be selected by the ISIZE bit in the CRC control register (CRC_CR). When the input is 8 bits, the value range of the CRC data register (CRC_DR) is [0x00, 0xFF]; when the input is 16 bits, the value range of the CRC data register (CRC_DR) is [0x0000, 0xFFFF]; when the input is 32 bits, the value range of the CRC data register (CRC_DR) is [0x00000000, 0xFFFFFFFF].
- The IES bit in the CRC control register (CRC_CR) can be used to configure the input to be big-endian or little-endian while the OES bit to configure the output to be big-endian or little-endian.
- Interleaved CRC calculation between multiple strings of data can be conducted by using the CRC middle data register (CRC_MIR).

6.3.3 Usage method

6.3.3.1 CRC calculation procedures

- Enable the CRC module clock ;
- Reset the CRC module ;
- Configuration the CRC_CR register:
- Select the algorithm: CRC-32 or CRC-32/MPEG-2;
- Select the bit width for input: 8, 16 or 32 bits;

(Example: if a value is 0x00000012, CRC_DR should assign 0x12 for 8-bit input, 0x0012 for 16-bit input, 0x00000012 for 32-bit input)

- Select the input/output to be big-endian or little-endian.

(Example: for 0x12345678, it is 0x78563412 as big-endian, 0x12345678 as little-endian)

- Restore the CRC to its initial state by configuring the RST bit in the CRC control register (CRC_CR);
- Write data to the CRC data register (CRC_DR) consecutively.

- The CRC calculation result is gained by reading the CRC data register (CRC_DR).

6.3.3.2 Interleaved CRC calculation procedures

To insert a second dataset before the first dataset has been calculated:

- Read the intermediate result of the first calculation from the CRC middle data register (CRC_MIR) and store the result;
- Reset the CRC data register (CRC_DR) by setting the RST bit in the CRC control register (CRC_CR).
- Write the second dataset to the CRC data register (CRC_DR) sequentially. After the CRC calculation is complete, read the CRC calculation result of the second dataset from the CRC data register (CRC_DR) and store the result;
- Again, reset the CRC data register (CRC_DR) by setting the RST bit in the CRC control register (CRC_CR), and set the CRC calculation mode to the same mode as that of the first dataset;
- Write the intermediate result of the first storage to the CRC middle data register (CRC_MIR), and then write the remaining data in the first dataset to the CRC data register (CRC_DR) sequentially. The CRC will continue to calculate the unfinished data of the first dataset automatically. When the calculation is completed, the CRC calculation result of the first dataset from the CRC data register (CRC_DR) can be read.

6.4 Register

6.4.1 Overview of registers

Table 6-1 Overview of CRC registers

Offset	Acronym	Register Name	Reset
0x00	CRC_DR	CRC Data Register	0xFFFFFFFF
0x04	CRC_IDR	CRC Independent Data Register	0x00000000
0x08	CRC_CR	CRC Control Register	0x00000000
0x0C	CRC_MIR	CRC Middle Data Register	0xFFFFFFFF

6.4.2 CRC_DR CRC Data Register

Address offset : 0x00

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR															
rw															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR															
rw															

Bit	Field	Description
31:0	DR	Data Register This register is used to write new data to the CRC calculator. It holds the previous CRC calculation result when it is read.

6.4.3 CRC_IDR CRC Independent Data Register

Address offset : 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								IDR							
								rw							

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value
7:0	IDR	General-purpose 8-bit Data Register Can be used as a temporary storage location for one byte. This register is not affected by the RST bit in the CRC control register (CRC_CR).

6.4.4 CRC_CR CRC Control Register

Address offset : 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.										OES	IES	ISIZE		AS	RST
										rw	rw	rw		rw	w

Bit	Field	Description
31:6	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
5	OES	Output Endian Selection 0: Little-endian 1: Big-endian
4	IES	Input Endian Selection 0: Little-endian 1: Big-endian
3:2	ISIZE	Input Size 00: 32-bit input 01: 16-bit input 10: 8-bit input 11: Reserved
1	AS	CRC Algorithm Selection 0: CRC-32/MPEG-2 algorithm 1: CRC-32 algorithm
0	RST	CRC Reset The CRC data register (CRC_DR) and CRC middle data register (CRC_MIR) can be reset to 0xFFFF FFFF. This bit can only be set to '1', and it is automatically cleared to '0' by hardware.

6.4.5 CRC_MIR CRC Middle Data Register

Address offset : 0x0C

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MIR															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIR															
rw															

Bit	Field	Description
31:0	MIR	Middle Data Register It is used to read the intermediate result of the calculation. Both the write and read accesses are in the 32-bit mode. If the read value is rewritten, the previous CRC calculation would continue. If the CRC calculations for multiple sequences of data are conducted, the calculation results can be retrieved and saved; when the other CRC calculation is finished, you can write the stored data to the register and continue with the unfinished calculation.

7 GPIO General-Purpose I/Os

7.1 Overview

Each general-purpose IO (GPIO) port, can be individually configured by two 32-bit control registers (GPIOx_CRL/GPIOx_CRH) and two 32-bit multiplexed control registers (GPIOx_AFRL, GPIOx_AFRH) in eight modes: analog input、input floating、input pull-up、input pull-down、output push-pull、output open-drain、alternate function push-pull and alternate function open-drain.

Each I/O port bit is freely programmable. All registers can be accessed as 32-bit (words), 16-bit (half-words) or 8-bit (bytes). The GPIOx_BSRR and GPIOx_BRR control registers in the GPIO register group are able to modify the GPIOx_ODR bit to output 0 or 1 independently through write accesses.

7.2 Main characteristics

- A single AHB write access changes one or multiple bits for GPIOx_ODR
- All I/Os support programmable EXTI configuration register to output externally triggered interrupts
- Support GPIO locking mechanism
- Supported input states: floating, pull-up/down, analog
- Supported output states: push-pull or open drain with pull-up/down
- Input floating as default, configurable input/output direction
- Configurable input/output speed

7.3 Functional description

7.3.1 Functional block diagram

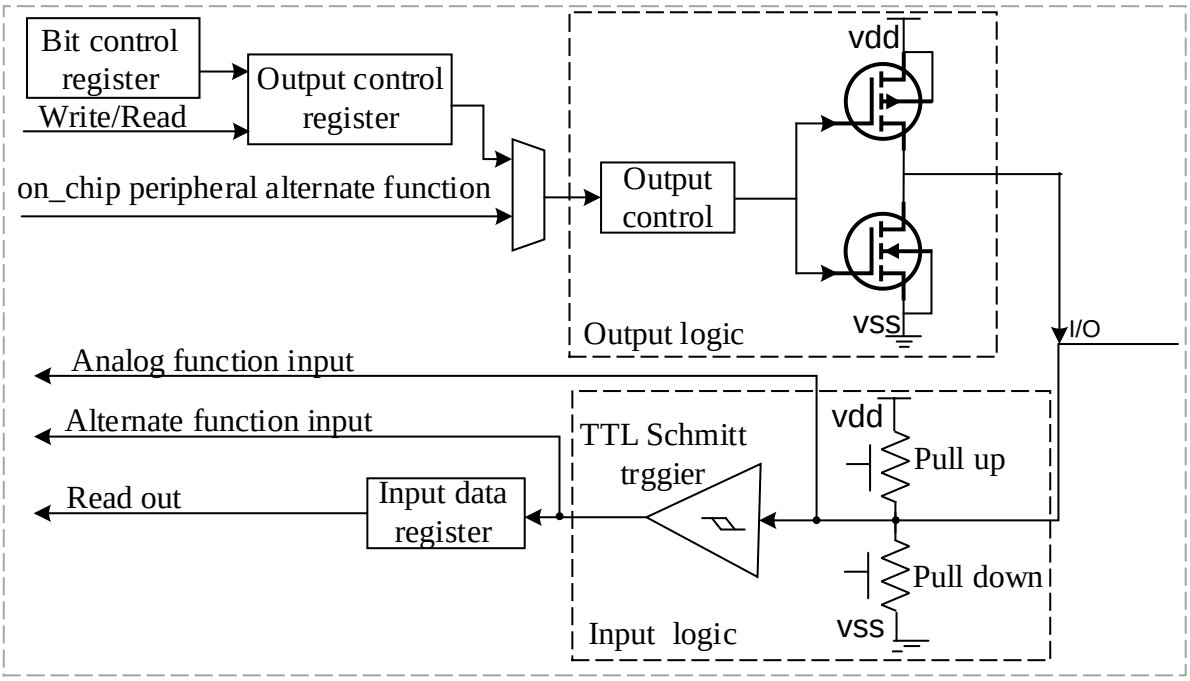


Figure 7-1 Standard I/O port

7.3.2 GPIO port configuration

Table 7-1 Port bit configuration table (take port0 as an example)

Pin_mode		Pull up/down	DCR[1:0]		CNF0		MODE0	ODRx
Analog input		x	x	x	0	0	00	x
General input Alternate input		Floating	x	x	0	1		x
		Pull-up	x	x	1	0		1
		Pull-down	x	x	1	0		0
General output	Push-pull	x	x	x	0	0	01 10 11	0 or 1
	Open-drain	Floating	x	0	0	1		0 or 1
		Pull-up	1	1	0	1		0 or 1
		Pull-down	0	1	0	1		0 or 1
Alternate output	Push-pull	x	x	x	1	0		x
	Open-drain	Floating	x	0	1	1		x
		Pull-up	1	1	1	1		x
		Pull-down	0	1	1	1		x

Note: x represents “don’t care” for I/Os in the corresponding mode. ODR0 represents the bit 0 of the output data register.

Input and output follow these configurations:

- General-purpose input:

The user should configure the CNF0 in the GPIOx_CRL to choose the Input mode.

- General-purpose output:

Push-Pull output: The user configures MODE0 to choose the output speed and sets CNF0=00;

Open-Drain output: The user configures MODE0 to choose the output speed and sets CNF0=01. To activate the pull-up/pull-down feature for pins, the GPIOx_DCR register should be configured individually. This feature is invalid if it is not an output open-drain mode.

- Alternate function:

Configure the AFRLx[3:0] and AFRHx[3:0] registers to choose the alternate function:

Alternate push-pull output: The user configures MODE0 to choose the output speed and sets CNF0=10;

Alternate open-drain output: The user configures MODE0 to choose the output speed and sets CNF0=11.

To activate the pull-up/pull-down feature for IO in the output mode, the GPIOx_DCR register should be configured individually. This feature is invalid if it is not an output open-drain mode.

During and just after reset, the GPIO ports are configured in Input Floating mode. The Serial-Wired Debug pins default to input pull-up (PU)/pull-down (PD).

When configured in general purpose output mode, the value of the output data register (GPIOx_ODR) is output on the corresponding I/O pin. The Input Data register (GPIOX_IDR) captures the data present on the I/O pin at every AHB clock cycle.

Note: Not all chips contain the JTAG and SWD ports. For specific chip configuration, refer to the chip datasheet.

- PA14: SWCLK in PD
- PA13: SWDIO in PU

7.3.3 Alternate function

The corresponding IO alternate functions are enabled by setting the alternate function register.

- When an IO is configured as Alternate Function Input, the port should choose pull-up, pull-down or floating input.
- When an IO is configured as Alternate Function Output, the port should choose output push-pull or open-drain mode.

- When an IO is configured with bidirectional alternate function, the port should choose output push-pull or open-drain mode. In this case the input is configured in input floating mode. In the open-drain mode, the GPIOx_DCR register is configured to choose weak pull-up or weak pull-down resistor.

If a port bit is configured as Alternate Function Output, this connects the port to the output signal of an on-chip peripheral. If a GPIO pin is configured as Alternate Function Output solely via software, but peripheral is not activated, its output is not specified.

7.3.4 GPIO locking mechanism

It is possible to freeze the IO configuration by applying the GPIO locking mechanism. When the LOCK mechanism has been applied on a port, it is no longer possible to modify the port configuration until the next reset. The write sequence of the LOCK key:

- GPIOx_LCKR[16]='1'+LCKR[15:0].
- GPIOx_LCKR[16]='0'+LCKR[15:0].
- GPIOx_LCKR[16]='1'+LCKR[15:0].

To lock the PA[0] port of GPIOA, follow these configurations:

- GPIOA->GPIOA_LCKR=0x10001.
- GPIOA->GPIOA_LCKR=0x00001.
- GPIOA->GPIOA_LCKR=0x10001.

After three steps above are finished, the bit 16 of the GPIOA_LCKR register is set. The GPIOA_LCKR register can no longer be written until the next soft reset. The bit 16 of the GPIOA_LCKR register remains 1 and PA[0] keeps its configuration before the locking.

When the port is locked, the value of the port bit can no longer be modified until the soft reset. Each lock bit in the GPIOx_LCKR register freezes 4 bits in the port configuration registers (GPIOx_CRL and GPIOx_CRH).

Notes:

Only the value of PA[0] is locked due to the proceeding configuration. It is still possible to configure the values of PA[15:1] and other GPIO control registers.

7.3.5 Input configuration

When the I/O port is programmed as Input:

- The Schmitt Trigger Input is activated.
- The Output Buffer is disabled.
- Floating, pull-up or pull-down input mode can be used.

- The data present on the I/O pin is sampled into the Input Data register every AHB clock cycle.
- A read access to the Input Data register gets the I/O state.

The figure below shows the input configuration of the I/O port:

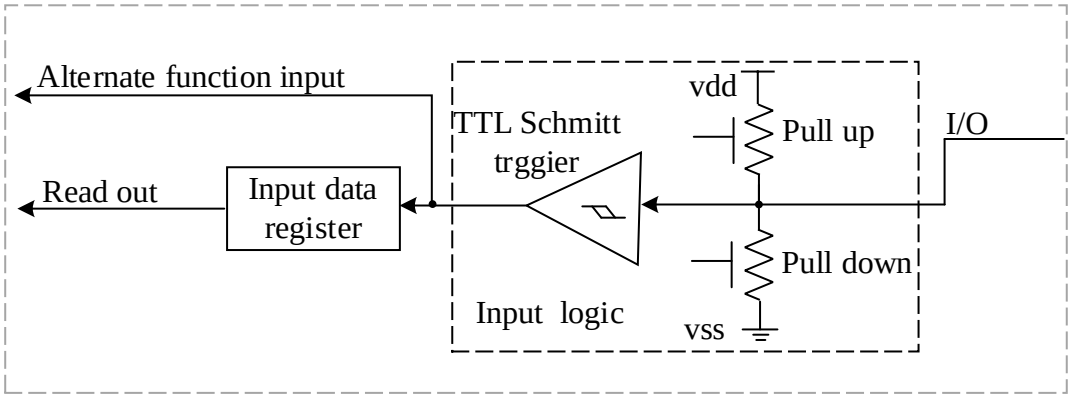


Figure 7-2 Floating/pull-up/pull-down input configuration

To configure the PA[0] as pull-up input in the GPIOA, refer to the following:

- GPIOA->GPIOA_ODR=0x0001.
- GPIOA->GPIOA_CRL=0x00000008.

To configure the PA[0] as pull-down input in the GPIOA, refer to the following:

- GPIOA->GPIOA_ODR=0x0000.
- GPIOA->GPIOA_CRL=0x00000008.

Notes:

To configure the port as pull-up input, the corresponding bit in the GPIO_ODR register should output 1 firstly.

To configure the port as pull-down input, the corresponding bit in the GPIO_ODR register should output 0 firstly.

7.3.6 Output configuration

When the GPIO pin is configured as output:

- The Schmitt Trigger Input is activated.
- The Output Buffer is enabled.
- In general purpose output mode, the weak pull-up and pull-down resistors are disabled.
- Open Drain Mode: When the Port Output Data Register is set to 0, the corresponding pin outputs low level; when the Port Output Data Register is set to 1, the corresponding pin is in Hi-Z.
- Push-Pull Mode: When the output register is set to 0, the corresponding pin outputs low level; when the output register is set to 1, the corresponding pin outputs high level.
- A read access to the Port Output Data register gets the last written value.

- A read access to the Port Input Data register gets the current I/O state.

The figure below shows the output configuration of the I/O port:

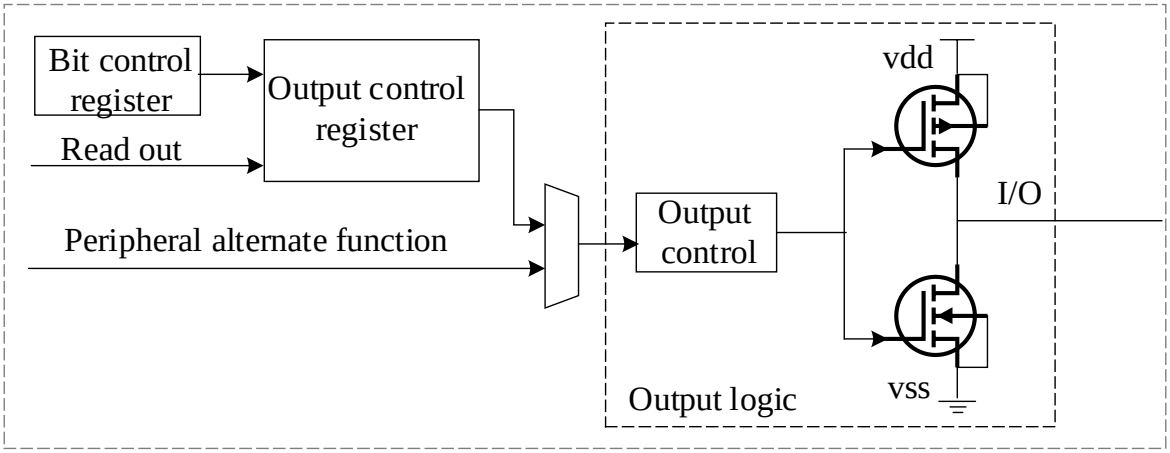


Figure 7-3 Output configuration

7.3.7 Alternate function configuration

When the pin is configured as alternate function:

- The Schmitt Trigger Input is activated.
- The Output Buffer can be configured as Open Drain or Push-Pull.
- In the output open-drain mode, the GPIOx_DCR register is configured to choose weak pull-up or weak pull-down resistor.
- The weak pull-up or weak pull-down resistor can be chosen when the pin is configured as Input.
- The data present on the I/O pin is sampled into the Input Data register every AHB clock cycle.

The figure below shows the alternate function configuration of the I/O port. Refer to AFRL and AFRH registers as well as the datasheet for further information.

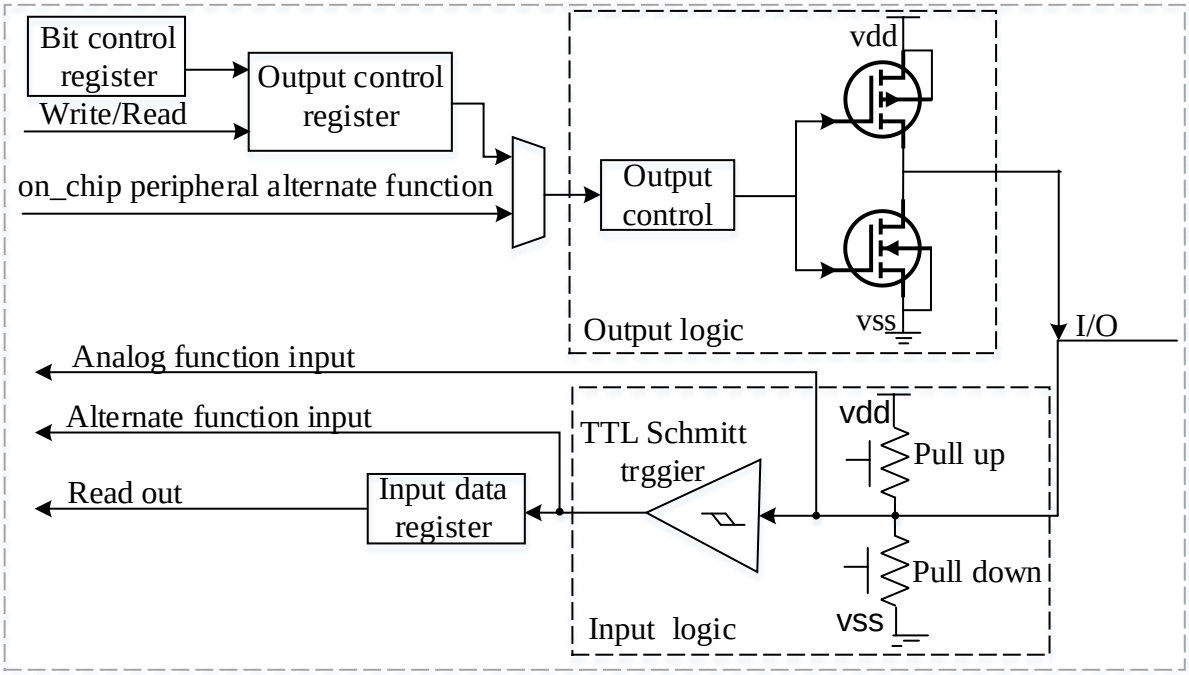


Figure 7-4 Alternate function configuration

7.3.8 Analog input configuration

When the I/O port is configured as analog input configuration:

- The Output Buffer is disabled.
- The Schmitt Trigger Input is de-activated.
- The weak pull-up and pull-down resistors are disabled.
- The Port Input Data Register remains 0.

The figure below shows the analog input configuration of the I/O port:

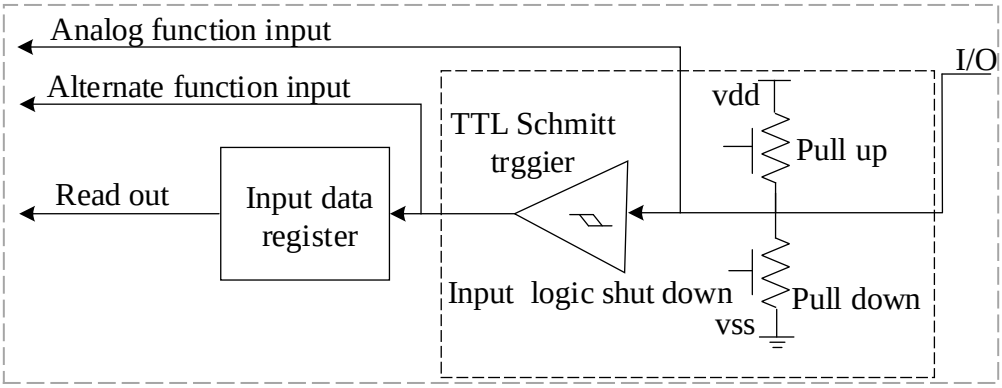


Figure 7-5 Analog input

7.3.9 External clock used as alternate GPIO port

The external HSE/LSE clock can be used as alternate GPIO. When the corresponding clock PAD is used as the GPIO, the user should turn off the external clock first and then set the GPIO function as normal. For specific mapping, please refer to the chip datasheet.

7.3.10 NRST alternate function remapping

For GPIO with NRST function, it is used as NRST function by default. When it needs to be used as GPIO function, it should be first configure the flash option byte register (FLASH_OBR) with the OBR_RSTN bit equal to 0, and then after the system reset, the configuration takes effect.

For GPIO with NRST function, its chip is connected with a pull resistor by default. When it is used for NRST function, it is forbidden to configure GPIO controller register with this pin to avoid affecting reset function. Keep the default Settings.

7.3.11 SWD alternate function remapping

The SWD interface signals are mapped on the GPIO ports as shown in the table below:

Table 7-2 SWD alternate function remapping

Alternate function	GPIO port
SWDIO	PA13
SWCLK	PA14

7.4 GPIO register description

7.4.1 Overview of registers

Table 7-3 Overview of GPIO registers

Offset	Acronym	Register Name	Reset
0x00	GPIOx_CRL	Port configuration register low	0x44444444
0x04	GPIOx_CRH	Port configuration register high	0x44444444
0x08	GPIOx_IDR	Port input data register	0x0000XXXX
0x0C	GPIOx_ODR	Port output data register	0x00000000
0x10	GPIOx_BSRR	Port bit set/reset register	0x00000000
0x14	GPIOx_BRR	Port bit reset register	0x00000000
0x18	GPIOx_LCKR	Port configuration lock register	0x00000000
0x1C	GPIOx_DCR	Port output open drain control register	0x00000000
0x20	GPIOx_AFRL	Port alternate function register low	0x00000000
0x24	GPIOx_AFRH	Port alternate function register high	0x00000000

Note: The possible range of "X" in GPIOx is A to I, but not all chips include all GPIOA to GPIOI groups. The specific configuration of each chip can be referred to the data manual of each chip.

7.4.2 GPIOx_CRL Port Configuration Register Low

Address offset: 0x00

Reset value: 0x4444 4444

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF7		MODE7		CNF6		MODE6		CNF5		MODE5		CNF4		MODE4	
rw		rw		rw		rw		rw		rw		rw		rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF3		MODE3		CNF2		MODE2		CNF1		MODE1		CNF0		MODE0	
rw		rw		rw		rw		rw		rw		rw		rw	

Bit	Field	Description
31:30	CNF7	Port configuration bits (y=7..0)
27:26	CNF6	Set MODEy to be 0 and the port as input mode while configuring the CNFy bit to choose the input mode:
23:22	CNF5	00: Analog input mode
19:18	CNF4	01: Input floating mode
15:14	CNF3	10: Input pull-up/pull-down mode
11:10	CNF2	11: Reserved
7:6	CNF1	Set MODEy not to be 0 and the port as output mode while configuring the CNFy bit to choose the output mode:
3:2	CNF0	00: General purpose output push-pull
29:28	MODE7	01: General purpose output open-drain
25:24	MODE6	10: Alternate function output push-pull
21:20	MODE5	11: Alternate function output open-drain
17:16	MODE4	Port input/output configuration (MODEy)(y = 0..7)
13:12	MODE3	The corresponding I/O port is configured via software; refer to Port bit configuration table.
9:8	MODE2	Set MODEy is not equal to 0, different configurations result in different rates:
5:4	MODE1	00: input mode;
1:0	MODE0	As for the other configuration output speeds, please refer to the datasheet.

7.4.3 GPIOx_CRH Port Configuration Register High

Address offset: 0x04

Reset value: 0x4444 4444

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF15		MODE15		CNF14		MODE14		CNF13		MODE13		CNF12		MODE12	

rw		rw		rw		rw		rw		rw		rw		rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF11		MODE11		CNF10		MODE10		CNF9		MODE9		CNF8		MODE8	
rw		rw		rw		rw		rw		rw		rw		rw	

Bit	Field	Description
31:30	CNF15	Port configuration bits (y=15..8) Set MODEy to be 0 and the port as input mode while configuring the CNFy bit to choose the input mode: 00: Analog input mode 01: Input floating mode 10: Input pull-up/pull-down mode 11: Reserved Set MODEy not to be 0 and the port as output mode while configuring the CNFy bit to choose the output mode: 00: General purpose output push-pull 01: General purpose output open-drain 10: Alternate function output push-pull 11: Alternate function output open-drain Port input/output configuration (MODEy)(y = 15..8) The corresponding I/O port is configured via software; refer to Port bit configuration table. Set MODEy is not equal to 0, different configurations result in different rates: 00: input mode; As for the other configuration output speeds, please refer to the datasheet.
27:26	CNF14	
23:22	CNF13	
19:18	CNF12	
15:14	CNF11	
11:10	CNF10	
7:6	CNF9	
3:2	CNF8	
29:28	MODE15	
25:24	MODE14	
21:20	MODE13	
17:16	MODE12	
13:12	MODE11	
9:8	MODE10	
5:4	MODE9	
1:0	MODE8	

7.4.4 GPIOx_IDR Port Input Data Register

Address offset: 0x08

Reset value: 0x0000 XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDRY(y=15~0)															
r															

Bit	Field	Description
31:16	Reserved	Always read as 0
15:0	IDRY	Port input data (y=15..0) The reads represent the corresponding I/O states.

7.4.5 GPIOx_ODR Port Output Data Register

Address offset: 0xC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODRY(y=15~0)															
rw															

Bit	Field	Description
31:16	Reserved	Always read as 0
15:0	ODRY	Port output data (y=15..0) When configured in general purpose output mode, the written value is output to the corresponding I/O. Note: The ODR bits can be individually set and cleared by setting the GPIOx_BSRR (x=A..H) Register.

7.4.6 GPIOx_BSRR Port Bit Set/Reset Register

Address offset: 0x10

Reset value: 0x0000 000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BRy(y=15~0)															
w															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BSy(y=15~0)															
w															

Bit	Field	Description
31:16	BRy	Port Reset bit y (y=15..0) Writing it to 0 will make the corresponding ODRY bit unchanged Writing it to 1 will clear the corresponding ODRY bit
15:0	BSy	Port Set bit y (y=15..0) Writing it to 0 will make the corresponding ODRY bit unchanged Writing it to 1 will set the corresponding ODRY bit to 1 Note: When both BSy and BRy bits are written to 1, BSy has priority over BRy.

7.4.7 GPIOx_BRR Port Bit Reset Register

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BRy(y=15~0)															
w															

Bit	Field	Description
31:16	Reserved	Always read as 0
15:0	BRy	Port Reset bit y (y=15..0) Writing it to 0 will make the corresponding ODRY bit unchanged Writing it to 1 will clear the corresponding ODRY bit

7.4.8 GPIOx_LCKR Port Configuration Lock Register

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res															LCKK
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCKy(y=15~0)															
w															

Bit	Field	Description
31:17	Reserved	Always read as 0
16	LCKK	Lock key This bit can be read anytime. It can only be modified using the Lock Key Writing Sequence. 0: port configuration lock key not active 1: port configuration lock key active. GPIOx_LCKR register is locked until the next soft reset. LOCK key sequence: write 1 -> write 0 -> write 1

Bit	Field	Description
15:0	LCKy	Port x Lock bit y (y = 15..0) These bits are read write but can only be written when the LCKK bit is 0. 0: port configuration not locked 1: port configuration locked

7.4.9 GPIOx_DCR Port Output Open Drain Control Register

Address offset: 0x1C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PX15		PX14		PX13		PX12		PX11		PX10		PX9		PX8	
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PX7		PX6		PX5		PX4		PX3		PX2		PX1		PX0	
rw															

Bit	Field	Description
31:2	PX15-PX1	See PX0
1:0	PX0	PX0[1:0]: 11: output open-drain mode with port pull-up 01: output open-drain mode with port pull-down x0: output open-drain mode without port pull-up/pull-down

7.4.10 GPIOx_AFRL Port Alternate Function Register Low

Address offset: 0x20

Reset value: 0xFFFF FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFR7				AFR6				AFR5				AFR4			
rw				rw				rw				rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFR3				AFR2				AFR1				AFR0			
rw				rw				rw				rw			

Bit	Field	Description
31:0	AFRy	Port x alternate function bit y (y = 0..7), which can be accessed by software to configure the function. 0000: AF0 0001: AF1 0010: AF2 0011: AF3 0100: AF4 0101: AF5 0110: AF6 0111: AF7 1000: AF8 1001: AF9 1010: AF10 1011: AF11 1100: AF12 1101: AF13 1110: AF14 1111: AF15

7.4.11 GPIOx_AFRH Port Alternate Function Register High

Address offset: 0x24

Reset value: GPIOx_AFRH (x = B..I): 0xFFFF FFFF，GPIOA_AFRH: 0xF00F FFFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFR15				AFR14				AFR13				AFR12			
rw				rw				rw				rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFR11				AFR10				AFR9				AFR8			
rw				rw				rw				rw			

Bit	Field	Description
31:0	AFRy	Port x alternate function bit y (y = 8..15), which can be accessed by software to configure the function. 0000: AF0 0001: AF1 0010: AF2 0011: AF3 0100: AF4 0101: AF5 0110: AF6 0111: AF7 1000: AF8 1001: AF9 1010: AF10 1011: AF11 1100: AF12 1101: AF13 1110: AF14 1111: AF15

8 EXTI Interrupt and Event

8.1 Overview

Nestled vector interrupt controller (NVIC) connects the processor core, and manages the anomaly and interrupt processing with low latency. NVIC includes 4-bit interrupt priority levels, which can thus provide 16 interrupt priority levels. For other more anomalies and NVIC programming details, please refer to the Cortex®-M0 Technology Reference Manual.

The EXTI module includes the edge detection circuit, which can produce the interrupt request or wake-up event, and the edge detection supports the rising edge, falling edge or any edge configuration. Each edge detection circuit supports the independent enable and mask.

8.2 Main characteristics

- Independently trigger and mask each interrupt
- Configure interrupt/event output by software
- Produce the wake up event to wake up the low power mode
- The pending register saves the status of each corresponding interrupt line
- All GPIOs supports the trigger source configured as EXTI
- Support the rising edge trigger, falling edge trigger and any edge trigger

8.3 Functional description

8.3.1 Function block diagram

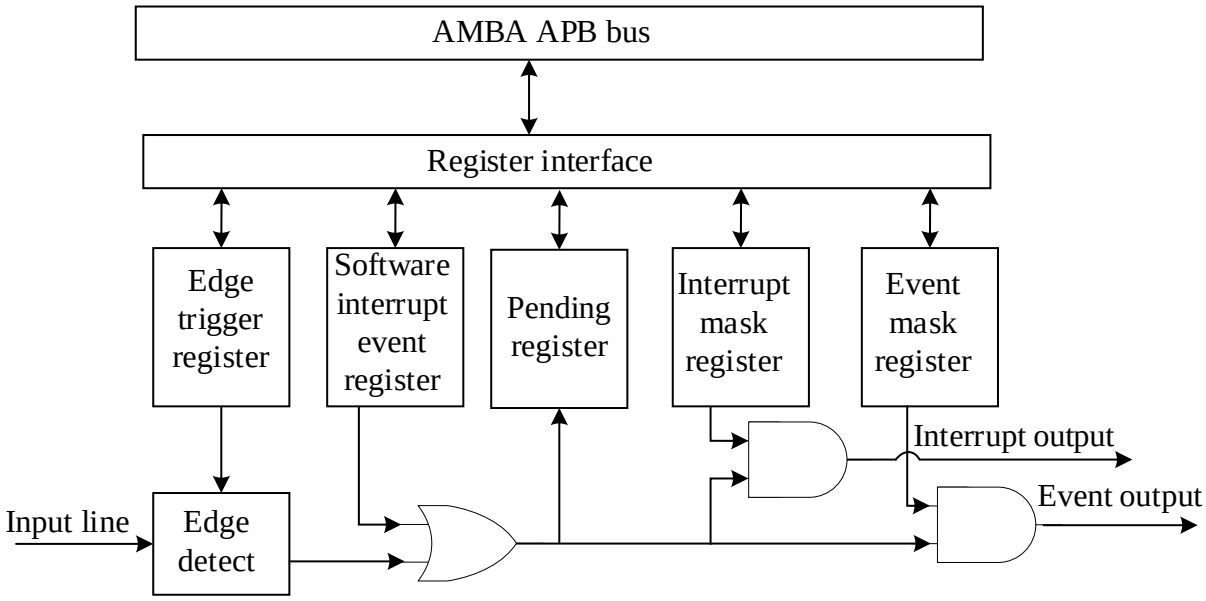


Figure 8-1 EXTI structure block diagram

8.3.2 Interrupt and anomaly vector

Under Handler mode, Cortex®-M0 processor and nested interrupt vector control (NVIC) can process all anomalies based on priority. In case of anomaly, the system will pop the current processing work stack after interrupt service.Simultaneous processing of vector and current work status improves the interrupt efficiency. The table below presents the anomaly type and interrupt vector.

Table 8-1 Abnormal vector

Position	Priority	Priority type	Name	Description	Address
				Reserved	0x0000 0000
	-3	Fixed	Reset	Reset	0x0000 0004
	-2	Fixed	NMI	Unmaskable interrupt RCC Clock security system (CSS) connected to NMI	0x0000 0008
	-1	Fixed	Hardware fault	All types of faults	0x0000 000C

Table 8-2 Interrupt vector

Position	Priority	Priority type	Name	Description	Address
	3	Settable	SVCall	System service call via SWI instruction	0x0000 002C
	-	-		Reserved	0x0000 0030

Position	Priority	Priority type	Name	Description	Address
	-	-		Reserved	0x0000 0034
	6	Settable	PendSV	Pending system service	0x0000 0038
	7	Settable	SysTick	System tick timer	0x0000 003C
0	8	Settable	WWDG_IWDG	Window watchdog interrupt and independent watchdog interrupt connected to EXTI17	0x0000_0040
1	9	Settable	PVD_VDT	Power voltage detection (PVD) connected to EXTI16	0x0000_0044
2	10	-		Reserved	0x0000_0048
3	11	Settable	FLASH	Flash global interrupt	0x0000_004C
4	12	Settable	RCC	RCC global interrupt	0x0000_0050
5	13	Settable	EXTI0_1	EXTI line [1:0] interrupt	0x0000_0054
6	14	Settable	EXTI2_3	EXTI line [3:2] interrupt	0x0000_0058
7	15	Settable	EXTI4_15	EXTI line [15:4] interrupt	0x0000_005C
8	16	-		Reserved	0x0000_0060
9	17	-		Reserved	0x0000_0064
10	18	-		Reserved	0x0000_0068
11	19	-		Reserved	0x0000_006C
12	20	Settable	ADC1	ADC1 global interrupt	0x0000_0070
13	21	Settable	TIM1_BRK_UP_TRG_COM	TIM1 brake refresh trigger COM interrupt	0x0000_0074
14	22	Settable	TIM1_CC	TIM1 capture compare interrupt	0x0000_0078
15	23	-		Reserved	0x0000_007C
16	24	Settable	TIM3	TIM3 global interrupt	0x0000_0080
17	25	-		Reserved	0x0000_0084
18	26	-		Reserved	0x0000_0088
19	27	Settable	TIM14	TIM14 global interrupt	0x0000_008C
20	28	-		Reserved	0x0000_0090
21	29	-		Reserved	0x0000_0094
22	30	-		Reserved	0x0000_0098
23	31	Settable	I2C1	I2C1 global interrupt	0x0000_009C
24	32	-		Reserved	0x0000_00A0
25	33	Settable	SPI1	SPI1 global interrupt	0x0000_00A4
26	34	-		Reserved	0x0000_00A8
27	35	Settable	UART1	UART1 global interrupt	0x0000_00AC
28	36	Settable	UART2	UART2 global interrupt	0x0000_00B0
29	37	-		Reserved	0x0000_00B4
30	38	-		Reserved	0x0000_00B8

8.3.3 Wake-up event management

All EXTI lines support the generation of interrupt or the event is used to wake the system from the low power mode. The user executes the WFE instruction to enter the corresponding low power mode. Configure the EXTI line event output to wake up the system. The user executes the WFI to enter the low power mode, and configures EXTI line interrupt output to wake up the system. For the detailed configuration, please refer to the section of power control.

8.3.4 Interrupt functional description

To enable interrupt function and produce interrupt, configure the edge detection trigger register as the required trigger type, and open the corresponding bit of interrupt mask register allowing interrupt request. When the corresponding external interrupt line detects the configured trigger condition, it produces an interrupt request and the corresponding pending position of register is 1. Write the corresponding bit of pending register as 1 and clear the interrupt.

To configure the generation event, configure the edge detection trigger register as the required trigger type, and open the corresponding bit of interrupt mask register allowing interrupt request. When the corresponding external interrupt line detects the configured trigger condition, it produces an interrupt request.

Enabling the corresponding bit of the software interrupt/event register can also produce the interrupt/event request.

8.3.5 Hardware interrupt output

Specific steps to configure hardware interrupt source:

- Open the mask bit of the corresponding interrupt line (EXTI IMR), enable interrupt.
- Configure the trigger register bit (EXTI_RISE/EXTI_FALL) of the corresponding interrupt line;
- Open the NVIC connected interrupt channel so that the interrupt request can be transmitted to CPU and correctly responded.

When configuring EXTIx (x=31~0) line and produce interrupt output, the corresponding bit of EXTI_PR register will be set 1. It's required to clear the corresponding pending bit of EXTI_PR register before again detecting the rolling of EXTIx (x=31~0) line, and producing the interrupt.

There are three ways to clear the pending bit of EXTI_PR register:

- Write the pending bit of EXTI_PR register as 1.
- When configuring the rising edge trigger selection register (EXTI_RTSR), writing 0 to the corresponding bit will clear the pending bit. When configuring the falling edge trigger selection register (EXTI_FTSR), writing 0 in the corresponding bit will clear the pending bit.

- Clear by changing the edge detection polarity of EXTI line.

8.3.6 Hardware event output

Specific steps to configure hardware event source:

- Open the mask bit (EXTI_EMR) of the corresponding event line;
- Configure the trigger register bit (EXTI_RISE/EXTI_FALL) of the corresponding event line.

8.3.7 Software interrupt and event output

Specific steps to support configuration of interrupt and event by software:

- Enable event or interrupt enable bit (EXTI IMR, EXTI EMR) .
- Configure the corresponding bit of the software interrupt event register as 1 (EXTI SWIER).

8.3.8 External interrupt mapping

All GPIOs are used as the EXTI trigger source for producing the interrupt or event request. Configure SYSCFG_EXTICRx register, and meanwhile support the internal module (including PVD, RTC, USB, comparator, IWDG) trigger. The specific connection relationships are shown as below:

Table 8-3 EXTI trigger source

External interrupt line	IO mapping	Control bit
EXTI0	PA0;PB0	SYSCFG_EXTICR1 register EXTI0
EXTI1	PA1;PB1	SYSCFG_EXTICR1 register EXTI1
EXTI2	PA2	SYSCFG_EXTICR1 register EXTI2
EXTI3	PA3	SYSCFG_EXTICR1 register EXTI3
EXTI4	PA4	SYSCFG_EXTICR2 register EXTI4
EXTI5	PA5	SYSCFG_EXTICR2 register EXTI5
EXTI6	PA6	SYSCFG_EXTICR2 register EXTI6
EXTI7	PA7	SYSCFG_EXTICR2 register EXTI7
EXTI8	PA8	SYSCFG_EXTICR3 register EXTI8
EXTI9	PA9	SYSCFG_EXTICR3 register EXTI9
EXTI10	PA10	SYSCFG_EXTICR3 register EXTI10
EXTI11	PA11	SYSCFG_EXTICR3 register EXTI11
EXTI12	PA12	SYSCFG_EXTICR4 register EXTI12
EXTI13	PA13	SYSCFG_EXTICR4 register EXTI13
EXTI14	PA14	SYSCFG_EXTICR4 register EXTI14
EXTI15	PA15	SYSCFG_EXTICR4 register EXTI15

Other external interrupt/event controller connection:

- EXTI line16 connected to PVD output

- EXTI line17 connected to IWDG output

8.4 Register description

8.4.1 Register overview

Table 8-4 EXTI register overview

Offset	Acronym	Register Name	Reset
0x00	EXTI_IMR	Interrupt mask register	0x00000000
0x04	EXTI_EMR	Event mask register	0x00000000
0x08	EXTI_RTSR	Rising edge trigger selection register	0x00000000
0x0C	EXTI_FTSR	Falling edge trigger selection register	0x00000000
0x10	EXTI_SWIER	Software interrupt event register	0x00000000
0x14	EXTI_PR	Pending register	0x00000000

8.4.2 EXTI_IMR Interrupt Mask Register

Address offset:0x0

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IMRx(x=31~16)															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IMRx(x=15~0)															
rw															

Bit	Field	Description
31:0	IMRx	Line x interrupt enable bit 1: Configure the bit as 1, enable line x corresponding interrupt 0: Configure the bit as 0, disable line x corresponding interrupt

8.4.3 EXTI_EMR Event Mmask Register

Address offset:0x04

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EMRx(x=31~16)															

rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EMRx(x=15~0)															
rw															

Bit	Field	Description
31:0	EMRx	Line x event enable bit 1: Configure the bit as 1, enable line x corresponding event 0: Configure the bit as 0, disable line x corresponding event

8.4.4 EXTI_RTSR Rising Edge Trigger Selection Register

Address offset:0x08

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TRx(31~16)															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRx(15~0)															
rw															

Bit	Field	Description
31:0	TRx	Line x corresponding interrupt or event trigger polarity 1: Configure the bit as 1, enable line x corresponding rising edge trigger interrupt or event 0: Configure the bit as 0, disable line x corresponding rising edge trigger interrupt or event

8.4.5 EXTI_FTSR Falling Edge Trigger Selection Register

Address offset:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TRx(31~16)															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRx(15~0)															
rw															

Bit	Field	Description
31:0	TRx	Line x corresponding interrupt or event trigger polarity 1: Configure the bit as 1, enable line x corresponding falling edge trigger interrupt or event 0: Configure the bit as 0, disable line x corresponding falling edge trigger interrupt or event

8.4.6 EXTI_SWIER Software Interrupt Event Register

Address offset:0x10

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWIERx(31~16)															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIERx (15~0)															
rw															

Bit	Field	Description
31:0	SWIERx	Line x software interrupt or event enable Writing 1 will set the corresponding pending bit of EXTI_PR register. Meanwhile, configuring the corresponding bit of EXTI_IMR or EXTI_EMR as 1 can produce interrupt or event.
		Note: Writing 1 to the corresponding bit of EXTI_PR register may clear the bit

8.4.7 EXTI_PR Software Interrupt Event Pending Register

Address offset:0x14

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRx(x=31~16)															
rc_w1															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRx(x=15~0)															
rc_w1															

Bit	Field	Description
31:0	PRx	Line x trigger pending bit 1: Selected trigger request 0: No trigger request In case of the selected edge event on the external interrupt line, the bit is set as 1. Writing 1 clears the bit, or clear by changing the edge detection polarity.

9 ADC Analog-to-digital Converter

9.1 Overview

The 12-bit ADC is a successive approximation analog-to-digital converter (SAR), which converts an analog signal to a digital signal.

ADC has 9 channels allowing it measure signals from external and internal signal sources. Among them, ADC has 8 external input channels and 1 internal channel. These ADC channels are available for single-shot, one-cycle and continuous conversions. Depending on the method, a normal channel conversion, any channel conversion, or injected channel conversion can be selected.

The ADC input clock is generated from the APB clock (PCLK) divided by a prescaler and it must not exceed 16M.

9.2 Functional block diagram

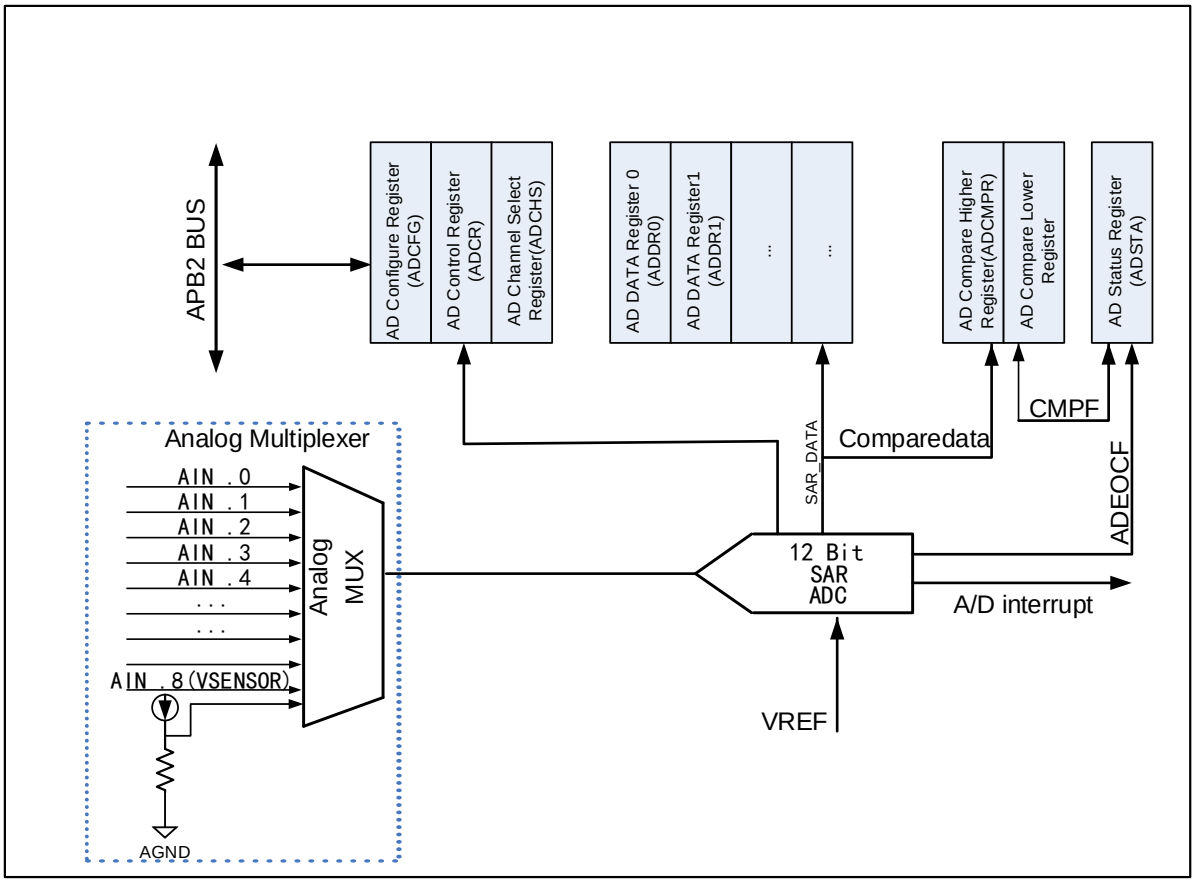


Figure 9-1 ADC system block diagram

Note: V_SENSOR (internal reference voltage) is in the ADC's AIN8 channel.

9.3 Main characteristics

- Conversion rate up to 1Msps
- Normal channel conversion supported
  - Single conversion mode: one conversion is performed in the specified channel
  - One-cycle scan mode: one-cycle conversion is performed in all specified channels (from low sequence number channel to high sequence number channel, or from high sequence number channel to low sequence number channel)
  - Continuous scan mode: continuous one-cycle scan conversions are performed until the software stops A/D conversion. A/D conversion must stop if the conversion channel needs to be changed. The conversion can restart after configuring the corresponding register
- Any channel conversion supported
  - Single conversion mode: one conversion is performed in the specified channel
  - One-cycle scan mode: one-cycle conversion is performed in all specified channels according to the channel configuration
  - Continuous scan mode: continuous one-cycle scan conversions are performed until the software stops A/D conversion. The user doesn't have to stop the conversion if the channel needs to be changed during that period. The conversion in a new configured channel will start in the next scan cycle after configuring the corresponding channel register
- Injected channel conversion supported
  - Auto-injection: in any channel conversion mode, the injected channels are automatically converted after the operation of any channel is complete;
- Programmable channel sampling time
- Up to 12-bit programmable resolution SAR
- A/D conversion start condition
  - Software start
  - Trigger start with configurable trigger delay
- Analog watchdog function. The conversion result is compared with the specified threshold; when the conversion result exceeds the specified threshold, an interrupt is generated if the ADC_ADCR.AWDIE is set

9.4 Interrupt

ADC interrupts and flag bits are shown in the following table. When an event occurs, the corresponding interrupt is generated if the interrupt enable bit is set.

Table 9-1 Interrupts

Interrupt	Flag Bit
Analog watchdog comparator interrupt	AWDIF
End of sequential conversion interrupt	EOSIF
End of conversion interrupt	EOCIF
End of sample interrupt	EOSMPIF
End of injected sequence interrupt	JEOSIF
End of injected conversion interrupt	JEOCIF
End of injected sample interrupt	JEOSMPIF

9.5 Functional description

9.5.1 Clock

The ADC input clock is synchronous with PCLK. ADC clock control bit in the RCC controller should be set before using the ADC.

9.5.2 Data offset

The data from a non-injected channel doesn't have a data offset value.

The converted value from an injected channel decreased by the offset defined in the ADC_JOFRn register is then stored in the injected channel data registers ADC_JADDATA and ADC_JDRn. The value decreased by the offset may be a negative value. Therefore, the converted result from the injected channel is a signed value (the SEXT bit in the data alignment represents the extended sign value).

If ADC_JOFRn.JOFFSET is not set to 0, then the data in the injected channel data registers ADC_JADDATA and ADC_JDRn are signed values, otherwise they are unsigned values.

The following table describes the data resolution in relation to the left aligned data offset:

Table 9-2 Data resolution in relation to the left aligned data offset

Resolution option	Analog conversion data SAR_DATA, left shift	Offset value	Result	Description
12bit	SAR_DATA[11:0]	ADC_JOFRn.JOFFESET	12-bit signed data	-
11bit	SAR_DATA[11:1],0	ADC_JOFRn.JOFFESET	11-bit signed data	User must set ADC_JOFRn.JOFFSET[0] to '0'
10bit	SAR_DATA[11:2],00	ADC_JOFRn.JOFFESET	10-bit signed data	User must set ADC_JOFRn.JOFFSET[1:0] to '00'

Resolution option	Analog conversion data SAR_DATA, left shift	Offset value	Result	Description
9bit	SAR_DATA[11:3],000	ADC_JOFRn.JOFFESET	9-bit signed data	User must set ADC_JOFRn.JOFFSET[2:0] to ‘000’
8bit	SAR_DATA[11:4],0000	ADC_JOFRn.JOFFESET	8-bit signed data	User must set ADC_JOFRn.JOFFSET[3:0] to ‘0000’

9.5.3 Data alignment

ADC_ADCR.ALIGN bit selects the data after conversion is stored in a left-alignment or right-alignment manner.

See the diagram below:

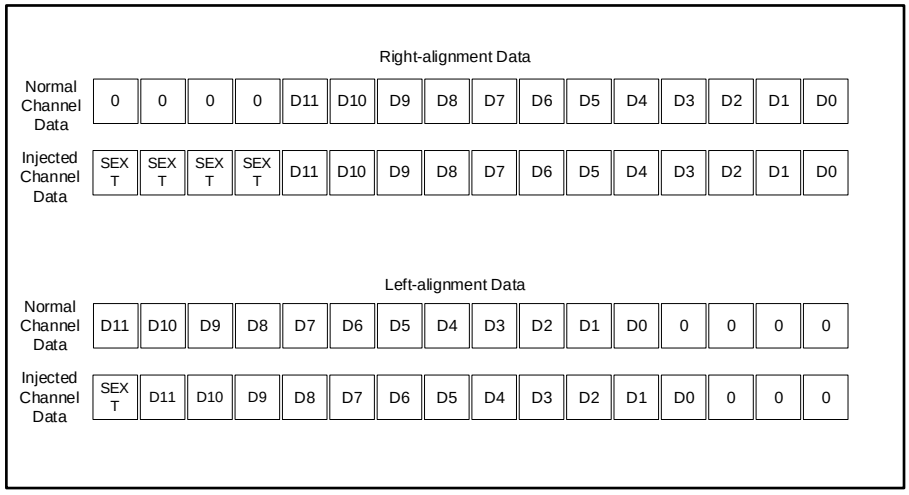


Figure 9-2 Data alignment style

9.5.4 Programmable resolution

The ADC conversion significance can be configured using the ADC_ADCFG.RSLTCTL[2:0] bits. The significant data bits are aligned from the right bits of 12-bit data by default. Low resolution configuration may accelerate the data conversion speed.

9.5.5 Programmable sampling time

The sampling time of ADC conversion channels can be configured by SAMPCTL0~SAMPCTL8 in ADC_SMPR1 and ADC_SMPR2 registers.

The ADC conversion clock ADC_CLK is generated from the PCLK divided by a prescaler. The prescale coefficient can be determined by setting the ADC_ADCFG. ADCPRE bit, i.e. PCLK/ (ADCPRE

+2) divided by a prescaler to be the ADC conversion clock. ADC samples the input voltage for a number of ADC_CLK cycles. The number of sampling cycles (m) can be modified using the ADC_SMPR1 and ADC_SMPR2 registers. Set the ADC resolution as n bits (n=8, 9, 10, 11, 12). The sampling cycles of each channel is m*T (T is the clock cycle of ADC module).

The conversion frequency is calculated as follows:

$F_{sample} = F_{ADC_CLK} /(m+n+0.5)$.

Suppose the resolution (n) is set to 12 bits, and the sampling cycle of each channel (m) is 3.5T, then $F_{sample}=F_{ADC_CLK}/16$.

The total conversion time is calculated as follows:

T_{CONV} = Sampling time + 12.5 conversion cycles

Example: With an ADC_CLK = 16MHz and a sampling time of 3.5T, $T_{CONV}=(3.5+12.5)*T=16*T$, the total conversion time is 1μs.

9.5.6 Data channel register

After the ADC conversion is complete, the conversion result from a non-injected channel is stored in the ADC_ADDATA register, and ADC_ADDATA.CHANNELSEL indicates the channel number corresponding to the current data.

The conversion result from an injected channel is stored in the ADC_JADDATA register, and ADC_JADDATA.JCHANNELSEL indicates the injected channel number corresponding to the current data.

9.5.7 Channel selection

ADC has 8 external input channels 0~7 and internal 1.2V reference voltage channel 8.

Different registers can be used to enable channels in different operating modes: ADC_ADCHS register can be set for normal channel conversion. ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 registers can be set for any channel conversion; the sequence of any channel conversion is organized from highest to lowest priority among CHANY_SEL0 to 15 bits in the register. JSQR register can be set for injection channel conversion; the sequence of injection channel conversion is organized from highest to lowest priority among JSQ0 to 3 bits in the register.

9.6 ADC on-off control

The ADC_ADCFG.ADEN bit controls the A/D converter. When the ADEN bit is set to 0, the analog converter enters Power Down mode. The A/D converter wakes up from the Power Down mode if the ADEN is set to 1. The ADC_ADCFG.ADEN bit can be cleared to stop the conversion and then the ADC enters Power Down mode, after which the ADC consumes almost no power.

9.6.1 Normal channel conversion

9.6.1.1 Single sampling

A single conversion is selected by configuring ADC_ADCR.ADMODE. The configured channel enters the idle state after one conversion. The specific procedures are as follows:

- The A/D conversion is started by setting the ADC_ADCR.ADST bit via external trigger input, or timer trigger or software.
- By default, the sampling output SAMPL is cleared after 2.5 ADC clocks since it is set.
- After A/D conversion, the SAR conversion result will be stored in the data registers ADC_ADDATA and ADC_ADDRn.
- At that time, the ADC_ADSTA.EOSIF bit in the status register is set to 1; meanwhile, an AD EOC (End Of Conversion) interrupt request is generated if the ADC_ADCR.EOSIE bit in the control register is set to 1.
- ADC_ADCR.ADST bit remains 1 during the A/D conversion period. Once the A/D channel conversion is complete, the ADC_ADCR.ADST bit is automatically cleared by hardware and the A/D converter enters the idle state.

Note: In Single conversion mode, the channel of lowest sequence is converted while others are ignored, if more than one channel is enabled via software.

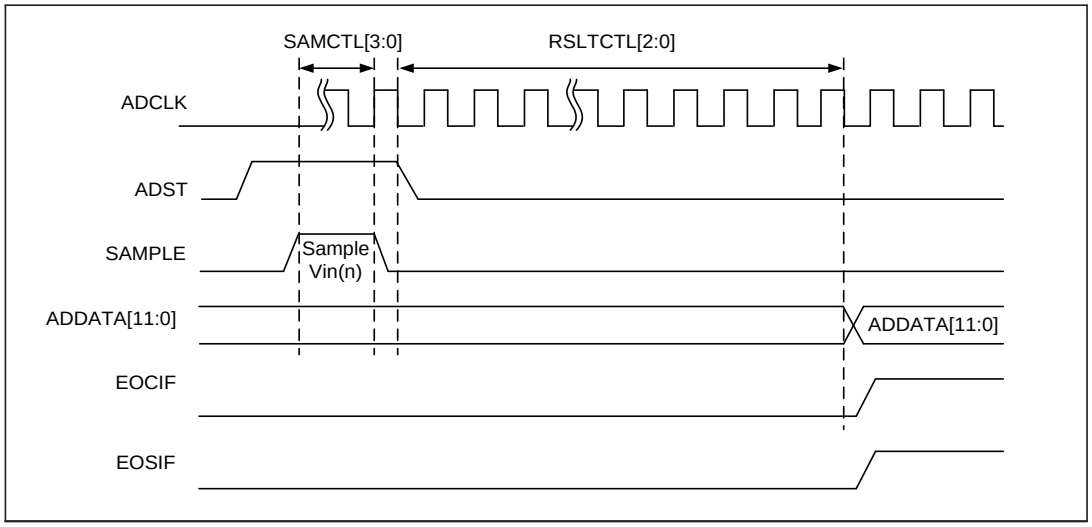


Figure 9-3 Single conversion mode timing diagram

9.6.1.2 One-cycle scan mode

ADC_ADCR.ADMODE is configured as one-cycle conversion. In One-cycle scan mode, scan channel direction can be selected by setting ADC_ADCR.SCANDIR. The A/D conversion is performed in sequence for channels enabled by ADC_ADCHS.CHENy(y=0~8), with the following operation steps:

- Start the A/D conversion either by trigger with the configuration of delay or by setting the ADC_ADCR.ADST bit via software. The default direction is from the lowest sequence number channel to the highest sequence number channel.
- The channel outputs are organized according to the conversion channel configuration.
- After the end of A/D conversion in each channel, the converted result SAR_DATA will be loaded to the data register (ADC_ADDATA and ADC_ADDRn) of corresponding channel in order. The ADC_ADSTA_EXT.EOCIF flag is set when the conversion in the current channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt bit is set. At the end of sequential conversion, the ADC_ADSTA.EOSIF (end of sequential conversion interrupt) flag is set. If the end of sequential conversion interrupt is enabled, then the corresponding request will be generated.
- Once the final A/D conversion is complete, the ADC_ADCR.ADST bit is automatically cleared by hardware and the A/D converter enters the idle state.

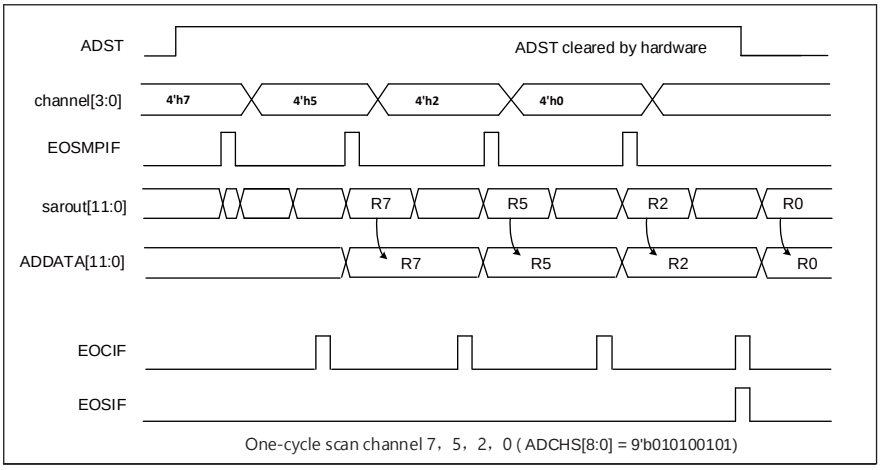


Figure 9-4 Enable channel conversion timing diagram in One-cycle scan mode (channel direction: from high to low)

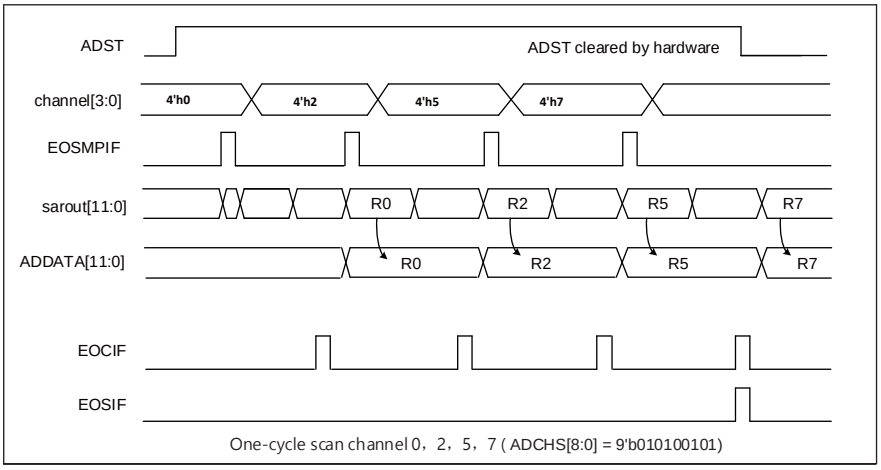


Figure 9-5 Enable channel conversion timing diagram in One-cycle scan mode (channel direction: from low to high)

9.6.1.3 Continuous scan mode

ADC_ADCR.ADMODE is configured as continuous conversion. In continuous scan mode, scan channel direction can be selected by setting ADC_ADCR.SCANDIR. The A/D conversion is performed in sequence for channels enabled by ADC_ADCHS.CHENy(y=0~8), with the following operation steps:

- Start the A/D conversion either by trigger with the configuration of delay or by setting the ADC_ADCR.ADST bit via software. The default direction is from the lowest sequence number channel to the highest sequence number channel.
- The channel outputs are organized according to the conversion channel configuration.
- After the end of A/D conversion in each channel, the converted result SAR_DATA will be loaded to the data register (ADC_ADDATA and ADC_ADDRn) of corresponding channel in order. The ADC_ADSTA_EXT.EOCIF flag is set when the conversion in the current channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt bit is set. At the end of sequential conversion, the ADC_ADSTA.EOSIF (end of sequential conversion interrupt) flag is set. If the end of sequential conversion interrupt is enabled, then the corresponding request will be generated.
- The A/D conversion continues as long as the ADC_ADCR.ADST bit remains 1. Once the ADC_ADCR.ADST bit is cleared and the conversion stops, the converter enters the idle state. Once the ADC_ADCR.ADST bit is cleared, the A/D converter will put an end to the current conversion.

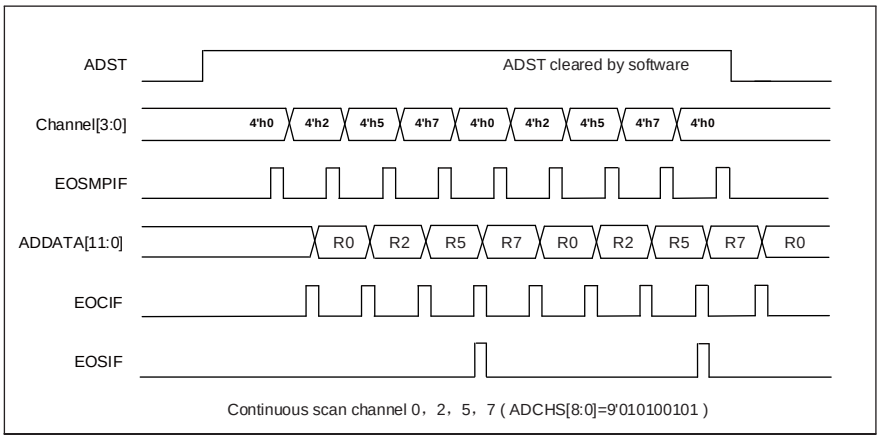


Figure 9-6 Enable channel conversion timing diagram in continuous scan mode (channel direction: from low to high)

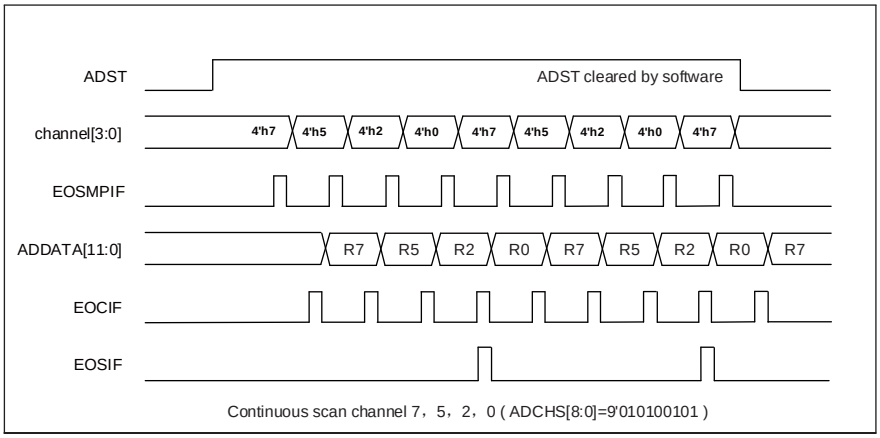


Figure 9-7 Enable channel conversion timing diagram in continuous scan mode (channel direction: from high to low)

9.6.2 Any channel conversion

9.6.2.1 Single conversion mode

A single conversion is selected by configuring ADC_ADCR.ADMODE. The configured channel enters the idle state after one conversion. The specific procedures are as follows:

- Set by software the ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 registers, desired channels, and the ADC_ANY_CR.CHANY_MDEN bit.
- Set the ADC_ADCR.ADST bit by trigger with the configuration of trigger delay or by software. Conversion channels can be arbitrarily configured through registers ADC_CHANY0 and ADC_CHANY1.
- By default, the sampling output SAMPL is cleared after 2.5 ADC clocks since it is set.
- After the end of A/D conversion, the converted result SAR_DATA will be loaded to ADC_ADDATA and ADC_ADDR0. The ADC_ADSTA.EOSIF flag is set when the conversion in the channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt is enabled.
- Once the final A/D channel conversion is complete, the ADST bit is automatically cleared by hardware and the A/D converter enters the idle state.

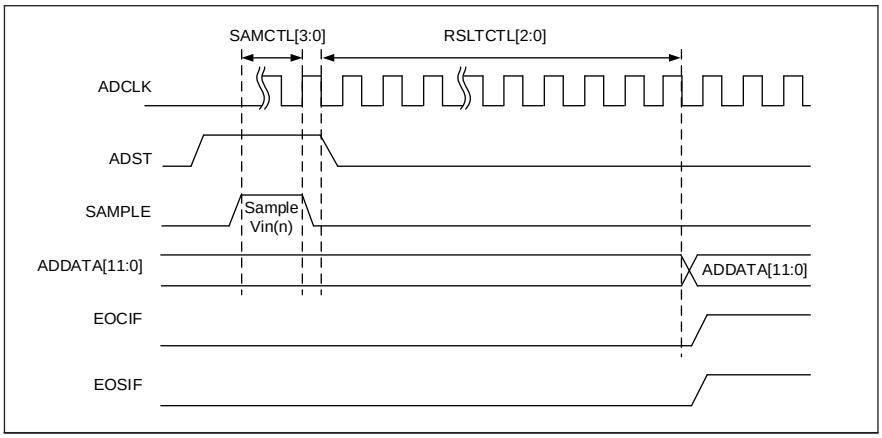


Figure 9-8 Channel conversion timing diagram in Single conversion mode

9.6.2.2 One-cycle scan mode

A one-cycle conversion is selected by configuring ADC_ADCR.ADMODE. In One-cycle scan mode the A/D converter does one conversion according to the software configuration. The specific procedures are as follows:

- Set by software the ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 registers, desired channels and quantity, and the ADC_ANY_CR.CHANY_MDEN bit.
- Set the ADC_ADCR.ADST bit by trigger with the configuration of trigger delay or by software. The number of A/D conversion channels is configured by ADC_ANY_CFG.CHANY_NUM. The input channels for each conversion can be arbitrarily configured through CHANY_SEL0 to CHANY_SEL15 bits in the register, which can be identical or not.
- The channel outputs are organized according to the conversion channel configuration.
- After the end of A/D conversion in each channel, the converted result SAR_DATA will be loaded to the data register (ADC_ADDATA and ADC_ADDRn) of corresponding channel in order. The ADC_ADSTA_EXT.EOCIF flag is set when the conversion in the current channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt bit is set. At the end of sequential conversion, the ADC_ADSTA.EOSIF (end of sequential conversion interrupt) flag is set. If the end of sequential conversion interrupt is enabled, then the corresponding request will be generated.
- Once the final A/D channel conversion is complete, the ADST bit is automatically cleared by hardware and the A/D converter enters the idle state.
- If the software updates registers ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 during the A/D conversion period, these settings will not be applied immediately. They will be applied when the conversion of the final channel is complete. The next conversion is started by setting the ADC_ADCR.ADST bit via software.

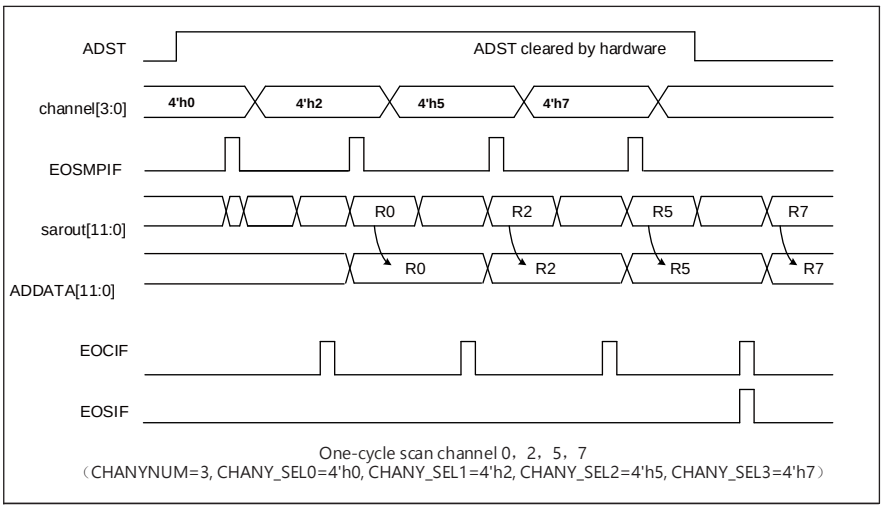


Figure 9-9 Channel conversion timing diagram In One-cycle scan mode

9.6.2.3 Continuous scan mode

Continuous conversions are selected by configuring ADC_ADCR.ADMODE. In continuous scan mode the A/D converter does continuous conversions according to the software configuration until the software stops conversion. The specific procedures are as follows:

- Set by software the ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 registers, desired channels and quantity, and the ADC_ANY_CR.CHANY_MDEN bit.
- Set the ADC_ADCR.ADST bit by trigger with the configuration of trigger delay or by software. The number of A/D conversion channels is configured by ADC_ANY_CFG.CHANY_NUM. The input channels for each conversion can be arbitrarily configured through CHANY_SEL0 to CHANY_SEL15 bits in the register, which can be identical or not.
- The channel outputs are organized according to the conversion channel configuration.
- After the end of A/D conversion in each channel, the converted result SAR_DATA will be loaded to the data register (ADC_ADDATA and ADC_ADDRn) of corresponding channel in order. The ADC_ADSTA_EXT.EOCIF flag is set when the conversion in the current channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt bit is set. At the end of sequential conversion, the ADC_ADSTA.EOSIF flag is set. If the end of sequential conversion interrupt is enabled, then the corresponding request will be generated.
- The A/D conversion continues as long as the ADC_ADCR.ADST bit remains 1. Once the ADC_ADCR.ADST bit is cleared by software, the current A/D conversion stops after it is complete, and the A/D converter enters the idle state.
- If the user updates registers ADC_ANY_CFG, ADC_CHANY0, and ADC_CHANY1 during the A/D conversion period, these settings will not be applied immediately. They will be applied when the conversion of the final channel is complete, i.e. the time a conversion in the new channel

starts in the next scan cycle.

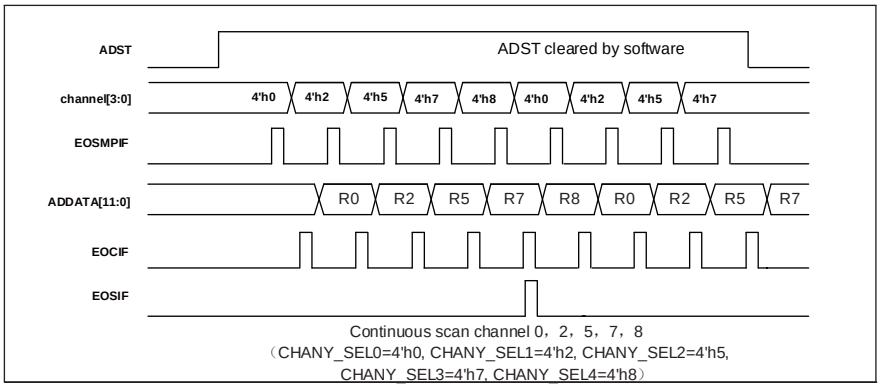


Figure 9-10 Channel conversion timing diagram in Continuous scan mode

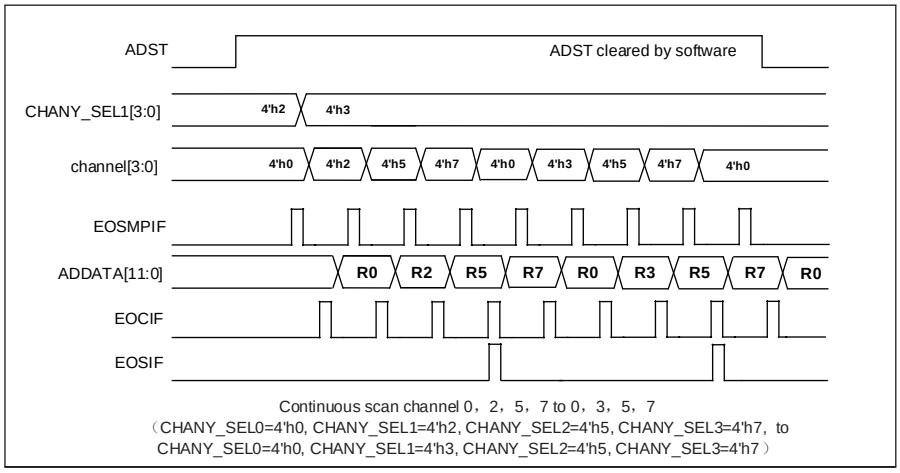


Figure 9-11 Timing diagram with dynamically updated configuration in Continuous scan mode

9.6.3 Injected channel conversion

9.6.3.1 Auto-injection conversion

In the any channel conversion mode, the injected channels are automatically converted after the operation of any channel is complete, if the auto-injection conversion is enabled; If any channel is in continuous scan mode, the A/D conversion can only be stopped by resetting the ADC_ADCR.ADST bit.

- Set by software the ADC_ANY_CFG, ADC_CHANY0, ADC_CHANY1, and ADC_JSQR registers, the ADC_ANY_CFG.JAUTO bit, desired channels and quantity, and finally the ADC_ANY_CR.JCEN and ADC_ANY_CR.CHANY_MDEN bits.
- Set the ADC_ADCR.ADST bit by trigger with the configuration of trigger delay or by software. The number of any A/D conversion channels is configured by ADC_ANY_CFG.CHANY_NUM. The input channels for each conversion can be arbitrarily configured through CHANY_SEL0 to CHANY_SEL15 bits in the register. The injected channels are converted after the operation of any channel is complete. The number of conversion channels is configured by JCHANY_NUM.

The input channels for each conversion can be configured through JSQ0 to JSQ3 bits in the register.

- The channel outputs are organized according to the conversion channel configuration.
- After the end of A/D conversion in each channel, the converted result SAR_DATA will be loaded to the data register (ADC_ADDATA and ADC_ADDRn, ADC_JDATA and ADC_JDRn) of corresponding channel in order. The ADC_ADSTA_EXT.EOCIF and ADC_ADSTA_EXT.JEOCIF flags are set when the conversion in the current channel is complete. At this time, an end of conversion interrupt request is generated if the EOC interrupt is enabled. At the end of continuous conversion, the ADC_ADSTA.EOSIF and ADC_ADSTA_EXT.JEOSIF flags are set. If the end of sequential conversion interrupt is enabled, then the corresponding request will be generated.
- If the scan is in the discontinuous mode, the ADC_ADCR.ADST bit is automatically cleared by hardware once the conversion is complete; otherwise, the A/D conversion continues as long as the ADC_ADCR.ADST bit remains 1. Once the ADC_ADCR.ADST bit is cleared, the current A/D conversion stops after it is complete, and the A/D converter enters the idle state.

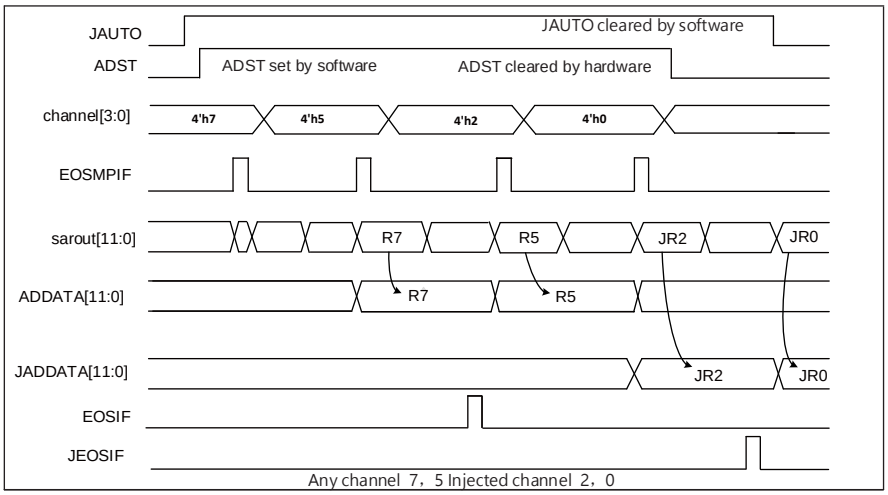


Figure 9-12 Auto-injection conversion timing diagram in One-cycle scan mode

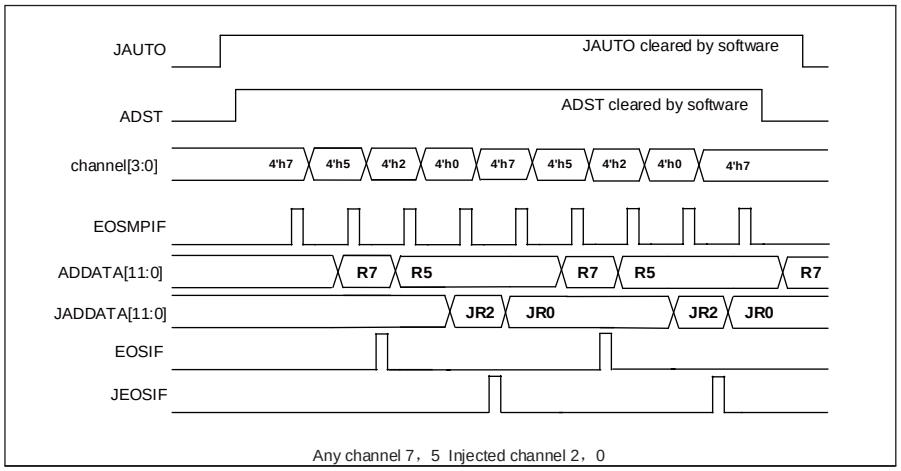


Figure 9-13 Auto-injection conversion timing diagram in Continuous scan mode

9.6.4 ADC trigger signal

In addition to software, the trigger sources for ADC conversion include timers and external events.

After a trigger signal is generated, the sampling starts after a delay of N PCLK clock cycles (configurable). In case of trigger scan mode, only the sampling in the first channel is delayed and the rest start immediately after the last conversion is complete.

If the ADC_ADCR.TRGEN bit is set then external events are able to trigger a conversion of non-injected channel.

An external trigger source of non-injected conversion can be selected by setting the ADC_ADCR.TRGSEL bit.

Refer to the description of ADC_ADCR.TRGSEL in the A/D control register for details of external trigger source selection. Refer to relevant description of ADC_ADCR.TRGSHIFT in the AD control register for details of external trigger delay control.

9.6.5 Analog watchdog

The watchdog compares the data before the data alignment. An upper limit comparison register and a lower limit comparison register are provided by the watchdog comparator. The monitoring channel can be selected by setting the ADC_ADCR.CMPCH bit in the register. Providing ADC_ADCMPR.CPMHDATA is greater than or equal to ADC_ADCMPR.CPMLDATA, and if analog watchdog is enabled on non-injected channels (ADC_ADCFG.AWDEN), and the result from non-injected channels is greater than or equal to the ADC_ADCMPR.CMPHDATA value or less than the ADC_ADCMPR.CMPLDATA value, the ADC_ADSTA.AWDIF bit in the status register is set to 1; if analog watchdog is enabled on injected channels (ADC_ADCFG.JAWDEN), and the result from injected channels is greater than or equal to the ADC_ADCMPR.CMPHDATA value or less than the ADC_ADCMPR.CMPLDATA value, the ADC_ADSTA.AWDIF bit in the status register is also set to 1.

Providing ADC_ADCMPR.CPMHDATA is less than ADC_ADCMPR.CPMLDATA, and if analog watchdog is enabled on non-injected channels (ADC_ADCFG.AWDEN), and the result from non-injected channels is equal to the ADC_ADCMPR.CPMHDATA value or between two specified values, the ADC_ADSTA. AWDIF bit in the status register is set to 1; if analog watchdog is enabled on injected channels (ADC_ADCFG.JAWDEN), and the result from injected channels is equal to the ADC_ADCMPR.CPMHDATA value or between two specified values, the ADC_ADSTA. AWDIF bit in the status register is also set to 1.

An interrupt request is generated if the ADC_ADCR.AWDIE bit in the control register is set.

9.6.6 Internal voltage sensor

The internal signal source channel of ADC is connected to an internal reference voltage (Vref) of 1.2V. This channel converts the 1.2V reference voltage output to a digital value. Then the internal reference voltage can be obtained by calculation.

Example: if the voltage VDD of the 12bit ADC is 3.3V, the converted value of V_SENSOR is ADC_ADDR(VSensor). The formula is as below:

3.3/4096 = Vref/ ADC_ADDR(VSensor)

Vref= (ADC_ADDR*3.3)/ 4096

The internal reference voltage is calibrated at the factory and the calibration value is stored in the flash space 0x1FFFF7E0, which allows the user to deduce the current system analog reference voltage value based on the calibration value. The formula is as below:

Vref_calc / ADC_ADDR(VSensor) = VDDA / 4096

VDDA = (Vref_calc * 4096) / ADC_ADDR(VSensor)

Vref_calc: Internal reference voltage factory-calibrated value in mv

The internal reference voltage has an independent enable bit that can be enabled or disabled by setting the ADC_ADCFG.VSEN bit in the register.

9.7 Register description

9.7.1 Overview of registers

Table 9-3 Overview of ADC register

Offset	Acronym	Register Name	Reset
0x00	ADC_ADDATA	Data Register	0x00000000
0x04	ADC_ADCFG	Configuration Register	0x00000000
0x08	ADC_ADCR	Control Register	0x00000000
0x0C	ADC_ADCHS	Channel Selection Register	0x00000000
0x10	ADC_ADCMPR	Analog Watchdog Compare Register	0x00000000
0x14	ADC_ADSTA	Status Register	0x00000000
0x18~0x38	ADC_ADDR 0~8	Channel Data Register	0x00000000
0x58	ADC_ADSTA_EXT	Extended State Register	0x00000000
0x5C	ADC_CHANY0	Any Channel Selection Register 0	0x00000000
0x60	ADC_CHANY1	Any Channel Selection Register 1	0x00000000
0x64	ADC_ANY_CFG	Any Channel Configuration Register	0x00000000
0x68	ADC_ANY_CR	Any Channel Control Register	0x00000000
0x70	ADC_SMPR1	Sample Configuration Register 1	0x00000000
0x74	ADC_SMPR2	Sample Configuration Register 2	0x00000000

Offset	Acronym	Register Name	Reset
0x7C~0x88	ADC_JOFR0~3	Injected Channel Data Offset Register	0x00000000
0x8C	ADC_JSQR	Injected Sequence Register	0x00000000
0x90	ADC_JADDATA	Injected Data Register	0x00000000
0xB0~0xBC	ADC_JDR0~3	Injected Channel Data Register	0x00000000

9.7.2 ADC_ADDATA Data Register

Address offset: 0x00

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.										VAILD	OVERRUN	CHANNELSEL			
										r	r	r			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA															
r															

Bit	Field	Description
31:22	Reserved	Reserved, must be kept at reset value.
21	VALID	Valid flag for data (read-only) 1: DATA[11:0] bits valid 0: DATA[11:0] bits invalid This bit is set by hardware after the conversion is complete. It is cleared automatically after reading the ADC_ADDATA register.
20	OVERRUN	Data overrun flag (read-only) 1: DATA[11:0] for last conversion is overrun 0: DATA[11:0] for last conversion is not overrun Before the data is loaded to the DATA[11:0] bits in the data register, OVERRUN is set to '1' if the DATA[11:0] for last conversion is not read. This bit is cleared automatically after reading the ADC_ADDATA register.

Bit	Field	Description
19:16	CHANNELSEL	4 bits show the channel corresponding to the data in the current data register (Channel Selection) 0000: indicates that the corresponding channel of the current data register is channel 0 0001: indicates that the corresponding channel of the current data register is channel 1 0010: indicates that the corresponding channel of the current data register is channel 2 0011: indicates that the corresponding channel of the current data register is channel 3 0100: indicates that the corresponding channel of the current data register is channel 4 0101: indicates that the corresponding channel of the current data register is channel 5 0110: indicates that the corresponding channel of the current data register is channel 6 0111: indicates that the corresponding channel of the current data register is channel 7 1000: indicates that the corresponding channel of the current data register is channel 8 (corresponding to the conversion data of internal reference voltage) Others: invalid
15:0	DATA	12-bit A/D conversion result from the current channel (Conversion Data) The alignment style is selected via software.

9.7.3 ADC_ADCFG Configuration Register

Address offset: 0x04

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															JAWDEN
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	ADCPREL	Res.				RSLTCTL			ADCPREH			VSEN	Res.	AWDEN	ADEN
	rw					rw			rw			rw		rw	rw

Bit	Field	Description
31:17	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
16	JAWDEN	A/D Analog Watchdog Enable for Injected Channels 1: A/D analog watchdog enabled for injected channels 0: A/D analog watchdog disabled for injected channels
15	Reserved	Reserved, must be kept at reset value
14	ADCPREL	ADC Prescaler Low Bits Prescale coefficient ADCPRE={ADCPREH, ADCPREL}
13: 10	Reserved	Reserved, must be kept at reset value
9: 7	RSLTCTL	ADC conversion data resolution selection 000: 12 significant bits 001: 11 significant bits 010: 10 significant bits 011: 9 significant bits 100: 8 significant bits Others: reserved
6: 4	ADCPREH	ADC Prescaler High Bits Prescale coefficient ADCPRE={ADCPREH, ADCPREL} ADC clock division: div= (ADCPRE+2)
3	VSEN	Internal reference voltage enable (Voltage Sensor enable) 1: Internal voltage sensor enabled 0: Internal voltage sensor disabled
2	Reserved	Reserved, must be kept at reset value
1	AWDEN	Analog watchdog enable on non-injected channels 1: analog watchdog enabled on non-injected channels 0: analog watchdog disabled on non-injected channels
0	ADEN	A/D conversion enable (ADC enable) 1: enabled 0: disabled

9.7.4 ADC_ADCR Control Register

Address offset: 0x08

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.				EOCIE	EOSMPIE	TRG_EDGE		Res.		TRGSHIFT			TRGSELH		SCANDIR
				rw	rw	rw				rw			rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMPCH				ALIGN	ADMD		ADST	Res.	TRGSELL			Res.	TRGEN	AWDIE	EOSIE
rw				rw	rw		rw		rw				rw	rw	rw

Bit	Field	Description
31:28	Reserved	Reserved, must be kept at reset value
27	EOCIE	Interrupt Enable for End of Conversion 1: interrupt enabled for end of single A/D conversion 0: interrupt disabled for end of single A/D conversion An A/D EOC interrupt request is generated if the EOCIF bit is set.
26	EOSMPIE	Interrupt Enable for End of Sample 1: interrupt enabled for end of A/D sampling 0: interrupt disabled for end of A/D sampling An A/D end of sample interrupt request is generated if the EOSMPIF bit is set.
25 : 24	TRG_EDGE	Trigger Sources Edge Selection 00: Double-edge trigger 01: Falling edge trigger 10: Rising edge trigger 11: Reserved
23:22	Reserved	Reserved, must be kept at reset value.
21:19	TRGSHIFT	External Trigger Sources Shift For Sample After a trigger event is generated, the sampling starts after a delay of N PCLK clock cycles (N=1, 2, 3…). 000: no delay 001: 4 cycles 010: 16 cycles 011: 32 cycles 100: 64 cycles 101: 128 cycles 110: 256 cycles 111: 512 cycles In case of trigger scan, the sampling in the rest channels starts immediately after the last conversion is complete.
18 :17	TRGSELH	External Trigger Sources Select For High Bits TRGSEL={TRGSELH,TRGSELL}
16	SCANDIR	ADC Channel Scan Direction Only valid for normal channel conversion 1: Backward scan from high sequence channel to low sequence channel by the ADC channel selection register 0: Upward scan from low sequence channel to high sequence channel by the ADC channel selection register

Bit	Field	Description
15:12	CMPCH	Compare Channel Selection For Analog Watchdog 0000: select to compare the conversion result from channel 0 0001: select to compare the conversion result from channel 1 0010: select to compare the conversion result from channel 2 0011: select to compare the conversion result from channel 3 0100: select to compare the conversion result from channel 4 0101: select to compare the conversion result from channel 5 0110: select to compare the conversion result from channel 6 0111: select to compare the conversion result from channel 7 1000: select to compare the conversion result from channel 8 (ie. select to compare the conversion result from the internal voltage sensor) 1111: select to compare the conversion result from all scan channels Others: invalid
11	ALIGN	Data Alignment style 1: left aligned 0: right aligned
10 : 9	ADMD	A/D conversion mode (ADC Mode) 00: single conversion 01: one-cycle conversion 10: continuous scan 11: Reserved When the conversion mode is changed, the software has to clear the ADST bit.
8	ADST	A/D conversion start 1: conversion start 0: end of conversion or idle state There are two ways of clearing ADST: In the single or one-cycle mode, ADST is automatically cleared by hardware after conversion. In continuous scan mode, the A/D converter does continuous conversions until the software writes ‘0’ to ADST or the system is reset.
7	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
6: 4	TRGSELL	External Trigger sources Select for Low bits TRGSEL={TRGSELH,TRGSELL} ADC selects external trigger source TRGSEL: 00000: TIM1_CC1 00001: TIM1_CC2 00010: TIM1_CC3 00100: TIM3_TRGO 00110: TIM3_CC1 00111: EXTI 11 01000: TIM1_TRGO 01100: TIM3_CC4 01111: EXTI 15 10000: TIM1_CC4 10001: TIM1_CC5 Others: invalid
3	Reserved	Reserved, must be kept at reset value.
2	TRGEN	External Hardware Trigger Sources Enable 1: external trigger enabled for A/D conversion 0: external trigger disabled for A/D conversion
1	AWDIE	Interrupt Enable of Analog Watchdog 1: A/D analog watchdog interrupt enabled 0: A/D analog watchdog interrupt disabled
0	EOSIE	Interrupt Enable for End of Sequence on non-injected channels 1: ADC interrupt enabled 0: ADC interrupt disabled An A/D EOC interrupt request is generated if the EOSIF bit is set.

9.7.5 ADC_ADCHS Channel Selection Register

Address offset: 0x0C

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.							CHEN8	CHEN7	CHEN6	CHEN5	CHEN4	CHEN3	CHEN2	CHEN1	CHEN0
							rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31:9	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
8	CHEN8	ADC: Voltage Sensor Enable 1: enabled 0: disabled
7	CHEN7	Analog Input Channel 7 Enable 1: enabled 0: disabled
6	CHEN6	Analog Input Channel 6 Enable 1: enabled 0: disabled
5	CHEN5	Analog Input Channel 5 Enable 1: enabled 0: disabled
4	CHEN4	Analog Input Channel 4 Enable 1: enabled 0: disabled
3	CHEN3	Analog Input Channel 3 Enable 1: enabled 0: disabled
2	CHEN2	Analog Input Channel 2 Enable 1: enabled 0: disabled
1	CHEN1	Analog Input Channel 1 Enable 1: enabled 0: disabled
0	CHEN0	Analog Input Channel 0 Enable 1: enabled 0: disabled

Note: 1) If all channels are disabled, the default conversion channel is 0 or 0xF when the conversion starts according to SCANDIR configuration.

2) This register can only be operated when ADC_ADSTA.BUSY is invalid.

9.7.6 ADC_ADCMPR Analog Watchdog Compare Register

Address offset: 0x10

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.				CMPHDATA											
				rw											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				CMPLDATA											
				rw											

Bit	Field	Description
31:28	Reserved	Reserved, must be kept at reset value
27: 16	CMPHDATA	Compare High Threshold Data For Analog Watchdog The high threshold of comparison for the watchdog
15:12	Reserved	Reserved, must be kept at reset value.
11 : 0	CMPLDATA	Compare Low Threshold Data For Analog Watchdog The low threshold of comparison for the watchdog

9.7.7 ADC_ADSTA Status Register

Address offset: 0x14

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.			OVERRUN									Res.			VALID
			r												r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALID								CHANNEL				Res.	BUSY	AWDIF	EOSIF
r								r					r	rc_w1	rc_w1

Bit	Field	Description
31:29	Reserved	Reserved, must be kept at reset value
28:20	OVERRUN	Data overrun flag for channel 0 to channel 8 (Overrun Flag)
19:17	Reserved	Reserved, must be kept at reset value
16:8	VALID	Valid flag for channel 0 to channel 8 (Valid Flag)
7:4	CHANNEL	Current Convert Channel Note: these bits indicate the channel in conversion when BUSY=1; these bits indicate the channel for the next conversion when BUSY=0.
3	Reserved	Reserved, must be kept at reset value
2	BUSY	Non-Injected Conversion Busy/Idle 1 = A/D converter is busy 0 = A/D converter is idle
1	AWDIF	Analog Watchdog Flag This flag bit is cleared by writing ‘1’. 1: Analog watchdog event occurs 0: Analog watchdog event doesn’t occur

Bit	Field	Description
0	EOSIF	End of Sequential Conversion Flag This bit is set by hardware at the end of sequential conversion, and cleared by software. 1: A/D conversion complete 0: A/D conversion incomplete This flag bit is cleared by writing '1'.

9.7.8 ADC_ADDRn Channel Data Register (n=0~8)

Address offset: 0x18~0x38

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.										VAILD	OVERRUN	Res.			
										r	r				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA															
r															

Bit	Field	Description
31:22	Reserved	Reserved, must be kept at reset value.
21	VALID	Valid flag (read-only) 1: DATA[11:0] bits valid 0: DATA[11:0] bits invalid This bit is set by hardware after the conversion in the corresponding channel is complete. It is cleared automatically after reading the ADC_ADDRn register.
20	OVERRUN	Data overrun flag (read-only) 1: DATA [11:0] data is overrun 0: DATA [11:0] for last conversion result OVERRUN is set to '1' if the DATA[11:0] data for last conversion is not read before it is loaded to the DATA[11:0] bits in the data register. This bit is cleared by hardware automatically after reading the ADC_ADDRn register.
19:16	Reserved	Reserved, must be kept at reset value.
15 : 0	DATA	12-bit A/D conversion result from the channel n (Conversion Data) The alignment style is selected via software.

9.7.9 ADC_ADSTA_EXT Extended State Register

Address offset: 0x58

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.										JBUSY	JEOSIF	JEOCIF	JEOSMPIF	EOCIF	EOSMPIF
										r	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value.
21	JBUSY	Injected Conversion Busy/Idle Flag 1: Injected channel of A/D converter is busy 0: Injected channel of A/D converter is idle
20	JEOSIF	End of Injected Sequence Flag This bit is set by hardware at the end of sequential conversion, and cleared by software. 1: A/D conversion complete 0: A/D conversion incomplete This flag bit is cleared by writing ‘1’.
19	JEOCIF	End of Injected Conversion Flag This bit is set by hardware at the end of conversion, and cleared by software. 1: A/D conversion complete 0: A/D conversion incomplete This flag bit is cleared by writing ‘1’.
18	JEOSMPIF	End of Injected Sample Flag This bit is set by hardware at the end of sampling, and cleared by software. 1: A/D sampling complete 0: A/D sampling incomplete This flag bit is cleared by writing ‘1’.
17	EOCIF	End of Conversion Flag This bit is set by hardware at the end of conversion, and cleared by software. 1: A/D conversion complete 0: A/D conversion incomplete This flag bit is cleared by writing ‘1’.
16	EOSMPIF	End of Sample Flag This bit is set by hardware at the end of sampling, and cleared by software. 1: A/D sampling complete 0: A/D sampling incomplete This flag bit is cleared by writing ‘1’.
15:0	Reserved	Reserved, must be kept at reset value.

9.7.10 ADC_CHANY0 Any Channel Selection Register 0

Address offset: 0x5C

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CHANY_SEL7				CHANY_SEL6				CHANY_SEL5				CHANY_SEL4			
rw				rw				rw				rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHANY_SEL3				CHANY_SEL2				CHANY_SEL1				CHANY_SEL0			
rw				rw				rw				rw			

Bit	Field	Description
31:28	CHANY_SEL7	7th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
27:24	CHANY_SEL6	6th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
23:20	CHANY_SEL5	5th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
19:16	CHANY_SEL4	4th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
15:12	CHANY_SEL3	3th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
11:8	CHANY_SEL2	2th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
7:4	CHANY_SEL1	1th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
3:0	CHANY_SEL0	0th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel

Note: In the one-cycle or continuous scan modes, the ADC_CHANY0 shadow register is started by hardware. When ADC is working, only the shadow register is updated if the ADC_CHANY0 value is changed. Only when ADC starts to convert the last channel, the value of the shadow register will update to ADC_CHANY0, so as to implement the dynamic channel switch.

9.7.11 ADC_CHANY1 Any Channel Selection Register 1

Address offset: 0x60

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CHANY_SEL15				CHANY_SEL14				CHANY_SEL13				CHANY_SEL12			
rw				rw				rw				rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHANY_SEL11				CHANY_SEL10				CHANY_SEL9				CHANY_SEL8			
rw				rw				rw				rw			

Bit	Field	Description
31:28	CHANY_SEL15	15th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
27:24	CHANY_SEL14	14th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
23:20	CHANY_SEL13	13th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
19:16	CHANY_SEL12	12th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
15:12	CHANY_SEL11	11th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
11:8	CHANY_SEL10	10th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
7:4	CHANY_SEL9	9th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel
3:0	CHANY_SEL8	8th Conversion Select for Any Channel sequence 0000~1000: configure any channel from channel 0 to channel 8 as an input channel in the operation of any channel

Note: In the one-cycle or continuous scan modes, the ADC_CHANY1 shadow register is started by hardware. When ADC is working, only the shadow register is updated if the ADC_CHANY1 value is changed. Only when ADC starts to convert the last channel, the value of the shadow register will update to ADC_CHANY1, so as to implement the dynamic channel switch.

9.7.12 ADC_ANY_CFG Any Channel Configuration Register

Address offset: 0x64

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												CHANY_NUM			
												rw			

Bit	Field	Description
31:4	Reserved	Reserved, must be kept at reset value
3: 0	CHANY_NUM	Number of Any Channel Mode: 0000: CHANY_SEL0 channel 0001: CHANY_SEL0~CHANY_SEL1 channels 0010: CHANY_SEL0~CHANY_SEL2 channels 0011: CHANY_SEL0~CHANY_SEL3 channels 0100: CHANY_SEL0~CHANY_SEL4 channels 0101: CHANY_SEL0~CHANY_SEL5 channels 0110: CHANY_SEL0~CHANY_SEL6 channels 0111: CHANY_SEL0~CHANY_SEL7 channels 1000: CHANY_SEL0~CHANY_SEL8 channels Others: invalid

Note: In the one-cycle or continuous scan modes, the ADC_NUM shadow register is started by hardware. When ADC is working, only the shadow register is updated if the ADC_ NUM value is changed. Only when ADC starts to convert the last channel, the value of the shadow register will update to ADC_ NUM, so as to implement the dynamic channel switch.

9.7.13 ADC_ANY_CR Any Channel Control Register

Address offset: 0x68

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.														JTRG_EDGE	
														rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JTRGSHIFT			JTRGSEL					JTRGEN	JADST	JAUTO	JEOSIE	JEOCIE	JEOSMPIE	JCEN	CHANY_MDEN
rw			rw					rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31:18	Reserved	Reserved, must be kept at reset value
17:16	JTRG_EDGE	External Trigger sources Edge Selection for Injected Channels 00: Double-edge trigger 01: Falling edge trigger 10: Rising edge trigger 11: Trigger blocked
15:13	JTRGSHIFT	External Trigger Shift Sample for Injected Channels After a trigger signal is generated, the sampling starts after a delay of N PCLK clock cycles. In case of trigger scan, the sampling in the rest channels starts immediately after the last sampling is complete. 000: no delay 001: 4 cycles 010: 16 cycles 011: 32 cycles 100: 64 cycles 101: 128 cycles 110: 256 cycles 111: 512 cycles
12:8	JTRGSEL	ADC External Trigger Sources Selection For Injected Channel 00000: TIM1_CC1 00001: TIM1_CC2 00010: TIM1_CC3 00100: TIM3_TRGO 00110: TIM3_CC1 00111: EXTI 11 01000: TIM1_TRGO 01100: TIM3_CC4 01111: EXTI 15 10000: TIM1_CC4 10001: TIM1_CC5 Others: invalid
7	JTRGEN	External Hardware Trigger Enable For Injected Channels 1: external trigger enabled for A/D conversion 0: external trigger disabled for A/D conversion

Bit	Field	Description
6	JADST	ADC Start for Injected Channels: enable JCEN first 1: Injected channel conversion start 0: Injected channel end of conversion or entering the idle state There are two ways of clearing JADST: JADST is automatically cleared by hardware after the conversion of injected channel is complete; JADST is cleared if JCEN is 0; it is also cleared after a system reset.
5	JAUTO	Automatic Injected Conversion 1: Automatic injected conversion enabled 0: Automatic injected conversion disabled
4	JEOSIE	Interrupt Enable for JEOS 1: end of A/D sequential conversion interrupt enabled 0: end of A/D sequential conversion interrupt disabled An A/D EOS interrupt request is generated if the JEOSIE bit is set.
3	JEOCIE	Interrupt Enable for JEOC 1: end of A/D conversion interrupt enabled 0: end of A/D conversion interrupt disabled An A/D EOC interrupt request is generated if the JEOCIF bit is set.
2	JEOSMPIE	Interrupt Enable for JEOSAMP 1: interrupt enabled for end of A/D sampling 0: interrupt disabled for end of A/D sampling An A/D end of conversion interrupt request is generated if the JEOSMPIF bit is set.
1	JCEN	Conversion Enable for Injected Channels 1: Injected conversion enabled 0: Injected conversion disabled
0	CHANY_ MDEN	Any Channel Mode Enable: 1: any channel mode enabled 0: any channel mode disabled

Note: In the any channel mode plus the one-cycle/continuous scan mode, wait until the ADC_ADCR.ADST and ADC_ADSTA.BUSY bits turn to 0 before closing the ADC. Then clear the ADC_ANY_CR.CHANY_MDEN bit.

9.7.14 ADC_SMPR1 Sample Configuration Register 1

Address offset: 0x70

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SAMP7				SAMP6				SAMP5				SAMP4			
rw				rw				rw				rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

SAMP3	SAMP2	SAMP1	SAMP0
rw	rw	rw	rw

Bit	Field	Description
31 : 0	SAMP7~SAMP0	Channel 0~7 Sample Time Selection These bits are used to independently select the sample time of each channel. In a sample cycle, the channel selection bit should remain unchanged. 0000: 2.5 cycles 0100: 42.5 cycles 0001: 8.5 cycles 0101: 56.5 cycles 0010: 14.5 cycles 0110: 72.5 cycles 0011: 29.5 cycles 0111: 240.5 cycles 1000: 3.5 cycles 1001: 4.5 cycles 1010: 5.5 cycles 1011: 6.5 cycles 1100: 7.5 cycles Others: reserved

9.7.15 ADC_SMPR2 Sample Configuration Register 2

Address offset: 0x74

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												SAMP8			
												rw			

Bit	Field	Description
31:4	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
3: 0	SAMP8	Channel 8 Sample Time Selection These bits are used to independently select the sample time of each channel. In a sample cycle, the channel selection bit should remain unchanged. 0000: 2.5 cycles 0100: 42.5 cycles 0001: 8.5 cycles 0101: 56.5 cycles 0010: 14.5 cycles 0110: 72.5 cycles 0011: 29.5 cycles 0111: 240.5 cycles 1000: 3.5 cycles 1001: 4.5 cycles 1010: 5.5 cycles 1011: 6.5 cycles 1100: 7.5 cycles Others: reserved

9.7.16 ADC_JOFRn Injected Channel Data Offset Register (n=0~3)

Address offset: 0x7C~0x88

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				JOFFSET[11:0]											
				rw											

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11: 0	JOFFSET	Data Offset Register for 12-bit AD Injected Channel n The converted value from the injected channel n decreased by the JOFFSET offset value is then stored in the registers JADDATA and ADC_JDRn. Note: Writing to this bit is permitted when JADST=0.

9.7.17 ADC_JSQR Injected Sequence Register

Address offset: 0x8C

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.										JNUM		JSQ3			
										rw		rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JSQ3.	JSQ2					JSQ1				JSQ0					
rw	rw					rw				rw					

Bit	Field	Description
23:22	Reserved	Reserved, must be kept at reset value.
21:20	JNUM	Channel Number for Injected Sequence 00: JSQ0 channel 01: JSQ0~JSQ1 channels 10: JSQ0~JSQ2 channels 11: JSQ0~JSQ3 channels
19:15	JSQ3	3th Conversion for Injected Sequence 00000~01000: configure any channel from channel 0 to channel 8 as an injected channel Others: reserved
14:10	JSQ2	2th Conversion for Injected Sequence 00000~01000: configure any channel from channel 0 to channel 8 as an injected channel Others: reserved
9:5	JSQ1	1th Conversion for Injected Sequence 00000~01000: configure any channel from channel 0 to channel 8 as an injected channel Others: reserved
4:0	JSQ0	0th Conversion for Injected Sequence 00000~01000: configure any channel from channel 0 to channel 8 as an injected channel Others: reserved

Note: In the one-cycle or continuous scan modes, the ADC_JSQR shadow register is started by hardware. When ADC is working, only the shadow register is updated if the ADC_JSQR value is changed. Only when ADC starts to convert the last channel, the value of the shadow register will update to ADC_JSQR, so as to implement the dynamic channel switch.

9.7.18 ADC_JADDATA Injected Data Register

Address offset: 0x90

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Res.									JVAILD	JOVERRUN	Res.	JCHANNELSEL			
									r	r		r			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA															
r															

Bit	Field	Description
31:23	Reserved	Reserved, must be kept at reset value
22	JVALID	Valid Flag for Injected Channels (read-only) 1: JDATA[11:0] bits valid 0: JDATA[11:0] bits invalid This bit is set by hardware after the corresponding conversion is complete. It is cleared automatically after reading the ADC_JADDATA register.
21	JOVERRUN	Overrun Flag for Injected Channels (read-only) 1: JDATA [11:0] data is overrun 0: JDATA [11:0] for last conversion result Before the new conversion result is loaded to the register, JOVERRUN is set to '1' if the JDATA[11:0] is not read. This bit is cleared by hardware automatically after reading the ADC_JADDATA register.
20	Reserved	Reserved, must be kept at reset value.
19:16	JCHANNELSEL	4 bits show the injected channel corresponding to the current data (Injected Channel Selection) 0000 = conversion data of channel 0 0001 = conversion data of channel 1 0010 = conversion data of channel 2 0011 = conversion data of channel 3 0100 = conversion data of channel 4 0101 = conversion data of channel 5 0110 = conversion data of channel 6 0111 = conversion data of channel 7 1000 = conversion data of channel 8 (Conversion data of internal reference voltage) Others: invalid
15:0	JDATA	12-bit A/D conversion result of current injected channel (Transfer Data for Injected Channels) The alignment style is selected via software.

9.7.19 ADC_JDRn Injected Channel Data Register (n=0~3)

Address offset: 0xB0 ~ 0xBC

Reset value: 0x00000000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.									JVALID	JOVERRUN	Res.				
									r	r					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA															
r															

Bit	Field	Description
31:23	Reserved	Reserved, must be kept at reset value
22	JVALID	Valid Flag for Injected Channels 1: JDATA[11:0] bits valid 0: JDATA[11:0] bits invalid This bit is set by hardware after the corresponding conversion is complete. It is cleared automatically after reading the ADC_JADR register.
21	JOVERRUN	Overrun Flag for Injected Channels 1: JDATA [11:0] data is overrun 0: JDATA [11:0] for last conversion result Before the new conversion result is loaded to the register, JOVERRUN is set to ‘1’ if the JDATA[11:0] is not read. This bit is cleared by hardware automatically after reading the JDRn register.
20: 16	Reserved	Reserved, must be kept at reset value.
15: 0	JDATA	Conversion result of A/D injected channel n (Transfer Data for Injected Channels) The alignment style is selected via software. Please refer to the data alignment section.

10 TIM1 Advanced Timer

10.1 Overview

TIM1 is composed of a 16 bit real-time programmable prescaler and a 16-bit auto reload counter with the programmable counting direction, and can conveniently provide with the counting and timing function. The counter is driven by a programmable prescaler. The advanced timer can be used for a variety of purposes, such as input capture function (measuring input signal pulse width or frequency, PWM input, etc), output compare function (PWM output, programmable dead-time inserted between the complementary outputs, one-pulse mode, etc.)

10.2 Function block diagram

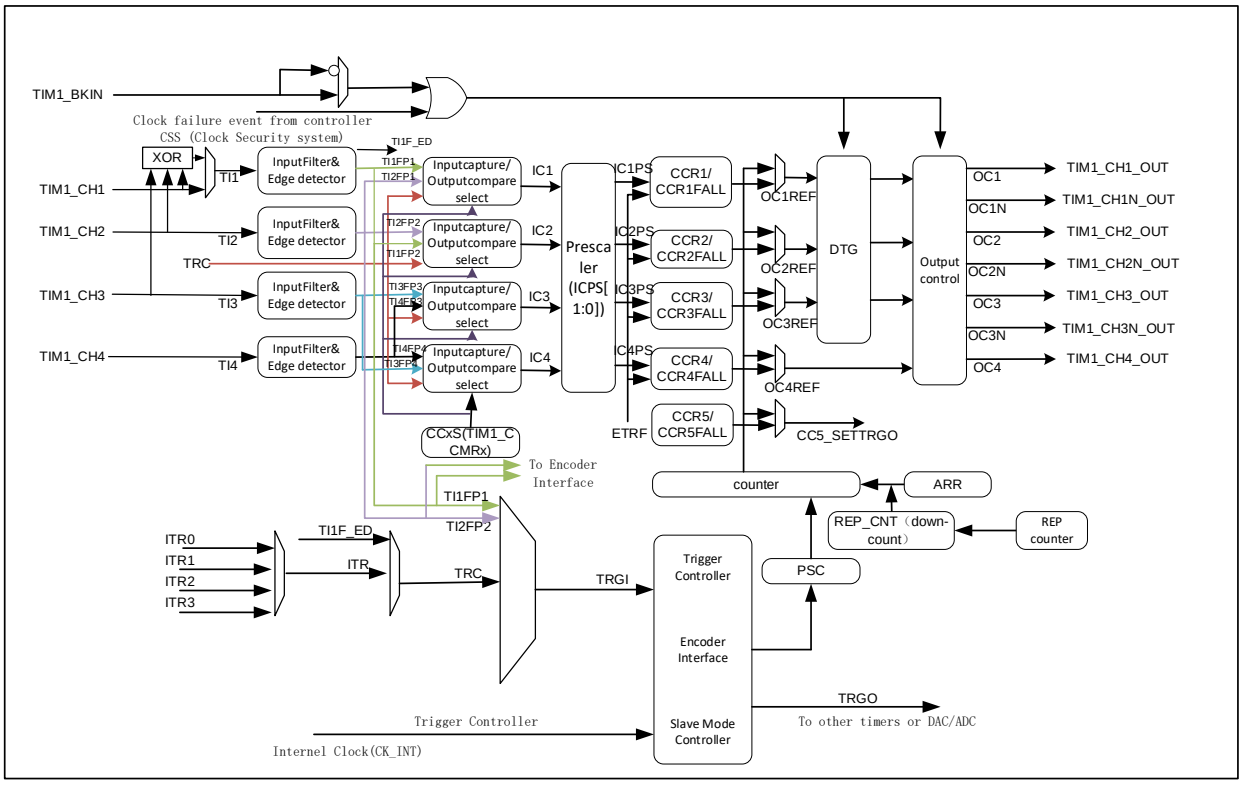


Figure 10-1 TIM1 block diagram

TIM1 structure block diagram is mainly composed of input capture unit, output compare unit, counter unit and break unit.

10.3 Main characteristics

- 16 bit real-time programmable prescaler, division factor: 1-65536.
- Clock source: Internal clock source, external clock input, internal trigger input.
- 16 bit auto reload counter (counting direction: up, down, up/down).
- 8 bit programmable repeat counting function, repeat counter can be auto reloaded (auto reload at update event)
- Synchronization circuit to control the timer with external signals and to interconnect several timers to each other.
- Input capture: measure input signal pulse width or period.
- Trigger input can serve as external clock or be managed cycle-by-cycle.
- Support interfaces such as encoder and hall-sensor.
- 4 output channels, channel 1/2/3 has complementary output channels and channel 4 doesn't have complementary output channel.
- Output compare (control output waveform or indicator counter already finishes timing).
- PWM output (dead-time programmable) (edge alignment or central alignment mode)
- Break input will put the timer output signal in a safe state (reset state or preset state, choice by user).
- One-pulse mode.
- Interrupt request event: update event, trigger event, input capture, output compare or break input.

10.4 Function description

10.4.1 Clock

10.4.1.1 Clock selection

Several types of clock sources of the counter below:

- Internal clock (INT_CK)
- External clock mode 1: External trigger input TRGI (including TIx, ITRx)
- Encoder mode

The diagram of several types of clock choice above:

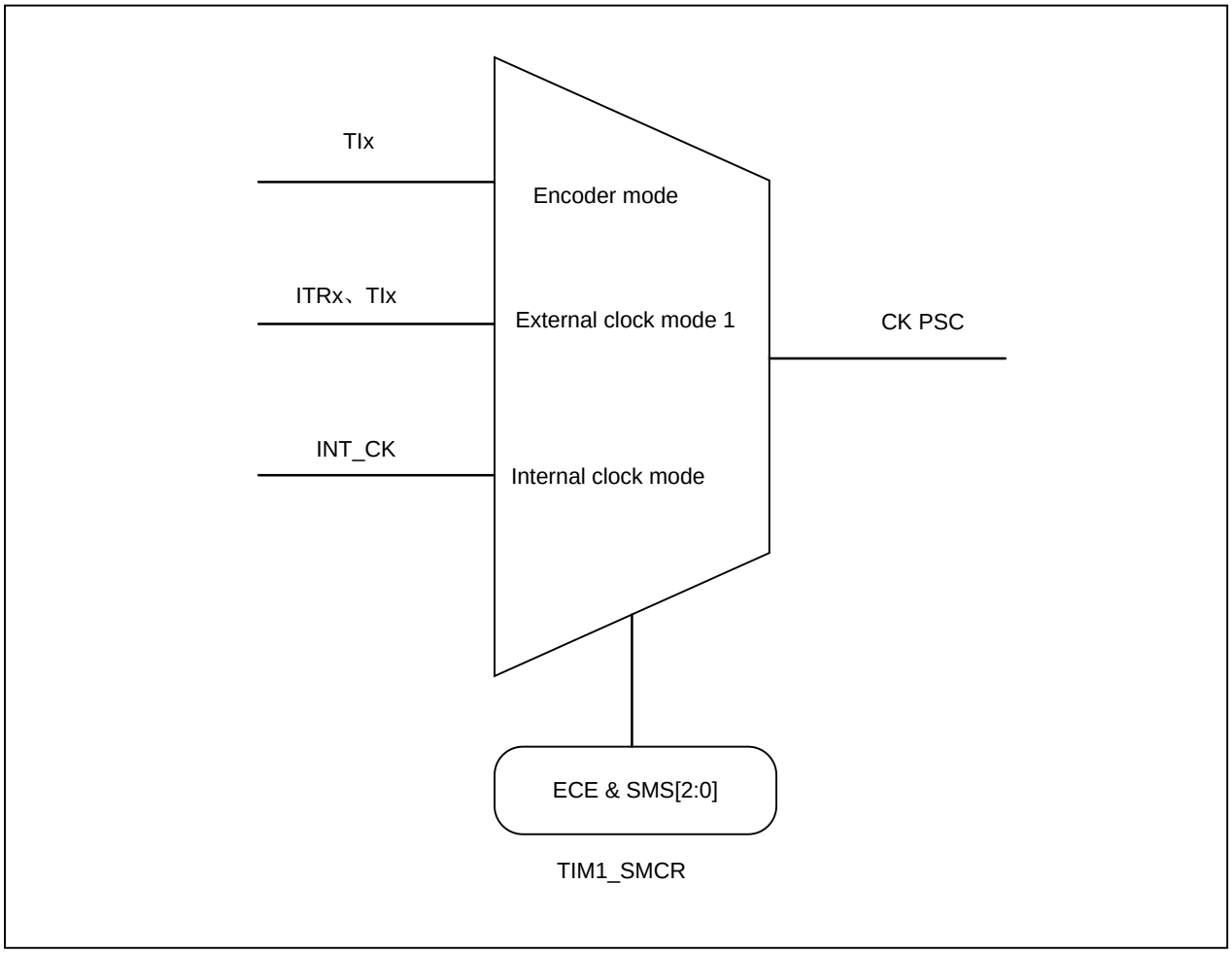


Figure 10-2 Clock selection

10.4.1.1.1 Internal clock source (INT _CK)

In case of SMS =000 of TIM1_SMCR register, disable the slave mode. Enable the counter, and the prescaler clock is driven by internal clock.

10.4.1.1.2 External clock mode 1(external trigger input TRGI, including TIx, ITRx)

In case of SMS = 111 of TIM1_SMCR register, select the external clock mode 1 (TRGI). The counter is driven by the rising edge or falling edge of the selected input signal.

For example: count up at the rising edge on TI1 input. Specific configuration:

1.Configure TIM1_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 is mapped on TI1; Configure TIM1_CCMR1 register IC1F[3： 0], set input filter bandwidth; configure TIM1_CCER register CC1P=0, and select the rising edge as the active edge.

2. Configure TIM1_SMCR register TS=101, select TI1 as the trigger input source; configure TIM1_SMCR register SMS=111, select the external clock mode 1.

3. Configure TIM1_CR1 register DIR=0, select up counting mode; configure TIM1_CR1 register CEN=1 and start the counter.

TI1 input the active edge, count up once. The delay between the active edge of TI1 and actual clock of counter depends on the TI1 input pin sync circuit.

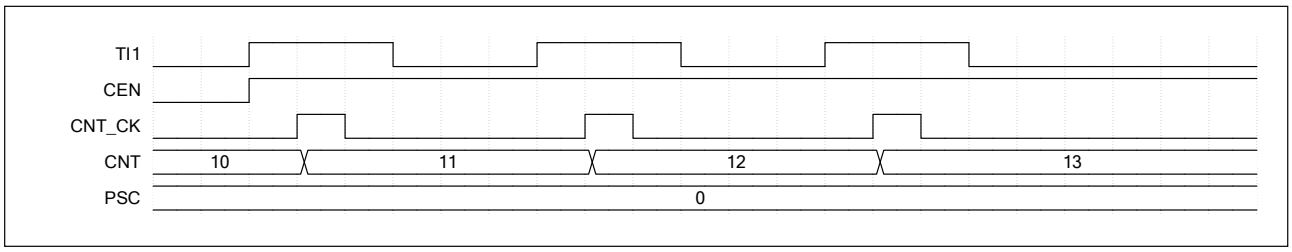


Figure 10-3 Control circuit in external clock mode 1

10.4.1.1.3 Encoder mode

Refer to the chapter of slave mode-encoder interface for details.

10.4.1.2 Time base unit

TIM1 time base unit mainly includes: Counter register (TIM1_CNT), prescaler register (TIM1_PSC), auto reload register (TIM1_ARR) and repetition counter register (TIM1_RCR).

The counter unit is composed of a 16 bit counter with its related auto reload register, The counter can count up, down or both up and down.

The counter clock is provided by prescaler. The prescaler is composed of the prescaler counter with its related register. The division factor is 1-65536. It can write any time, and gets active at the next update.

The auto preload register has a 16 bit shadow register with preload function. Set TIM1_CR1 register ARPE bit to select the content of ARR register are transferred into the shadow register permanently or at each update event.

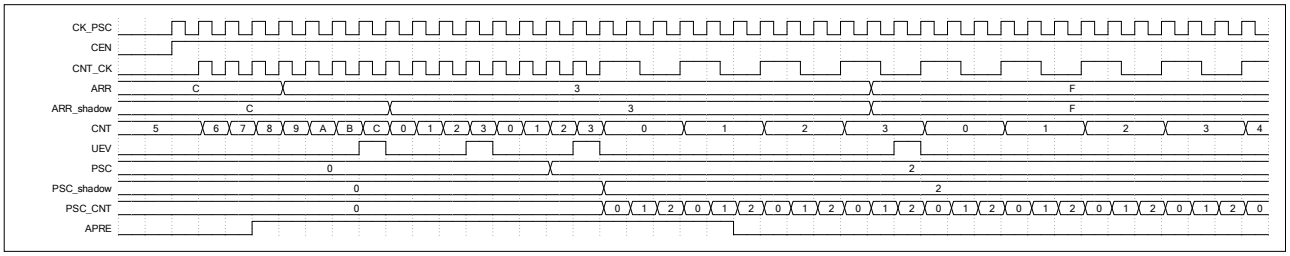


Figure 10-4 Auto preload

10.4.1.3 Counter modes

Configure DIR bit and CMS bit of TIM1_CR1 register to select the count mode. There are three count

modes, up counting mode, down counting mode and central alignment mode (up/ down counting mode). Detailed descriptions are provided for each count mode.

10.4.1.3.1 Up counting mode

Configure TIM1_CR1 register　CMS=0 , DIR＝0, select up counting mode.

In up counting mode, enable TIM1_CR1 register CEN, and the counter begins up count from 0 to TIM1_ARR value and generates a counter overflow event (update). The counter begins up count again from 0. When the user enables the repeat count function, the repeat counter conducts down count at each overflow event. When the repeat counter down to 0 from the set value, generate a update event. Set TIM1_EGR register UG =1, also generate an update event.

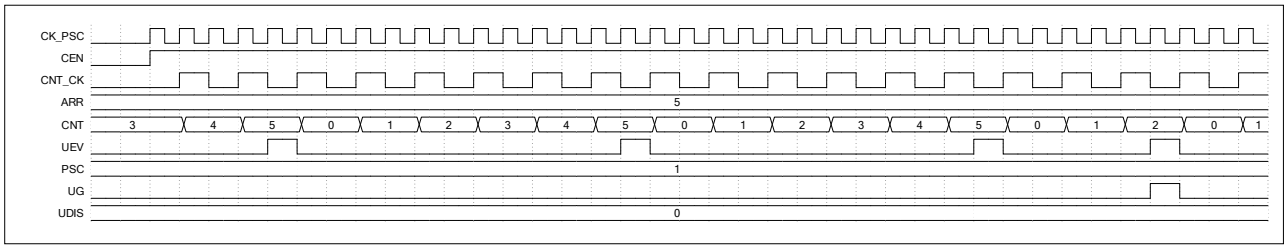


Figure 10-5 Up counting mode（UDIS=0）

Configuring TIM1_CR1 register UDIS=1, disable update event. In case of counter overflow event, the update event is not generated. Configuring UG=1, the update event is not generated. However, the counter and prescaler counter will initialize, and begin up count from 0.

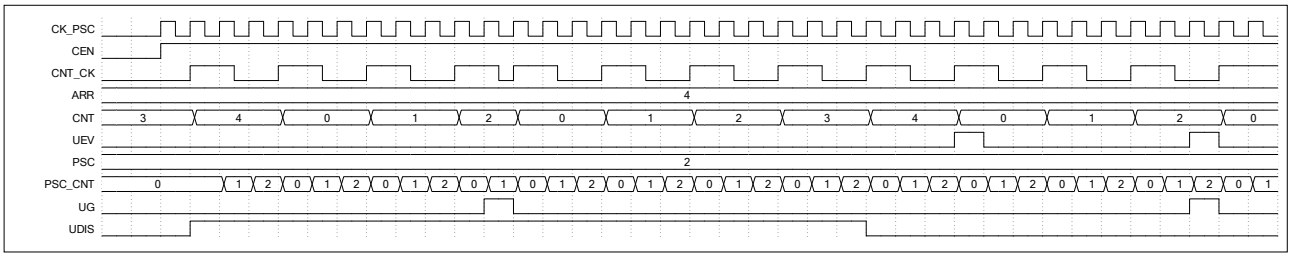


Figure 10-6 Up counting mode (UDIS =1, disable update event)

Note: In case of update event:

- Repeat counter is loaded with RCR register value, and down count again.
- ARR register value is loaded into ARR shadow register.
- Prescaler preload value becomes active.

10.4.1.3.2 Down counting mode

Configure TIM1_CR1 register CMS=0, DIR＝1, select down counting mode.

In down counting mode, the counter begins down count from the auto preload value TIM1_ARR. When counting to 0, generate an underflow event (update event).When the user enables the repeat count

function, the repeat counter conducts down count at each underflow event. When the repeat counter down to 0 from the set value, generate the update event. Set TIM1_EGR register UG =1, also generate a update event. After the update event, begin down count (TIM1_CR1 register UDIS=0) from the auto load value TIM1_ARR.

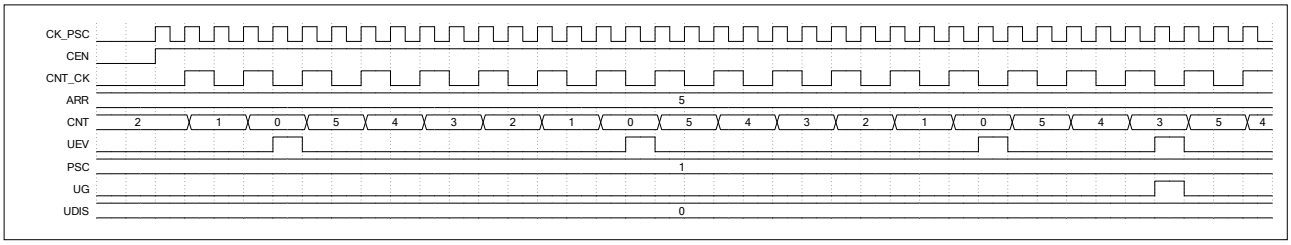


Figure 10-7 Down counting mode（UDIS＝0）

Configuring TIM1_CR1 register UDIS=1, disable update event. When the counter generates the underflow event, it doesn't generate update event. Configuring UG=1 also doesn't generate update event. However, the counter and prescaler counter will initialize, and count begins from TIM1_ARR.

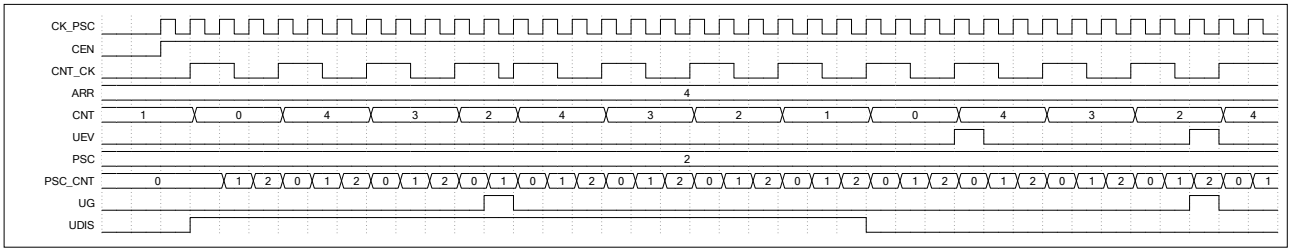


Figure 10-8 Down counting mode (UDIS=1 disables update event)

10.4.1.3.3 Central count mode (up/down count mode)

Configure TIM1_CR1 register CMS ≠ 0 (write DIR failure), and select central alignment count mode.

Central alignment count mode, up count and down count run alternatively. In case of up count to ARR-1, generate an overflow event. Begin down count to 1 from ARR, generate an underflow event, and begin up count from 0.

When the user enables repeat count function, the repeat counter count down once at each overflow or underflow event. When the repeat counter down to 0 from the set value, generate a update event; setting TIM1_EGR register UG =1 also generate a update event. After update event, the counter begins up count again from 0 (TIM1_CR1 register UDIS=0).

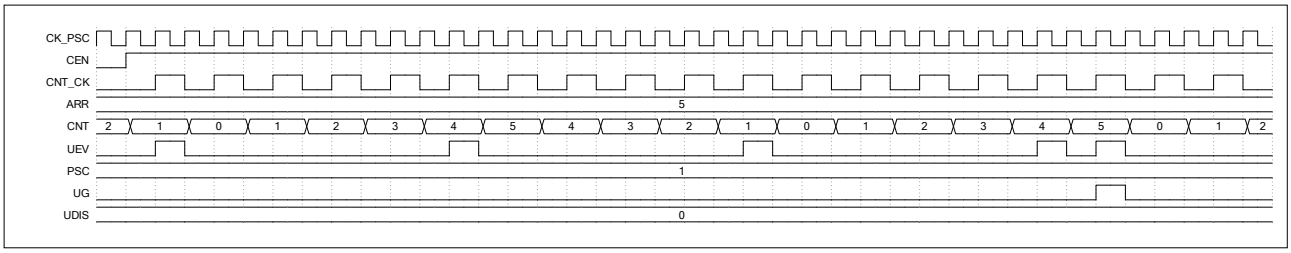


Figure 10-9 Central count mode（UDIS=0）

When configuring TIM1_CR1 register UDIS=1, disable update event. When the counter generates

the overflow or underflow event, it doesn't generate update event. Configuring UG=1 also doesn't generate update event. However, the counter and prescaler counter will initialize, and the count begins from 0.

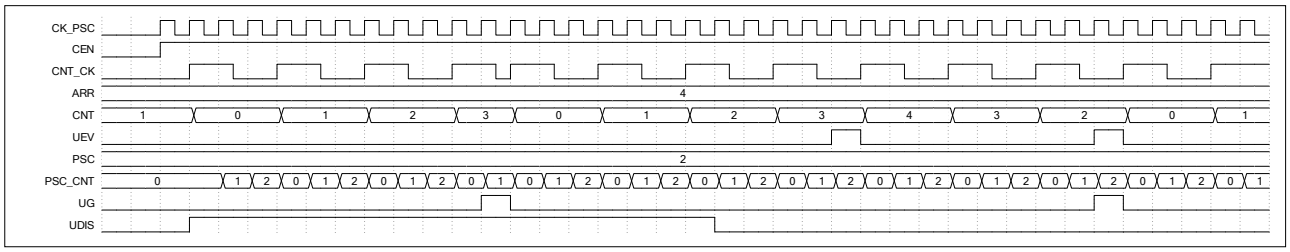


Figure 10-10 Central count mode (UDIS=1 disables update event)

10.4.2 Repeat counter

The repeat counter may adjust the frequency of the update event. During up count in the edge alignment mode, the repeat counter counting down at each overflow of the counter; during down count, the repeat counter counting down at each underflow of the counter. In the central alignment mode, the repeat counter counting down during the counter overflow and underflow. Configuring REP of the TIM1_RCR register to adjust the frequency of the update event. The repeat counter generates the update event after REP+1 count cycle. In the central alignment mode, the writing REP value decides whether the update event generates at overflow or underflow.

In case of the update event, REP value will be updated to the real-time repeat counter REP_CNT. To enable the real-time writing of REP_CNT achieves the flexible adjustment of the occurrence time of the update event.

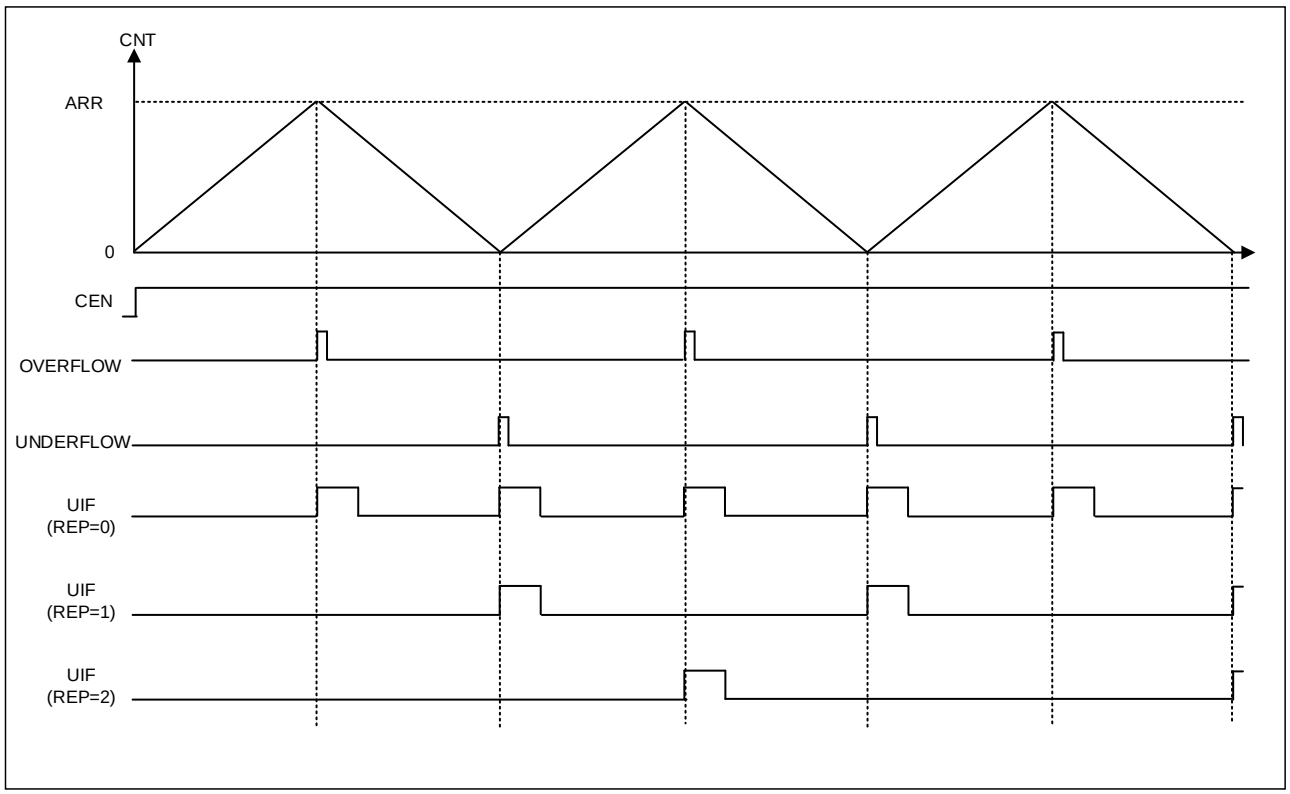


Figure 10-11 Repeat count time sequence in the central alignment mode

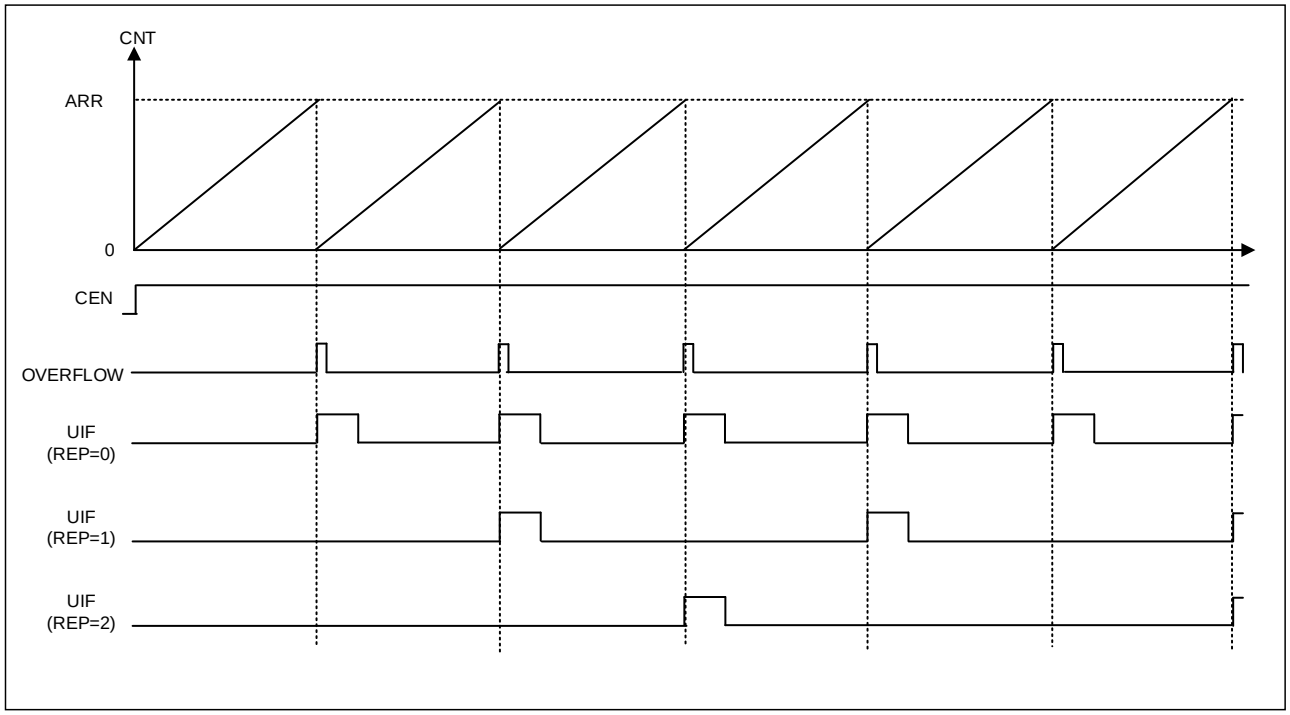


Figure 10-12 Up count time sequence in the edge alignment mode

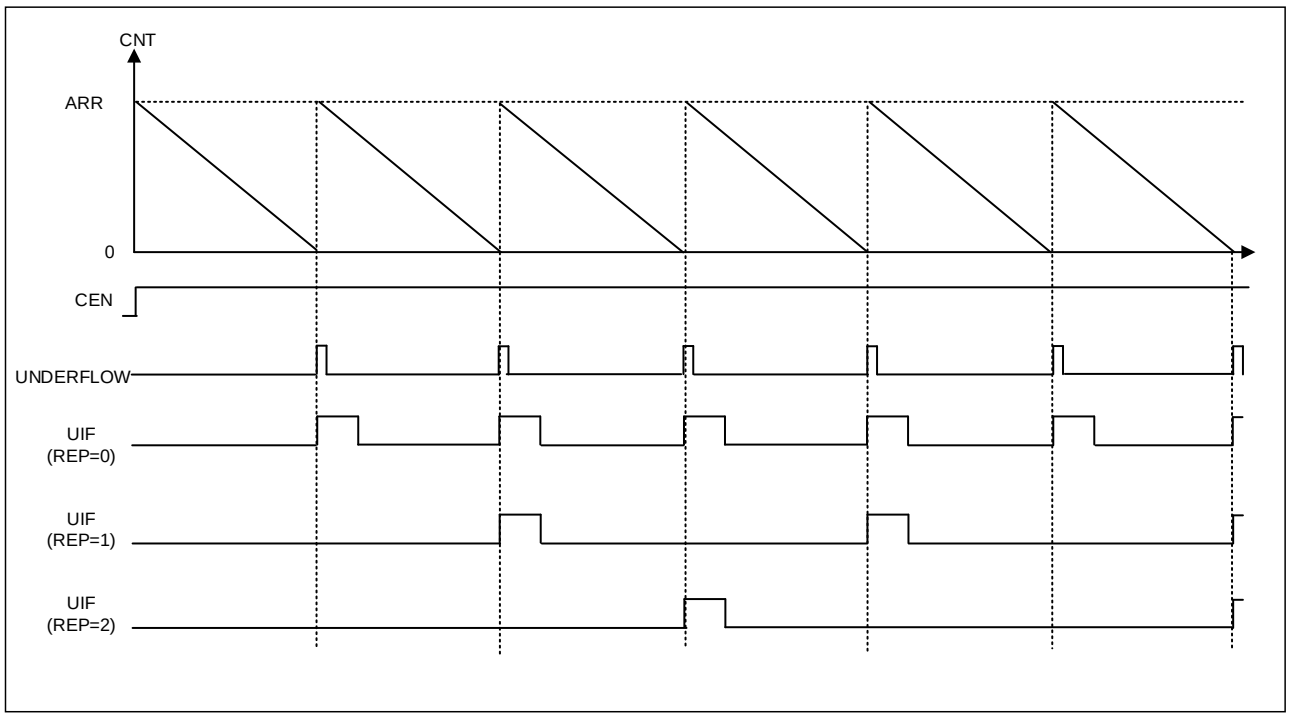


Figure 10-13 Down count time sequence in the edge alignment mode

10.4.3 Input capture

10.4.3.1 Input capture

The input capture is composed of digital filter, multiplex, prescaler, etc. Its block diagram:

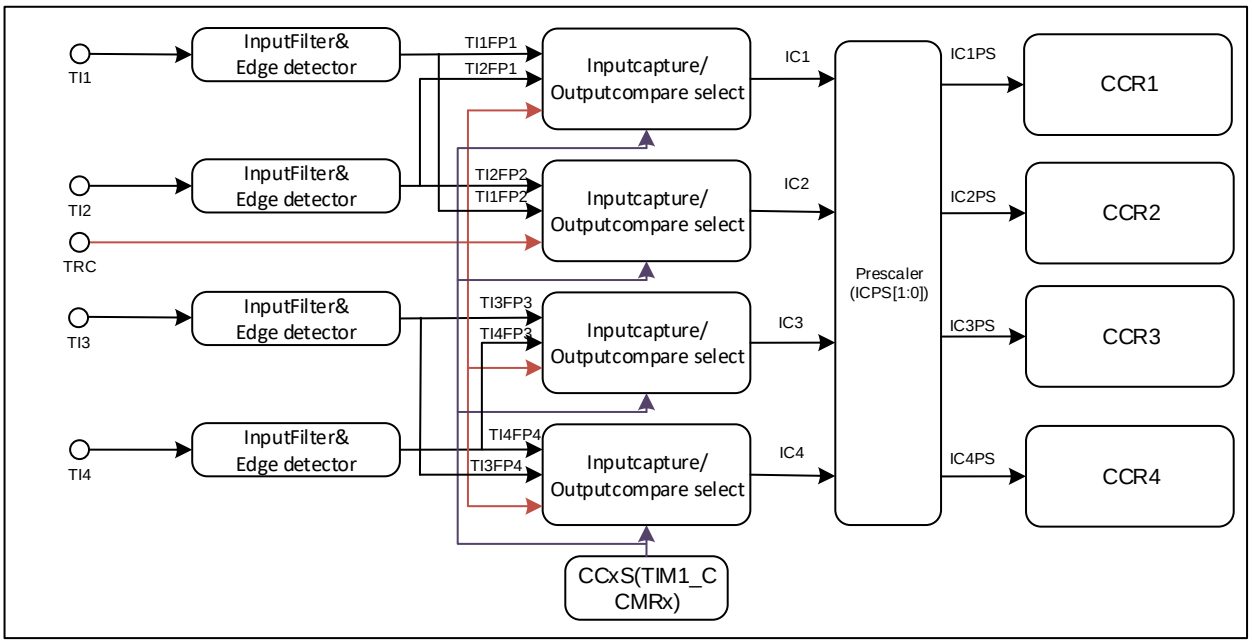


Figure 10-14 TIM1 input capture block diagram

Configure TIM1_CCMRx register ICxF, set the filter width of a digital filter (filter sampling frequency

and digital filter width are shown in the table below). When the input signal width is greater than the filter width, the input signal is active. When the digital filter samples the input signal of the input pin TIx, generate a filtered signal TIxF. Through the edge detector with the optional polarity, generate an effective signal TIxFPx, this signal can serve as the trigger input signal of the slave mode counter. This signal generates ICxPS via the prescaler for trigger input capture event.

Table 10-1 Digital filter width and ICxF correlation

IC1F[3：0]	Sample frequency and filter length	IC1F[3：0]	Sample frequency and filter length
0000	No filter, sample by f_{DTS}	1000	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=6
0001	Sampling frequency fsampling=f_{INT_CK}, N=2	1001	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=8
0010	Sampling frequency fsampling=f_{INT_CK}, N=4	1010	Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=5
0011	Sampling frequency fsampling=f_{INT_CK}, N=8	1011	Sampling frequency $f_{sampling}=f_{DTS}$/16, N=6
0100	Sampling frequency fsampling=f_{DTS} /2, N=6	1100	Sampling frequency $f_{sampling}=f_{DTS}$ /16, N=8
0101	Sampling frequency fsampling=f_{DTS}/2, N=8	1101	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=5
0110	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=6	1110	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=6
0111	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=8	1111	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=8

In the input capture mode, when detecting the active edge of signal ICx, the counter's current value is locked in the corresponding shadow register, and then loaded into the corresponding capture/compare register. When enabling interrupt, and producing the capture event, generate the corresponding interrupt. At the capture event, the capture flag bit CCxIF of the status register (TIM1_SR) will be set. Configure CCxIF =0 or read TIM1_CCRx data, clear CCxIF flag bit. When CCxIF is not cleared, next input capture event generated, the repeat capture flag CCxOF will be set. Configure CCxOF =0, can clear CCxOF flag bit.

For example, Sample TI1 input signal's active edge. At TI1 rising edge, capture the current counter value and lock it in the TIM1_CCR1 register, procedure steps as follows:

1. Configure TIM1_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 is mapped on TI1.
2. Configure TIM1_CCMR1 register IC1F[3:0] , configure the filter width of digital filter (configure upon need).
3. Configure TIM1_CCER register CC1P=0, select capture occurs in the rising edge of TI1 signal.
4. Configure TIM1_CCMR1 register IC1PSC[1:0], select the prescaler factor.
5. Configure TIM1_CCER register CC1E = 1, channel 1 capture enable.

6. Configure TIM1_DIER register CC1IE=1, enable channel 1 interrupt request.

Note：

- When the channel is configured as input mode, TIM1_CCRx register becomes read only.
- In case of more than two continuous captures and CCxIF flag is not cleared, the repeat capture flag CCxOF is set. In order to avoid losing the capture information that may be generated before the repeat capture flag CCxOF is set. It is recommended to read data before reading the repeat capture flag.
- Set the corresponding CCxG bit of TIM1_EGR register, generate the input capture interrupt request through software.

10.4.3.2 PWM capture

Differences of PWM input mode configuration with ordinary input capture:

- Two ICx signals are active on edges with opposite polarity and mapped on the same TIx input.
- Configure the slave mode counter at reset mode, set one TIxFP as trigger input signal.

For example: Measure TI1 PWM signal period (TIM1_CCR1 register) and duty cycle (TIM1_CCR2 register), the measuring value depends on the internal clock INT_Ck frequency and prescaler value. Specific steps:

1.Configure TIM1_CR1 register DIR =0, select counter as up counting mode.

2.Configure TIM1_CCMR1 register CC1S = 01, map IC1 on TI1, and select active input of TIM1_CCR1.

3. Configure TIM1_CCER register CC1P =0, select active polarity of TI1FP1 (active on rising edge) (capture the counter value to TIM1_CCR1, and clear the counter).

4.Configure TIM1_CCMR1 register CC2S =10, map IC2 on TI1, select active input of TIM1_CCR2.

5. Configure TIM1_CCER register CC2P =1, and select active polarity of TI2FP2 (active on falling edge) (capture counter value to TIM1_CCR2).

6.Configure TIM1_SMCR register TS = 101, select TI1FP1 as the active trigger input signal.

7.Configure TIM1_SMCR SMS = 100, set the slave mode counter as reset mode.

8.Configure TIM1_CCER register CC1E=1 and CC2E = 1. Enable CC1 channel and CC2 channel capture.

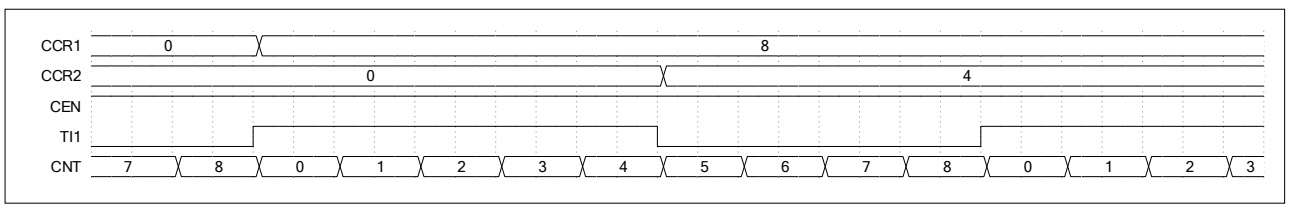


Figure 10-15 PWM input mode sequence

Note: The slave mode counter only connects TI1FP1 and TI2FP2, PWM input mode is only applicable to TIM1_CH1/TIM1_CH2 port input signal.

10.4.4 Output compare

The compare output of the capture/compare channel is composed of the comparator, output control circuit and capture/compare register. Its structure is shown below:

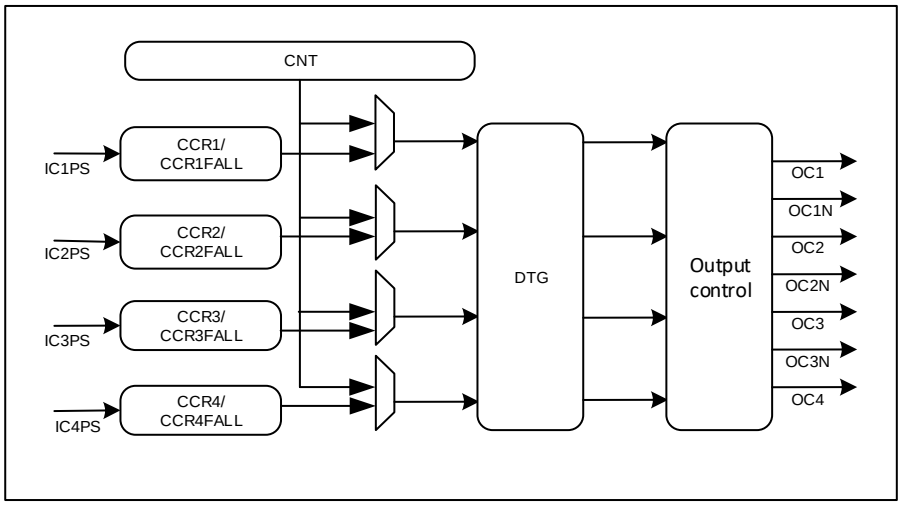


Figure 10-16 Output compare block diagram

In output compare mode, the content of capture/compare register is loaded into the shadow register, and then compare the content of shadow register with the current value of counter. The capture/compare module includes a capture/compare register (preload register) and a shadow register. When read and write, only programmed the capture/compare register.

10.4.4.1 Force output

Configure TIM1_CCMRx register CCxS = 00 and set the channel CCx as output mode. Configure TIM1_CCMRx register OCxM bit, and directly enforce the output compare signal as active or inactive level, independently of the output compare result. Configure TIM1_CCMRx register OCxM = 100, and enforce output compare signal as inactive level. And OCxREF is enforced as low level. Configure TIM1_CCMRx register OCxM = 101, and enforce output compare signal as active level. And the OCxREF is enforced as high level (OCxREF always active as high level).

Note: In force output mode, TIM1_CCRx shadow register and counter output compare is still performed. The corresponding flag bit of compare result will also be changed. When enable the corresponding interrupt, it will also generate the corresponding interrupt request.

10.4.4.2 Output compare

In the output compare mode, when the values of counter and capture compare register are matches, different waveform can be exported according to the OCxM bit of TIM1_CCMRx register.

For example, When the value of counter and capture/compare register matches, the functions in the output compare mode:

1.In compare match, OCxM value is different, output channel x signal OCx run differently:

- OCxM = 000: OCx signal keeps its level.
- OCxM = 001: OCx signal is set as active level.
- OCxM = 010: OCx signal is set as inactive level.
- OCxM = 011: OCx signal toggle.

2. When match, set the flag bit of status register (CCxIF bit of TIM1_SR register).
3. Configure TIM1_DIER register CCxIE =1, generate an interrupt when match.

The output compare mode can also be used to export a single pulse (one-pulse mode).

Configuration steps of channel 1 output compare mode:

1. Configure counter clock (select clock source, configure prescaler factor).
2. Configure TIM1_ARR and TIM1_CCR1 register.
3. Configure TIM1_DIER register CC1IE =1, enable capture/compare 1 interrupt.
4. Configure output mode:
   - Configure TIM1_CCMR1 register OC1M = 011, toggle when OC1 compare match.
   - Configure TIM1_CCMR1 register OC1PE = 0, disable preload function of TIM1_CCR1 register.
   - Configure TIM1_CCMR1 register CC1P = 1, OC1 is active low polarity.
   - Configure TIM1_CCER register CC1E = 1, enable channel 1 output, OC1 signal is exported to the corresponding output pin.
5. Configure TIM1_CR1 register CEN =1, enable the counter.

Configure TIM1_CCMRx register OCxPE=0, disable TIM1_CCRx register preload function, write TIM1_CCRx register any time, and the write value updated immediately. Configure TIM1_CCMRx register OCxPE=1, enable TIM1_CCRx register preload function, Updated only at the next update event. The figure below is an example.

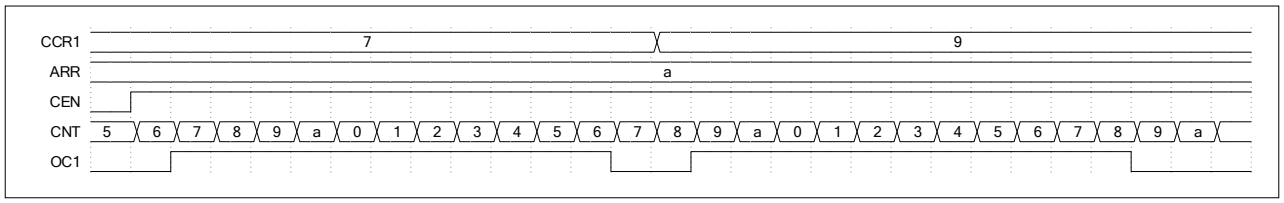


Figure 10-17 Output compare mode, OC1 signals toggle when match

Note: In the output compare mode, the update event has no effect on output result. In the force output mode, the compare output between TIM1_CCRx shadow register and counter still performed. The corresponding flag bit of compare result is also be changed. When enabling the corresponding interrupt request, it still generates the corresponding interrupt request.

10.4.4.3 PWM output

In the PWM mode, according to the value of TIM1_ARR register and TIM1_CCRx register, generate a frequency and duty cycle controllable PWM waveform.

Configure OCxM=110 or OCxM=111 of TIM1_CCMRx register corresponding to the channel x, select channel x to PWM mode 1 or PWM mode 2. In the PWM mode, the counter will always compare with CCRx. According to configuration and compare results, the channel x exports different signals. Therefore, TIM1 can generate 4 PWM output signals with independent duty under the same frequency. In PWM mode, enable the preload function of TIM1_CCRx and TIM1_ARR. The written value of TIM1_CCRx preload register and TIM1_ARR preload register will become effective at the next update event, and load the corresponding shadow register. Therefore, in the PWM mode, before enabling counter, it's required to set TIM1_EGR UG=1, and generate the update event to initialize all registers.

Configure TIM1_CCER register CCxP, select OCx's active polarity. Configure TIM1_CCER register CCxE, CCxNE bit and TIM1_BDTR register MOE, OSSI, OSSR bit and control OCx output enable. Configure TIM1_CR1 register CMS bit, and select edge alignment or central alignment PWM signal.

- CMS=00, edge alignment mode, configure DIR, select count mode (up or down).
- CMS=01, central alignment mode 1.
- CMS=10, central alignment mode 2.
- CMS=11, central alignment mode 3.

10.4.4.3.1 PWM edge alignment mode - up counting mode

Based on up counting mode configuration, configure TIM1_CCMRx register CCxS =00, select output mode, OCxM =110, select PWM mode 1. In case of TIM1_CNT ＜TIM1_CCRx, channel x (OCxREF) is in active level. Otherwise, it's inactive level. If TIM1_CCRx compare value is greater than the auto reload value (TIM1_ARR), OCxREF stays in active level. When the compare value is 0, OCxREF is at inactive level. The figure below is the example of waveform in PWM mode 1 during edge alignment up count in case of CCR1 =1, CCR2 =4, CCR3 =7, CCR4 =b and ARR =a.

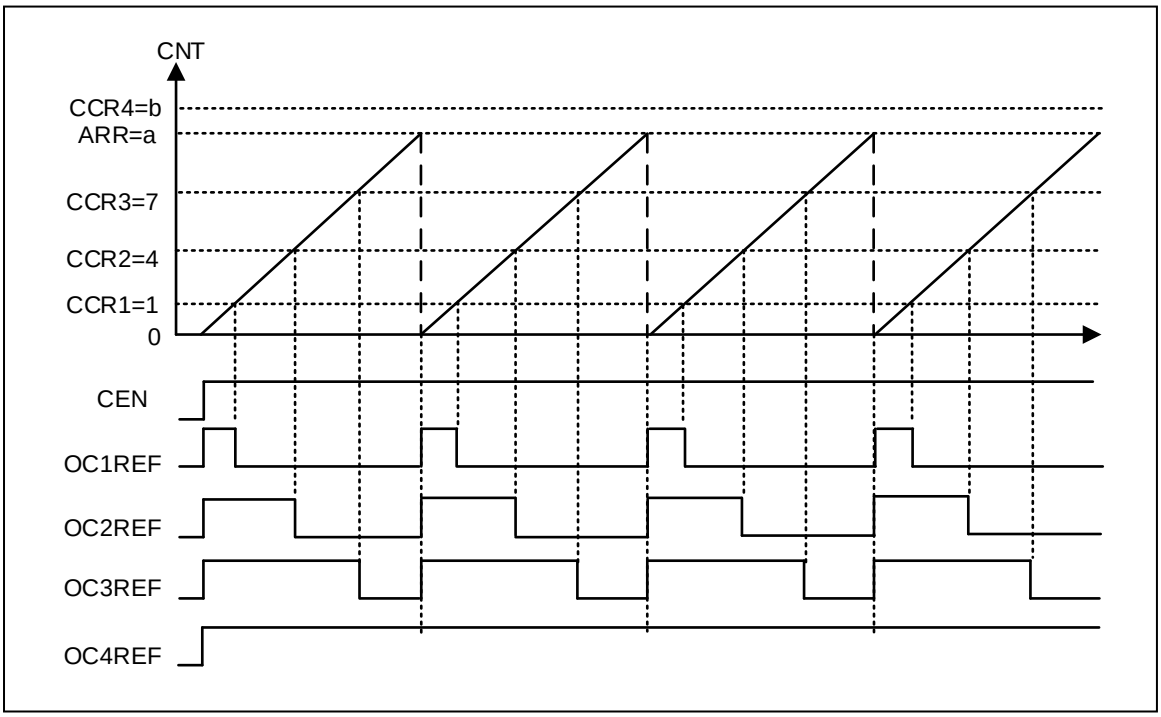


Figure 10-18 Waveform under PWM mode 1 in edge alignment up count

10.4.4.3.2 PWM edge alignment mode -down counting mode

Based on the down counting mode configuration, configure TIM1_CCMRx register CCxS=00, select output mode, OCxM=110, select PWM mode 1. In case of TIM1_CNT > TIM1_CCRx, channel x (OCxREF) is at inactive level. Or else, it's at active level. The figure is an example of waveform in PWM mode 1 during edge alignment down count in case of CCR1=4 , CCR2=6, CCR3=9 , CCR4=b and ARR=a.

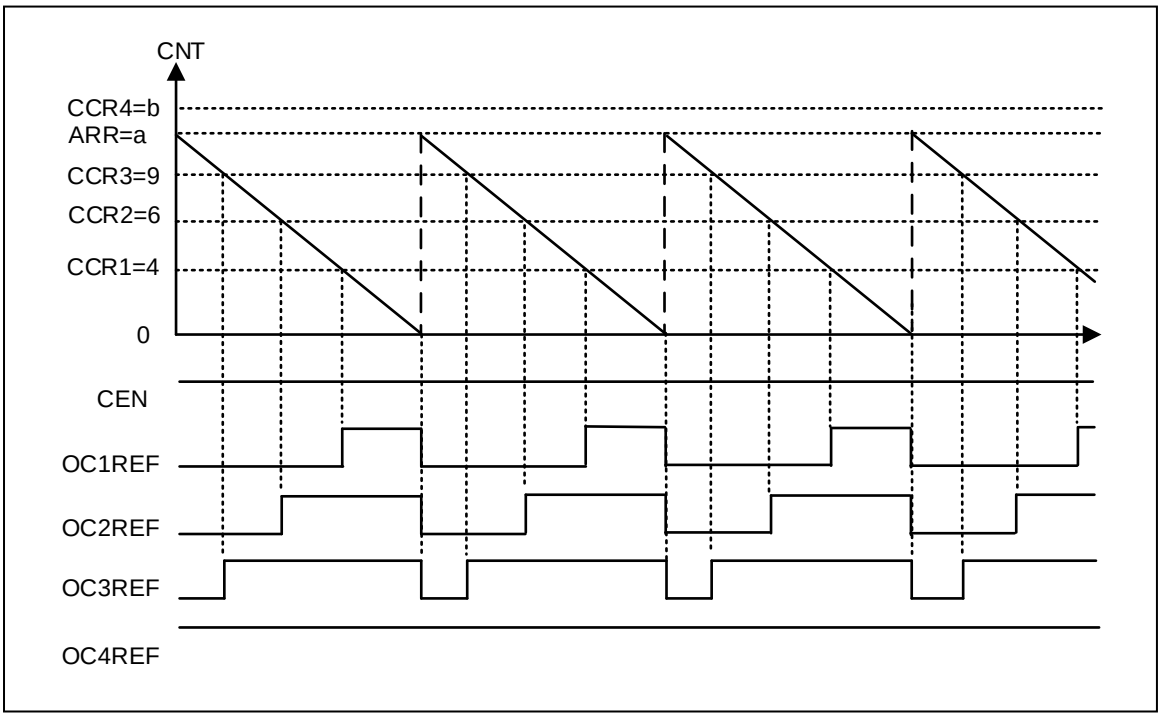


Figure 10-19 Waveform in PWM mode 1 during edge alignment down count

10.4.4.3.3 PWM central alignment mode

Configure TIM1 counter as central alignment count mode. Configure TIM1_CCMRx register CCxS=00 and select output mode. According to the CMS with different configurations, the output compare interrupt flag bit is set during up count (CMS =01), set during down count (CMS =10) or set during up or down count (CMS =11). The figure below is an example of waveform in central alignment PWM mode 1 (CMS =1) during CCR1=4, CCR2=6, CCR3=9, CCR4=b, ARR=a.

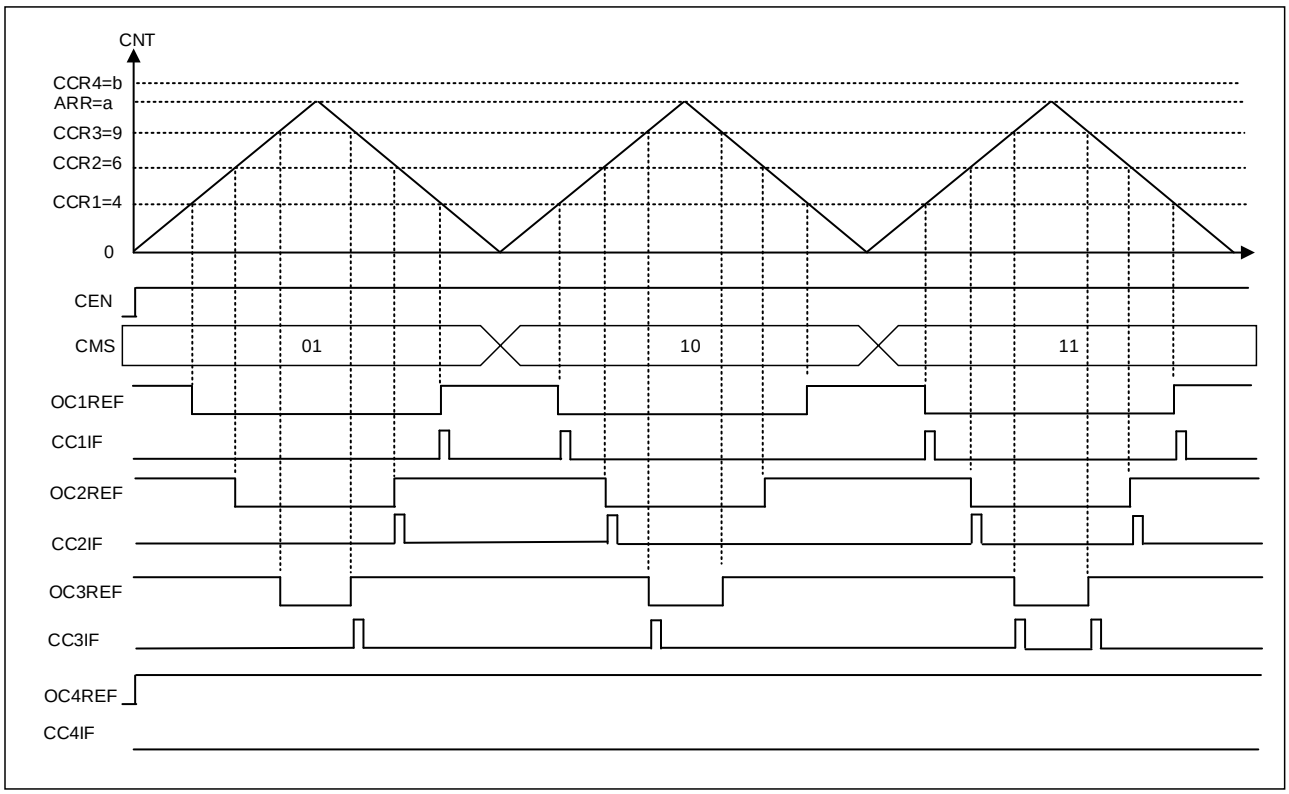


Figure 10-20 Waveform under central alignment PWM mode 1

10.4.4.3.4 Phase move function under PWM central alignment mode

Configure PDER (channel x output PWM phase shift enable bit) and CCRxFALL register (channel x capture/compare value in PWM central alignment down count mode), set the channel output PWM phase shift function. Configure CCRxFALL and CCRx, PWM output programmable phase shift waveform, and it can shift left or right.

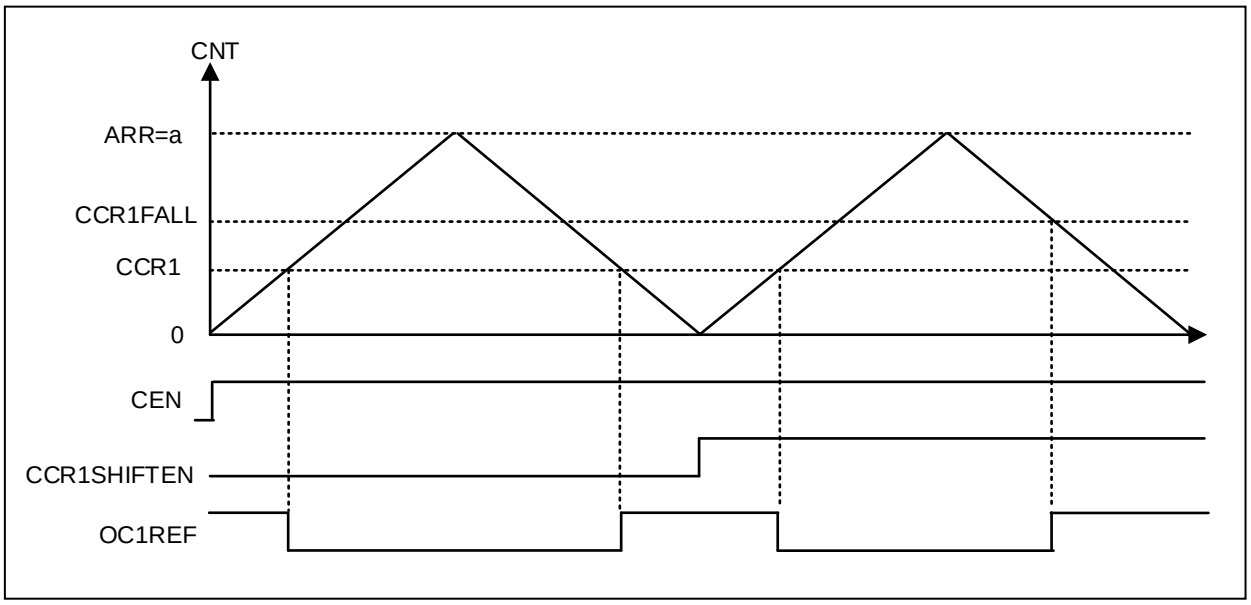


Figure 10-21 Move phase function

Note：

- In central alignment mode, use the current up/down count configuration, and the count direction depends on the current DIR value.
- In the central alignment mode, it's better not to change the counter value or it may generate unexpected result. When the counter is in up counting, write the counter value > TIM1_ARR, and the counter will continue up count. Writing 0 or ARR will immediately update the count direction. However, it won't generate the update event.
- In the central alignment mode, before enabling counter, configure TIM1_EGR register UG =1, generate a software update. Update all registers, don't change the counter value after enabling the counter.

10.4.4.3.5 6-step PWM output

Configure OCxM and select output mode. CCxE=1 and CCxNE=1, Open channel x and complementary channel output enable. In channel x, generate complementary output. These functions are in preload bit. In case of COM commutation event, these preload bits will be loaded in the corresponding shadow register. Thus, writing these bits won't influence the current output, and can change the configuration of all the channels at the same time. Configure TIM1_EGR register COMG =1 or generate COM event at TRGI rising edge.

In case of COM event, COM interrupt flag will be set by hardware. Configure TIM1_DIER register COMIE=1 and COMDE = 1, generate COM event will generate one COM interrupt request at the same time.

OCx and OCxN output under different configurations in case of COM event.

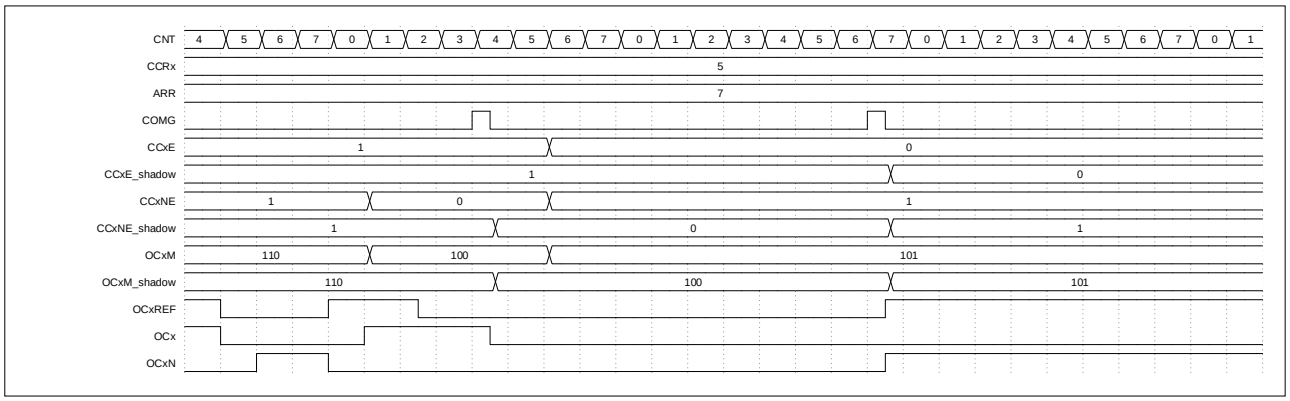


Figure 10-22 Example of six step PWM with COM (OSSR = 1)

10.4.4.4 Complementary output and dead-time insertion

OCx and OCxN are a pair of complementary output channel. TIM1 channel 1/2/3 can export three complementary signals with switching-off and the switching-on instants of the outputs. It also has the

programmable dead-time area. The user adjusts the dead time according to the connected output devices and their characteristics (intrinsic delays of level-shifters, delays due to power switches).

TIM1_BDTR register DTG[7:0] bit defines the dead-time when inserting the complementary output. The specific calculation mode is given in the table below:

Table 10-1 Dead-time calculation

DTG[7：5]	DT
0xx	DT= DTG[7：0] x T_{dtg}（T_{dtg}=T_{DTS}）
10x	DT=（64+DTG[5：0]）x T_{dtg}（T_{dtg} =2xT_{DTS}）
110	DT=（32+DTG[4：0]）x T_{dtg}（T_{dtg} =8xT_{DTS}）
111	DT=（32+DTG[4：0]）x T_{dtg}（T_{dtg} =16xT_{DTS}）

For example, in case of T_{DTS}=125ns, possible dead time:

- In case of step time 125ns, dead time 0-15875ns.
- In case of step time 250ns, dead time 16μs -31750ns.
- In case of step time 1μs, dead time 32μs -63μs.
- In case of step time 2us, dead time 64μs -126μs.

In case of no break circuit, configure CCxE=1 and CCxNE=1, enable dead-time insert. Otherwise, configure MOE =1.

Configure TIM1_CCER register CCxP and CCxNP, and independently select polarity for each output (main output OCx or complementary output OCxN).

Configure TIM1_CCER register CCxE and CCxNE bit. Different combinations of MOE, OISx , OISxN, OSSI and OSSR in TIM1_BDTR and TIM1_CR2 register can control the complementary signal OCx and OCxN output. The specific combination control configuration is given in the complementary output channel OCx and OCxN control bit in table 3, table 4, table 5 and table 6 in this chapter.

For example: OCx and OCxN are active high. In PWM mode, in case of match, export reference signal OCxREF signal toggle and output signal OCx and reference signal are same. However, OCx signal’s rising edge has a latency with response to the reference signal; complementary output signal OCxN and reference signal are opposite. OCxN signal’s rising edge has a latency with response to the reference signal’s falling edge.

Note: Dead time cannot be more than or equal to the duty cycle of OCx or OCxN signal (active output). Or else, OCx or OCxN signal are always inactive values.

The figures below show the relation of dead-time generator output signal and current reference signal OCxREF.

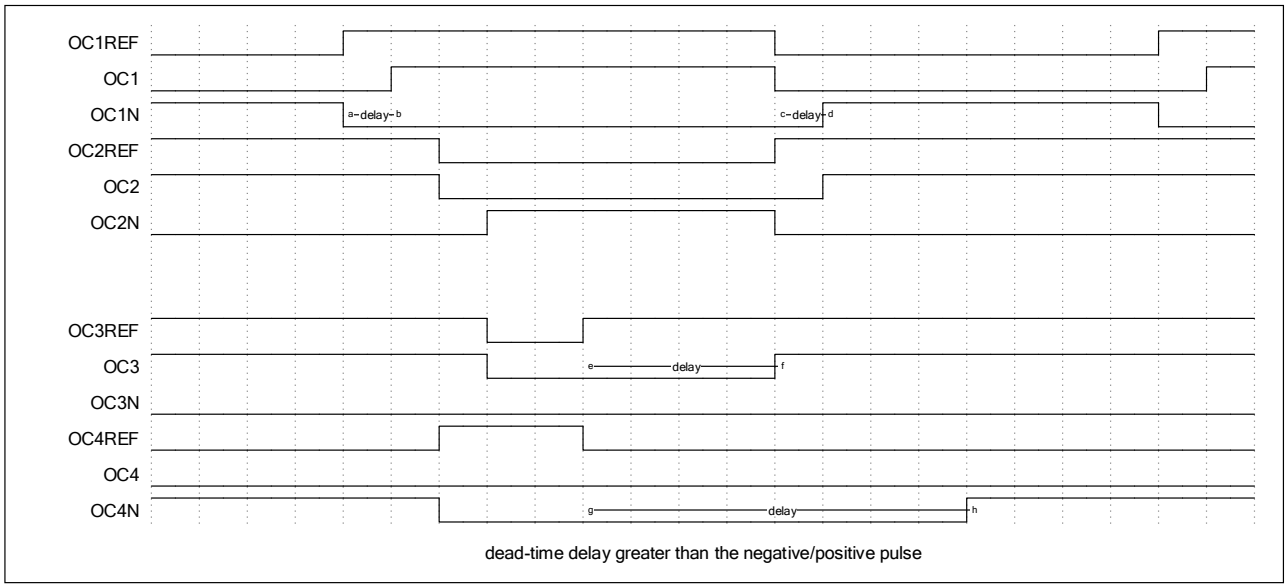


Figure 10-23 Dead-time insertion

10.4.4.5 Break function

Break input pin, clock failure event and comparator output may serve as TIM1 break source. The clock failure event is generated by the clock safety system of the reset clock counter.

During the use of break function, OCx and OCxN output signal levels are controlled by the function level combination: TIM1_BDTR register MOE, OSSI and OSSR bit, and TIM1_CR2 register OISx and OISxN bit. In case of break event, OCx and OCxN output cannot be active at the same time. The output state of complementary output channel OCx and OCxN with the break function is shown in the table below.

Table 10-2 In case of MOE=1 , OSSI=0/1 , OSSR=0:

CCxE	CCxNE	OCx	OCxN
0	0	OCx=0，OCx_EN=0	OCxN=0，OCxN_EN=0
0	1	OCx=0，OCx_EN=0	OCxN=OCxREF+Polarity，OCxN_EN=1
1	0	OCx=OCxREF+Polarity，OCx_EN=1	OCxN=0，OCxN_EN=0
1	1	OCx=OCxREF+Polarity+dead-time, OCx_EN=1	OCxN=OCxREF+reverse phase +Polarity+dead-time, OCxN_EN=1

Table 10-3 In case of MOE=1, OSSI=0/1, OSSR=1

CCxE	CCxNE	OCx	OCxN
0	0	OCx=0，OCx_EN=0	OCxN=0，OCxN_EN=0
0	1	OCx=CCxP，OCx_EN=1	OCxN=OCxREF+Polarity，OCxN_EN=1
1	0	OCx=OCxREF+Polarity, OCx_EN=1	OCxN=CCxNP，OCxN_EN=1

CCxE	CCxNE	OCx	OCxN
1	1	OCx=OCxREF+Polarity+dead-time, OCx_EN=1	OCxN=OCxREF reverse phase+Polarity+dead-time, OCxN_EN=1

Table 10-4 In case of MOE=0, OSSI=0, OSSR=0/1

CCxE	CCxNE	OCx	OCxN
0	0	OCx_EN=0，OCxN_EN=0 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don’t correspond to the active level of OCx and OCxN	
0	1	OCx_EN=0, OCxN_EN=0 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don’t correspond to the active level of OCx and OCxN	
1	0	OCx_EN=0，OCxN_EN=0 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don’t correspond to the active level of OCx and OCxN	
1	1	OCx_EN=0，OCxN_EN=0 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don’t correspond to the active level of OCx and OCxN	

Table 10-5 In case of MOE=0, OSSI=1, OSSR=0/1

CCxE	CCxNE	OCx	OCxN
0	0	OCx_EN=1，OCxN_EN=1 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don’t correspond to the active level of OCx and OCxN	
0	1	OCx_EN=1, OCxN_EN=1 Insync：OCx=CCxP, OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN	

CCxE	CCxNE	OCx	OCxN
		OISx and OISxN don't correspond to the active level of OCx and OCxN	
1	0	OCx_EN=1，OCxN_EN=1 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time, OCx=OISx，OCxN=OISxN OISx and OISxN don't correspond to the active level of OCx and OCxN	
1	1	OCx_EN=1，OCxN_EN=1 Insync：OCx=CCxP，OCxN=CCxNP In case the clock is available: After a dead-time,OCx=OISx，OCxN=OISxN OISx and OISxN don't correspond to the active level of OCx and OCxN	

Note: When the channel output and complementary output are closed, OISx, OISxN, CCxP and CCxNP must be configured as 0.

After system reset, MOE =0, disable break function. Configure TIM1_BKINF register BKIN_SEL and select break channels, It supports the selection of multiple break channels, it will enter the break state if break signal from any selected channel is active. Configure BKINFE of TIM1_BKINF register, enable the break signal filtering function. Configure BKINF of TIM1_BKINF register, select the sampling frequency of break digital filter. Before changing the sampling frequency of break digital filter, turn off the break filter function. Configure TIM1_BDTR register BKE =1 and enable break function. Configure TIM1_BDTR register BKP bit and select break input signal polarity. BKP and BKE can also be written at the same time, and become active in the next clock period.

Because MOE is asynchronous cleared, insert a resynchronous circuit in between the actual signal and synchronous control, and generate latency between the synchronous and asynchronous signal (write MOE =1 when the state is 0; input an dummy instruction for latency after write and before read. Otherwise, correct reading cannot be guaranteed).

In case of break event, MOE is asynchronous cleared. According to OSSI configuration OCx/OCxN output, put it in inactive state, idle state and reset state; in case of MOE=0, the output is determined by TIM1_CR2 register OISx bit. In case of OSSI =0, the timer disables the output. Otherwise, it enables the output. When using the complementary output, the output is first put in the reset state. The dead-time is generated again. After the dead-time, the output level is determined by OISx and OISxN.

Configure TIM1_DIER register BIE =1. In case of a break event, generate a break interrupt. Configure TIM1_BDTR register AOE =1, auto set MOE bit when the next update event is occurred.

Note: The break input is acting on level. When the break input is active, cannot (auto or via software) set MOE, and the status flag BIF cannot be cleared.

In the break circuit, enable write protection and safeguard application safety. Allow the user to lock

the dead-time duration, OCx/OCxN polarity and disabled state, OCxM configuration, break enable and polarity. ConfigureTIM1_BDTR register LOCK bit, select lock levels (three levels of lock). Lock can only be written once after system reset.

The figure below shows the example of response break output:

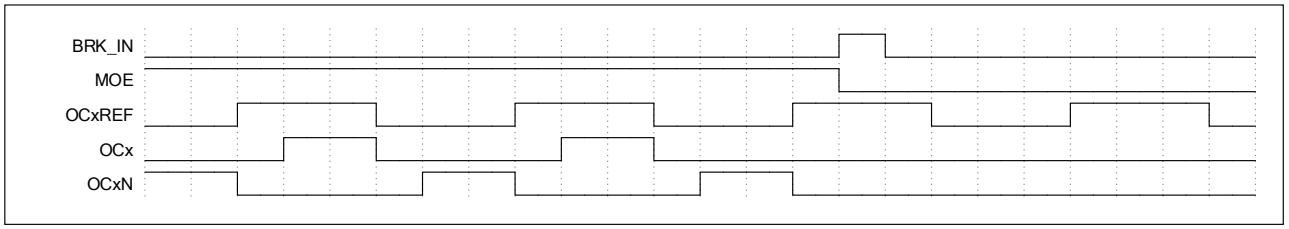


Figure 10-24 Response break output（OISx=0 , OISxN=0）

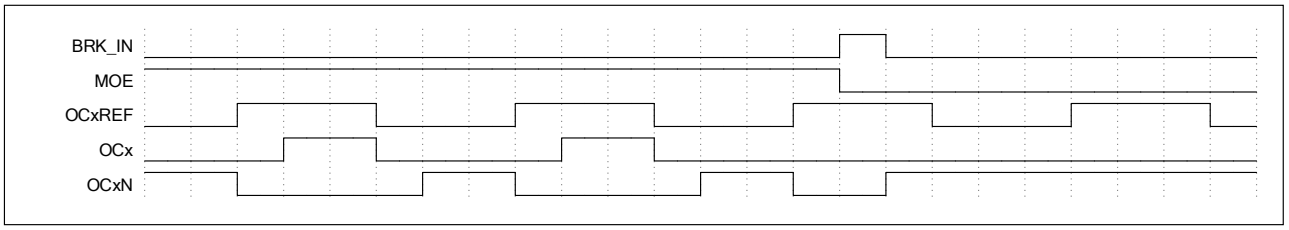


Figure 10-25 Response break output (OISx=0, OISxN=1)

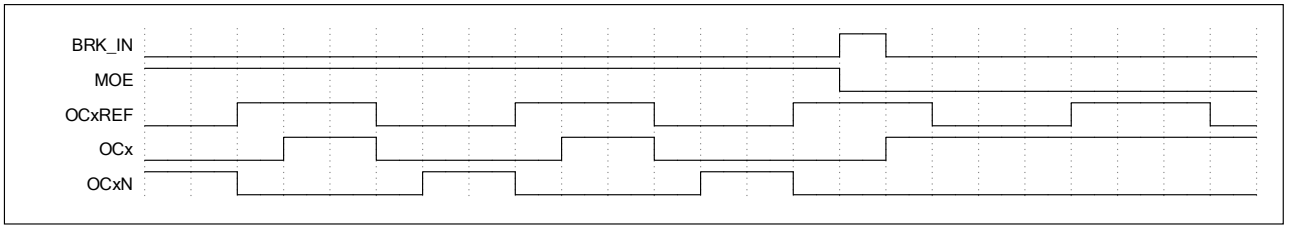


Figure 10-26 Response break output (OISx=1,OISxN=0）

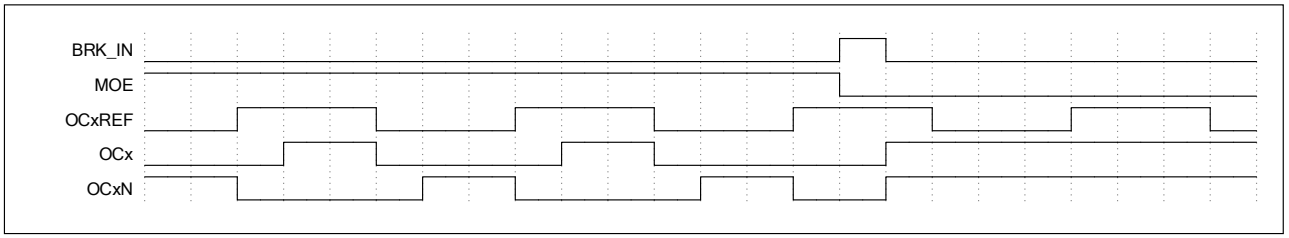


Figure 10-27 Response break output (OISx=1, OISxN=1)

10.4.4.6 One-pulse output

In the one-pulse mode (OPM), the counter responds an stimulus, and generate a pulse with width programmable. Configure TIM1_CR1 register OPM =1. Select one-pulse mode, trigger signal's active edge or configuring CEN =1 can enable the counter. CEN =1 remains until the next update event or configure CEN =0.

The necessary condition to generate pulse is the difference of compare value and initial value of the counter. Necessary configuration prior to counter enable:

- Up counting mode: Counter CNT < CCRx ≤ ARR.
- Down counting mode: Counter CNT > CCRx.

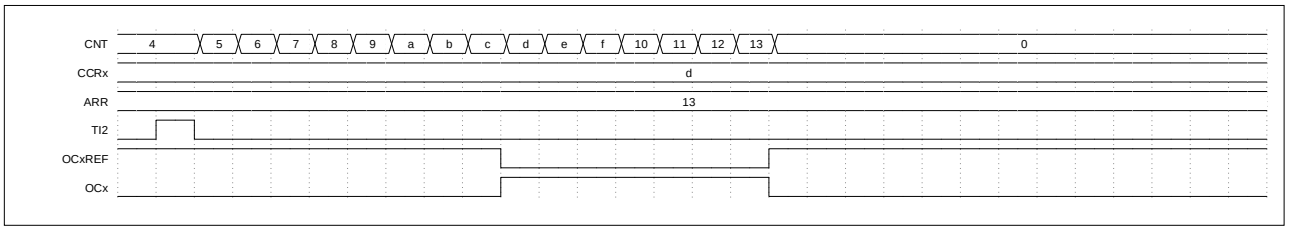


Figure 10-28 one-pulse mode

For example, when TI2 detects the rising edge, and after latency t_{DELAY}, OC2 generate a positive pulse with the length of t_{PULSE}.

Configure TI2FP2 serves as the trigger source:

1. Configure TIM1_CCMR1 register CC2S = 01, map TI2FP2 in TI2.
2. Configure TIM1_CCER register CC2P = 0, detect TI2FP2 rising edge.
3. Configure TIM1_SMCR register TS = 110, TI2FP2 as the slave mode counter trigger (TRGI).
4. Configure TIM1_SMCR register SMS = 110, select trigger mode, TI2FP2 enables the counter.

OPM waveform is determined by TIM1_ARR and TIM1_CCR1 (consider clock frequency and counter prescaler): t_{DELAY} is determined by TIM1_CCR1 register value and CNT initial value; TIM1_ARR - TIM1_CCR1 value is t_{PULSE}.

Next example is generate a negative pulse. In case of compare and match, generate a waveform from 1 to 0. When the counter preload value is achieved, generate a waveform from 0 to 1:

1. Configure TIM1_CCMR1 register OC1M = 111, select PWM mode 2.
2. Configure TIM1_CCER register CC1P = 1, export active low level.
3. Configure TIM1_CCMR1 OC1PE = 1 and TIM1_CR1 register ARPE, enable preload register.
4. Configure TIM1_CCR1 register and TIM1_ARR register.
5. Configure TIM1_EGR register UG =1, generate a update event.
6. Wait for an external trigger event on TI2.

In this case, TIM1_CR1 register DIR=0 , CMS=0 , OPM= 1, stop counting in the next update event (when the counter toggles to 0 from the auto load value).

10.4.4.6.1 OCx fast enable

OCx fast enable is a special scenario of one-pulse mode. In the one-pulse mode, set TIM1_CCMR register OCxFE=1, and enforce OCxREF to directly respond to stimulus without relying on the compare result of counter and compare value. The output waveform is the same as the waveform when compare and match. Thus, it can remove the compare time, and fast export the compare result. OCx rapid output enable only becomes active in the PWM mode.

10.4.5 Slave mode

10.4.5.1 Encoder interface

The encoder interface mode means the counter count under interaction of orthogonal signals of TI1 and TI2. At the change of input source, the count direction is auto modified by hardware. Configure TIM1_SMCR register SMS bit and select input source. According to input source difference, divide the encoder interface into 3 modes: SMS =001, encoder interface mode 1; SMS =010, encoder interface mode 2; SMS =011, encoder interface mode 3; the specific count in the three modes are expressed below. Two input TI1 and TI2 are used as the interface of orthogonal encoder.

In the encoder mode, prior to counter enable, configure the ARR register. In the encoder interface mode, it's like using an external clock with direction selection. The counter counts continuously between 0 to the auto load value of TIM1 _ARR register (up count and down count are controlled by external clock).

Note: The encoder mode doesn't support the external clock mode 2.

In the encoder interface code, the counter is auto changed according to the speed and direction of increment encoder. Therefore, the counter content always points to the encoder position. The count direction corresponds to the direction of the connected sensor. The table below presents all possible combinations, assuming that TI1 and TI2 are not exchanged at the same time.

Table 10-6 Relations of count direction and encoder signal

Count mode	Relative level (TI1FP1 relative to TI2, TI2FP2 relative to TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Encoder interface mode 1 (only count at TI2)	High level	-	-	up count	down count
Encoder interface mode 1 (only count at TI2)	Low level	-	-	down count	up count
Encoder interface mode 2 (only count at TI1)	High level	down count	up count	-	-
Encoder interface mode 2 (only count at TI1)	Low level	up count	down count	-	-

Count mode	Relative level (TI1FP1 relative to TI2, TI2FP2 relative to TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Encoder interface mode 3 (only count at TI1 and TI2)	High level	down count	up count	up count	down count
Encoder interface mode 3 (only count at TI1 and TI2)	Low level	up count	down count	down count	up count

The counter configuration and sequence figure in the encoder interface mode is below. We can see the count signal generation and direction control in the figure. Specific configuration:

1. Configure TIM1_CCMR register CC1S=01, map IC1FP1 on TI1.
2. Configure TIM1_CCMR register CC2S=01, map IC2FP2 on TI2.
3. Configure TIM1_CCER register CC1P =0 , IC1 non-inverted, IC1=TI1.
4. Configure TIM1_CCER register CC2P =0, IC2 non-inverted, IC1=TI2.
5. Configure TIM1_SMCR register SMS =011, select encoder mode 3. According to the input level of another signal, the counter counts at the edge of TI1FP1 and TI2FP2.
6. Configure TIM1_CR1 register CEN =1, enable the counter.

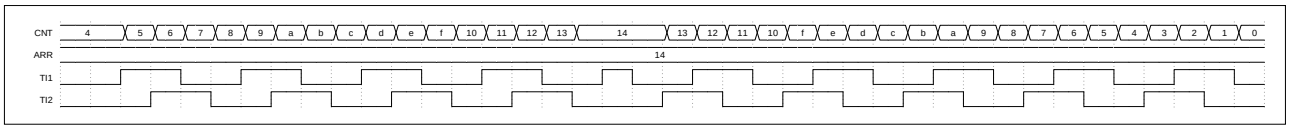


Figure 10-29 Counter time sequence in the encoder code

Counter sequence when the IC1FP1 polarity inverted (CC1P = 1, other configuration unchanged).

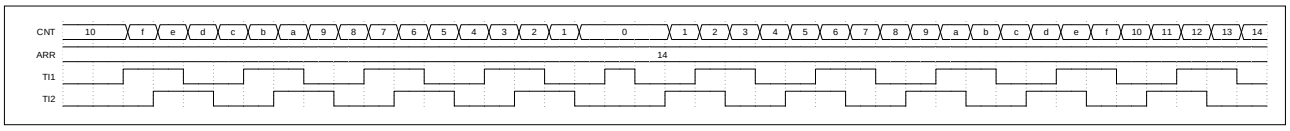


Figure 10-30 IC1FP1 polarity inverted sequence in encoder interface mode

In the encoder interface mode, the counter can provide the current position information of the sensor. Use another timer in the capture mode to measure the interval period of two encoder events and acquire dynamic information (speed, acceleration, deceleration). According to the interval period of two encoder events, regularly read the counter. Lock the counter value in the third input capture register (the acquire signal should be cyclic and generated by another timer).

10.4.5.2 Reset mode

Configure TIM1_SMCR register SMS=100, select slave reset mode. In this mode, the TRGI input

event will clear and restart the counter.

For example, TI2 input terminal falling edge trigger counter restart:

1. Configure TIM1_CCMR1 CC2S=01. CC2 channel is configured as input mode. IC2 is mapped on T12. Configure TIM1_CCER register CC2P=1, detect the falling edge.

2. Configure TIM1_SMCR register SMS = 100, and select the slave mode as reset mode. Configure TIM1_SMCR register TS =110, select timer input 2 after filter (TI2FP2) as the trigger input of sync timer.

3. Configure TIM1_CR1 register DIR=0, select the count direction as the up count; configure PSC=0, no division; configure CEN=1, enable the counter.

The counter clock source is provided by the internal clock. When detecting T12 falling edge, clear and restart the counter. The trigger interrupt flag is set by hardware.

TIM1_ARR = 0x13 time sequence in reset mode

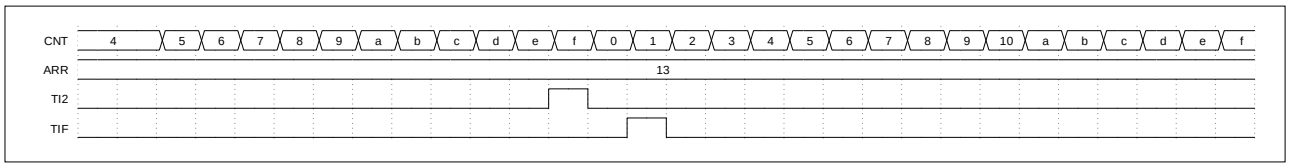


Figure 10-31 Control time sequence in reset mode

10.4.5.3 Gate mode

Configure TIM1_SMCR register SMS=101, select slave gate mode. In this mode, select active level (0: high level active, 1: low level active) according to the TIM1_CCER register CCxP value. When TRGI input is active level, the counter always starts. Otherwise, the counter stops (but without reset operation), and the counter start and stop are countable.

For example, the counter only counts when TI1 is high:

1. Configure CC1S=01 of TIM1_CCMR1 register. CC1 channel is configured as input mode. IC1 is mapped on T11. Configure TIM1_CCER register CC1P=0, detect the high level of TI1.

2. Configure TIM1_SMCR register SMS=101, select slave gate mode; configure TIM1_SMCR register TS=101, select the timer input after filter 1 (TI1FP1) as the sync counter trigger input.

3. Configure TIM1_CR1 register DIR=0, select count direction as the up count; configure PSC=0, no division; configure CEN=1, enable the counter.

The counter clock source is provided by the internal clock. When detecting T11 high level, the counter count start. When TI1 is low level, the counter stops. When the counter stop, set TIF as 1.

TIM1_ARR=0xf time sequence in gate mode.

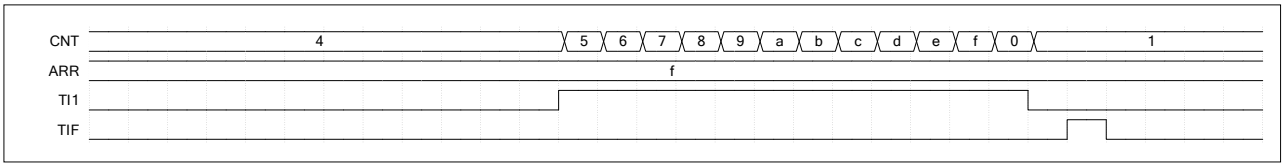


Figure 10-32 Control time sequence in gate mode

10.4.5.4 Trigger mode

Configure TIM1_SMCR register SMS=110, and select slave trigger mode. Select active edge (0: rising edge active, 1: falling edge active) according to the TIM1_CCER register CCxP value. When TRGI input is active edge, the counter starts counting. The timer startup is controlled and the stop is not controlled.

For example, the counter count starts at the rising edge of TI1 input:

1. Configure CC1S=01 of TIM1_CCMR1 register. CC1 channel is configured as input mode. IC1 is mapped on TI1. Configure TIM1_CCER register CC1P=0, and detect the rising edge.

2. Configure TIM1_SMCR register SMS = 110, select slave trigger mode; configure TIM1_SMCR register TS=101, select the timer input 1 (TI1FP1) after filter as the trigger input of counte.

3. Configure TIM1_CR1 register DIR =0, select count direction as the up count; configure PSC=0, no division.

The counter clock source is provided by internal clock. When detecting TI1 rising edge, the counter count starts.

TIM1_ARR =0xf sequence in the trigger mode.

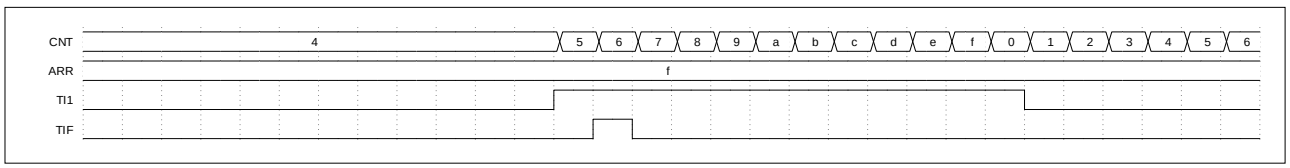


Figure 10-33 Control time sequence in trigger mode

10.4.6 Timer synchronization

Different timers are internally connected, achieve timer cascading or synchronization.

Timer synchronization connectivity:

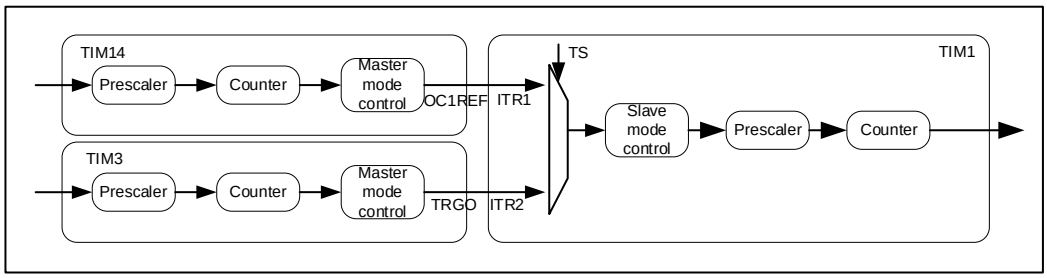


Figure 10-34 Timer connectivity

Several typical applications of connectivity.

Use one timer as the prescaler of another timer

For example, use TIM3 as the timer 1 prescaler, the time sequence diagram is shown above:

1. Configure TIM3_CR2 register MMS=010, TIM3 as main mode, TIM3 update event as trigger input (TRGO), TIM3 exports a cyclic signal at each update event.

2. Configure TIM3_ARR register, set TIM3 output period.

3. Configure TIM1_SMCR register TS=010, select TIM1 trigger source as TIM3_TRGO;

4. Configure TIM1_SMCR register SMS=111, select slave external trigger mode 1.

5. Configure TIM3_CR1 register CEN=1, start TIM3.

6. Configure TIM1_CR1 register CEN=1, start TIM1.

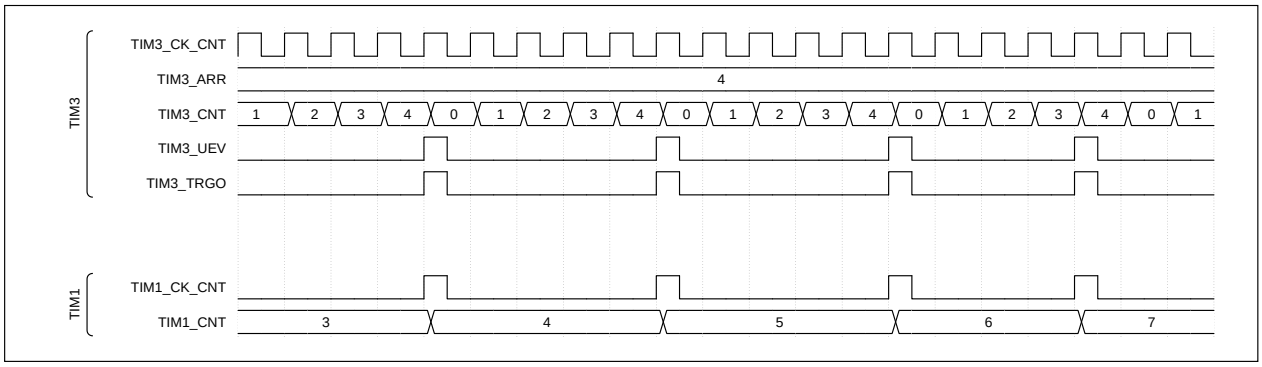


Figure 10-35 Use TIM3 as TIM1 prescaler

Use one timer to enable the other timer

For example: Use TIM1 to enable TIM3, and the time sequence diagram is shown below:

When TIM1 OC1REF is high, TIM3 counter count begins. The clock frequency of both TIM is provided by CK_CNT/3. Specific configuration:

1. Configure TIM1_CR2 register MMS=100, select TIM1 as main mode; configure TIM1 output compare reference signal (OC1REF) as trigger output (TRGO).

2. Configure TIM1_CCR1 register, TIM1_ARR register, TIM1_CCMR1 register OC1M bit, and CC1S bit; configure the output waveform of TIM1 output signal TRGO.

3. Configure TIM3_SMCR register TS=000, configure TIM1 OC1REF as TIM3 trigger output.

4. Configure TIM3_SMCR register SMS=101, configure TIM3 as gate mode.

5. Configure TIM3_CR1 register CEN=1, enable TIM3.

6. Configure TIM1_CR1 register CEN =1, enable TIM1.

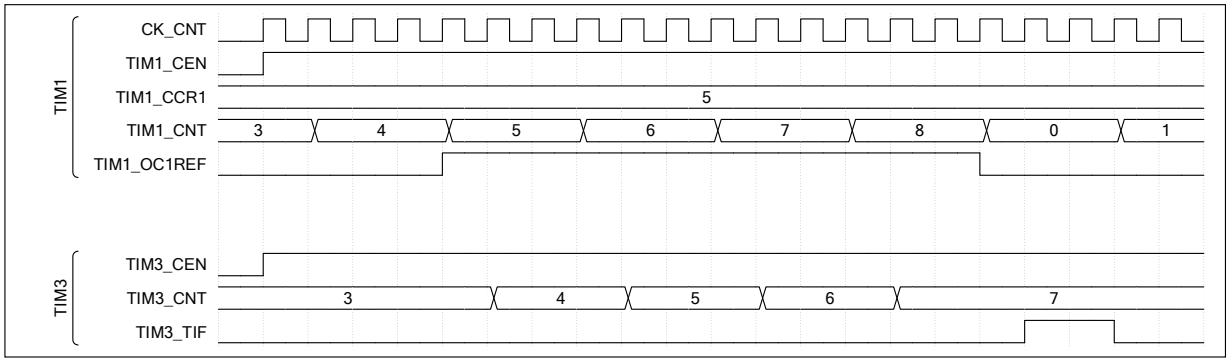


Figure 10-36 Use TIM1 to enable TIM3

Use one timer to start another timer

For example: Use TIM1 update event to start TIM3, and the time sequence is shown in the figure below:

When TIM1 generate the update event, TIM3 receives the trigger signal, TIM3 CEN is auto set by hardware. TIM3 counter count begins. The clock frequency of two TIMs is provided by CK_CNT/3. The specific configuration:

1. Configure TIM1_CR2 register MMS=010, select TIM1 update event as trigger output (TRGO);
2. Configure TIM1_ARR register, configure the period of update event;
3. Configure TIM3_SMCR register TS=000, configure TIM1 TRGO as TIM3 trigger input;
4. Configure TIM3_SMCR register SMS=110, configure TIM3 as trigger mode;
5. Configure TIM1_CR1 register CEN=1, enable TIM1.

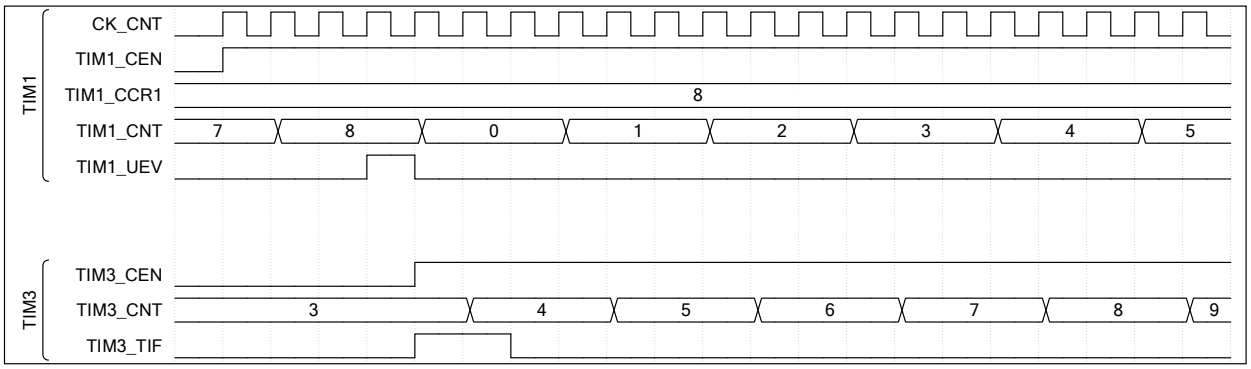


Figure 10-37 Use TIM1 update event to start TIM3

Use one external trigger to start two timers in step

For example: Use TIM1 TI1 rising edge start TIM1 and TIM3, the time sequence is shown in the figure below:

In order to ensure simultaneous start two timers, TIM1 must be configured in master/slave mode. Specific configuration:

1. Configure TIM1_CR2 register MMS=001, configure TIM1 as main mode, make TIM1 enable signal CEN as trigger output (TRGO).
2. Configure TIM1_SMCR register TS=100, configure TIM1 as slave mode, make TI1 as trigger

input.

3. Configure TIM1_SMCR register SMS=110, configure TIM1 as trigger mode.
4. Configure TIM3_SMCR register TS=000, configure TIM1 trigger output as TIM3 trigger input.
5. Configure TIM3_SMCR register SMS=110, configure TIM3 as trigger mode.

When TIM1 TI1 generate the rising edge, two timers start simultaneously (according to internal clock). The counter count begins, and two TIF flags are set.

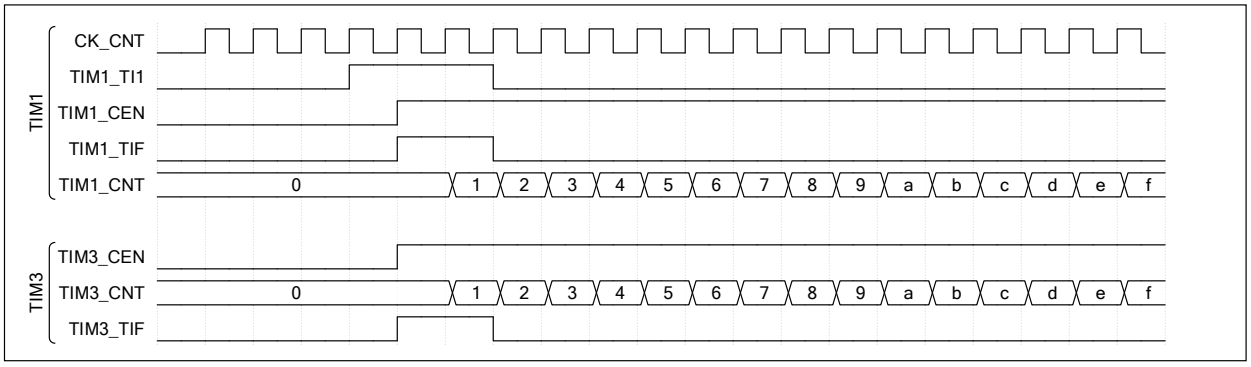


Figure 10-38 TIM1 TI1 start TIM1 TIM3 in step

10.4.7 Timer XOR function

Configure TIM1_CR2 registerTI1S =1, connect the TIM1_CH1 , TIM1_CH2 and TIM1_CH3 pin to TI1 input terminal via XOR for all input modes of the timer.

For example, TIM1_CH1 , TIM1_CH2 and TIM1_CH3 pins are connected to TI1 input terminal via XOR. Sample TI1 input signal's active edge. When TI1 rising edge is active, capture the current counter value, and lock it in the TIM1_CCR1 register. Specific configuration:

1. Configure TIM1_CR2 register TI1S=1, configure three inputs of the timer and connect it to TI1 input channel via XOR.
2. Configure TIM1_CCMR1 register CC1S=01, CC1 channel is configured as input, IC1 is mapped on TI1.
3. Configure TIM1_CCMR1 register IC1F[3:0], configure digital filter width (configure upon need).
4. Configure TIM1_CCER register CC1P=0, select capture in the rising edge of TI1 signal.
5. Configure TIM1_CCMR1 register IC1PS, select prescaler factor.
6. Configure TIM1_CCER register CC1E = 1, enable input/capture channel 1 capture.
7. Configure TIM1_CR1 register CEN=1, enable the counter.

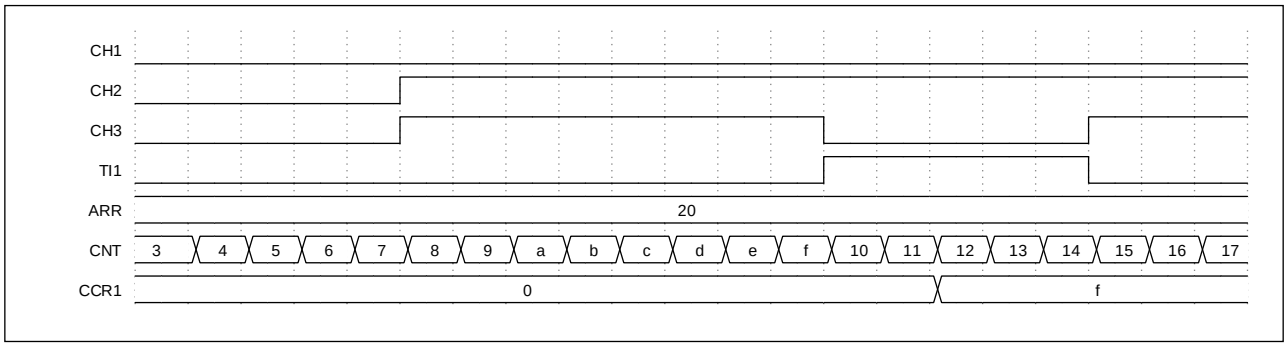


Figure 10-39（TI1 XOR input）input capture waveform

10.4.7.1.1 Hall interface circuit

The Hall sensor interface mode is an application example of XOR function, and can drive the motor. When using TIM1 to generate PWM signal and drive the motor, another counter (TIM3) may serve as the “interface timer” to connect Hall sensor. Connect 3 input pins of the “interface timer” (CH1, CH2 , CH3) to TI1 input channel via XOR. The “interface timer” receives the signal. Three Hall sensors are accordingly connected to the three input capture pins of the “interface timer”. Each sensor inputs one waveform to the input pin. The analysis input capture signal can calculate the motor speed.

In the output mode, the “interface timer” can generate a pulse to control TIM1_PWM output and drive the motor. Therefore, the “interface timer” generate a positive pulse at output compare or PWM mode latency, and is transmitted to TIM1 by TRGO output.

For example: Hall input is connected to TIM3 timer. Any change of the Hall input signal will change TIM1 PWM configuration.

1.Configure TIM3_CR2 register TI1S=1, configure three timer inputs and connect it to TI1 input channel via XOR.

2.Configure TIM3_ARR at the maximum value (the counter must be cleared via TI1 change). Configure PSC and set the count period greater than the time duration of two changes on the sensor.

4.Configure TIM3_CCMR1 register CC1S=01, select channel 1 as capture mode (enable TRC).

5.Configure TIM3_CCMR1 register CC2S=00 , OC2M=111, configure channel 2 as PWM2 mode with the required latency.

6.Configure TIM3_CR2 register MMS=101, select OC2REF as TRGO trigger output.

In TIM1, ITR input must be triggered by TIM3 TRGO. The timer is programmed as generation of PWM signal. Capture/compare counter is preloaded (TIM1_CR2 register CCPC = 1), and trigger input control COM event (TIM1_CR2 register CCUS = 1). After one COM event, write the next PWM control bit (CCxE, OCxM) to enable in the interrupt sub subroutine of OC2REF rising edge.

The actual case is shown in the figure below:

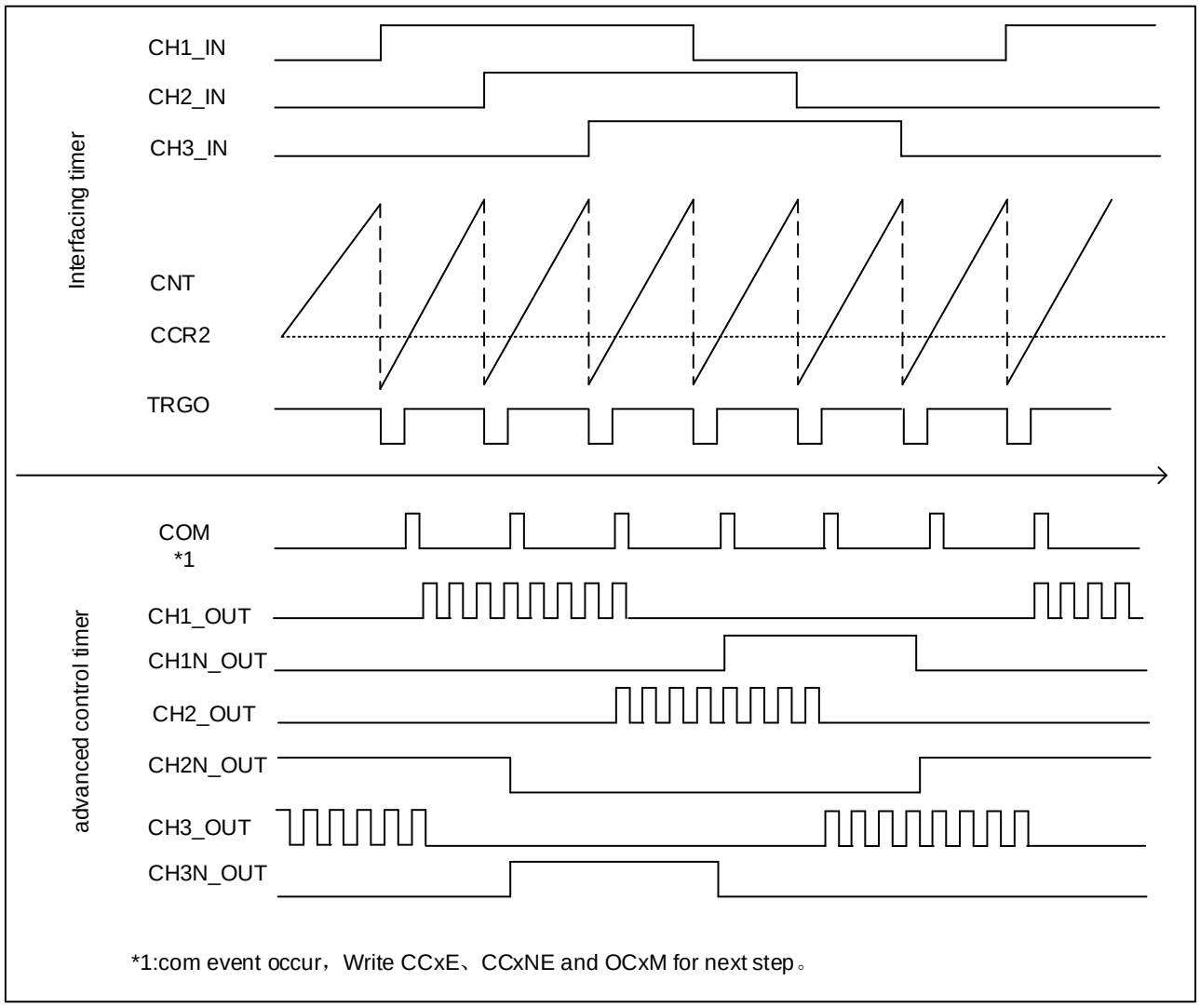


Figure 10-40 Example of Hall sensor interface

10.4.8 Debug mode

In the debug mode, configure DBG_CR register DBG_TIM1_STOP=1, TIM1 counter stops counting. (details in the debug section)

10.4.9 Interrupt

TIM1 interrupt includes: capture/compare 1 interrupt, capture/compare 2 interrupt, capture/compare 3 interrupt, capture/compare 4 interrupt, compare 5 interrupt, update interrupt, COM interrupt, trigger interrupt and break interrupt. When the corresponding interrupt enable bit is set, generate the corresponding event and interrupt.

Table 10-7 List of interrupt events

Interrupt event	Flag bit	Enable bit
Capture/compare 1 interrupt	CC1IF	CC1IE
Capture/compare 2 interrupt	CC2IF	CC2IE
Capture/compare 3 interrupt	CC3IF	CC3IE
Capture/compare 4 interrupt	CC4IF	CC4IE
Compare 5 interrupt	CC5IF	CC5IE
Update interrupt	UIF	UIE
COM interrupt	COMIF	COMIE
Trigger interrupt	TIF	TIE
Break interrupt	BIF	BIE

10.5 Register description

Table 10-8 TIM1 register overview

Offset	Acronym	Register Name	Reset
0x00	TIM1_CR1	Control Register 1	0x0000
0x04	TIM1_CR2	Control Register 2	0x0000
0x08	TIM1_SMCR	Slave Mode Control Register	0x0000
0x0C	TIM1_DIER	Interrupt Enable Register	0x0000 0000
0x10	TIM1_SR	Status Register	0x0000 0000
0x14	TIM1_EGR	Event Generation Register	0x0000 0000
0x18	TIM1_CCMR1	Capture/Compare Mode Register 1	0x0000
0x1C	TIM1_CCMR2	Capture/Compare Mode Register 2	0x0000
0x20	TIM1_CCER	Capture/Compare Enable Register	0x0000
0x24	TIM1_CNT	Counter	0x0000
0x28	TIM1_PSC	Prescaler	0x0000
0x2C	TIM1_ARR	Auto Reload Register	0x0000
0x30	TIM1_RCR	Repeat Count Register	0x0000
0x34	TIM1_CCR1	Capture/Compare Register 1	0x0000
0x38	TIM1_CCR2	Capture/Compare Register 2	0x0000
0x3C	TIM1_CCR3	Capture/Compare Register 3	0x0000
0x40	TIM1_CCR4	Capture/Compare Register 4	0x0000
0x44	TIM1_BDTR	Break And Dead-Time Register	0x0000 0000
0x54	TIM1_CCMR3	Capture/Compare Mode Register 3	0x0000
0x58	TIM1_CCR5	Capture/Compare Register 5	0x0000
0x5C	TIM1_PDER	PWM Phase Shift Register	0x0000
0x60~0x70	TIM1_CCRxFALL	PWM Phase Shift Count Down Capture/Compare Register	0x0000

Offset	Acronym	Register Name	Reset
0x74	TIM1_BKINF	Break Input Filter Register	0x0000 0000

10.5.1 TIM1_CR1 Control Register 1

Address offset:0x00

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						CKD		ARPE	CMS		DIR	OPM	URS	UDIS	CEN
						rw		rw	rw		rw	rw	rw	rw	rw

Bit	Field	Description
15：10	Reserved	Reserved, must be kept at reset value.
9：8	CKD	Clock division Division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock used by the dead-time generators and the digital filters (TIx). 00：$t_{DTS} = t_{INT_CK}$ 01：t_{DTS} =2x t_{INT_CK} 10：t_{DTS} =4x t_{INT_CK} 11：Reserved, do not program this value
7	ARPE	Auto reload preload enable 0: Disable the shadow register of TIM1_ARR register 1: Enable the shadow register of TIM1_ARR register
6：5	CMS	Center alignment mode selection 00: Edge alignment mode. Count direction depends on DIR bit 01: Central alignment mode 1. The counter alternatively conducts up and down count. The channel is in output mode. Only in down count, compare interrupt flag bit is set 10: Central alignment mode 2. The counter alternatively conducts up and down count. The channel is in output mode. Only in up count, compare interrupt flag bit is set 11: Central alignment mode 3. The counter alternatively conducts up and down count. The channel is in output mode. In up and down count, compare interrupt flag bit is set Note: During count, the alignment mode change is disabled.
4	DIR	Count direction 0: up count 1: down count Note: When the counter is configured as central alignment mode or encoder mode, the bit is read only.

Bit	Field	Description
3	OPM	one-pulse mode 0: Disable one-pulse mode. In case of update event, the counter count continues 1: Enable one-pulse mode. In case of the next update event (clear CEN bit), the counter count stops
2	URS	Update request source This bit is set and cleared by software, select update event source. 0: The event below may generate a update interrupt request: - Counter overflow/underflow - Set UG bit - Update generation through the slave mode controller 1: Only in counter overflow/underflow, generate update interrupt request
1	UDIS	Update disable This bit is used to enable or disable the update event 0: Update event (UEV) enabled. 1: Update event disabled. The Update event is not generated, shadow registers keep their value(ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the EGR.UG bit is set, the counter is reinitialized if a hardware reset is received from the slave mode controller.
0	CEN	Counter enable 0: Counter disabled 1: Counter enabled Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

10.5.2 TIM1_CR2 Control Register 2

Address offset:0x04

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS			Res.	CCUS	Res.	CCPC
	rw	rw	rw	rw	rw	rw	rw	rw	rw			rw	rw		rw

Bit	Field	Description
15	Reserved	Reserved, must be kept at reset value.
14	OIS4	Output idle state 4 (OC4 output). Refer to OIS1 bit.

Bit	Field	Description
13	OIS3N	Output idle state 3 (OC3N output). Refer to OIS1N bit.
12	OIS3	Output idle state 3 (OC3 output). Refer to OIS1 bit.
11	OIS2N	Output idle state 2 (OC2N output). Refer to OIS1N bit.
10	OIS2	Output idle state 2 (OC2 output). Refer to OIS1 bit.
9	OIS1N	(Output idle state 1) (OC1N output) 0: In case of MOE =0, OC1N =0 after dead-time 1: In case of MOE =0, OC1N =1 after dead-time Note: After setting LOCK (TIM1_BKR register) level 1, 2 or 3, this bit cannot be changed.
8	OIS1	(Output idle state 1) (OC1 output) 0: In case of MOE=0, if OC1N is implemented, OC1=0 after dead-time 1: In case of MOE=0, if OC1N is implemented, OC1 = 1 after dead-time Note: After setting LOCK (TIM1_BKR register) level 1, 2 or 3, this bit cannot be changed.
7	TI1S	TI1 selection 0: TIM1_CH1 pin is connected to TI1 input 1: TIM1_CH1, TIM1_CH2 and TIM1_CH3 pins pins are connected to the TI1 input after XOR combination

Bit	Field	Description
6：4	MMS	Master mode selection These bits control TRGO signal selection, used to select the sync information sent to the slave timers in master mode: 000: Reset TIM1_EGR register UG bit generate one pulse trigger output (TRGO). 001: Enable the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected. 010: Update Update event is selected as TRGO. 011: Capture/compare When a capture or a compare match occurred, send a positive pulse as TRGO. 100: Compare OC1REF signal is used as trigger output (TRGO) 101: Compare OC2REF signal is used as trigger output (TRGO) 110: Compare OC3REF signal is used as trigger output (TRGO) 111: Compare OC4REF signal is used as trigger output (TRGO)
3	Reserved	Reserved, must be kept at reset value.
2	CCUS	Capture/compare control update selection 0: CCPC=1, they are updated by setting the COMG=1 only. 1：CCPC=1, they are updated by setting the COMG=1 or when a rising edge occurs on TRGI. Note: This bit acts only on channels that have a complementary output.
1	Reserved	Reserved, must be kept at reset value.
0	CCPC	Capture/compare preloaded control bit 0: CCxE, CCxNE and OCxM bit preload disable 1: CCxE, CCxNE and OCxM bit preload enable Note: This bit acts only on channels that have a complementary output.

10.5.3 TIM1_SMCR Slave Mode Control Register

Address offset:0x08

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								MSM	TS			Res.	SMS		
								rw	rw				rw		

Bit	Field	Description
15：8	Reserved	Reserved, must be kept at reset value.
7	MSM	Master/slave mode 0: No action 1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.
6：4	TS	Trigger selection Trigger input source selection. 000：Internal trigger 0 (ITR0) 001：Internal trigger 1 (ITR1) 010：Internal trigger 2 (ITR2) 011：Internal trigger 3 (ITR3) 100：TI1 edge detector (TI1F_ED) 101：Timer input after filter 1 (TI1FP1) 110：Timer input after filter 2 (TI2FP2) 111：Reserved More details on ITRx are given in the table below. Note: After the slave mode enable, these bits cannot be changed
3	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
2：0	SMS	Slave mode selection When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input. 000: Close slave mode - In case of CEN =1, the prescaler is directly driven by the internal clock. 001: Encoder mode 1 - Based on TI1FP1 level, the counter conducts up/down count along TI2FP2 edge. 010: Encoder mode 2 - Based on TI2FP2 level, the counter conducts up/down count along TI1FP1edge. 011: Encoder mode 3- Based on another input level, the counter conducts up/down count along TI1FP1 and TI2FP2 edge. 100: Reset mode -The rising edge of the selected trigger input (TRGI) reinitialize the counter and generate a update register signal. 101: Gate mode - When the trigger input (TRGI) is high, the counter count begins. When the trigger input turns low, the counter count stops (but without reset) and generate a update event. The counter start and stop are controlled. 110: Trigger mode -The counter starts in the rising edge of the trigger input TRGI (but without reset) and generate a update event. Only the counter start is controlled. 111: External clock mode 1 -The rising edge of the selected trigger input (TRGI) drives the counter. Note: If TI1F_EN is selected as trigger input (TS =100), don't use the gate mode. This is because that EI1F_ED exports a pulse at each TI1F change. However, in the gate mode, it's required to check the trigger input level.

Table 10-9 TIM1 internal trigger connection

Slave timer	ITR0	ITR1	ITR2	ITR3
TIM1	-	-	TIM3_TRGO	TIM14_OC1REF
TIM3	TIM1_TRGO	-	-	TIM14_OC1REF

10.5.4 TIM1_DIER Interrupt Enable Register

Address offset:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															CC5IE
															rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE
								rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31：17	Reserved	Reserved, must be kept at reset value.
16	CC5IE	Compare 5 interrupt enable 0: Compare 5 interrupt disable 1: Compare 5 interrupt enable
15：8	Reserved	Reserved, must be kept at reset value.
7	BIE	Break interrupt enable 0: Break interrupt disable 1: Break interrupt enable
6	TIE	Trigger interrupt enable 0: Break interrupt disable 1: Break interrupt enable
5	COMIE	Enable COM interrupt 0: COM interrupt disable 1: COM interrupt enable
4	CC4IE	Enable capture/compare 4 interrupt 0: Capture/compare interrupt 4 disable 1: Capture/compare interrupt 4 enable
3	CC3IE	Enable capture/compare 3 interrupt 0: Capture/compare interrupt 3 disable 1: Capture/compare interrupt 3 enable
2	CC2IE	Enable capture/compare 2 interrupt 0: Capture/compare interrupt 2 disable 1: Capture/compare interrupt 2 enable
1	CC1IE	Enable capture/compare 1 interrupt 0: Capture/compare interrupt 1 disable 1: Capture/compare interrupt 1 enable
0	UIE	Enable update interrupt 0: Update event interrupt disable 1: Update event interrupt enable

10.5.5 TIM1_SR Status Register

Address offset:0x10

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															CC5IF
															rw0c

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			CC4OF	CC3OF	CC2OF	CC1OF	Res.	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			r_w0c					r_w0c							

Bit	Field	Description
31：17	Reserved	Reserved, must be kept at reset value.
16	CC5IF	Compare 5 interrupt flag Refer to CC1IF description.
15：13	Reserved	Reserved, must be kept at reset value.
12	CC4OF	Capture/compare 4 over capture flag Refer to CC1OF description.
11	CC3OF	Capture/compare 3 over capture flag Refer to CC1OF description.
10	CC2OF	Capture/compare 2 over capture flag Refer to CC1OF description.
9	CC1OF	Capture/compare 1 over capture flag Only when the channel 1 is configured as input capture mode and CC1F is configured as 1, and the capture event occurs again. The flag is set by hardware. Write 0 may clear the bit. 0: No overcapture 1: Overcapture
8	Reserved	Reserved, must be kept at reset value.
7	BIF	Break interrupt flag When the break input is active, the bit is set by hardware. If the Break input is inactive, the bit is set as 0 by software 0: No break event occurred 1: Detect active level on Break input
6	TIF	Trigger interrupt flag In case of trigger event (When the slave mode counter is in any other mode except for the gate mode and external clock mode，detect the active edge in TRGI input terminal, or detect the stop edge in gate mode), the bit is set by hardware. It’s cleared by software. 0: No trigger event occurred 1: Trigger interrupt pending
5	COMIF	COM interrupt flag In case of COM event (capture/compare control bit: CCxE, CCxNE, OCxM updated), the bit is set by hardware. It is cleared by software. 0：No COM event occurred 1: COM interrupt pending
4	CC4IF	Capture/compare 4 interrupt flag Refer to CC1IF description

Bit	Field	Description
3	CC3IF	Capture/compare 3 interrupt flag Refer to CC1IF description
2	CC2IF	Capture/compare 2 interrupt flag Refer to CC1IF description.
1	CC1IF	Capture/compare 1 interrupt flag Channel 1 in output mode: When the counter value and compare value match, the bit is set by hardware except for the central alignment mode (the bit is set in the central alignment mode according to TIM1_CR1.CMS[1:0]). It's cleared by software. 0: No match 1: TIM1_CNT value and TIM1_CCR1 value match Channel 1 as output mode: In case of capture event, the bit is set by hardware. It's cleared by software or reading TIM1_CCR1 register 0: No input capture occurred 1: Counter value is captured to TIM1_CCR1
0	UIF	Update interrupt flag In case of update event, the bit is set by hardware. It's cleared by software. 0: No update interrupt occurred 1: update interrupt pending This bit is set by hardware when the registers are updated: - In case of TIM1_CR1 register UDIS=0 and REP_CNT=0, when the counter generates overflow/underflow. -In case of TIM1_CR1 register UDIS=0, URS=0, when TIM1_EGR register UG =1. - In case of TIM1_CR1 register UDIS=0, URS=0, when update generation through the slave mode controller.

10.5.6 TIM1_EGR Event Generation Register

Address offset:0x14

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															CC5G
															w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG
								w	w	w	w	w	w	w	w

Bit	Field	Description
31：17	Reserved	Reserved, must be kept at reset value.
16	CC5G	Compare 5 generation Refer to CC1G description.
15：8	Reserved	Reserved, must be kept at reset value.
7	BG	Break generation 0: No action 1: generate a break event. MOE =0, BIF =1; when producing the corresponding interrupt, generate the corresponding interrupt. It's cleared by hardware.
6	TG	Trigger generation 0: No action 1: generate a trigger event. TIM1_SR register TIF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. It's auto cleared by hardware.
5	COMG	Capture/compare control update generation 0: No action 1:Capture/compare event control update. It's auto cleared by hardware. In case of CCPC =1, enable update CCxE, CCxNE and OCxM bit. Note: This bit is only active for the complementary output channel.
4	CC4G	Capture/compare 4 generation Refer to CC1G description.
3	CC3G	Capture/compare 3 generation Refer to CC1G description.
2	CC2G	Capture/compare 2 generation Refer to CC1G description.
1	CC1G	Capture/compare 1 generation Refer to CC1G description. This bit is set by software. It generates a capture/compare event, and is auto cleared by hardware. 0: No action 1：Generate a capture/compare event in channel CC1: When the channel CC1 is configured as output: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. When the channel CC1 is configured input: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. When CC1IF is set, set CC1OF =1.

Bit	Field	Description
0	UG	Update event generation 0：No action 1: Initialize the counter, and generate a update event. It's auto cleared by hardware. When selecting the central alignment or up counting mode, the counter is clear; otherwise (down counting mode) the counter auto reload value is loaded. The prescaler counter is cleared at the same time.

10.5.7 TIM1_CCMR1 Capture/Compare Mode Register 1

Address offset:0x18

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OC2M			OC2PE	OC2FE	CC2S		Res.	OC1M			OC1PE	OC1FE	CC1S	
IC2F				IC2PSC		CC2S		IC1F				IC1PSC		CC1S	
rw	rw			rw	rw	rw		rw	rw			rw	rw	rw	

The channel may be used for input (capture mode) or output (compare mode). The channel direction is defined by the corresponding CCxS. The register's bit other than CCxS functions differently in the input mode and output mode. OCxx describes the channel function in the output mode. ICxx describes the channel function in the input mode.

Output compare mode:

Bit	Field	Description
15	Reserved	Reserved, must be kept at reset value.
14：12	OC2M	Channel 2 output and compare mode Refer to OC1M description.
11	OC2PE	Channel 2 output and compare preload enable Refer to OC1PE description.
10	OC2FE	Channel 2 output and compare quick enable Refer to OC1FE description.
9：8	CC2S	Channel 2 capture/compare 2 selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 2 is configured as output 01：Channel 2 is configured as input, IC2 is mapped on TI2 10：Channel 2 is configured as input, IC2 is mapped on TI1 11：Channel 2 is configured as input, IC2 is mapped on TRC. The mode only works when the internal generator input is selected (TIM1_SMCR register TS bit selection)
7	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
6：4	OC1M	Channel 1 output compare mode The bit has defined the output reference signal OC1REF action. OC1REF has determined OC1 and OC1N value. OC1REF is active at high level. The active level of OC1, OC1N depends on CC1P and CC1NP bit. 000：Freeze. TIM1_CCR1 and TIM1_CNT compare results has no effect on OC1REF. 001: Set as high when configuration. When TIM1_CNT value and TIM1_CCR1 value are same, enforce OC1REF as high level 010: Set as low when configuration. When TIM1_CNT value and TIM1_CCR1 value are same, enforce OC1REF as low level 011: Toggle when match. when TIM1_CCR1=TIM1_CNT, OC1REF toggle. 100: Enforce as low. Enforce OC1REF at low level 101: Enforce as high. Enforce OC1REF at high level 110 ： PWM mode 1. During up count, in case of TIM1_CNT<TIM1_CCR1, enforce OC1REF is at high level. Or else, it's at low level. During down count, in case of TIM1_CNT > TIM1_CCR1, enforce OC1REF is at low level. Or else, it's at high level. 111 ： PWM mode 2. During up count, in case of TIM1_CNT<TIM1_CCR1, channel 1 enforce OC1REF is at low level. Or else, it's at high level. During down count, in case of TIM1_CNT > TIM1_CCR1, enforce OC1REF is at high level. Or else, it's at low level. Note 1: In case of LOCK level 3 (TIM1_BDTR register LOCK bit) and CC1S = 00 (the channel is configured as output), the bit cannot be changed. Note 2: In PWM mode 1 or PWM mode 2, only when the compare result changes or it changes over from the freeze mode to PWM mode in the output compare mode, OC1REF level may change.

Bit	Field	Description
3	OC1PE	Channel 1 output compare preload enable 0: Disable TIM1_CCR1 register preload function. The value written into TIM1_CCR1 register becomes active immediately 1: Enable TIM1_CCR1 register preload function. Only read and write the preload register. TIM1_CCR1 preload value becomes active at the update event Note 1: When the LOCK level is 3 (TIM1_BDTR register LOCK bit) and CC1S = 00 (when the channel is configured as output), the bit cannot be changed. Note 2: Only in the one-pulse mode (TIM1_CR1 register OPM= 1), it has no influence whether the preload register is set. Under other scenarios, it's required to set the preload register. Otherwise, the follow up action is not certain.
2	OC1FE	Channel 1 output compare quick enable In case of the bit is set 1, when the channel is configured as PWM mode, it speeds up response of the capture/compare output to the trigger time. The output channel deems the active edge of the trigger input signal as one compare match. Therefore, OC is set as compare level, but is irrelevant to the compare result. 0: channel 1 output compare fast enable. 1: channel 1 output compare fast disable.
1：0	CC1S	Channel 1 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 1 is configured as output 01：Channel 1 is configured as input, IC1 is mapped on TI1 10：Channel 1 is configured as input, IC1 is mapped on TI2 11：Channel 1 is configured as input, IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)

Input capture mode:

Bit	Field	Description
15：12	IC2F	Input capture 2 filter Refer to IC1F description
11：10	IC2PSC	Input/capture 2 prescaler Refer to IC1PSC description

Bit	Field	Description
9：8	CC2S	Channel 2 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 2 is configured as output 01：Channel 1 is configured as input, IC2 is mapped on TI2 10：Channel 2 is configured as input, IC2 is mapped on TI1 11：Channel 2 is configured as input, IC2 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)
7：4	IC1F	Channel 1 input capture filter The digital filter is composed of an event counter. It records an output jump after N input events. These bits define IC1 input signal sampling frequency and digital filter length. 0000: No filter, f_{DTS} sampling 001: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =2 0010: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =4 0011: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =8 0100: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0111: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =8 1000: Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =6 1001:Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =8 1010: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =5 1011: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =6 1100: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =8 1101: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =5 1110: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =6 1111: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =8
3：2	IC1PSC	Channel 1 input/capture prescaler This bit defines the ratio of the prescaler acting on IC1.The prescaler is reset as soon as CC1E=0(TIM1_CCER register). 00: No prescaler, the capture input detects each edge to trigger one capture 01: Trigger one capture for every 2 events 10: Trigger one capture for every 4 events 11: Trigger one capture for every 8 events

Bit	Field	Description
1：0	CC1S	Channel 1 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 1 is configured as output 01: Channel 1 is configured as input, IC1 is mapped on TI1 10: Channel 1 is configured as input, IC1 is mapped on TI2 11: Channel 1 is configured as input, IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)

10.5.8 TIM1_CCMR2 Capture/Compare Mode Register 2

Address offset:0x1C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	OC4M			OC4PE	OC4FE	CC4S		Res.	OC3M			OC3PE	OC3FE	CC3S	
IC4F				IC4PSC		CC4S		IC3F				IC3PSC		CC3S	
rw	rw			rw	rw	rw		rw	rw			rw	rw	rw	

Compare output mode:

Bit	Field	Description
15	Reserved	Reserved, must be kept at reset value.
14：12	OC4M	Channel 4 output compare mode Refer to OC3M description
11	OC4PE	Channel 4 output compare preload enable Refer to OC3PE description
10	OC4FE	Channel 4 output compare rapid enable Refer to OC3FE description
9：8	CC4S	Channel 4 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 4 is configured as output 01: Channel 4 is configured as input, IC4 is mapped on TI4 10: Channel 4 is configured as input, IC4 is mapped on TI3 11: Channel 4 is configured as input, IC4 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection
7	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
6：4	OC3M	Channel 3 output compare mode The bit has defined the output reference signal OC3REF action. OC3REF has determined OC3 and OC3N value. OC3REF is active at high level. The active level of OC3, OC3N depends on CC3P and CC3NP bit. 000：Freeze. TIM1_CCR3 and TIM1_CNT compare results has no effect on OC3REF. 001: Set as high when configuration. When TIM1_CNT value and TIM1_CCR3 value are same, enforce OC3REF as high level 010: Set as low when configuration. When TIM1_CNT value and TIM1_CCR3 value are same, enforce OC3REF as low level 011: Toggle when match. When TIM1_CCR3=TIM1_CNT,OC3REF toggle. 100: Enforce as low. Enforce OC3REF at low level 101: Enforce as high. Enforce OC3REF at high level 110: PWM mode 1. During up count, in case of TIM1_CNT<TIM1_CCR3, enforce OC3REF as high level, otherwise it's low level; during down count, in case of TIM1_CNT > TIM1_CCR3, enforce OC3REF as low level, otherwise it's high level. 111: PWM mode 2. During up count, in case of TIM1_CNT<TIM1_CCR3, enforce OC3REF as low level, otherwise it's high level; during down count, in case of TIM1_CNT > TIM1_CCR3, enforce OC3REF as high level, otherwise it's low level. Note 1: In case of LOCK level 3 (TIM1_BDTR register LOCK bit) and CC3S = 00 (the channel is configured as output), the bit cannot be changed. Note 2: In PWM mode 1 or PWM mode 2, only when the compare result changes or it changes over from the freeze mode to PWM mode in the output compare mode, OC3REF level may change.

Bit	Field	Description
3	OC3PE	Channel 3 output compare preload enable 0: Disable TIM1_CCR3 register preload function, the value written into TIM1_CCR3 register becomes active immediately 1: Enable TIM1_CCR3 register preload function. The read and write only works on the preload register. TIM1_CCR3 preload value becomes active when the update event is available Note 1: When the LOCK level is set 3 (TIM1_BDTR register LOCK bit) and CC3S =0 (the channel is configured as output), the bit cannot be changed. Note 2: Only in the one-pulse mode (TIM1_CR1 register OPM= 1), it's not required to set the preload register. Under other scenarios, it's required to set the preload register. Otherwise, the follow up action is not certain.
2	OC3FE	Channel 3 output compare quick enable In case of the bit 1, when the channel is configured as PWM mode, it speeds up response of the capture/compare output to the trigger time. The output channel deems the active edge of the trigger input signal as one compare match. Therefore, OC is set as compare level, but is irrelevant to the compare result. 0: channel 3 output compare fast disable. 1: channel 3 output compare fast enable.
1：0	CC3S	Channel 3 capture/compare selection The bit has defined the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 3 is configured as input 01: Channel 3 is configured as input, IC3 is mapped on TI3 10: Channel 3 is configured as input, IC3 is mapped on TI4 11: Channel 3 is configured as input, IC3 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)

Input capture mode:

Bit	Field	Description
15：12	IC4F	Input capture 4 filter Refer to IC3F description
11：10	IC4PSC	Input/capture 4 prescaler Refer to IC3PSC description

Bit	Field	Description
9：8	CC4S	Channel 4 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 4 is configured as output 01：Channel 4 is configured as input, IC4 is mapped on TI4 10：Channel 4 is configured as input, IC2 is mapped on TI3 11：Channel 4 is configured as input, IC4 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)
7：4	IC3F	Channel 3 input capture filter The digital filter is composed of an event counter. It records an output jump after N input events. These bits define IC1 input signal sampling frequency and digital filter length. 0000: No filter, f_{DTS} sampling 001: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =2 0010: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =4 0011: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =8 0100: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0111: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =8 1000: Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =6 1001:Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =8 1010: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =5 1011: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =6 1100: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =8 1101: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =5 1110: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =6 1111: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =8
3：2	IC3PSC	Channel 3 input/capture prescaler This bit defines the ratio of the prescaler acting on IC3.The prescaler is reset as soon as CC3E=0(TIM1_CCER register). 00: No prescaler, the capture input detects each edge to trigger one capture 01: Trigger one capture for every 2 events 10: Trigger one capture for every 4 events 11: Trigger one capture for every 8 events

Bit	Field	Description
1：0	CC3S	Channel 3 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 3 is configured as output 01: Channel 3 is configured as input, IC3 is mapped on TI3 10: Channel 3 is configured as input, IC3 is mapped on TI4 11: Channel 3 is configured as input, IC3 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM1_SMCR register TS bit selection)

10.5.9 TIM1_CCER Capture/Compare Enable Register

Address offset:0x20

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC 4P	CC 4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E
rw		rw													

Bit	Field	Description
15	CC4NP	Channel 4 input/capture complementary output polarity Refer to CC1NP description
14	Reserved	Reserved, must be kept at reset value.
13	CC4P	Channel 4 capture/compare output polarity Refer to CC1P description.
12	CC4E	Channel 4 capture/compare enable Refer to CC1E description
11	CC3NP	Channel 3 input/capture complementary output polarity Refer to CC1NP description
10	CC3NE	Channel 3 input/capture complementary output enable Refer to CC1NE description
9	CC3P	Channel 3 input capture output polarity Refer to CC1P description
8	CC3E	Channel 3 input/capture output enable Refer to CC1E description
7	CC2NP	Channel 2 input/capture complementary output polarity Refer to CC1NP description
6	CC2NE	Channel 2 input/capture complementary output enable Refer to CC1NE description
5	CC2P	Channel 2 input capture output polarity Refer to CC1P description

Bit	Field	Description
4	CC2E	Channel 2 input/capture output enable Refer to CC1E description
3	CC1NP	Channel 1 input/capture complementary output polarity When channel 1 is configured as output, this bit has defined the input signal polarity: 0: OC1N is active at high level 1: OC1N is active at low level When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the ICx polarity/level selection table. Note: When LOCK level (TIM1_BDTR register LCCK bit) is set 3 or 2 and CC1S = 00 (channel is configured as output), the bit cannot be changed.
2	CC1NE	Channel 1 input/capture complementary output enable 0: Close channel 1 complementary output. OC1N output disable. 1: OC1N signal exports to the corresponding output pin. The output level relies on the values in MOE, OSSI, OSSR, OIS1, OIS1N and CC1E.
1	CC1P	Channel 1 input/capture output polarity When channel 1 is configured as output, the bit defines the output signal polarity: 0: OC1 is active at high level 1: OC1 is active at low level When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the ICx polarity/level selection table. Note: When LOCK level (TIM1_BDTR register LCCK) is set 3 or 2, this bit cannot be changed.
0	CC1E	Channel 1 input/capture output enable When channel 1 is configured as output: 0: Close. OC1 output disable 1: Enable. The output level relies on the values in MOE, OSSI, OSSR, OIS1, OIS1N and CC1NE. CC1channel is configured as input: This bit determines whether the input capture function is enabled. 0：Capture disable 1：Capture enable

In input mode, ICx polarity/level selection is given in the Table below:

Table 10-10 ICx polarity/level selection

CCxP	CCxNP	ICx polarity/level
0	0	Rising edge active/high level active
1	0	Falling edge active/low level active
1	1	Rising edge or falling edge active/high level active
0	1	Reserved

10.5.10 TIM1_CNT Counter

Address offset:0x24

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
15：0	CNT	Counter value

10.5.11 TIM1_PSC Prescaler

Address offset:0x28

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC															
rw															

Bit	Field	Description
15：0	PSC	Prescaler value Counter clock frequency (ck_cnt) =$f_{CK_PSC/（PSC+1）}$ In case of update event, PSC value is loaded into the current prescaler register.

10.5.12 TIM1_ARR Auto Reload Register

Address offset:0x2C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR															
rw															

Bit	Field	Description
15：0	ARR	Auto reload value These bits define the auto reload value of the counter. When auto reload value is 0, the counter doesn't work.

10.5.13 TIM1_RCR Repeat Count Register

Address offset:0x30

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REP_CNT								REP							
rw								rw							

Bit	Field	Description
15：8	REP_CNT	Real-time value written by the repeat counter In the repeat count mode, it's to change real-time phase move of the detection point of update interrupt flag bit (UIF) Note: After the update event, write these bits. Write REP_CNT before the update event will make the bit move inactive.
7：0	REP	Repeat counter value The repeat counter value defines the generation speed of the update event. When the repeat counter value reduces to 0, it generate the update event. If the update interrupt is enabled, it will simultaneously affect the update interrupt generation speed. The written REP value becomes active at the next update event. In PWM mode, (REP+1) corresponds to: In the edge alignment mode, the number of PWM period In the central alignment mode, the number of PWM half period

10.5.14 TIM1_CCR1 Capture/Compare Register 1

Address offset:0x34

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1															
rw															

Bit	Field	Description
15：0	CCR1	Channel 1 capture/compare value When channel 1 is configured as input: CCR1 value determines the counter value of the previous capture event (this register is only readable). When channel 1 is configured as output: If the preload function is not selected in TIM1_CCMR1 register (OC1PE bit), the write value is immediately transmitted to the current capture/compare shadow register. Or else, the write value is only loaded into the capture/compare register in case of update event. The current capture/compare shadow register is involved in the compare with the counter TIM1_CNT, and the compare result is reflected to the output signal of OC1 port.

10.5.15 TIM1_CCR2 Capture/Compare Register 2

Address offset:0x38

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2															
rw															

Bit	Field	Description
15：0	CCR2	Channel 2 capture/compare value Refer to CCR1 description.

10.5.16 TIM1_CCR3 Capture/Compare Register 3

Address offset:0x3C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3															
rw															

Bit	Field	Description
15：0	CCR3	Channel 3 capture/compare value Refer to CCR1 description.

10.5.17 TIM1_CCR4 Capture/Compare Register 4

Address offset:0x40

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4															
rw															

Bit	Field	Description
15：0	CCR4	Channel 4 capture/compare value Refer to CCR1 description.

10.5.18 TIM1_BDTR Break And Dead-Time Register

Address offset:0x44

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															DOE
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK		DTG							
rw	rw	rw	rw	rw	rw	rw		rw							

Note: According to the lock setting, DOE, AOE, BKP, BKE, OSSI, OSSR and DTG bits can be write protected; it's necessary to configure them during the first write into TIM1_BDTR register.Details are given in the chapter of complementary output and dead-time insert.

Bit	Field	Description
31：17	Reserved	Reserved, must be kept at reset value.
16	DOE	Direct output enable When the break is active, MOE is set 0, active. 0：After break output, wait for a dead time and export the idle state (output disable). 1：Immediately export the idle state (output disable). Note: When LOCK level 1 (TIM1_BDTR register LOCK bit) is set, the bit cannot be changed.

Bit	Field	Description
15	MOE	Main output enable When the channel x is configured as output, according to AOE bit setting value, the bit can be cleared or auto set by software. When the break input is active, the bit is cleared by hardware. 0：Disable OCx and OCxN output or enforce as the idle state (output disable). 1：When setting the corresponding enable bit (TIM1_CCER register CCxE, CCxNE), OCx and OCxN output enable
14	AOE	Auto output enable 0: MOE cannot be set by hardware 1: MOE can be set by software or auto set by hardware at the next update event when the break is inactive Note: When LOCK level 1 (TIM1_BDTR register LOCK bit) is set, the bit cannot be changed.
13	BKP	Break input polarity 0: Break input is active low 1: Break input is active high Note: When LOCK level 1 (TIM1_BDTR register LOCK bit) is set, the bit cannot be changed.
12	BKE	Break function enable 0：Break input disabled 1：Break input enabled Note 1: When LOCK level 1 (TIM1_BDTR register LOCK bit) is set, the bit cannot be changed. Note 2: Break input includes pin input, comparator compare result input and CSS clock failure event. Before break function is enabled, configure BKIN_SEL bit in TIM1_BKINF register to select break source.
11	OSSR	Off state selection in the run mode This bit only applies in case of MOE =1 and the channel is in complementary output. 0: When the timer inactive, disable OC/OCN output 1: When the timer inactive, in case of CCxE = 1 or CCxNE = 1, first enable OC/OCN and export inactive level, and then set OC/OCN enable output signal Note: When LOCK level 2 (TIM1_BDTR register LOCK bit) is set , the bit cannot be changed.

Bit	Field	Description
10	OSSI	Off state selection in the idle mode This bit only applies in case of MOE =0 and the channel is in output. 0: When the timer inactive, disable OC/OCN output 1: When the timer inactive, in case of CCxE = 1 or CCxNE = 1, first OC/OCN exports inactive level, and then set OC/OCN enable output signal. Note: When LOCK level 2 (TIM1_BDTR register LOCK bit) is set, the bit cannot be changed.
9：8	LOCK	Lock configuration This bit defines the register write protection function. 00: Write protection function closes, the register doesn't have the write protection 01: Lock level 1, unable to write TIM1_BDTR register DOE, DTG, BKE, BKP and AOE bit and TIM1_CR2 register OISx/OISxN bit 10: Lock level 2, unable to write lock level 1 bit, or CC polarity level (when relevant channel is set as output via CCxS bit, CC polarity level TIM1_CCER register CCxP/CCxNP bit) as well as OSSR/OSSI bit 11：Lock level 3, unable to write lock level 2 bit, or CC control bit (when relevant channel is set as output via CCxS bit, CC control level TIM1_CCER register CCxM/CCxPE bit) Note: The LOCK bits can be written only once after the reset. Once the TIM1_BDTR register has been written, their content is frozen until the next reset.
7：0	DTG	Dead-time generator setup adjustment These bits define the dead-time duration of the complementary output. Note: When LOCK level (TIM1_BDTR register LOCK bit) is set 1, 2 or 3, the bit cannot be changed.

10.5.19 TIM1_CCMR3 Capture/Compare Mode Register 3

Address offset:0x54

Reset value:0x0000

The channel only applies to the output compare mode

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												OC5PE	Res.		
												rw			

Bit	Field	Description
15：4	Reserved	Reserved, must be kept at reset value.
3	OC5PE	Output compare 5 preload enable 0: Disable TIM1_CCR5 register preload function, and the value written into TIM1_CCR5 register will become effective immediately. 1: Enable TIM1_CCR5 register preload function. Only read and write the preload register. TIM1_CCR5 preload value becomes active at the update event Note 1: When LOCK level 3 is set (TIM1_BDTR register LOCK level), this bit cannot be changed.
2：0	Reserved	Reserved, must be kept at reset value.

10.5.20 TIM1_CCR5 Capture/Compare Register 5

Address offset:0x58

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR5															
rw															

Bit	Field	Description
15：0	CCR5	Compare 5 value CC5 channel is only configured as output: If the preload function is not selected in the TIM1_CCMR3 register (OC5PE bit), the write value will immediately be transmitted to the corresponding capture/compare shadow register. Or else, only in case of the update event, the preload value will be transmitted to the corresponding capture/compare shadow register. The capture/compare shadow register is involved in the counter TIM1_CNT compare. Because CC5 channel is the internal channel and cannot be transmitted to the pin, the compare result is used for the internal trigger event.

10.5.21 TIM1_PDER PWM Phase Shift Enable Register

Address offset:0x5C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Res.	CCR5_SHIFT_EN	CCR4_SHIFT_EN	CCR3_SHIFT_EN	CCR2_SHIFT_EN	CCR1_SHIFT_EN	Res.
	rw	rw	rw	rw	rw	

Bit	Field	Description
15：6	Reserved	Reserved, must be kept at reset value.
5	CCR5_SHIFT_EN	Enable channel 5 output PWM phase shift bit 0: Channel 5 output PWM phase shift disable 1: Channel 5 output PWM phase shift enable Details given in CCRxFALL register phase shift operation
4	CCR4_SHIFT_EN	Enable channel 4 output PWM phase shift bit 0: Channel 4 output PWM phase shift disable 1: Channel 4 output PWM phase shift enable Details given in CCRxFALL register phase shift operation
3	CCR3_SHIFT_EN	Enable channel 3 output PWM phase shift bit 0: Channel 3 output PWM phase shift disable 1: Channel 3 output PWM phase shift enable Details given in CCRxFALL register phase shift operation
2	CCR2_SHIFT_EN	Enable channel 2 output PWM phase shift bit 0: Channel 2 output PWM phase shift disable 1: Channel 2 output PWM phase shift enable Details given in CCRxFALL register phase shift operation
1	CCR1_SHIFT_EN	Enable channel 1 output PWM phase shift bit 0: Channel 1 output PWM phase shift disable 1: Channel 1 output PWM phase shift enable Details given in CCRxFALL register phase shift operation
0	Reserved	Reserved, must be kept at reset value.

10.5.22 TIM1_CCRxFALL PWM Phase Shift Count Down Capture/Compare Register

Address offset:0x60 ~ 0x70

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCRxFALL															
rw															

Bit	Field	Description
15：0	CCRxFALL	Channel x capture/compare value during count down in PWM central alignment mode PWM phase shift function: Enable PDER register PWM phase shift. According to the required phase shift, configure CCRxFALL and CCRx, PWM exports programmable phase shift waveform, and can shift left or right.

10.5.23 TIM1_BKINF Break Input Filter Register

Address offset:0x74

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															BKIN_SEL
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BKIN_SEL											BKINF				BKINFE
rw											rw				rw

Bit	Field	Description
31：17	Reserved	Reserved, must be kept at reset value.
16：5	BKIN_SEL	Break input channel selects xxxxxxxxxx1x: Set bit1, PA[4] break input enable xxxxxxxxxxx1: Set bit0, CSS break input enable

Bit	Field	Description
4：1	BKINF	Break input filter 0000: 2 cycles 0001: 4 cycles 0010: 8 cycles 0011: 16 cycles 0100: 32 cycles 0101: 64 cycles 0110: 128 cycles 0111: 256 cycles 1000: 384 cycles 1001: 512 cycles 1010: 640 cycles 1011: 768 cycles 1100: 896 cycles 1101: 1024 cycles 1110: 1152 cycles 1111: 1280 cycles Note: Before changing the sampling frequency of break digital filter, turn off the break filter function.
0	BKINFE	Break input filter enable 1: Break input filter enable 0: Break input filter disable Note: Enable the filter after the break condition is configured. Break filter is used for level active signal.

11 TIM3 General-Purpose Timer

11.1 Overview

TIM3 is composed of a 16 bit real-time programmable prescaler and a 16-bit auto reload counter with the programmable counting direction, and can conveniently provide with the counting and timing function. The counter is driven by a programmable prescaler. The general-purpose timer can be used for a variety of purposes, such as input capture function (measuring input signal pulse width or frequency, PWM input, etc), output compare function (PWM output, one-pulse mode, etc.)

11.2 Function block diagram

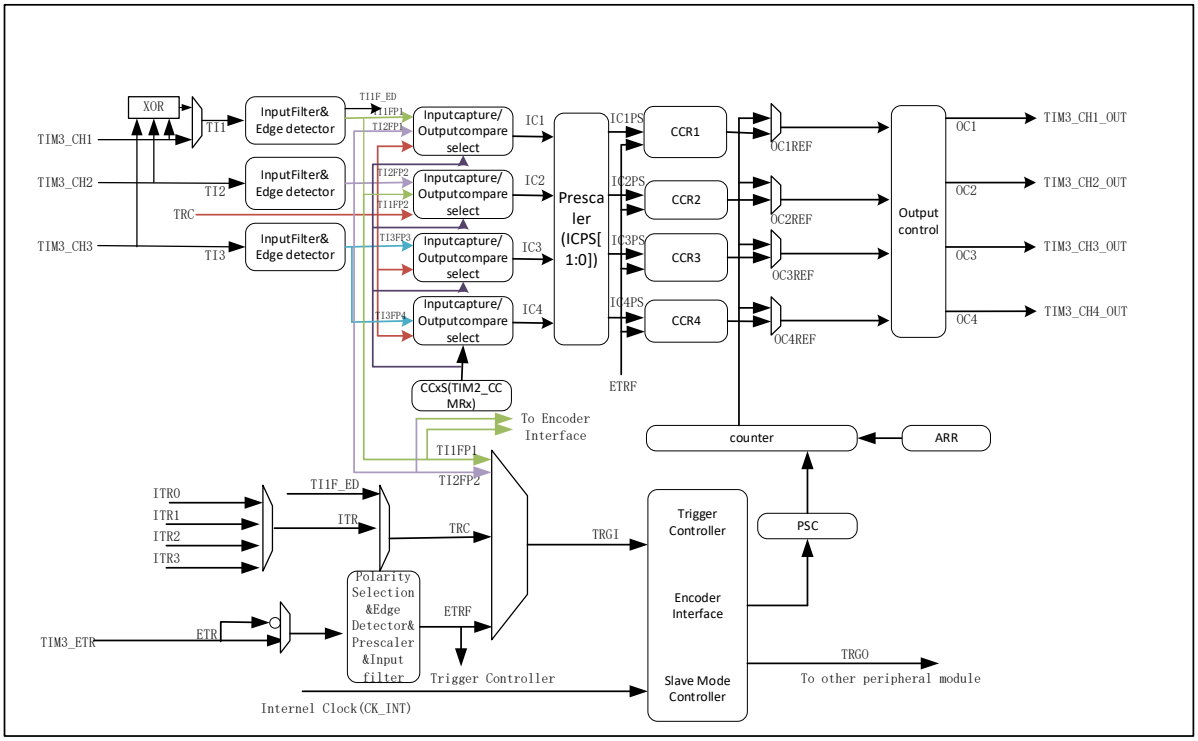


Figure 11-1 TIM3 block diagram

TIM3 structure block diagram is mainly composed of input capture unit, output compare unit, counter unit.

11.3 Main characteristics

- 16 bit real-time programmable prescaler, division factor: 1-65536.
- Clock source: Internal clock source, external clock input (TIx, ETRx), internal trigger input (ITRx).

- 16 bit auto reload counter (counting direction: up, down, up/down).
- Synchronization circuit to control the timer with external signals and to interconnect several timers to each other.
- Input capture: measure input signal pulse width or period.
- Trigger input can serve as external clock or be managed cycle-by-cycle.
- Support interfaces such as encoder and hall-sensor.
- 4 output channels
- Output compare (control output waveform or indicator counter already finishes timing).
- PWM output (edge alignment or central alignment mode)
- One-pulse mode.
- Interrupt request event: update event, trigger event, input capture, output compare.

11.4 Function description

11.4.1 Clock

11.4.1.1 Clock selection

Several types of clock sources of the counter below:

- Internal clock (INT_CK)
- External clock mode 1: External trigger input TRGI (including TIx, ITRx, ETRx)
- External clock mode 2: External trigger input ETR (including ETRx)
- Encoder mode

The diagram of several types of clock choice above:

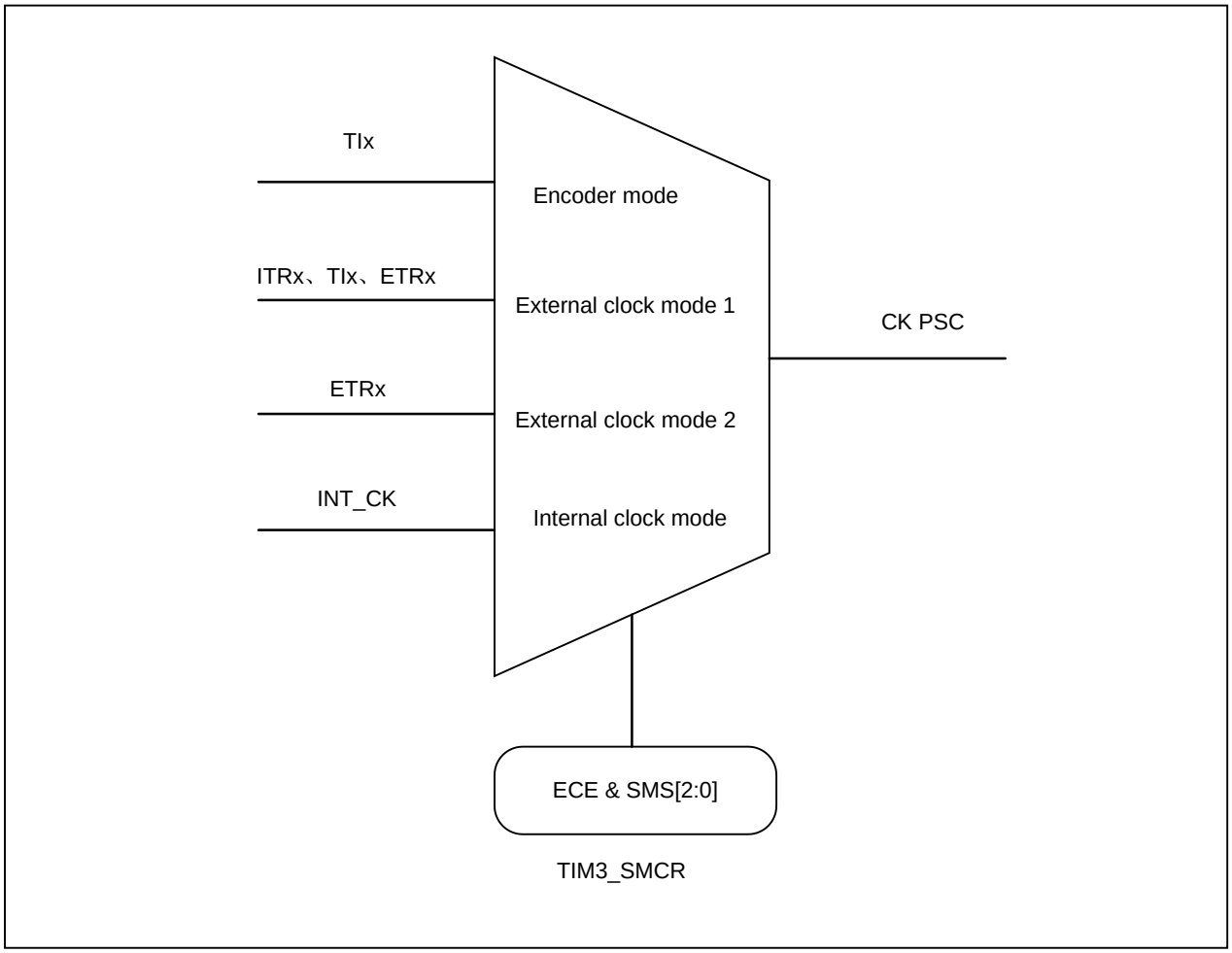


Figure 11-1 Clock selection

11.4.1.1.1 Internal clock source (INT _CK)

In case of SMS =000 of TIM3_SMCR register, disable the slave mode. Enable the counter, and the prescaler clock is driven by internal clock.

11.4.1.1.2 External clock mode 1(external trigger input TRGI, including TIx, ITRx, ETRx)

In case of SMS = 111 of TIM3_SMCR register, select the external clock mode 1 (TRGI). The counter is driven by the rising edge or falling edge of the selected input signal.

For example: count up at the rising edge on TI1 input. Specific configuration:

1.Configure TIM3_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 is mapped on TI1; Configure TIM3_CCMR1 register IC1F[3： 0], set input filter bandwidth; configure TIM3_CCER register CC1P=0, and select the rising edge as the active edge.

2. Configure TIM3_SMCR register TS=101, select TI1 as the trigger input source; configure TIM3_SMCR register SMS=111, select the external clock mode 1.

3. Configure TIM3_CR1 register DIR=0, select up counting mode; configure TIM3_CR1 register CEN=1 and start the counter.

TI1 input the active edge, count up once. The delay between the active edge of TI1 and actual clock of counter depends on the TI1 input pin sync circuit.

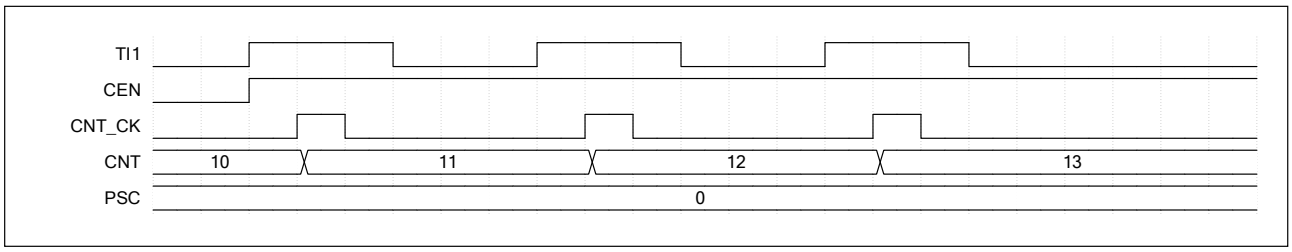


Figure 11-2 Control circuit in external clock mode 1

11.4.1.1.3 External clock mode 2(external trigger input ETR, including ETRx)

In case of TIM3_SMCR register ECE=1, enable external clock mode 2, and the counter is driven by the active edge of ETR signal.

For example: count up once of each 4 falling edge on ETR Specific configuration:

1.Configure TIM3_SMCR register ETF[3:0] =0010, count up once of each 4 active edges on ETR; configure TIM3_SMCR register ETP=1, select falling edge as active edge; configure TIM3_SMCR register ECE=1, select external clock mode 2.

2.Configure TIM3_CR1 register DIR=0, select up counting mode; Configure TIM3_CR1 register CEN=1, start the counter.

The delay between the falling edge of ETR and actual clock of counter depends on the sync circuit of ETR signal terminal.

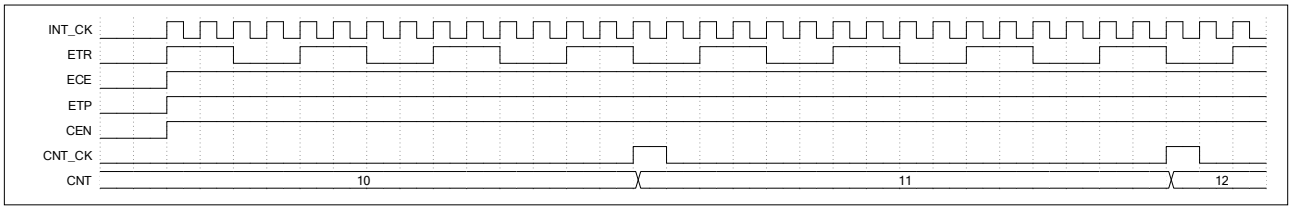


Figure 11-3 Control circuit in external clock mode 2

11.4.1.1.4 Encoder mode

Refer to the chapter of slave mode-encoder interface for details.

11.4.1.2 Time base unit

TIM3 time base unit mainly includes: Counter register (TIM3_CNT), prescaler register (TIM3_PSC), auto reload register (TIM3_ARR).

The counter unit is composed of a 16 bit counter with its related auto reload register, The counter can count up, down or both up and down.

The counter clock is provided by prescaler. The prescaler is composed of the prescaler counter with its related register. The division factor is 1-65536. It can write any time, and gets active during the next update.

The auto preload register has a 16 bit shadow register with preload function. Set TIM3_CR1 register ARPE bit to select the content of ARR register are transferred into the shadow register permanently or at each update event.

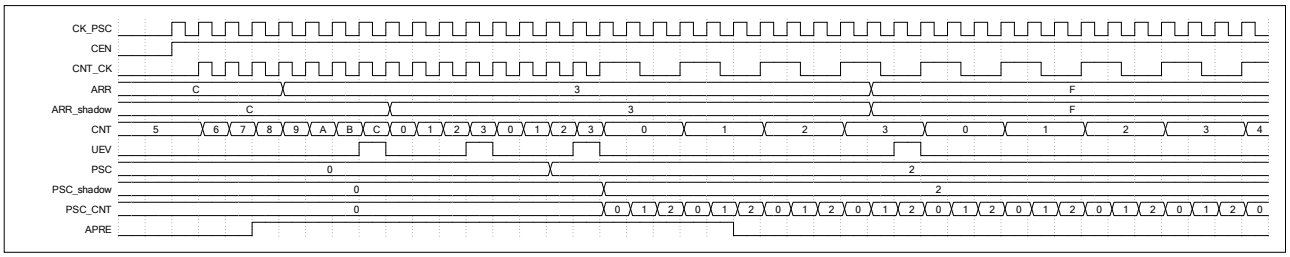


Figure 11-4 Auto preload

11.4.1.3 Counter modes

Configure DIR bit and CMS bit of TIM3_CR1 register to select the count mode. There are three count modes, up counting mode, down counting mode and central alignment mode (up/ down counting mode). Detailed descriptions are provided for each count mode.

11.4.1.3.1 Up counting mode

Configure TIM3_CR1 register　CMS=0 , DIR＝0, select up counting mode.

In up counting mode, enable TIM3_CR1 register CEN, and the counter begins up count from 0 to TIM3_ARR value and generates a counter overflow event (update). The counter begins up count again from 0. Set TIM3_EGR register UG =1, also generate an update event.

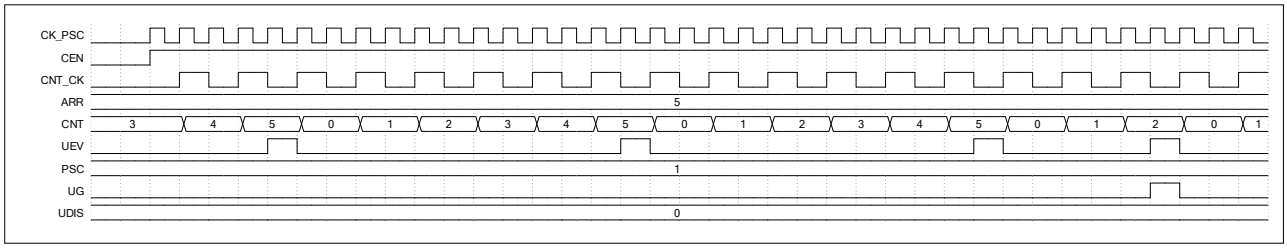


Figure 11-5 Up counting mode（UDIS=0）

Configuring TIM3_CR1 register UDIS=1, disable update event. In case of counter overflow event, the update event is not generated. Configuring UG=1, the update event is not generated. However, the counter and prescaler counter will initialize, and begin up count from 0.

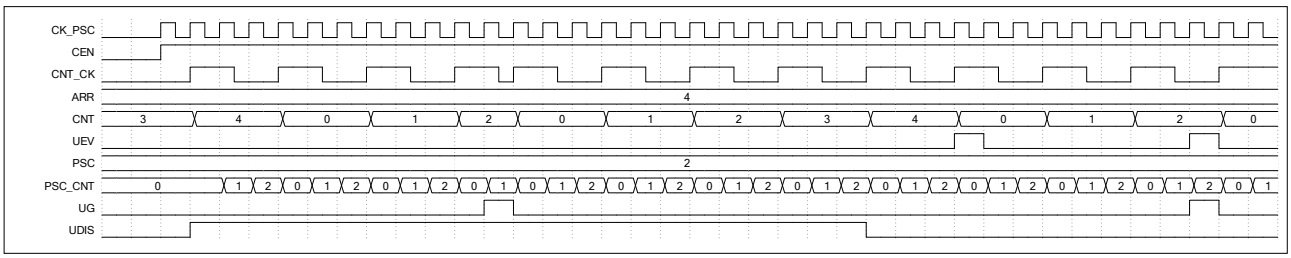


Figure 11-6 Up counting mode (UDIS =1, disable update event)

Note: In case of update event:

- ARR register value is loaded into ARR shadow register.
- Prescaler preload value becomes active.

11.4.1.3.2 Down counting mode

Configure TIM3_CR1 register CMS=0, DIR＝1, select down counting mode.

In down counting mode, the counter begins down count from the auto preload value TIM3_ARR. When counting to 0, generate an underflow event (update event). Set TIM3_EGR register UG =1, also generate a update event. After the update event, begin down count (TIM3_CR1 register UDIS=0) from the auto load value TIM3_ARR.

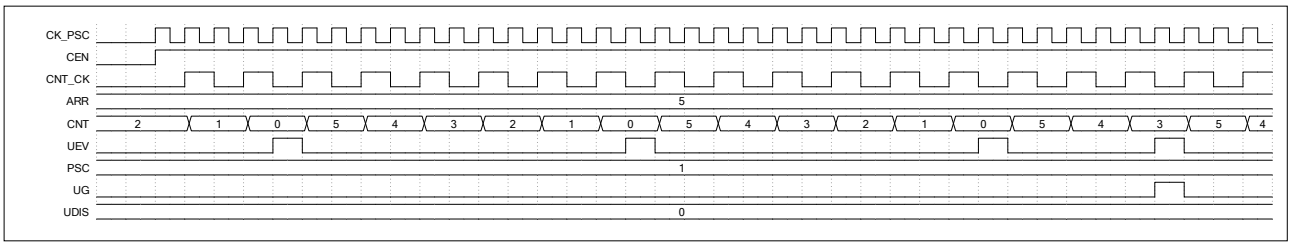


Figure 11-2 Down counting mode（UDIS＝0）

Configuring TIM3_CR1 register UDIS=1, disable update event. When the counter generates the underflow event, it doesn’t generate update event. Configuring UG=1 also doesn’t generate update event. However, the counter and prescaler counter will initialize, and count begins from TIM3_ARR.

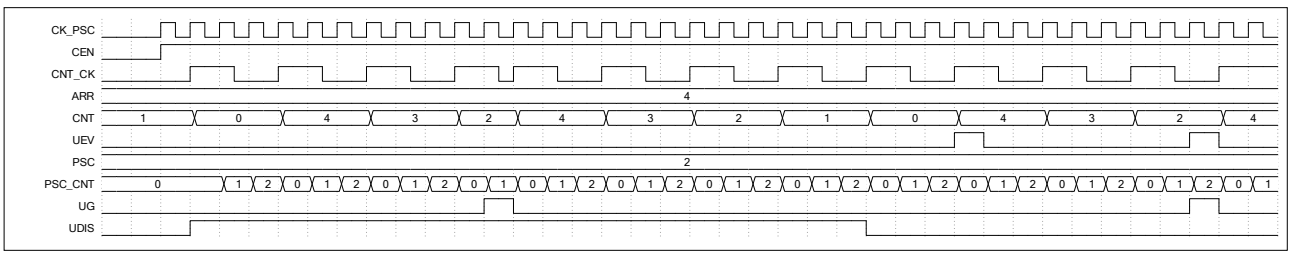


Figure 11-7 Down counting mode (UDIS=1 disables update event)

11.4.1.3.3 Central count mode (up/down count mode)

Configure TIM3_CR1 register CMS ≠ 0 (write DIR failure), and select central alignment count mode.

Central alignment count mode, up count and down count run alternatively. In case of up count to ARR-1, generate an overflow event. Begin down count to 1 from ARR, generate an underflow event, and begin up count from 0.

setting TIM3_EGR register UG =1 also generate a update event. After update event, the counter begins up count again from 0 (TIM3_CR1 register UDIS=0).

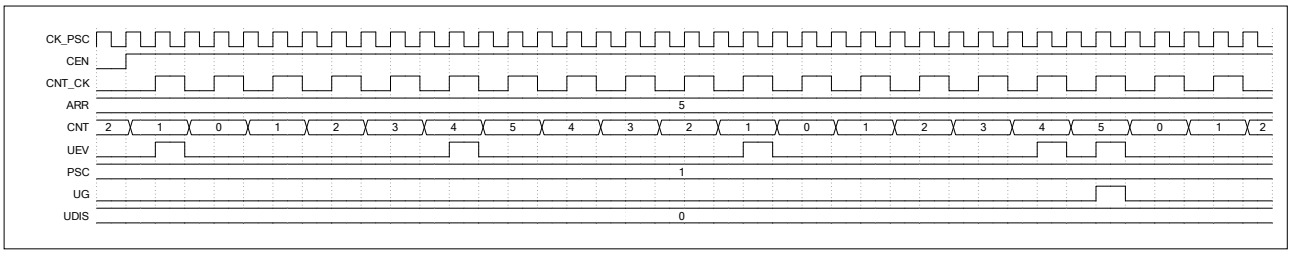


Figure 11-8 Central count mode（UDIS=0）

When configuring TIM3_CR1 register UDIS=1, disable update event. When the counter generates the overflow or underflow event, it doesn't generate update event. Configuring UG=1 also doesn't generate update event. However, the counter and prescaler counter will initialize, and the count begins from 0.

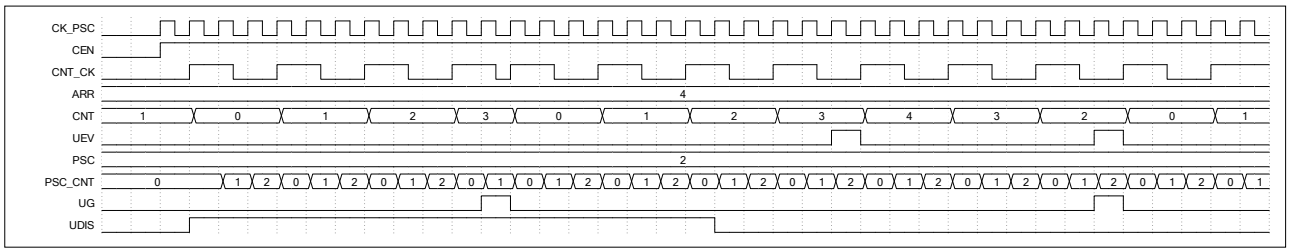


Figure 11-9 Central count mode (UDIS=1 disables update event)

11.4.2 Input capture

11.4.2.1 Input capture

The input capture is composed of digital filter, multiplex, prescaler, etc. Its block diagram:

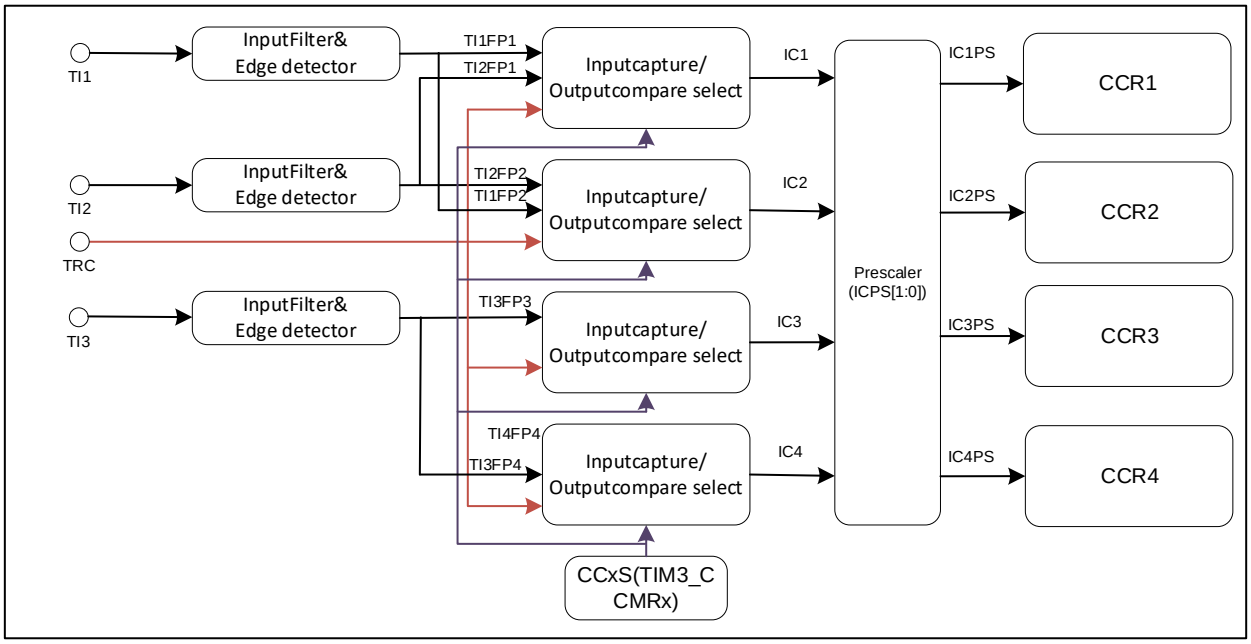


Figure 11-10 TIM3 input capture block diagram

Configure TIM3_CCMRx register ICxF, set the filter width of a digital filter (filter sampling frequency and digital filter width are shown in the table below). When the input signal width is greater than the filter width, the input signal is active. When the digital filter samples the input signal of the input pin TIx, generate a filtered signal TIxF. Through the edge detector with the optional polarity, generate an effective signal TIxFPx, this signal can serve as the trigger input signal of the slave mode counter. This signal generates ICxPS via the prescaler for trigger input capture event.

Table 11-1 Digital filter width and ICxF correlation

IC1F[3：0]	Sample frequency and filter length	IC1F[3：0]	Sample frequency and filter length
0000	No filter, sample by f_{DTS}	1000	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=6
0001	Sampling frequency fsampling=f_{INT_CK}, N=2	1001	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=8
0010	Sampling frequency fsampling=f_{INT_CK}, N=4	1010	Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=5
0011	Sampling frequency fsampling=f_{INT_CK}, N=8	1011	Sampling frequency $f_{sampling}=f_{DTS}$/16, N=6
0100	Sampling frequency fsampling=f_{DTS} /2, N=6	1100	Sampling frequency $f_{sampling}=f_{DTS}$ /16, N=8
0101	Sampling frequency fsampling=f_{DTS}/2, N=8	1101	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=5
0110	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=6	1110	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=6
0111	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=8	1111	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=8

In the input capture mode, when detecting the active edge of signal ICx, the counter’s current value

is locked in the corresponding shadow register, and then loaded into the corresponding capture/compare register. When enabling interrupt, and producing the capture event, generate the corresponding interrupt request. At the capture event, the capture flag bit CCxIF of the status register (TIM3_SR) will be set. Configure CCxIF =0 or read TIM3_CCRx data, clear CCxIF flag bit. When CCxIF is not cleared, next input capture event generated, the repeat capture flag CCxOF will be set. Configure CCxOF =0, can clear CCxOF flag bit.

For example, Sample TI1 input signal's active edge. At TI1 rising edge, capture the current counter value and lock it in the TIM3_CCR1 register, procedure steps as follows:

1. Configure TIM3_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 is mapped on TI1.
2. Configure TIM3_CCMR1 register IC1F[3:0] , configure the filter width of digital filter (configure upon need).
3. Configure TIM3_CCER register CC1P=0, select capture occurs in the rising edge of TI1 signal.
4. Configure TIM3_CCMR1 register IC1PSC[1:0], select the prescaler factor.
5. Configure TIM3_CCER register CC1E = 1, channel 1 capture enable.
6. Configure TIM3_DIER register CC1IE=1, enable channel 1 interrupt request.

Note：

- When the channel is configured as input mode, TIM3_CCRx register becomes read only.
- In case of more than two continuous captures and CCxIF flag is not cleared, the repeat capture flag CCxOF is set. In order to avoid losing the capture information that may be generated before the repeat capture flag CCxOF is set. It is recommended to read data before reading the repeat capture flag.
- Set the corresponding CCxG bit of TIM3_EGR register, generate the input capture interrupt request through software.

11.4.2.2 PWM capture

Differences of PWM input mode configuration with ordinary input capture:

- Two ICx signals are active on edges with opposite polarity and mapped on the same TIx input.
- Configure the slave mode counter at reset mode, set one TIxFP as trigger input signal.

For example: Measure TI1 PWM signal period (TIM3_CCR1 register) and duty cycle (TIM3_CCR2 register), the measuring value depends on the internal clock INT_Ck frequency and prescaler value. Specific steps:

1.Configure TIM3_CR1 register DIR =0, select counter as up counting mode.

2.Configure TIM3_CCMR1 register CC1S = 01, map IC1 on TI1, and select active input of TIM3_CCR1.

3. Configure TIM3_CCER register CC1P =0, select active polarity of TI1FP1 (active on rising edge) (capture the counter value to TIM3_CCR1, and clear the counter).

4.Configure TIM3_CCMR1 register CC2S =10, map IC2 on TI1, select active input of TIM3_CCR2.

5. Configure TIM3_CCER register CC2P =1, and select active polarity of TI2FP2 (active on falling edge) (capture counter value to TIM3_CCR2).

6.Configure TIM3_SMCR register TS = 101, select TI1FP1 as the active trigger input signal.

7.Configure TIM3_SMCR SMS = 100, set the slave mode counter as reset mode.

8.Configure TIM3_CCER register CC1E=1 and CC2E = 1. Enable CC1 channel and CC2 channel capture.

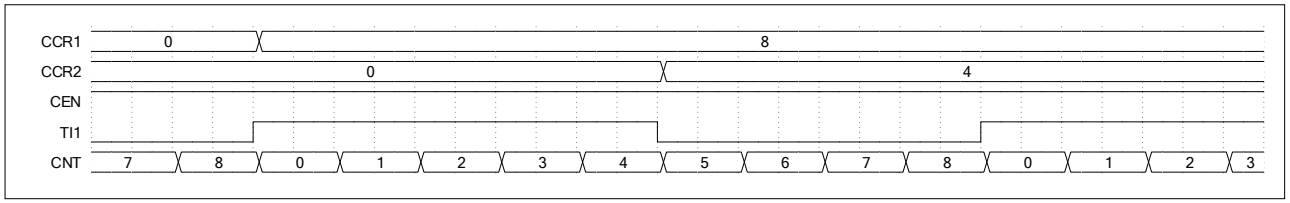


Figure 11-11 PWM input mode sequence

Note: The slave mode counter only connects TI1FP1 and TI2FP2, PWM input mode is only applicable to TIM3_CH1/TIM3_CH2 port input signal.

11.4.3 Output compare

The compare output of the capture/compare channel is composed of the comparator, output control circuit and capture/compare register. Its structure is shown below:

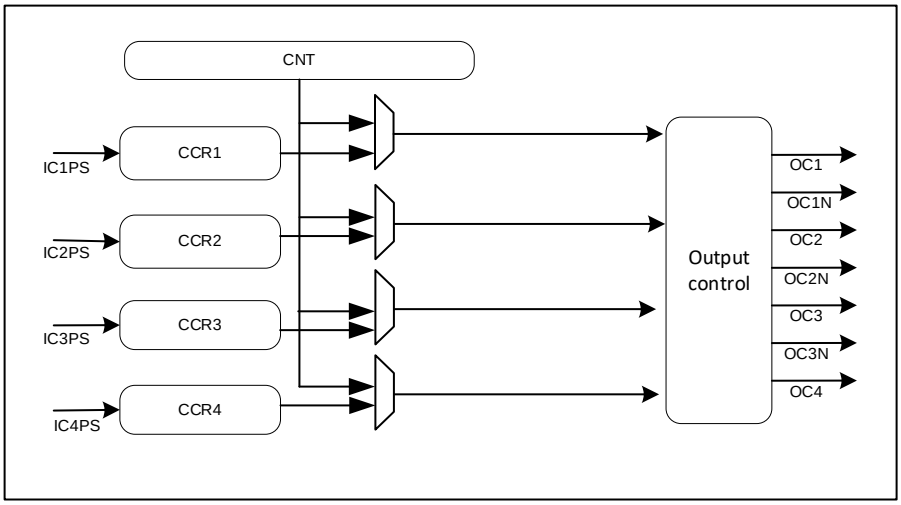


Figure 11-12 Output compare block diagram

In output compare mode, the content of capture/compare register is loaded into the shadow register, and then compare the content of shadow register with the current value of counter. The capture/compare module includes a capture/compare register (preload register) and a shadow register. When read and write, only programmed the capture/compare register.

11.4.3.1 Force output

Configure TIM3_CCMRx register CCxS = 00 and set the channel CCx as output mode. Configure TIM3_CCMRx register OCxM bit, and directly enforce the output compare signal as active or inactive level, independently of the output compare result. Configure TIM3_CCMRx register OCxM = 100, and enforce output compare signal as inactive level. And OCxREF is enforced as low level. Configure TIM3_CCMRx register OCxM = 101, and enforce output compare signal as active level. And the OCxREF is enforced as high level (OCxREF always active as high level).

Note: In force output mode, TIM3_CCRx shadow register and counter output compare is still performed. The corresponding flag bit of compare result will also be changed. When enable the corresponding interrupt request, it will also generate the corresponding interrupt request.

11.4.3.2 Output compare

In the output compare mode, when the values of counter and capture compare register matches, different waveform can be exported according to the OCxM bit of TIM3_CCMRx register.

For example, When the value of counter and capture/compare register matches, the functions in the output compare mode:

1.When compare match, OCxM value is different, output channel x signal OCx run differently:

- OCxM = 000: OCx signal keeps its level.
- OCxM = 001: OCx signal is set as active level.
- OCxM = 010: OCx signal is set as inactive level.
- OCxM = 011: OCx signal toggle.

2. When match, set the flag bit of status register (CCxIF bit of TIM3_SR register).
3. Configure TIM3_DIER register CCxIE =1, generate an interrupt when match.

The output compare mode can also be used to export a single pulse (one-pulse mode).

Configuration steps of channel 1 output compare mode:

1. Configure counter clock (select clock source, configure prescaler factor).
2. Configure TIM3_ARR and TIM3_CCR1 register.
3. Configure TIM3_DIER register CC1IE =1, enable capture/compare 1 interrupt.
4. Configure output mode:
   - Configure TIM3_CCMR1 register OC1M = 011, toggle when OC1 compare match.
   - Configure TIM3_CCMR1 register OC1PE = 0, disable preload function of TIM3_CCR1 register.
   - Configure TIM3_CCMR1 register CC1P = 1, OC1 is active low polarity.

- Configure TIM3_CCER register CC1E = 1, enable channel 1 output, OC1 signal is exported to the corresponding output pin.

5. Configure TIM3_CR1 register CEN =1, enable the counter.

Configure TIM3_CCMRx register OCxPE=0, disable TIM3_CCRx register preload function, write TIM3_CCRx register any time, and the write value updated immediately. Configure TIM3_CCMRx register OCxPE=1, enable TIM3_CCRx register preload function, Updated only at the next update event. The figure below is an example.

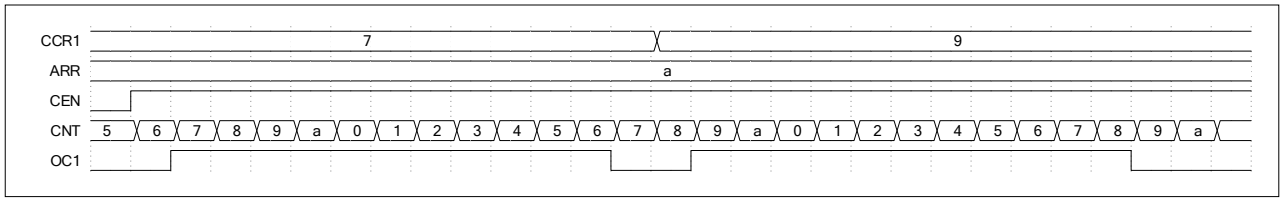


Figure 11-13 Output compare mode, OC1 signals toggle when match

Note: In the output compare mode, the update event has no effect on output result. In the force output mode, the compare output between TIM3_CCRx shadow register and counter still performed. The corresponding flag bit of compare result is also be changed. When enabling the corresponding interrupt request, it still generates the corresponding interrupt request.

11.4.3.3 PWM output

In the PWM mode, according to the value of TIM3_ARR register and TIM3_CCRx register, generate a frequency and duty cycle controllable PWM waveform.

Configure OCxM=110 or OCxM=111 of TIM3_CCMRx register corresponding to the channel x, select channel x to PWM mode 1 or PWM mode 2. In the PWM mode, the counter will always compare with CCRx. According to configuration and compare results, the channel x exports different signals. Therefore, TIM3 can generate 4 PWM output signals with independent duty under the same frequency. In PWM mode, enable the preload function of TIM3_CCRx and TIM3_ARR. The written value of TIM3_CCRx preload register and TIM3_ARR preload register will become effective at the next update event, and load the corresponding shadow register. Therefore, in the PWM mode, before enabling counter, it's required to set TIM3_EGR UG=1, and generate the update event to initialize all registers.

Configure TIM3_CCER register CCxP, select OCx's active polarity. Configure TIM3_CCER register CCxE bit and control OCx output enable. Configure TIM3_CR1 register CMS bit, and select edge alignment or central alignment PWM signal.

- CMS=00, edge alignment mode, configure DIR, select count mode (up or down).
- CMS=01, central alignment mode 1.
- CMS=10, central alignment mode 2.
- CMS=11, central alignment mode 3.

11.4.3.3.1 PWM edge alignment mode - up counting mode

Based on up counting mode configuration, configure TIM3_CCMRx register CCxS =00, select output mode, OCxM =110, select PWM mode 1. In case of TIM3_CNT ＜TIM3_CCRx, channel x (OCxREF) is in active level. Otherwise, it's inactive level. If TIM3_CCRx compare value is greater than the auto reload value (TIM3_ARR), OCxREF stays in active level. When the compare value is 0, OCxREF is at inactive level. The figure below is the example of waveform in PWM mode 1 during edge alignment up count in case of CCR1 =1, CCR2 =4, CCR3 =7, CCR4 =b and ARR =a.

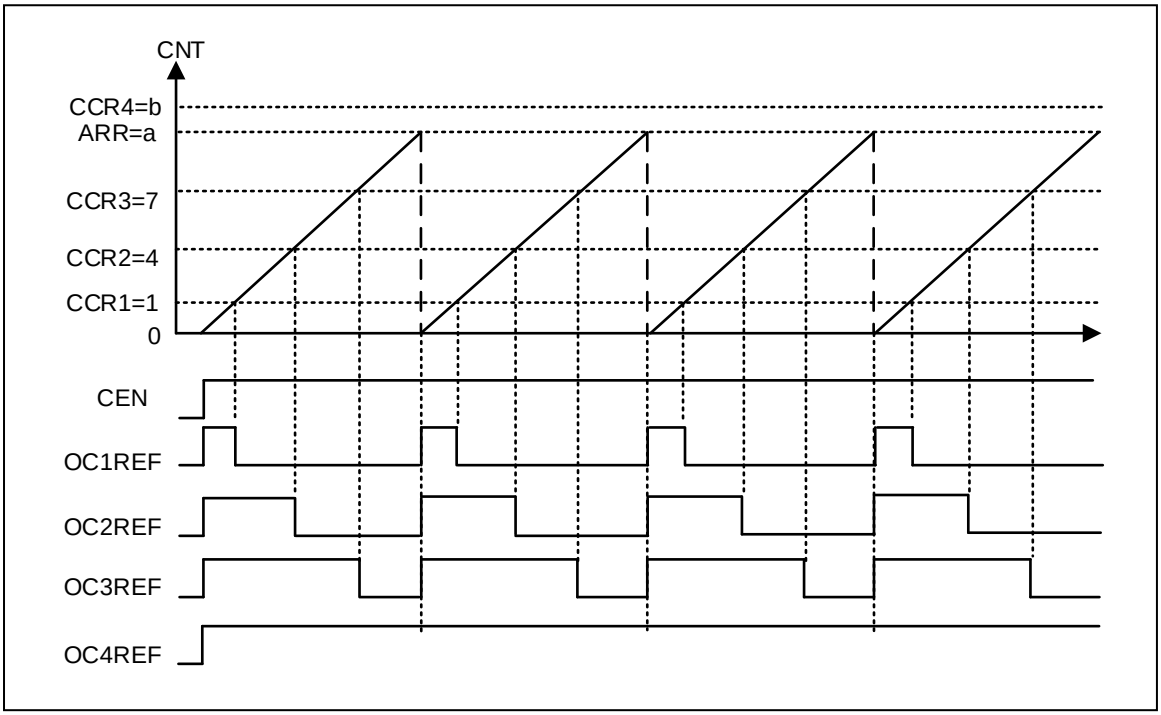


Figure 11-14 Waveform in PWM mode 1 during edge alignment up count

11.4.3.3.2 PWM edge alignment mode -down counting mode

Based on the down counting mode configuration, configure TIM3_CCMRx register CCxS=00, select output mode, OCxM=110, select PWM mode 1. In case of TIM3_CNT > TIM3_CCRx, channel x (OCxREF) is at inactive level. Or else, it’s at active level. The figure is an example of waveform in PWM mode 1 during edge alignment down count in case of CCR1=4 , CCR2=6, CCR3=9 , CCR4=b and ARR=a.

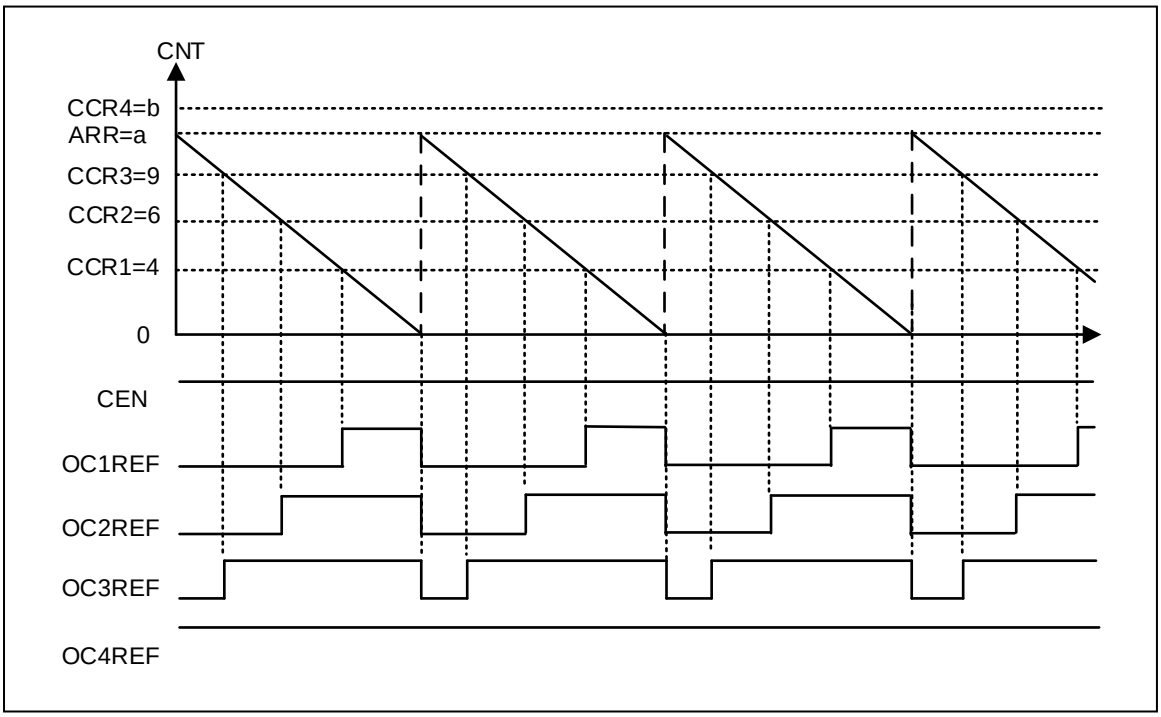


Figure 11-15 Waveform in PWM mode 1 during edge alignment down count

11.4.3.3.3 PWM central alignment mode

Configure TIM3 counter as central alignment count mode. Configure TIM3_CCMRx register CCxS=00 and select output mode. According to the CMS with different configurations, the output compare interrupt flag bit is set during up count (CMS =01), set during down count (CMS =10) or set during up or down count (CMS =11). The figure below is an example of waveform in central alignment PWM mode 1 (CMS =1) during CCR1=4, CCR2=6, CCR3=9, CCR4=b, ARR=a.

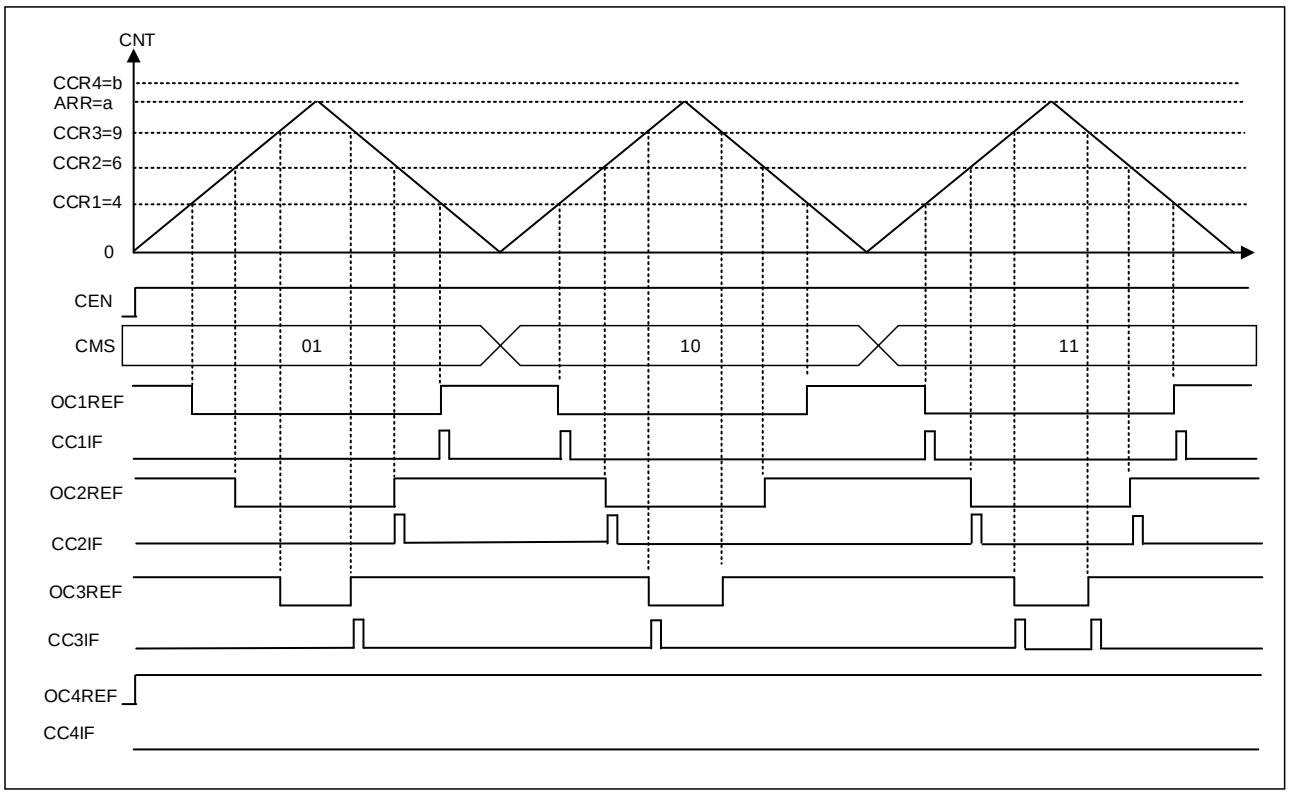


Figure 11-16 Waveform in central alignment PWM mode 1

11.4.3.4 External event clear OCxREF

When configuring TIM3_CCMR register OCxCE=1, OCxREF can be driven low by the active level of ETRF input terminal till the next update event (UEV). The function can only be used in the output mode and PWM mode, does not work in the force output mode.

For example, when OCxREF signal is connected to an external input, ETR configuration:

1. Configure TIM3_SMCR register ETPS[1:0]=00, close the external trigger prescaler.

2. Configure TIM3_SMCR register ECE=0, disable external clock mode 2.

3.Configure TIM3_SMCR register ETF[3:0] and ETP, configure ETR signal trigger polarity and filter width.

In the figure below, when ETRF input change triggers ETRF as high, it corresponds to different OCxCE value, and OCxREF signal action (PWM mode).

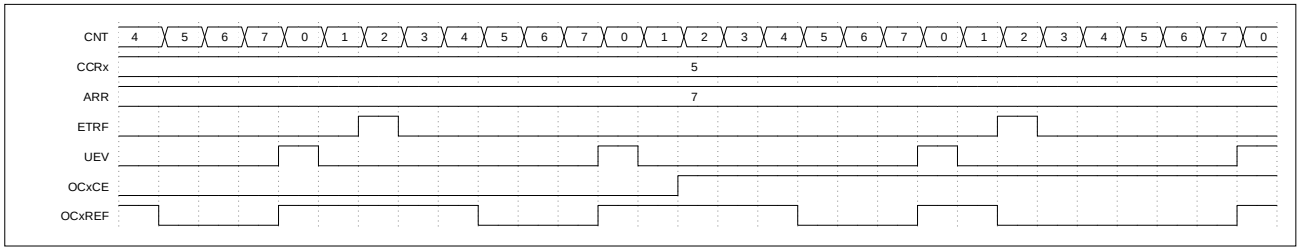


Figure 11-17 The external event clear OCxREF

11.4.3.5 One-pulse output

In the one-pulse mode (OPM), the counter responds an stimulus, and generate a pulse with width programmable. Configure TIM3_CR1 register OPM =1. Select one-pulse mode, trigger signal's active edge or configuring CEN =1 can enable the counter. CEN =1 remains until the next update event or configure CEN =0.

The necessary condition to generate pulse is the difference of compare value and initial value of the counter. Necessary configuration prior to counter enable:

- Up counting mode: Counter CNT < CCRx ≤ ARR.
- Down counting mode: Counter CNT > CCRx.

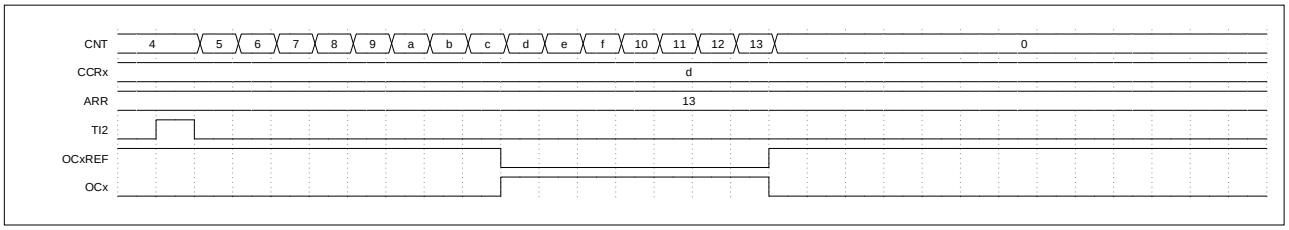


Figure 11-18 one-pulse mode

For example, when TI2 detects the rising edge, and after latency t_{DELAY}, OC2 generate a positive pulse with the length of t_{PULSE}.

Configure TI2FP2 serves as the trigger source:

1. Configure TIM3_CCMR1 register CC2S = 01, map TI2FP2 in TI2.
2. Configure TIM3_CCER register CC2P = 0, detect TI2FP2 rising edge.
3. Configure TIM3_SMCR register TS = 110, TI2FP2 as the slave mode counter trigger (TRGI).
4. Configure TIM3_SMCR register SMS = 110, select trigger mode, TI2FP2 enables the counter.

OPM waveform is determined by TIM3_ARR and TIM3_CCR1 (consider clock frequency and counter prescaler): t_{DELAY} is determined by TIM3_CCR1 register value and CNT initial value; TIM3_ARR - TIM3_CCR1 value is t_{PULSE}.

Next example is generate a negative pulse. In case of compare and match, generate a waveform from 1 to 0. When the counter preload value is achieved, generate a waveform from 0 to 1:

1. Configure TIM3_CCMR1 register OC1M = 111, select PWM mode 2.
2. Configure TIM3_CCER register CC1P = 1, export active low level.
3. Configure TIM3_CCMR1 OC1PE = 1 and TIM3_CR1 register ARPE, enable preload register.
4. Configure TIM3_CCR1 register and TIM3_ARR register.
5. Configure TIM3_EGR register UG =1, generate a update event.
6. Wait for an external trigger event on TI2.

In this case, TIM3_CR1 register DIR=0 , CMS=0 , OPM= 1, stop counting in the next update event

(when the counter toggles to 0 from the auto load value).

11.4.3.5.1 OCx fast enable

OCx fast enable is a special scenario of one-pulse mode. In the one-pulse mode, set TIM3_CCMR register OCxFE=1, and enforce OCxREF to directly respond to stimulus without relying on the compare result of counter and compare value. The output waveform is the same as the waveform when compare and match. Thus, it can remove the compare time, and fast export the compare result. OCx rapid output enable only becomes active in the PWM mode.

11.4.4 Slave mode

11.4.4.1 Encoder interface

The encoder interface mode means the counter count under interaction of orthogonal signals of TI1 and TI2. At the change of input source, the count direction is auto modified by hardware. Configure TIM3_SMCR register SMS bit and select input source. According to input source difference, divide the encoder interface into 3 modes: SMS =001, encoder interface mode 1; SMS =010, encoder interface mode 2; SMS =011, encoder interface mode 3; the specific count in the three modes are expressed below. Two input TI1 and TI2 are used as the interface of orthogonal encoder.

In the encoder mode, prior to counter enable, configure the ARR register. In the encoder interface mode, it's like using an external clock with direction selection. The counter counts continuously between 0 to the auto load value of TIM3 _ARR register (up count and down count are controlled by external clock).

Note: The encoder mode doesn't support the external clock mode 2.

In the encoder interface code, the counter is auto changed according to the speed and direction of increment encoder. Therefore, the counter content always points to the encoder position. The count direction corresponds to the direction of the connected sensor. The table below presents all possible combinations, assuming that TI1 and TI2 are not exchanged at the same time.

Table 11-2 Relations of count direction and encoder signal

Count mode	Relative level (TI1FP1 relative to TI2, TI2FP2 relative to TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Encoder interface mode 1 (only count at TI2)	High level	-	-	up count	down count
Encoder interface mode	Low level	-	-	down count	up count

Count mode	Relative level (TI1FP1 relative to TI2, TI2FP2 relative to TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
1 (only count at TI2)					
Encoder interface mode 2 (only count at TI1)	High level	down count	up count	-	-
Encoder interface mode 2 (only count at TI1)	Low level	up count	down count	-	-
Encoder interface mode 3 (only count at TI1 and TI2)	High level	down count	up count	up count	down count
Encoder interface mode 3 (only count at TI1 and TI2)	Low level	up count	down count	down count	up count

The counter configuration and sequence figure in the encoder interface mode is below. We can see the count signal generation and direction control in the figure. Specific configuration:

1. Configure TIM3_CCMR register CC1S=01, map IC1FP1 on TI1.
2. Configure TIM3_CCMR register CC2S=01, map IC2FP2 on TI2.
3. Configure TIM3_CCER register CC1P =0 , IC1 non-inverted, IC1=TI1.
4. Configure TIM3_CCER register CC2P =0, IC2 non-inverted, IC1=TI2.
5. Configure TIM3_SMCR register SMS =011, select encoder mode 3. According to the input level of another signal, the counter counts at the edge of TI1FP1 and TI2FP2.
6. Configure TIM3_CR1 register CEN =1, enable the counter.

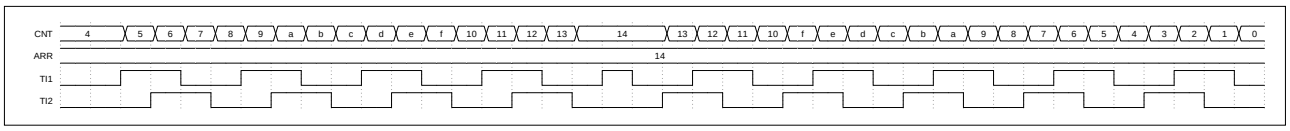


Figure 11-19 Counter time sequence in the encoder code

Counter sequence when the IC1FP1 polarity inverted (CC1P = 1, other configuration unchanged).

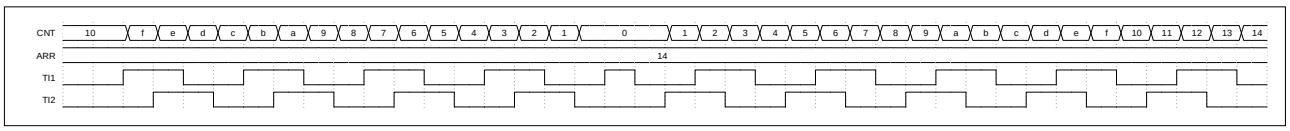


Figure 11-20 IC1FP1 polarity inverted sequence in encoder interface mode

In the encoder interface mode, the counter can provide the current position information of the sensor.

Use another timer in the capture mode to measure the interval period of two encoder events and acquire dynamic information (speed, acceleration, deceleration). According to the interval period of two encoder events, regularly read the counter. Lock the counter value in the third input capture register (the acquire signal should be cyclic and generated by another timer).

11.4.4.2 Reset mode

Configure TIM3_SMCR register SMS=100, select slave reset mode. In this mode, the TRGI input event will clear and restart the counter.

For example, TI2 input terminal falling edge trigger counter restart:

1. Configure TIM3_CCMR1 CC2S=01. CC2 channel is configured as input mode. IC2 is mapped on T12. Configure TIM3_CCER register CC2P=1, detect the falling edge.

2. Configure TIM3_SMCR register SMS = 100, and select the slave mode as reset mode. Configure TIM3_SMCR register TS =110, select timer input 2 after filter (TI2FP2) as the trigger input of sync timer.

3. Configure TIM3_CR1 register DIR=0, select the count direction as the up count; configure PSC=0, no division; configure CEN=1, enable the counter.

The counter clock source is provided by the internal clock. When detecting T12 falling edge, clear and restart the counter. The trigger interrupt flag is set by hardware.

TIM3_ARR = 0x13 time sequence in reset mode

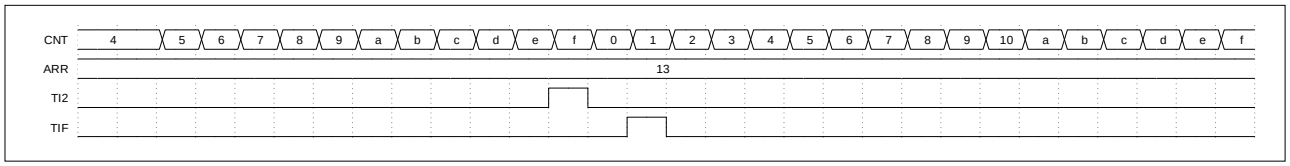


Figure 11-21 Control time sequence in reset mode

11.4.4.3 Gate mode

Configure TIM3_SMCR register SMS=101, select slave gate mode. In this mode, select active level (0: high level active, 1: low level active) according to the TIM3_CCER register CCxP value. When TRGI input is active level, the counter always starts. Otherwise, the counter stops (but without reset operation), and the counter start and stop are countable.

For example, the counter only counts when TI1 is high:

1. Configure CC1S=01 of TIM3_CCMR1 register. CC1 channel is configured as input mode. IC1 is mapped on T11. Configure TIM3_CCER register CC1P=0, detect the high level of TI1.

2. Configure TIM3_SMCR register SMS=101, select slave gate mode; configure TIM3_SMCR register TS=101, select the timer input after filter 1 (TI1FP1) as the sync counter trigger input.

3. Configure TIM3_CR1 register DIR=0, select count direction as the up count; configure PSC=0, no division; configure CEN=1, enable the counter.

The counter clock source is provided by the internal clock. When detecting T11 high level, the counter count start. When TI1 is low level, the counter stops. When the counter stop, set TIF as 1.

TIM3_ARR=0xf time sequence in gate mode.

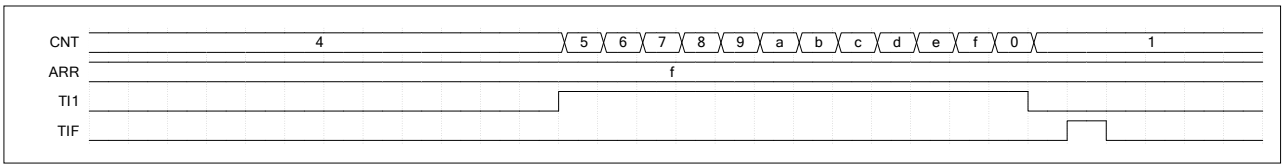


Figure 11-22 Control time sequence in gate mode

11.4.4.4 Trigger mode

Configure TIM3_SMCR register SMS=110, and select slave trigger mode. Select active edge (0: rising edge active, 1: falling edge active) according to the TIM3_CCER register CCxP value. When TRGI input is active edge, the counter starts counting. The timer startup is controlled and the stop is not controlled.

For example, the counter count starts at the rising edge of TI1 input:

1. Configure CC1S=01 of TIM3_CCMR1 register. CC1 channel is configured as input mode. IC1 is mapped on TI1. Configure TIM3_CCER register CC1P=0, and detect the rising edge.

2. Configure TIM3_SMCR register SMS = 110, select slave trigger mode; configure TIM3_SMCR register TS=101, select the timer input 1 (TI1FP1) after filter as the trigger input of counter.

3. Configure TIM3_CR1 register DIR =0, select count direction as the up count; configure PSC=0, no division.

The counter clock source is provided by internal clock. When detecting TI1 rising edge, the counter count starts.

TIM3_ARR =0xf sequence in the trigger mode.

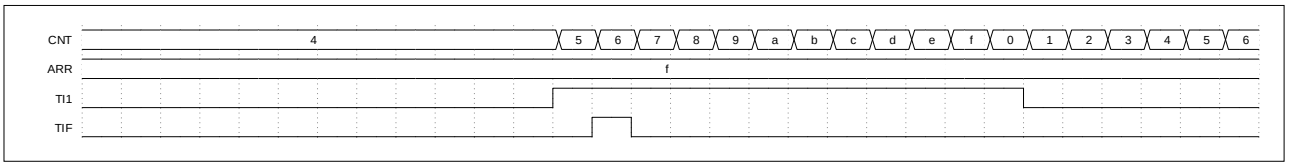


Figure 11-23 Control time sequence in trigger mode

11.4.4.5 External clock mode 2+ slave mode

When the clock source selects the external clock mode 2, and ETR signal is used as external clock input, it can be used together with the other slave mode. In this mode, the slave mode only supports the reset mode, gate mode, trigger mode, not supports the external clock mode 1 and encoder mode.

For example, in the trigger mode, the counter counts once at each rising edge of ETR:

1. Configure TIM3_SMCR register ETF = 0000, don't use the digital filter; configure TIM3_SMCR register ETPS = 00, close the prescaler; configure TIM3_SMCR register ETP =0, detect ETR rising edge; configureTIM3_SMCR register ECE = 1, enable external clock mode 2.

2. Configure TIM3_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 mapped on TI1 as the input capture source; configure TIM3_CCER register CC1P=0, selected rising edge is active.

3. Configure TIM3_SMCR register SMS = 110, select slave trigger mode. Configure TIM3_SMCR register TS = 101, select TI1 as input source.

4. Configure TIM3_CR1 register DIR=0, select the cont direction as up count. Configure PSC=0, no division.

The counter count begins at the rising edge of TI1, and set TIF. The latency between the ETR signal rising edge and actual reset time of counter depends on the sync circuit of ETR input terminal.

TIM3_ARR=13 time sequence in external clock mode 2 + slave mode (trigger mode).

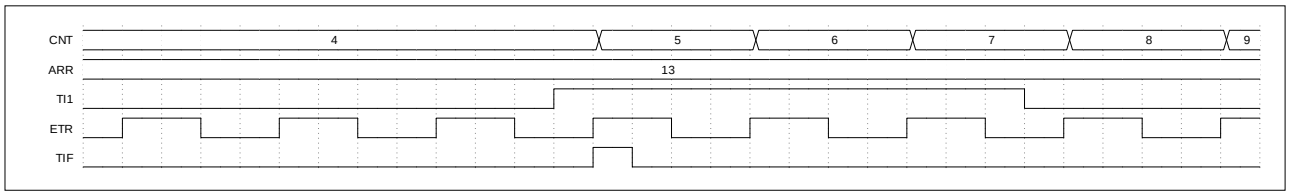


Figure 11-24 Control time sequence in external clock mode 2 +slave mode (trigger mode)

11.4.5 Timer synchronization

Different timers are internally connected, achieve timer cascading or synchronization.

Detailed descriptions are given in the relevant chapter of TIM1.

11.4.6 Timer XOR function

Configure TIM3_CR2 registerTI1S =1, connect the TIM3_CH1 , TIM3_CH2 and TIM3_CH3 pin to TI1 input terminal via XOR for all input modes of the timer.

For example, TIM3_CH1 , TIM3_CH2 and TIM3_CH3 pins are connected to TI1 input terminal via XOR. Sample TI1 input signal's active edge. When TI1 rising edge is active, capture the current counter value, and lock it in the TIM3_CCR1 register. Specific configuration:

1. Configure TIM3_CR2 register TI1S=1, configure three inputs of the timer and connect it to TI1 input channel via XOR.

2. Configure TIM3_CCMR1 register CC1S=01, CC1 channel is configured as input, IC1 is mapped on TI1.

3. Configure TIM3_CCMR1 register IC1F[3:0], configure digital filter width (configure upon need).
4. Configure TIM3_CCER register CC1P=0, select capture in the rising edge of TI1 signal.
5. Configure TIM3_CCMR1 register IC1PS, select prescaler factor.
6. Configure TIM3_CCER register CC1E = 1, enable input/capture channel 1 capture.
7. Configure TIM3_CR1 register CEN=1, enable the counter.

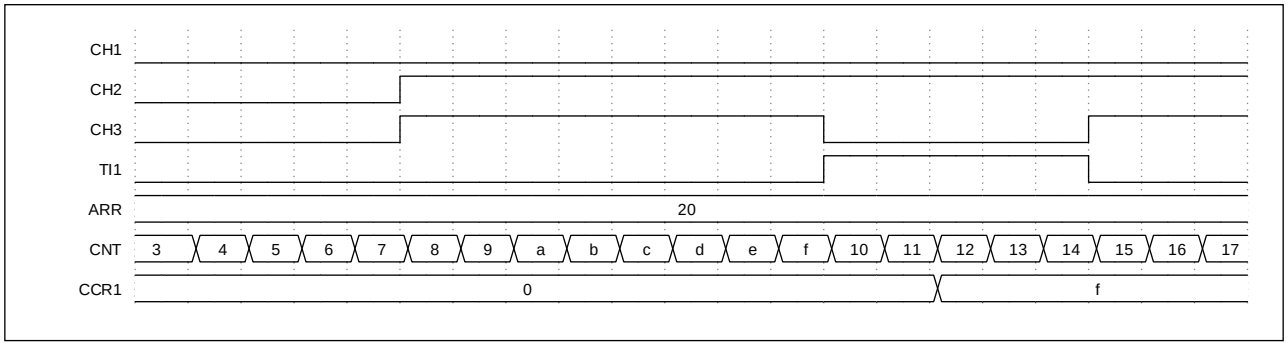


Figure 11-25（TI1 XOR input）input capture waveform

Hall interface circuit

Detailed descriptions are given in the relevant chapter of TIM1.

11.4.7 Debug mode

In the debug mode, configure DBG_CR register DBG_TIM3_STOP=1, TIM3 counter stops counting. (details in the debug section)

11.4.8 Interrupt

TIM3 interrupt includes: capture/compare 1 interrupt, capture/compare 2 interrupt, capture/compare 3 interrupt, capture/compare 4 interrupt, update interrupt, trigger interrupt. When the corresponding interrupt enable bit is set, generate the corresponding event and interrupt.

Table 11-1 List of interrupt events

Interrupt event	Flag bit	Enable bit
Capture/compare 1 interrupt	CC1IF	CC1IE
Capture/compare 2 interrupt	CC2IF	CC2IE
Capture/compare 3 interrupt	CC3IF	CC3IE
Capture/compare 4 interrupt	CC4IF	CC4IE
Update interrupt	UIF	UIE
Trigger interrupt	TIF	TIE

11.5 Register description

Table 11-3 TIM3 register overview

Offset	Acronym	Register Name	Reset
0x00	TIM3_CR1	Control Register 1	0x0000
0x04	TIM3_CR2	Control Register 2	0x0000
0x08	TIM3_SMCR	Slave Mode Control Register	0x0000
0x0C	TIM3_DIER	Interrupt Enable Register	0x0000
0x10	TIM3_SR	Status Register	0x0000
0x14	TIM3_EGR	Event Generation Register	0x0000
0x18	TIM3_CCMR1	Capture/Compare Mode Register 1	0x0000
0x1C	TIM3_CCMR2	Capture/Compare Mode Register 2	0x0000
0x20	TIM3_CCER	Capture/Compare Enable Register	0x0000
0x24	TIM3_CNT	Counter	0x0000
0x28	TIM3_PSC	Prescaler	0x0000
0x2C	TIM3_ARR	Auto Reload Register	0x0000
0x34	TIM3_CCR1	Capture/Compare Register 1	0x0000
0x38	TIM3_CCR2	Capture/Compare Register 2	0x0000
0x3C	TIM3_CCR3	Capture/Compare Register 3	0x0000
0x40	TIM3_CCR4	Capture/Compare Register 4	0x0000
0x50	TIM3_OR	Input Option Register	0x0000

11.5.1 TIM3_CR1 Control Register 1

Address offset:0x00

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						CKD		ARPE	CMS		DIR	OPM	URS	UDIS	CEN
						rw		rw	rw		rw	rw	rw	rw	rw

Bit	Field	Description
15：10	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
9：8	CKD	Clock division Division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock used by the dead-time generators and the digital filters (ETR, TIx). 00：$t_{DTS} = t_{INT_CK}$ 01：t_{DTS} =2x t_{INT_CK} 10：t_{DTS} =4x t_{INT_CK} 11：Reserved, do not program this value
7	ARPE	Auto reload preload enable 0: Disable the shadow register of TIM3_ARR register 1: Enable the shadow register of TIM3_ARR register
6：5	CMS	Center alignment mode selection 00: Edge alignment mode. Count direction depends on DIR bit 01: Central alignment mode 1. The counter alternatively conducts up and down count. The channel is in output mode. Only in down count, compare interrupt flag bit is set 10: Central alignment mode 2. The counter alternatively conducts up and down count. The channel is in output mode. Only in up count, compare interrupt flag bit is set 11: Central alignment mode 3. The counter alternatively conducts up and down count. The channel is in output mode. In up and down count, compare interrupt flag bit is set Note: During count, the alignment mode change is disabled.
4	DIR	Count direction 0: up count 1: down count Note: When the counter is configured as central alignment mode or encoder mode, the bit is read only.
3	OPM	one-pulse mode 0: Disable one-pulse mode. In case of update event, the counter count continues 1: Enable one-pulse mode. In case of the next update event (clear CEN bit), the counter count stops
2	URS	Update request source This bit is set and cleared by software, select update event source. 0: The event below may generate a update interrupt request: - Counter overflow/underflow - Set UG bit - Update generation through the slave mode controller 1: Only in counter overflow/underflow, generate update interrupt request

Bit	Field	Description
1	UDIS	Update disable This bit is used to enable or disable the update event 0: Update event (UEV) enabled. 1: Update event disabled. The Update event is not generated, shadow registers keep their value(ARR, PSC, CCRx). However the counter and the prescaler are reinitialized if the EGR.UG bit is set, the counter is reinitialized if a hardware reset is received from the slave mode controller.
0	CEN	Counter enable 0: Counter disabled 1: Counter enabled Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.

11.5.2 TIM3_CR2 Control Register 2

Address offset:0x04

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								TI1S	MMS			Res.			
								rw	rw						

Bit	Field	Description
15：8	Reserved	Reserved, must be kept at reset value.
7	TI1S	TI1 selection 0: TIM3_CH1 pin is connected to TI1 input 1: TIM3_CH1, TIM3_CH2 and TIM3_CH3 pins pins are connected to the TI1 input after XOR combination

Bit	Field	Description
6：4	MMS	Master mode selection These bits control TRGO signal selection, used to select the sync information sent to the slave timers in master mode: 000: Reset TIM3_EGR register UG bit generate one pulse trigger output (TRGO). 001: Enable the Counter Enable signal CNT_EN is used as trigger output (TRGO). It is useful to start several timers at the same time or to control a window in which a slave timer is enable. The Counter Enable signal is generated by a logic OR between CEN control bit and the trigger input when configured in gated mode. When the Counter Enable signal is controlled by the trigger input, there is a delay on TRGO, except if the master/slave mode is selected. 010: Update Update event is selected as TRGO. 011: Capture/compare When a capture or a compare match occurred, send a positive pulse as TRGO. 100: Compare OC1REF signal is used as trigger output (TRGO) 101: Compare OC2REF signal is used as trigger output (TRGO) 110: Compare OC3REF signal is used as trigger output (TRGO) 111: Compare OC4REF signal is used as trigger output (TRGO)
3：0	Reserved	Reserved, must be kept at reset value.

11.5.3 TIM3_SMCR Slave Mode Control Register

Address offset:0x08

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS		ETF				MSM	TS			OCCS	SMS		
rw	rw	rw		rw				rw	rw			rw	rw		

Bit	Field	Description
15	ETP	External trigger polarity The bit selects ETR signal polarity. 0: High level or rising edge is active 1: Low level or falling edge is active Note: It's only suitable for supporting the external trigger product
14	ECE	External clock enable The bit enables the external clock mode 2. 0: external clock mode 2 disabled 1: external clock mode 2 enabled, The counter is clocked by any active edge on the ETRFsignal. Note 1: It's only suitable for supporting the external trigger product. Note 2: Configuring ECE =1 has the same effect with configuring SMS =111 and TS =111. Note 3: TS≠111, reset mode, gate mode and trigger mode can be used together with the external clock mode 2. Note 4: If external clock mode 1 and external clock mode 2 are enabled at the same time, the external clock input is ETRF.
13：12	ETPS	External trigger prescaler The external trigger signal ETRP frequency must be lower than 1/4 of TIM3PCLK frequency. When input the quicker external clock, use the prescaler to reduce ETRP frequency. 00: Prescaler OFF 01: ETRP frequency divided by 2 10: ETRP frequency divided by 4 11: ETRP frequency divided by 8 Note: It's only suitable for supporting the external trigger product

Bit	Field	Description
11：8	ETF	External trigger filter These bits define ETRP signal sampling frequency and ETRP digital filter bandwidth. In fact, the digital filter is an event counter, which generate an output jump after N event. 0000: No filter, f_{DTS} sampling 001: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =2 0010: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =4 0011: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =8 0100: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0111: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =8 1000: Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =6 1001:Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =8 1010: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =5 1011: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =6 1100: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =8 1101: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =5 1110: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =6 1111: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =8 Note: It's only suitable for supporting the external trigger product
7	MSM	Master/slave mode 0: No action 1: The effect of an event on the trigger input (TRGI) is delayed to allow a perfect synchronization between the current timer and its slaves (through TRGO). It is useful if we want to synchronize several timers on a single external event.
6：4	TS	Trigger selection Trigger input source selection. 000：Internal trigger 0 (ITR0) 001：Internal trigger 1 (ITR1) 010：Internal trigger 2 (ITR2) 011：Internal trigger 3 (ITR3) 100：TI1 edge detector (TI1F_ED) 101：Timer input after filter 1 (TI1FP1) 110：Timer input after filter 2 (TI2FP2) 111：External trigger input (ETR) More details on ITRx are given in the table below. Note: After the slave mode enable, these bits cannot be changed

Bit	Field	Description
3	OCCS	Timer compare output signal(OCxREF) clear selection In PWM mode, clear the compare output signal(OCxREF) 0: External trigger signal as clear signal 1: Comparator output(COMP) as clear signal Note: It's only applicable to products that support external trigger or have a built-in comparator(COMP)
2：0	SMS	Slave mode selection When external signals are selected the active edge of the trigger signal (TRGI) is linked to the polarity selected on the external input. 000: Close slave mode - In case of CEN =1, the prescaler is directly driven by the internal clock. 001: Encoder mode 1 - Based on TI1FP1 level, the counter conducts up/down count along TI2FP2 edge. 010: Encoder mode 2 - Based on TI2FP2 level, the counter conducts up/down count along TI1FP1edge. 011: Encoder mode 3- Based on another input level, the counter conducts up/down count along TI1FP1 and TI2FP2 edge. 100: Reset mode -The rising edge of the selected trigger input (TRGI) reinitialize the counter and generate a update register signal. 101: Gate mode - When the trigger input (TRGI) is high, the counter count begins. When the trigger input turns low, the counter count stops (but without reset) and generate a update event. The counter start and stop are controlled. 110: Trigger mode -The counter starts in the rising edge of the trigger input TRGI (but without reset) and generate a update event. Only the counter start is controlled. 111: External clock mode 1 -The rising edge of the selected trigger input (TRGI) drives the counter. Note: If TI1F_EN is selected as trigger input (TS =100), don't use the gate mode. This is because that EI1F_ED exports a pulse at each TI1F change. However, in the gate mode, it's required to check the trigger input level.

Table 11-4 TIM3 internal trigger connection

Slave timer	ITR0	ITR1	ITR2	ITR3
TIM1	-	-	TIM3_TRGO	TIM14_OC1REF
TIM3	TIM1_TRGO	-	-	TIM14_OC1REF

11.5.4 TIM3_DIER Interrupt Enable Register

Address offset:0x0C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									TIE	Res.	CC4IE	CC3IE	CC2IE	CC1IE	UIE
									rw		rw	rw	rw	rw	rw

Bit	Field	Description
15：7	Reserved	Reserved, must be kept at reset value.
6	TIE	Trigger interrupt enable 0: Break interrupt disable 1: Break interrupt enable
5	Reserved	Reserved, must be kept at reset value.
4	CC4IE	Enable capture/compare 4 interrupt 0: Capture/compare interrupt 4 disable 1: Capture/compare interrupt 4 enable
3	CC3IE	Enable capture/compare 3 interrupt 0: Capture/compare interrupt 3 disable 1: Capture/compare interrupt 3 enable
2	CC2IE	Enable capture/compare 2 interrupt 0: Capture/compare interrupt 2 disable 1: Capture/compare interrupt 2 enable
1	CC1IE	Enable capture/compare 1 interrupt 0: Capture/compare interrupt 1 disable 1: Capture/compare interrupt 1 enable
0	UIE	Enable update interrupt 0: Update event interrupt disable 1: Update event interrupt enable

11.5.5 TIM3_SR Status Register

Address offset:0x10

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			CC4OF	CC3OF	CC2OF	CC1OF	Res.		TIF	Res.	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			r_w0c						r_w0c		r_w0c				

Bit	Field	Description
15：13	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
12	CC4OF	Capture/compare 4 over capture flag Refer to CC1OF description.
11	CC3OF	Capture/compare 3 over capture flag Refer to CC1OF description.
10	CC2OF	Capture/compare 2 over capture flag Refer to CC1OF description.
9	CC1OF	Capture/compare 1 over capture flag Only when the channel 1 is configured as input capture mode and CC1F is configured as 1, and the capture event occurs again. The flag is set by hardware. Write 0 may clear the bit. 0: No overcapture 1: Overcapture
8：7	Reserved	Reserved, must be kept at reset value.
6	TIF	Trigger interrupt flag In case of trigger event (When the slave mode counter is in any other mode except for the gate mode and external clock mode，detect the active edge in TRGI input terminal, or detect the stop edge in gate mode), the bit is set by hardware. It’s cleared by software. 0: No trigger event occurred 1: Trigger interrupt pending
5	Reserved	Reserved, must be kept at reset value.
4	CC4IF	Capture/compare 4 interrupt flag Refer to CC1IF description
3	CC3IF	Capture/compare 3 interrupt flag Refer to CC1IF description
2	CC2IF	Capture/compare 2 interrupt flag Refer to CC1IF description.
1	CC1IF	Capture/compare 1 interrupt flag Channel 1 in output mode: When the counter value and compare value match, the bit is set by hardware except for the central alignment mode (the bit is set in the central alignment mode according to TIM3_CR1.CMS[1:0]). It’s cleared by software. 0: No match 1: TIM3_CNT value and TIM3_CCR1 value match Channel 1 as output mode: In case of capture event, the bit is set by hardware. It’s cleared by software or reading TIM3_CCR1 register 0: No input capture occurred 1: Counter value is captured to TIM3_CCR1

Bit	Field	Description
0	UIF	Update interrupt flag In case of update event, the bit is set by hardware. It's cleared by software. 0: No update interrupt occurred 1: update interrupt pending This bit is set by hardware when the registers are updated: - In case of TIM3_CR1 register UDIS=0 and REP_CNT=0, when the counter generates overflow/underflow. -In case of TIM3_CR1 register UDIS=0, URS=0, when TIM3_EGR register UG =1. - In case of TIM3_CR1 register UDIS=0, URS=0, when update generation through the slave mode controller.

11.5.6 TIM3_EGR Event Generation Register

Address offset:0x14

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									TG	Res.	CC4G	CC3G	CC2G	CC1G	UG
									w		w	w	w	w	w

Bit	Field	Description
15：7	Reserved	Reserved, must be kept at reset value.
6	TG	Trigger generation 0: No action 1: generate a trigger event. TIM3_SR register TIF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. It's auto cleared by hardware.
5	Reserved	Reserved, must be kept at reset value.
4	CC4G	Capture/compare 4 generation Refer to CC1G description.
3	CC3G	Capture/compare 3 generation Refer to CC1G description.
2	CC2G	Capture/compare 2 generation Refer to CC1G description.

Bit	Field	Description
1	CC1G	Capture/compare 1 generation Refer to CC1G description. This bit is set by software. It generates a capture/compare event, and is auto cleared by hardware. 0: No action 1：Generate a capture/compare event in channel CC1: When the channel CC1 is configured as output: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. When the channel CC1 is configured input: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt When CC1IF is set, set CC1OF =1.
0	UG	Update event generation 0：No action 1: Initialize the counter, and generate a update event. It's auto cleared by hardware. When selecting the central alignment or up counting mode, the counter is clear; otherwise (down counting mode) the counter auto reload value is loaded. The prescaler counter is cleared at the same time.

11.5.7 TIM3_CCMR1 Capture/Compare Mode Register 1

Address offset:0x18

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M			OC2PE	OC2FE	CC2S		OC1CE	OC1M			OC1PE	OC1FE	CC1S	
IC2F				IC2PSC		CC2S		IC1F				IC1PSC		CC1S	
rw	rw			rw	rw	rw		rw	rw			rw	rw	rw	

The channel may be used for input (capture mode) or output (compare mode). The channel direction is defined by the corresponding CCxS. The register's bit other than CCxS functions differently in the input mode and output mode. OCxx describes the channel function in the output mode. ICxx describes the channel function in the input mode.

Output compare mode:

Bit	Field	Description
15	OC2CE	Channel 2 output and compare clear enable Refer to OC1CE description.
14：12	OC2M	Channel 2 output and compare mode Refer to OC1M description.

Bit	Field	Description
11	OC2PE	Channel 2 output and compare preload enable Refer to OC1PE description.
10	OC2FE	Channel 2 output and compare quick enable Refer to OC1FE description.
9：8	CC2S	Channel 2 capture/compare 2 selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 2 is configured as output 01：Channel 2 is configured as input, IC2 is mapped on TI2 10：Channel 2 is configured as input, IC2 is mapped on TI1 11：Channel 2 is configured as input, IC2 is mapped on TRC. The mode only works when the internal generator input is selected (TIM3_SMCR register TS bit selection)
7	OC1CE	Channel 1 capture/compare clear enable 0: OC1REF is not affected by ETR 1: When detecting ETR as active level, clear OC1REF

Bit	Field	Description
6：4	OC1M	Channel 1 output compare mode The bit has defined the output reference signal OC1REF action. OC1REF has determined OC1 and OC1N value. OC1REF is active at high level. The active level of OC1, OC1N depends on CC1P and CC1NP bit. 000：Freeze. TIM3_CCR1 and TIM3_CNT compare results has no effect on OC1REF. 001: Set as high when configuration. When TIM3_CNT value and TIM3_CCR1 value are same, enforce OC1REF as high level 010: Set as low when configuration. When TIM3_CNT value and TIM3_CCR1 value are same, enforce OC1REF as low level 011: Toggle when match. when TIM3_CCR1=TIM3_CNT, OC1REF toggle. 100: Enforce as low. Enforce OC1REF at low level 101: Enforce as high. Enforce OC1REF at high level 110 ： PWM mode 1. During up count, in case of TIM3_CNT<TIM3_CCR1, enforce OC1REF is at high level. Or else, it's at low level. During down count, in case of TIM3_CNT > TIM3_CCR1, enforce OC1REF is at low level. Or else, it's at high level. 111 ： PWM mode 2. During up count, in case of TIM3_CNT<TIM3_CCR1, channel 1 enforce OC1REF is at low level. Or else, it's at high level. During down count, in case of TIM3_CNT > TIM3_CCR1, enforce OC1REF is at high level. Or else, it's at low level. Note 1: In case of LOCK level 3 (TIM3_BDTR register LOCK bit) and CC1S = 00 (the channel is configured as output), the bit cannot be changed. Note 2: In PWM mode 1 or PWM mode 2, only when the compare result changes or it changes over from the freeze mode to PWM mode in the output compare mode, OC1REF level may change.

Bit	Field	Description
3	OC1PE	Channel 1 output compare preload enable 0: Disable TIM3_CCR1 register preload function. The value written into TIM3_CCR1 register becomes active immediately 1: Enable TIM3_CCR1 register preload function. Only read and write the preload register. TIM3_CCR1 preload value becomes active at the update event Note 1: When the LOCK level is 3 (TIM3_BDTR register LOCK bit) and CC1S = 00 (when the channel is configured as output), the bit cannot be changed. Note 2: Only in the one-pulse mode (TIM3_CR1 register OPM= 1), it has no influence whether the preload register is set. Under other scenarios, it's required to set the preload register. Otherwise, the follow up action is not certain.
2	OC1FE	Channel 1 output compare quick enable In case of the bit is set 1, when the channel is configured as PWM mode, it speeds up response of the capture/compare output to the trigger time. The output channel deems the active edge of the trigger input signal as one compare match. Therefore, OC is set as compare level, but is irrelevant to the compare result. 0: channel 1 output compare fast enable. 1: channel 1 output compare fast disable.
1：0	CC1S	Channel 1 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 1 is configured as output 01：Channel 1 is configured as input, IC1 is mapped on TI1 10：Channel 1 is configured as input, IC1 is mapped on TI2 11：Channel 1 is configured as input, IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)

Input capture mode:

Bit	Field	Description
15：12	IC2F	Input capture 2 filter Refer to IC1F description
11：10	IC2PSC	Input/capture 2 prescaler Refer to IC1PSC description

Bit	Field	Description
9：8	CC2S	Channel 2 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 2 is configured as output 01：Channel 1 is configured as input, IC2 is mapped on TI2 10：Channel 2 is configured as input, IC2 is mapped on TI1 11：Channel 2 is configured as input, IC2 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)
7：4	IC1F	Channel 1 input capture filter The digital filter is composed of an event counter. It records an output jump after N input events. These bits define IC1 input signal sampling frequency and digital filter length. 0000: No filter, f_{DTS} sampling 001: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =2 0010: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =4 0011: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =8 0100: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0111: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =8 1000: Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =6 1001:Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =8 1010: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =5 1011: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =6 1100: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =8 1101: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =5 1110: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =6 1111: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =8
3：2	IC1PSC	Channel 1 input/capture prescaler This bit defines the ratio of the prescaler acting on IC1.The prescaler is reset as soon as CC1E=0(TIM3_CCER register). 00: No prescaler, the capture input detects each edge to trigger one capture 01: Trigger one capture for every 2 events 10: Trigger one capture for every 4 events 11: Trigger one capture for every 8 events

Bit	Field	Description
1：0	CC1S	Channel 1 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 1 is configured as output 01: Channel 1 is configured as input, IC1 is mapped on TI1 10: Channel 1 is configured as input, IC1 is mapped on TI2 11: Channel 1 is configured as input, IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)

11.5.8 TIM3_CCMR2 Capture/Compare Mode Register 2

Address offset:0x1C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M			OC4PE	OC4FE	CC4S		OC3CE	OC3M			OC3PE	OC3FE	CC3S	
IC4F				IC4PSC		CC4S		IC3F				IC3PSC		CC3S	
rw	rw			rw	rw	rw		rw	rw			rw	rw	rw	

Compare output mode:

Bit	Field	Description
15	OC4CE	Channel 4 output compare clear enable Refer to OC3CE description
14：12	OC4M	Channel 4 output compare mode Refer to OC3M description
11	OC4PE	Channel 4 output compare preload enable Refer to OC3PE description
10	OC4FE	Channel 4 output compare rapid enable Refer to OC3FE description
9：8	CC4S	Channel 4 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 4 is configured as output 01: Channel 4 is configured as input, IC4 is mapped on TI4 10: Channel 4 is configured as input, IC4 is mapped on TI3 11: Channel 4 is configured as input, IC4 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection
7	OC3CE	Channel 3 output compare clear enable 0: OC3REF is not affected by ETR input 1: When detecting ETR input is at active level, clear OC3REF

Bit	Field	Description
6：4	OC3M	Channel 3 output compare mode The bit has defined the output reference signal OC3REF action. OC3REF has determined OC3 and OC3N value. OC3REF is active at high level. The active level of OC3, OC3N depends on CC3P and CC3NP bit. 000：Freeze. TIM3_CCR3 and TIM3_CNT compare results has no effect on OC3REF. 001: Set as high when configuration. When TIM3_CNT value and TIM3_CCR3 value are same, enforce OC3REF as high level 010: Set as low when configuration. When TIM3_CNT value and TIM3_CCR3 value are same, enforce OC3REF as low level 011: Toggle when match. When TIM3_CCR3=TIM3_CNT,OC3REF toggle. 100: Enforce as low. Enforce OC3REF at low level 101: Enforce as high. Enforce OC3REF at high level 110: PWM mode 1. During up count, in case of TIM3_CNT<TIM3_CCR3, enforce OC3REF as high level, otherwise it's low level; during down count, in case of TIM3_CNT > TIM3_CCR3, enforce OC3REF as low level, otherwise it's high level. 111: PWM mode 2. During up count, in case of TIM3_CNT<TIM3_CCR3, enforce OC3REF as low level, otherwise it's high level; during down count, in case of TIM3_CNT > TIM3_CCR3, enforce OC3REF as high level, otherwise it's low level. Note 1: In case of LOCK level 3 (TIM3_BDTR register LOCK bit) and CC3S = 00 (the channel is configured as output), the bit cannot be changed. Note 2: In PWM mode 1 or PWM mode 2, only when the compare result changes or it changes over from the freeze mode to PWM mode in the output compare mode, OC3REF level may change.

Bit	Field	Description
3	OC3PE	Channel 3 output compare preload enable 0: Disable TIM3_CCR3 register preload function, the value written into TIM3_CCR3 register becomes active immediately 1: Enable TIM3_CCR3 register preload function. The read and write only works on the preload register. TIM3_CCR3 preload value becomes active when the update event is available Note 1: When the LOCK level is set 3 (TIM3_BDTR register LOCK bit) and CC3S =0 (the channel is configured as output), the bit cannot be changed. Note 2: Only in the one-pulse mode (TIM3_CR1 register OPM= 1), it's not required to set the preload register. Under other scenarios, it's required to set the preload register. Otherwise, the follow up action is not certain.
2	OC3FE	Channel 3 output compare quick enable In case of the bit 1, when the channel is configured as PWM mode, it speeds up response of the capture/compare output to the trigger time. The output channel deems the active edge of the trigger input signal as one compare match. Therefore, OC is set as compare level, but is irrelevant to the compare result. 0: channel 3 output compare fast disable. 1: channel 3 output compare fast enable.
1：0	CC3S	Channel 3 capture/compare selection The bit has defined the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 3 is configured as input 01: Channel 3 is configured as input, IC3 is mapped on TI3 10: Channel 3 is configured as input, IC3 is mapped on TI4 11: Channel 3 is configured as input, IC3 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)

Input capture mode:

Bit	Field	Description
15：12	IC4F	Input capture 4 filter Refer to IC3F description
11：10	IC4PSC	Input/capture 4 prescaler Refer to IC3PSC description

Bit	Field	Description
9：8	CC4S	Channel 4 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 4 is configured as output 01：Channel 4 is configured as input, IC4 is mapped on TI4 10：Channel 4 is configured as input, IC2 is mapped on TI3 11：Channel 4 is configured as input, IC4 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)
7：4	IC3F	Channel 3 input capture filter The digital filter is composed of an event counter. It records an output jump after N input events. These bits define IC1 input signal sampling frequency and digital filter length. 0000: No filter, f_{DTS} sampling 001: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =2 0010: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =4 0011: Sampling frequency $f_{sampling}$ =f_{INT_CK}, N =8 0100: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =6 0101: Sampling frequency $f_{sampling}$ =$f_{DTS/2}$, N =8 0110: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =6 0111: Sampling frequency $f_{sampling}$ =$f_{DTS/4}$, N =8 1000: Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =6 1001:Sampling frequency $f_{sampling}$ =$f_{DTS/8}$, N =8 1010: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =5 1011: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =6 1100: Sampling frequency $f_{sampling}$ =$f_{DTS/16}$, N =8 1101: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =5 1110: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =6 1111: Sampling frequency $f_{sampling}$ =$f_{DTS/32}$, N =8
3：2	IC3PSC	Channel 3 input/capture prescaler This bit defines the ratio of the prescaler acting on IC3.The prescaler is reset as soon as CC3E=0(TIM3_CCER register). 00: No prescaler, the capture input detects each edge to trigger one capture 01: Trigger one capture for every 2 events 10: Trigger one capture for every 4 events 11: Trigger one capture for every 8 events

Bit	Field	Description
1：0	CC3S	Channel 3 capture/compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 3 is configured as output 01: Channel 3 is configured as input, IC3 is mapped on TI3 10: Channel 3 is configured as input, IC3 is mapped on TI4 11: Channel 3 is configured as input, IC3 is mapped on TRC. This mode only works when the internal trigger input is selected (TIM3_SMCR register TS bit selection)

11.5.9 TIM3_CCER Capture/Compare Enable Register

Address offset:0x20

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

Bit	Field	Description
15	CC4NP	Channel 4 input/capture complementary output polarity Refer to CC1NP description
14	Reserved	Reserved, must be kept at reset value.
13	CC4P	Channel 4 capture/compare output polarity Refer to CC1P description.
12	CC4E	Channel 4 capture/compare enable Refer to CC1E description
11	CC3NP	Channel 3 input/capture complementary output polarity Refer to CC1NP description
10	Reserved	Reserved, must be kept at reset value.
9	CC3P	Channel 3 input capture output polarity Refer to CC1P description
8	CC3E	Channel 3 input/capture output enable Refer to CC1E description
7	CC2NP	Channel 2 input/capture complementary output polarity Refer to CC1NP description
6	Reserved	Reserved, must be kept at reset value.
5	CC2P	Channel 2 input capture output polarity Refer to CC1P description
4	CC2E	Channel 2 input/capture output enable Refer to CC1E description

Bit	Field	Description
3	CC1NP	Channel 1 input/capture complementary output polarity When channel 1 is configured as output, this bit has defined the input signal polarity: 0: OC1N is active at high level 1: OC1N is active at low level When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the ICx polarity/level selection table. Note: When LOCK level (TIM3_BDTR register LCCK bit) is set 3 or 2 and CC1S = 00 (channel is configured as output), the bit cannot be changed.
2	Reserved	Reserved, must be kept at reset value.
1	CC1P	Channel 1 input/capture output polarity When channel 1 is configured as output, the bit defines the output signal polarity: 0: OC1 is active at high level 1: OC1 is active at low level When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the ICx polarity/level selection table. Note: When LOCK level (TIM3_BDTR register LCCK) is set 3 or 2, this bit cannot be changed.
0	CC1E	Channel 1 input/capture output enable When channel 1 is configured as output: 0: Close. OC1 output disable 1: Enable. The output level relies on the values in MOE, OSSI, OSSR, OIS1, OIS1N and CC1NE. CC1channel is configured as input: This bit determines whether the input capture function is enabled. 0：Capture disable 1：Capture enable

In input mode, ICx polarity/level selection is given in the Table below:

Table 11-5 ICx polarity/level selection

CCxP	CCxNP	ICx polarity/level
0	0	Rising edge active/high level active
1	0	Falling edge active/low level active
1	1	Rising edge or falling edge active/high level active
0	1	Reserved

11.5.10 TIM3_CNT Counter

Address offset:0x24

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
15：0	CNT	Counter value

11.5.11 TIM3_PSC Prescaler

Address offset:0x28

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC															
rw															

Bit	Field	Description
15：0	PSC	Prescaler value Counter clock frequency (ck_cnt) =$f_{CK_PSC/（PSC+1）}$ In case of update event, PSC value is loaded into the current prescaler register.

11.5.12 TIM3_ARR Auto Reload Register

Address offset:0x2C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR															
rw															

Bit	Field	Description
15：0	ARR	Auto reload value These bits define the auto reload value of the counter. When auto reload value is 0, the counter doesn’t work.

11.5.13 TIM3_CCR1 Capture/Compare Register 1

Address offset:0x34

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1															
rw															

Bit	Field	Description
15：0	CCR1	Channel 1 capture/compare value When channel 1 is configured as input: CCR1 value determines the counter value of the previous capture event (this register is only readable). When channel 1 is configured as output: If the preload function is not selected in TIM3_CCMR1 register (OC1PE bit), the write value is immediately transmitted to the current capture/compare shadow register. Or else, the write value is only loaded into the capture/compare register in case of update event. The current capture/compare shadow register is involved in the compare with the counter TIM3_CNT, and the compare result is reflected to the output signal of OC1 port.

11.5.14 TIM3_CCR2 Capture/Compare Register 2

Address offset:0x38

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2															
rw															

Bit	Field	Description
15：0	CCR2	Channel 2 capture/compare value Refer to CCR1 description.

11.5.15 TIM3_CCR3 Capture/Compare Register 3

Address offset:0x3C

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CCR3
rw

Bit	Field	Description
15：0	CCR3	Channel 3 capture/compare value Refer to CCR1 description.

11.5.16 TIM3_CCR4 Capture/Compare Register 4

Address offset:0x40

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4															
rw															

Bit	Field	Description
15：0	CCR4	Channel 4 capture/compare value Refer to CCR1 description.

11.5.17 TIM3_OR Input Option Register

Address offset:0x50

Reset value:0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														ETR_RMP	
														rw	rw

Bit	Field	Description
15:2	Reserved	Reserved, must be kept at reset value.
1:0	ETR_RMP	ETR multiplex 00：Reserved 01：LSI clock input 10：LSE clock input 11：HSE_CLK_DIV_128 clock input

12 TIM14 Basic Timer

12.1 Overview

TIM14 is composed of a 16 bit real-time programmable prescaler and a 16-bit auto reload counter, and can conveniently provide with the counting and timing function. The counter is driven by a programmable prescaler. The basic timer can be used for a variety of purposes, such as input capture function (measuring input signal pulse width or frequency, etc), output compare function (PWM output, compare output, etc.)

12.2 Function block diagram

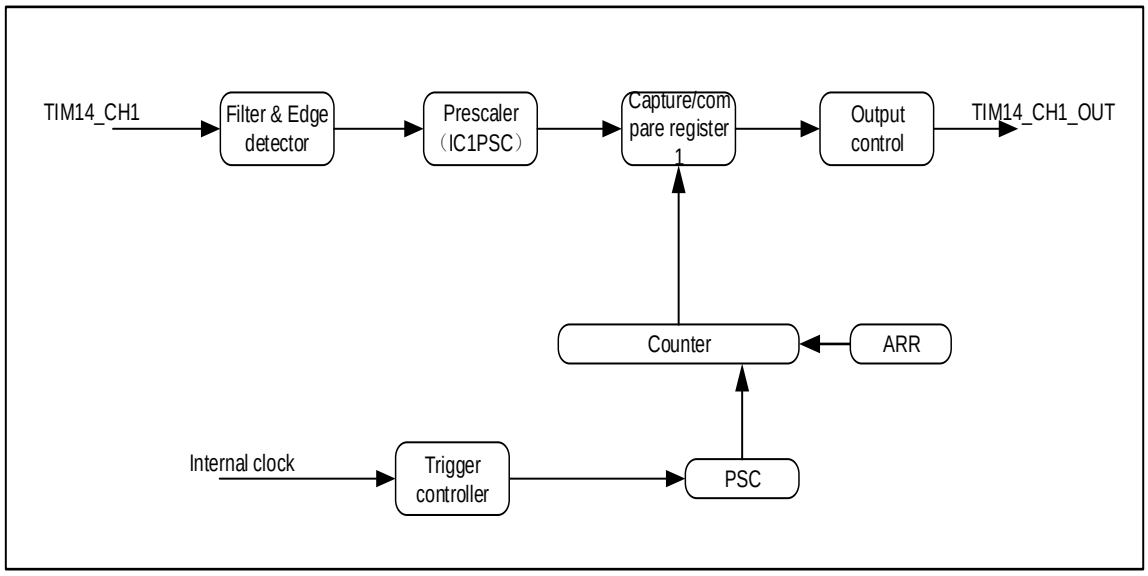


Figure 12-1 TIM14 block diagram

TIM14 structure block diagram is mainly composed of input capture unit, output compare unit, counter unit.

12.3 Main characteristics

- 16 bit real-time programmable prescaler, division factor: 1-65536.
- 16 bit auto reload counter (counting direction: up).
- Input capture: measure input signal pulse width or period.
- Output compare (control output waveform or indicator counter already finishes timing).

- One-pulse mode.
- PWM output (edge alignment mode)
- Interrupt request event: update event, input capture, output compare.

12.4 Function description

12.4.1 Clock

12.4.1.1 Clock selection

The clock of the counter is provided by the internal clock (INT_CK).

12.4.1.2 Time base unit

TIM14 time base unit mainly includes: Counter register (TIM14_CNT), prescaler register (TIM14_PSC), auto reload register (TIM14_ARR).

The counter unit is composed of a 16 bit counter with its related auto reload register, The counter can count up.

The counter clock is provided by prescaler. The prescaler is composed of the prescaler counter with its related register. The division factor is 1-65536. It can write any time, and gets active at the next update.

The auto preload register has a 16 bit shadow register with preload function. Set TIM14_CR1 register ARPE bit to select the content of ARR register are transferred into the shadow register permanently or at each update event.

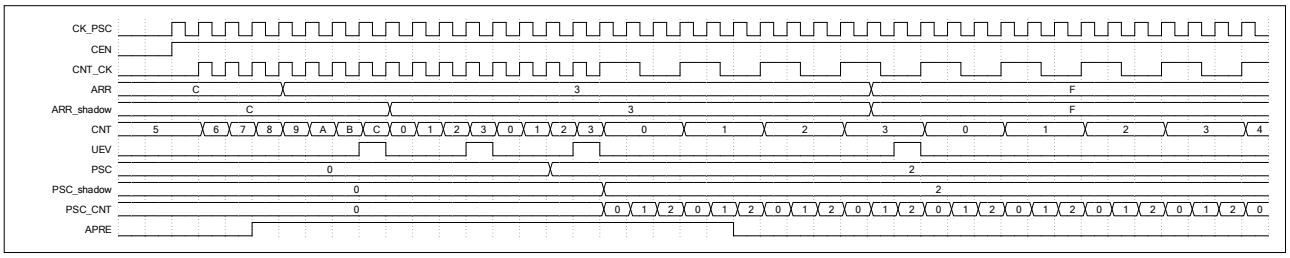


Figure 12-2 Auto preload

12.4.1.3 Counter modes

TIM14 supports only the up counting mode. Enable TIM14_CR1 register CEN, and the counter begins up count from 0 to TIM14_ARR value and generates a counter overflow event (update). The counter begins up count again from 0. Set TIM14_EGR register UG =1, also generate an update event.

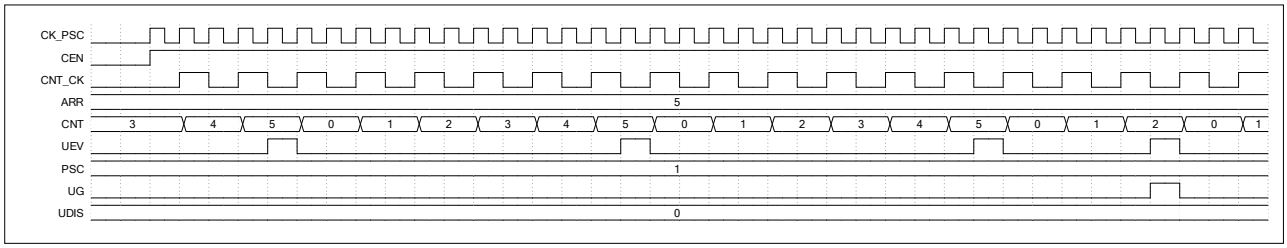


Figure 12-1 Up counting mode（UDIS=0）

Configuring TIM14_CR1 register UDIS=1, disable update event. In case of counter overflow event, the update event is not generated. Configuring UG=1, the update event is not generated. However, the counter and prescaler counter will initialize, and begin up count from 0.

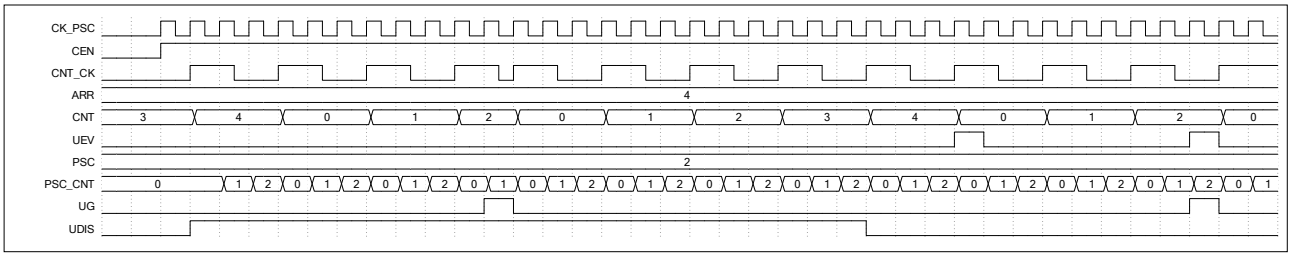


Figure 12-3 Up counting mode (UDIS =1, disable update event)

Note: In case of update event:

- ARR register value is loaded into ARR shadow register.
- Prescaler preload value becomes active.

12.4.2 Input capture

12.4.2.1 Input capture

The input capture is composed of digital filter, edge detector, prescaler, capture compare register, etc. Its block diagram:

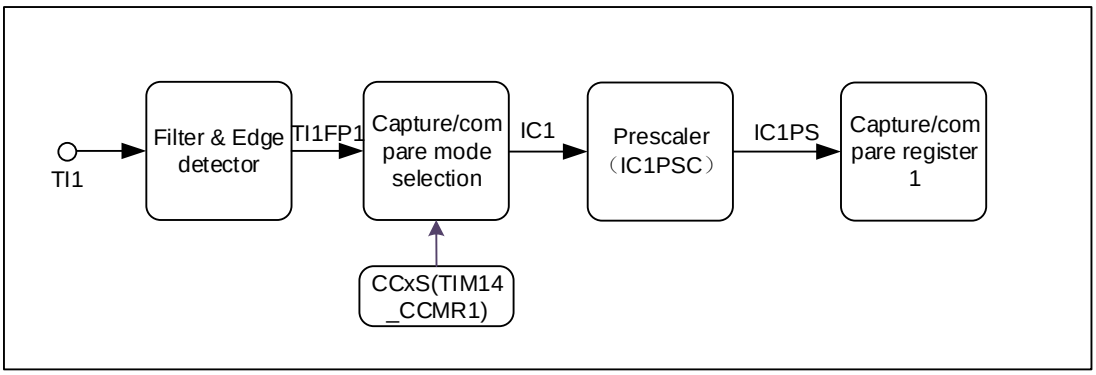


Figure 12-4 TIM14 input capture block diagram

Configure TIM14_CCMR1 register IC1F, set the filter width of a digital filter (filter sampling frequency

and digital filter width are shown in the table below). When the input signal width is greater than the filter width, the input signal is active. When the digital filter samples the input signal of the input pin TI1, generate a filtered signal TI1F. Through the edge detector with the optional polarity, generate an effective signal IC1. This signal generates IC1PS via the prescaler for trigger input capture event.

Table 12-1 Digital filter width and IC1F correlation

IC1F[3：0]	Sample frequency and filter length	IC1F[3：0]	Sample frequency and filter length
0000	No filter, sample by f_{DTS}	1000	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=6
0001	Sampling frequency fsampling=f_{INT_CK}, N=2	1001	Sampling frequency $f_{sampling}=f_{DTS}$ /8, N=8
0010	Sampling frequency fsampling=f_{INT_CK}, N=4	1010	Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=5
0011	Sampling frequency fsampling=f_{INT_CK}, N=8	1011	Sampling frequency $f_{sampling}=f_{DTS}$/16, N=6
0100	Sampling frequency fsampling=f_{DTS} /2, N=6	1100	Sampling frequency $f_{sampling}=f_{DTS}$ /16, N=8
0101	Sampling frequency fsampling=f_{DTS}/2, N=8	1101	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=5
0110	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=6	1110	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=6
0111	Sampling frequency $f_{sampling}=f_{DTS}$ /4, N=8	1111	Sampling frequency $f_{sampling}=f_{DTS}$ /32, N=8

In the input capture mode, when detecting the active edge of signal IC1, the counter's current value is locked in the corresponding shadow register, and then loaded into the corresponding capture/compare register. When enabling interrupt, and producing the capture event, generate the corresponding interrupt request. At the capture event, the capture flag bit CC1IF of the status register (TIM14_SR) will be set. Configure CC1IF =0 or read TIM14_CCR1 data, clear CC1IF flag bit. When CC1IF is not cleared, next input capture event generated, the repeat capture flag CC1OF will be set. Configure CC1OF =0, can clear CC1OF flag bit.

For example, Sample TI1 input signal's active edge. At TI1 rising edge, capture the current counter value and lock it in the TIM14_CCR1 register, procedure steps as follows:

1. Configure TIM14_CCMR1 register CC1S=01, CC1 channel is configured as input, and IC1 is mapped on TI1.
2. Configure TIM14_CCMR1 register IC1F[3:0] , configure the filter width of digital filter (configure upon need).
3. Configure TIM14_CCER register CC1P=0, select capture occurs in the rising edge of TI1 signal.
4. Configure TIM14_CCMR1 register IC1PSC[1:0], select the prescaler factor.
5. Configure TIM14_CCER register CC1E = 1, channel 1 capture enable.
6. Configure TIM14_DIER register CC1IE=1, enable channel 1 interrupt request.

Note：

- When the channel is configured as input mode, TIM14_CCR1 register becomes read only.
- In case of more than two continuous captures and CC1IF flag is not cleared, the repeat capture flag CC1OF is set. In order to avoid losing the capture information that may be generated before the repeat capture flag CC1OF is set. It is recommended to read data before reading the repeat capture flag.
- Set the corresponding CC1G bit of TIM14_EGR register, generate the input capture interrupt request through software.

12.4.3 Output compare

The compare output of the capture/compare channel is composed of the comparator, output control circuit and capture/compare register. Its structure is shown below:

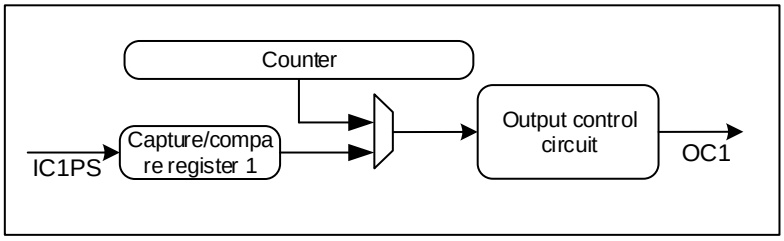


Figure 12-5 Output compare block diagram

In output compare mode, the content of capture/compare register is loaded into the shadow register, and then compare the content of shadow register with the current value of counter. The capture/compare module includes a capture/compare register (preload register) and a shadow register. When read and write, only programmed the capture/compare register.

12.4.3.1 Force output

Configure TIM14_CCMR1 register CC1S = 00 and set the channel CC1 as output mode. Configure TIM14_CCMR1 register OC1M bit, and directly enforce the output compare signal as active or inactive level, independently of the output compare result. Configure TIM14_CCMR1 register OC1M = 100, and enforce output compare signal as inactive level. And OC1REF is enforced as low level. Configure TIM14_CCMR1 register OC1M = 101, and enforce output compare signal as active level. And the OC1REF is enforced as high level (OC1REF always active as high level).

Note: In force output mode, TIM14_CCR1 shadow register and counter output compare is still performed. The corresponding flag bit of compare result will also be changed. When enable the corresponding interrupt request, it will also generate the corresponding interrupt request.

12.4.3.2 Output compare

In the output compare mode, when the values of counter and capture compare register matches, different waveform can be exported according to the OC1M bit of TIM14_CCMR1 register.

For example, When the value of counter and capture/compare register matches, the functions in the output compare mode:

1.When compare match, OC1M value is different, output channel 1 signal OC1 run differently:

- OC1M = 000: OC1signal keeps its level.
- OC1M = 001: OC1 signal is set as active level.
- OC1M = 010: OC1 signal is set as inactive level.
- OC1M = 011: OC1 signal toggle.

2. When match, set the flag bit of status register (CC1IF bit of TIM14_SR register).
3. Configure TIM14_DIER register CC1IE =1, generate an interrupt when match.

The output compare mode can also be used to export a single pulse (one-pulse mode).

Configuration steps of channel 1 output compare mode:

1. Configure counter clock (select clock source, configure prescaler factor).
2. Configure TIM14_ARR and TIM14_CCR1 register.
3. Configure TIM14_DIER register CC1IE =1, enable capture/compare 1 interrupt.
4. Configure output mode:
   - Configure TIM14_CCMR1 register OC1M = 011, toggle when OC1 compare match.
   - Configure TIM14_CCMR1 register OC1PE = 0, disable preload function of TIM14_CCR1 register.
   - Configure TIM14_CCMR1 register CC1P = 1, OC1 is active low polarity.
   - Configure TIM14_CCER register CC1E = 1, enable channel 1 output, OC1 signal is exported to the corresponding output pin.
5. Configure TIM14_CR1 register CEN =1, enable the counter.

Configure TIM14_CCMR1 register OC1PE=0, disable TIM14_CCR1 register preload function, write TIM14_CCR1 register any time, and the write value updated immediately. Configure TIM14_CCMR1 register OC1PE=1, enable TIM14_CCR1 register preload function, Updated only at the next update event. The figure below is an example.

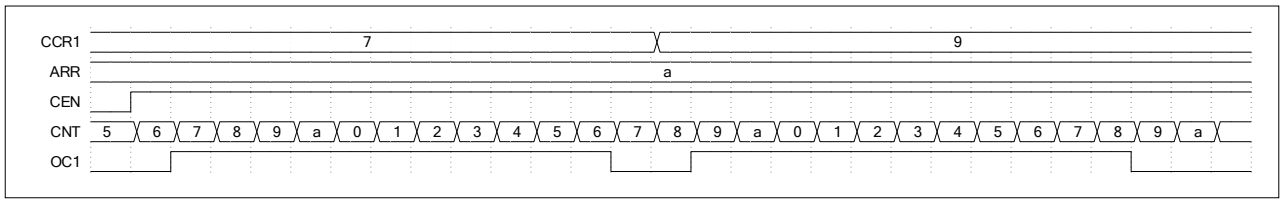


Figure 12-6 Output compare mode, OC1 signals toggle when match

Note: In the output compare mode, the update event has no effect on output result. In the force output mode, the compare output between TIM14_CCR1 shadow register and counter still performed. The corresponding flag bit of compare result is also be changed. When enabling the corresponding interrupt request, it still generates the corresponding interrupt request.

12.4.3.3 PWM output

In the PWM mode, according to the value of TIM14_ARR register and TIM14_CCR1 register, generate a frequency and duty cycle controllable PWM waveform.

Configure OC1M=110 or OC1M=111 of TIM14_CCMR1 register corresponding to the channel 1, select channel 1 to PWM mode 1 or PWM mode 2. In the PWM mode, the counter will always compare with CCR1. According to configuration and compare results, the channel x exports different signals. Therefore, TIM14 can generate 1 PWM output signals with independent duty under the same frequency. In PWM mode, enable the preload function of TIM14_CCR1 and TIM14_ARR. The written value of TIM14_CCR1 preload register and TIM14_ARR preload register will become effective at the next update event, and load the corresponding shadow register. Therefore, in the PWM mode, before enabling counter, it’s required to set TIM14_EGR UG=1, and generate the update event to initialize all registers.

Configure TIM14_CCER register CC1P, select OC1’s active polarity. Configure TIM14_CCER register CC1E to control OC1 output enable.

12.4.3.3.1 PWM edge alignment mode - up counting mode

Based on up counting mode configuration, configure TIM14_CCMR1 register CC1S =00, select output mode, OC1M =110, select PWM mode 1. In case of TIM14_CNT ＜TIM14_CCR1, channel 1 (OC1REF) is in active level. Otherwise, it's inactive level. If TIM14_CCR1 compare value is greater than the auto reload value (TIM14_ARR), OC1REF stays in active level. When the compare value is 0, OC1REF is at inactive level. The figure below is the example of waveform in PWM mode 1 during edge alignment up count in case of CCR1 =4, and ARR =a.

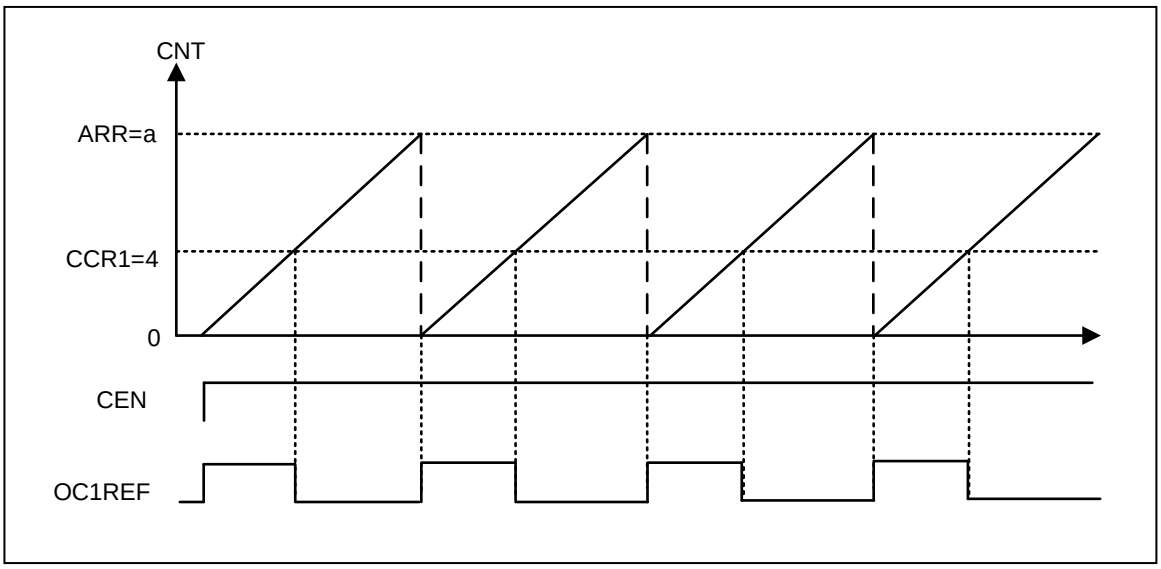


Figure 12-7 Waveform in PWM mode 1 during edge alignment up count

12.4.3.4 One-pulse output

In the one-pulse mode (OPM), the counter responds an stimulus, and generate a pulse with width programmable. Configure TIM14_CR1 register OPM =1. Select one-pulse mode, trigger signal's active edge or configuring CEN =1 can enable the counter. CEN =1 remains until the next update event or configure CEN =0.

The necessary condition to generate pulse is the difference of compare value and initial value of the counter. Necessary configuration prior to counter enable:

- Up counting mode: Counter CNT < CCR1 ≤ ARR.

The figure below is one pulse mode:

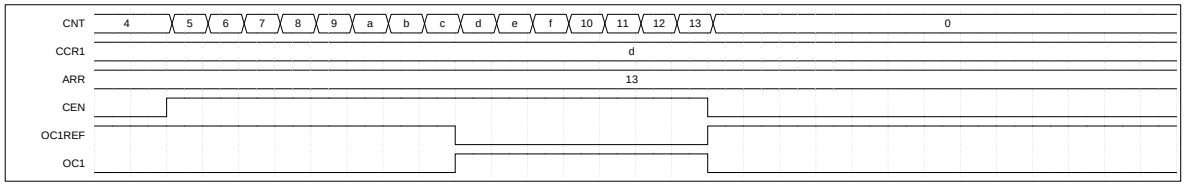


Figure 12-8 one-pulse mode

12.4.4 Debug mode

In the debug mode, configure DBG_CR register DBG_TIM14_STOP=1, TIM14 counter stops counting. (details in the debug section)

12.4.5 Interrupt

TIM14 interrupt includes: capture/compare 1 interrupt and update interrupt. When the corresponding

interrupt enable bit is set, generate the corresponding event and interrupt.

Table 12-2 List of interrupt events

Interrupt event	Flag bit	Enable bit
Capture/compare 1 interrupt	CC1IF	CC1IE
Update interrupt	UIF	UIE

12.5 Register description

Table 12-3 TIM14 register overview

Offset	Acronym	Register Name	Reset
0x00	TIM14_CR1	Control Register 1	0x0000
0x0C	TIM14_DIER	Interrupt Enable Register	0x0000
0x10	TIM14_SR	Status Register	0x0000
0x14	TIM14_EGR	Event Generation Register	0x0000
0x18	TIM14_CCMR1	Capture/Compare Mode Register 1	0x0000
0x20	TIM14_CCER	Capture/Compare Enable Register	0x0000
0x24	TIM14_CNT	Counter	0x0000
0x28	TIM14_PSC	Prescaler	0x0000
0x2C	TIM14_ARR	Auto Reload Register	0x0000
0x34	TIM14_CCR1	Capture/Compare Register 1	0x0000
0x44	TIM14_BDTR	Break And Dead-Time Register	0x0000

12.5.1 TIM14_CR1 Control Register 1

Address offset：0x00

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						CKD		ARPE	Res.			OPM	URS	UDIS	CEN
						rw		rw				rw	rw	rw	rw

Bit	Field	Description
15：10	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
9：8	CKD	Clock division Division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock used by the dead-time generators and the digital filters (TI1). 00：$t_{DTS} = t_{INT_CK}$ 01：t_{DTS} =2x t_{INT_CK} 10：t_{DTS} =4x t_{INT_CK} 11：Reserved, do not program this value
7	ARPE	Auto reload preload enable 0: Disable the shadow register of TIM14_ARR register 1: Enable the shadow register of TIM14_ARR register
6：4	Reserved	Reserved, must be kept at reset value.
3	OPM	one-pulse mode 0: Disable one-pulse mode. In case of update event, the counter count continues 1: Enable one-pulse mode. In case of the next update event (clear CEN bit), the counter count stops
2	URS	Update request source Software configures this bit, select update event source. 0: The event below may generate an update interrupt request: - Counter overflow - Set UG bit 1: Only in counter overflow, generate update interrupt request
1	UDIS	Update disable This bit is used to enable or disable the update event 0: Update event (UEV) enabled. 1: Update event disabled. The Update event is not generated, shadow registers keep their value(ARR, PSC, CCR1). However the counter and the prescaler are reinitialized if the EGR.UG bit is set.
0	CEN	Counter enable 0: Counter disabled 1: Counter enabled

12.5.2 TIM14_DIER Interrupt Enable Register

Address offset：0x0C

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														CC1IE	UIE
														rw	rw

Bit	Field	Description
15：2	Reserved	Reserved, must be kept at reset value.
1	CC1IE	Enable capture/compare 1 interrupt 0: Capture/compare interrupt 1 disable 1: Capture/compare interrupt 1 enable
0	UIE	Enable update interrupt 0: Update event interrupt disable 1: Update event interrupt enable

12.5.3 TIM14_SR Status Register

Address offset：0x10

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						CC1OF	Res.							CC1IF	UIF
						r_w0c								r_w0c	r_w0c

Bit	Field	Description
15：10	Reserved	Reserved, must be kept at reset value.
9	CC1OF	Capture/compare 1 over capture flag Only when the channel 1 is configured as input capture mode and CC1F is configured as 1, and the capture event occurs again. The flag is set by hardware. Write 0 may clear the bit. 0: No overcapture 1: Overcapture
8：2	Reserved	Reserved, must be kept at reset value.
1	CC1IF	Capture/compare 1 interrupt flag Channel 1 in output mode: When the counter value and compare value match, the bit is set by hardware, It's cleared by software. 0: No match 1: TIM14_CNT value and TIM14_CCR1 value match Channel 1 as output mode: In case of capture event, the bit is set by hardware. It's cleared by software or reading TIM14_CCR1 register 0: No input capture occurred 1: Counter value is captured to TIM14_CCR1

Bit	Field	Description
0	UIF	Update interrupt flag In case of update event, the bit is set 1 by hardware. It's cleared 0 by software. 0: No update interrupt occurred 1: update interrupt pending TIM1_EGR register UG =1 or counter overflow generate update event.

12.5.4 TIM14_EGR Event Generation Register

Address offset：0x14

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														CC1G	UG
														w	w

Bit	Field	Description
15：2	Reserved	Reserved, must be kept at reset value.
1	CC1G	Capture/compare 1 generation This bit is set by software. It generates a capture/compare event, and is auto cleared by hardware. 0: No action 1：Generate a capture/compare event in channel CC1: When the channel CC1 is configured as output: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. When the channel CC1 is configured input: Set CC1IF =1. When enabling the corresponding interrupt, generate the corresponding interrupt. When CC1IF is set, set CC1OF =1.
0	UG	Update generation 0：No action 1: Initialize the counter, and generate an update event. It's auto cleared by hardware.

12.5.5 TIM14_CCMR1 Capture/Compare Mode Register 1

Address offset：0x18

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								Res.	OC1M			OC1PE	Res.	CC1S	

	IC1F		IC1PSC		CC1S
	rw	rw	rw	rw	rw

The channel may be used for input (capture mode) or output (compare mode). The channel direction is defined by the corresponding CC1S. The register's other bits functions differently in the input mode and output mode. OC1x describes the channel function in the output mode. IC1x describes the channel function in the input mode.

Output compare mode:

Bit	Field	Description
15：7	Reserved	Reserved, must be kept at reset value.
6：4	OC1M	Output compare mode The bit has defined the output reference signal OC1REF action. OC1REF has determined OC1 value. OC1REF is active at high level. The active level of OC1 depends on CC1P bit. 000：Freeze. TIM14_CCR1 and TIM14_CNT compare results has no effect on OC1REF. 001: Set as high when configuration. When TIM14_CNT value and TIM14_CCR1 value are same, enforce OC1REF as high level 010: Set as low when configuration. When TIM14_CNT value and TIM14_CCR1 value are same, enforce OC1REF as low level 011: Toggle when match. when TIM14_CCR1=TIM14_CNT, OC1REF toggle. 100: Enforce as low. Enforce OC1REF at low level 101: Enforce as high. Enforce OC1REF at high level 110 ： PWM mode 1. During up count, in case of TIM14_CNT<TIM14_CCR1, enforce OC1REF is at high level. Or else, it's at low level. 111 ： PWM mode 2. During up count, in case of TIM14_CNT<TIM14_CCR1, channel 1 enforce OC1REF is at low level. Or else, it's at high level. Note: In PWM mode 1 or PWM mode 2, only when the compare result changes or it changes over from the freeze mode to PWM mode in the output compare mode, OC1REF level may change.
3	OC1PE	Output compare preload enable 0: Disable TIM14_CCR1 register preload function. The value written into TIM14_CCR1 register becomes valid immediately 1: Enable TIM14_CCR1 register preload function. Only read and write the preload register. TIM14_CCR1 preload value becomes valid during the update event Note: Only in the one-pulse mode (TIM14_CR1 register OPM= 1), it has no influence whether the preload register is set. Under other scenarios, it's required to set the preload register. Otherwise, the follow up action is not certain.
2	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
1：0	CC1S	Channel 1 Capture/Compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00：Channel 1 is configured as output 01：Channel 1 is configured as input 10：Reserved 11：Reserved

Input capture mode

Bit	Field	Description
15：8	Reserved	Reserved, must be kept at reset value.
7：4	IC1F	Channel 1 input capture filter The digital filter is composed of an event counter. It records an output jump after N input events. These bits define IC1 input signal sampling frequency and digital filter length. 0000：No filter, f_{DTS} sampling 0001：Sampling frequency $f_{sampling}=f_{INT_CK}$，N=2 0010：Sampling frequency $f_{sampling}=f_{INT_CK}$，N=4 0011：Sampling frequency $f_{sampling}=f_{INT_CK}$，N=8 0100：Sampling frequency $f_{sampling}=f_{DTS}$ /2，N=6 0101：Sampling frequency $f_{sampling}=f_{DTS}$ /2，N=8 0110：Sampling frequency $f_{sampling}=f_{DTS}$ /4，N=6 0111：Sampling frequency $f_{sampling}=f_{DTS}$ /4，N=8 1000：Sampling frequency $f_{sampling}=f_{DTS}$ /8，N=6 1001：Sampling frequency $f_{sampling}=f_{DTS}$ /8，N=8 1010：Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=5 1011：Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=6 1100：Sampling frequency $f_{sampling}=f_{DTS}$ /16，N=8 1101：Sampling frequency $f_{sampling}=f_{DTS}$ /32，N=5 1110：Sampling frequency $f_{sampling}=f_{DTS}$ /32，N=6 1111：Sampling frequency $f_{sampling}=f_{DTS}$ /32，N=8
3：2	IC1PSC	Channel 1 input capture prescaler This bit defines the ratio of the prescaler acting on IC1.The prescaler is reset as soon as CC1E=0(TIM14_CCER register). 00: No prescaler, the capture input detects each edge to trigger one capture 01: Trigger one capture for every 2 events 10: Trigger one capture for every 4 events 11: Trigger one capture for every 8 events

Bit	Field	Description
1：0	CC1S	Channel 1 capture/Compare selection This bit defines the channel direction and input signal selection. Only when the channel closes, these bits can be written: 00: Channel 1 is configured as output 01: Channel 1 is configured as input 10: Reserved 11: Reserved

12.5.6 TIM14_CCER Capture/Compare Enable Register

Address offset：0x20

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												CC1NP	Res.	CC1P	CC1E
												rw		rw	rw

Bit	Field	Description
15：4	Reserved	Reserved, must be kept at reset value.
3	CC1NP	Channel 1 input capture 1 polarity This bit is not valid when Channel 1 is configured as an output. When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the IC1 polarity/level selection table.
2	Reserved	Reserved, must be kept at reset value.
1	CC1P	Channel 1 capture/Compare output polarity When Channel 1 is configured as an output, this bit defines the output signal polarity. 0: OC1 active at high level 1: OC1 active at low level When channel 1 is configured as input, CC1P/CC1NP match use has defined the input signal polarity and level. Details are given in the IC1 polarity/level selection table.
0	CC1E	Channel 1 capture/compare output enable When channel 1 is configured as an output. 0: off. OC1 output is disabled 1: on. OC1 signal is output to the corresponding output pin CC1 Channel configured as input. This bit determines whether the input capture function is enabled. 0: Capture disable 1: Capture enable

The polarity/level selection of IC1 in the input mode is shown in the following table:

Table 12-4 IC1 polarity/level selection table

CC1P	CC1NP	IC1 polarity/level
0	0	Rising edge active/high level active
1	0	Falling edge active/low level active
1	1	Rising or falling edge active/high level active
0	1	Reserved

12.5.7 TIM14_CNT Counter

Address offset：0x24

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
15：0	CNT	Count value

12.5.8 TIM14_PSC Prescaler

Address offset：0x28

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC															
rw															

Bit	Field	Description
15：0	PSC	Prescaler value Counter clock frequency（ck_cnt）=f_{CK_PSC}/（PSC+1） In case of an update event, PSC value is loaded into the current prescaler register.

12.5.9 TIM14_ARR Auto Reload Register

Address offset：0x2C

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR															
rw															

Bit	Field	Description
15：0	ARR	Auto-reload value These bits define the auto reload value of the counter. When auto reload value is 0, the counter doesn't work.

12.5.10 TIM14_CCR1 Capture/Compare Register 1

Address offset：0x34

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1															
rw															

Bit	Field	Description
15：0	CCR1	Channel 1 capture/compare value When channel 1 is configured as input: CCR1 value determines the counter value of the previous capture event (this register is only readable). When channel 1 is configured as output: If the preload function is not selected in TIM14_CCMR1 register (OC1PE bit), the write value is immediately transmitted to the current capture/compare shadow register. Or else, the write value is only loaded into the capture/compare register in case of update event. The current capture/compare shadow register is involved in the compare with the counter TIM14_CNT, and the compare result is reflected to the output signal of OC1 port.

12.5.11 TIM14_BDTR Break And Dead-Time Register

Address offset：0x44

Reset value：0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	Res.														
rw															

Bit	Field	Description
15	MOE	Main output enable 0：Disable OC1 output or enforce as the idle state 1: If the corresponding enable bit (CC1E of TIM14_CCER register) is set, the OC1 output is turned on
14：0	Reserved	Reserved, must be kept at reset value.

13 WWDG Window Watchdog

13.1 Overview

The window watchdog is the watchdog where the feed time has the range of upper and lower limit. It's mainly to detect the software issue caused by the program runaway arising out of external interference and unpredictable logic condition.

According to the normal program time setting, to refresh a time window of the watchdog guarantees no advance or delayed refresh of the watchdog. It may detect whether the program runs according to the normal pathway or jumps some program sections.

13.2 Function block diagram

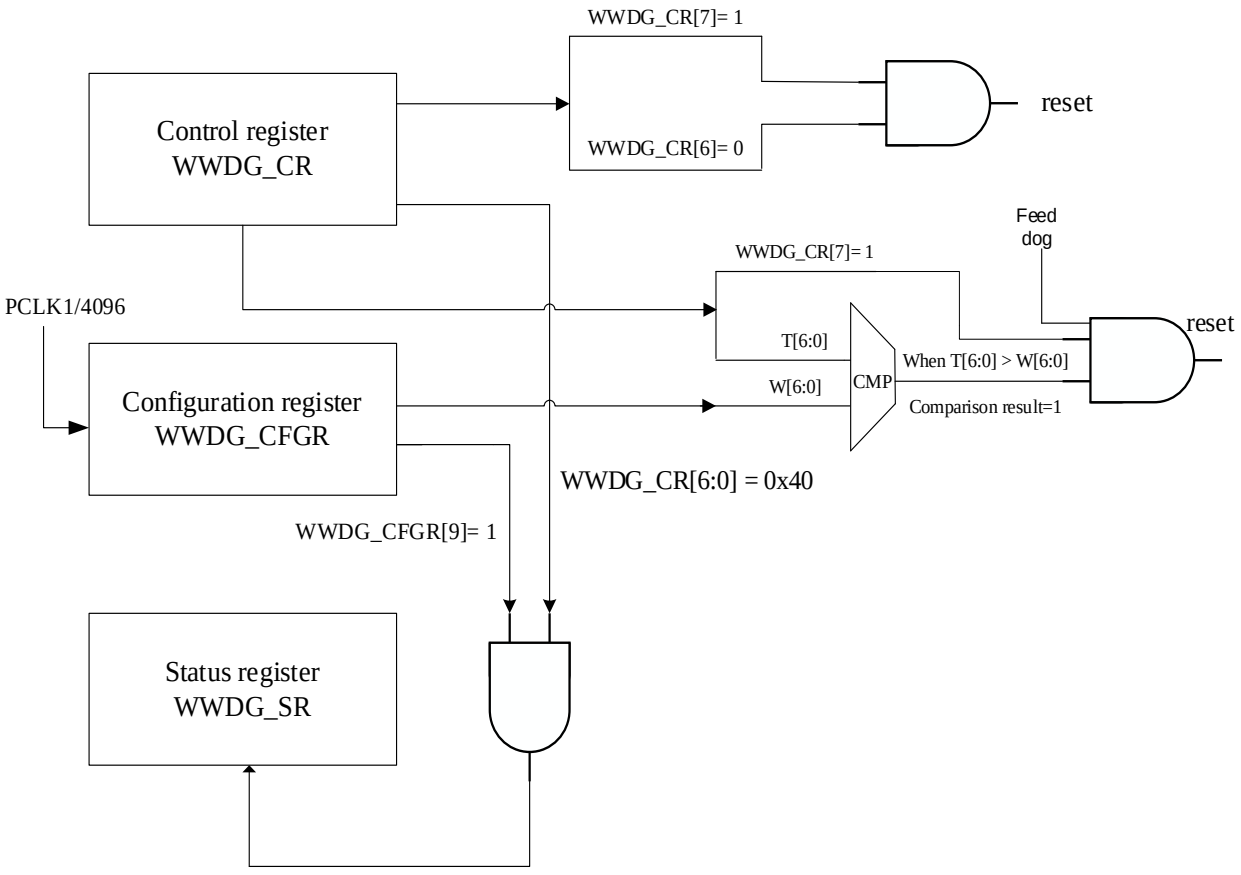


Figure 13-1 WWDG structure block diagram

13.3 Main characteristics

- Programmable free run 7-bit ramp down counter (0xFF and 0xC0);
- Reset during feed watchdog beyond the window

Upper limit: When the counter value is greater than the set value of the configuration register (WWDG_CFGR), produce a reset;

Lower limit: When the counter value is lower than 0x40, produce a reset;

- Interrupt:

When enabling the window watchdog and interrupt, and the ramp down counter reduces to 0x40, produce the earlier wake-up interrupt (EWI). During the interrupt processing function, reload the counter value to WWDG_CR to feed the watchdog and prevent reset.

13.4 Functional description

- When the 7th bit WDGA of the control register (WWDG_CR) is set “1”, enable the window watchdog. When the counter value rolls from the lower fixed value 0x40 to 0x3F (6th bit of WWDG_CR is cleared), produce a reset. Or when the counter value is greater than the configuration register (WWDG_CFGR) set value, feed the watchdog and produce a reset.
- Both the window watchdog and independent watchdog feed the watchdog when the application normally runs to prevent MCU reset. The difference: To feed the window watchdog, the counter value must be lower than the window register value. Only under such circumstance, the watchdog feed may reset. And the value written into the control register (WWDG_CR) must be within 0xFF and 0xC0.
- After system reset, the window watchdog won’t run automatically. Set the control register (WWDG_CR) WDGA bit and enable the window watchdog, because WDGA bit can only be reset and cleared by the hardware. Once the software is set, the window watchdog keeps opening. When the window watchdog opens, configure the 6th bit of the control register (WWDG_CR) as “1” to prevent an immediate reset. WWDG_CR[6:0] represents the count value before the watchdog reset. Because the prescaler value is configurable, the latency before reset, which is related to the configuration of prescaler value, changes between the min value and max value.
- Use early wake-up interrupt (EWI) to prevent reset and reload the counter. Configure EWI bit of the register (WWDG_CFG) to enable the interrupt. When the counter value reaches 0x40, produce the interrupt. Rewrite a value to the WWDG_CR in the interrupt function to feed the watchdog and prevent reset. In the status register (WWDG_SR), write “0” to clear the interrupt.

13.5 Window watchdog timeout

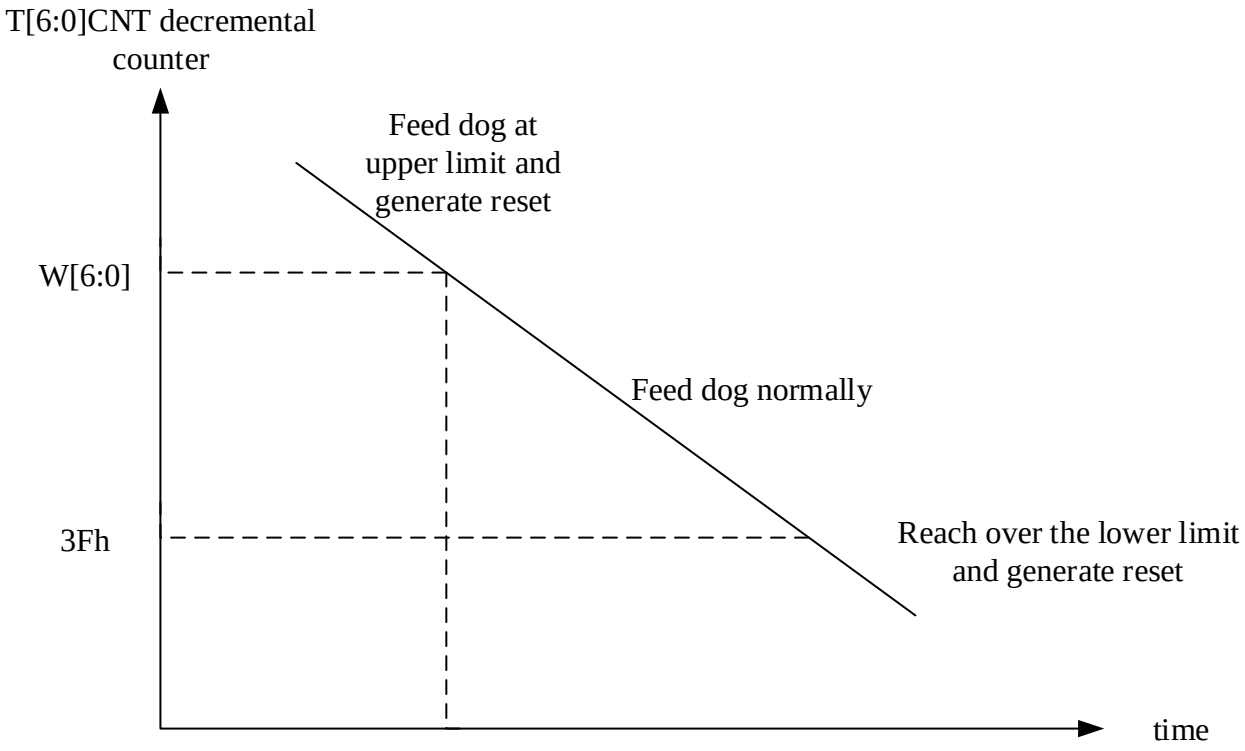


Figure 13-2 WWDG timeout coordinate

Table 13-1 Timeout

Time base (WDGTB)	Min time	Max time
0	56.5us	3.64ms
1	113.5us	7.28ms
2	227.5us	14.56ms
3	455us	29.12ms

Timeout calculation formula:

$$T = T_{pclk1} * 4096 * 2^{WDGTB} * (T[5:0]+1)$$

WDGTB: Divider time base

T_{pclk1}: APB1 clock interval

13.6 Register description

Table 13-2 WWDG register overview

Offset	Acronym	Register Name	Reset
0x00	WWDG_CR	Control register	0x0000007F
0x04	WWDG_CFGR	Configuration register	0x0000007F

Offset	Acronym	Register Name	Reset
0x08	WWDG_SR	Status register	0x0000000

13.6.1 Control register (WWDG_CR)

Offset address:0x00

Reset value:0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								WDGA	T[6:0]						
								rw	rw						

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value.
7	WDGA	Activation bit This bit is set 1 by software, and can only be cleared 0 after hardware reset. In case of WDGA = 1, the watchdog is started. 0: Disable watchdog 1: Enable watchdog
6:0	T[6:0]	7 bit count down counter (MSB to LSB) These bits are used to store the watchdog counter value. Reduce by 1 for every (4096x2^{WDGTB}) PCLK1 cycle. When the counter value turns from 40h to 3Fh (T[6] turns 0), produce the watchdog reset.

13.6.2 Configuration register (WWDG_CFGR)

Offset address:0x04

Reset value:0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						EWI	WDGTB		W[6:0]						
						rw	rw		rw						

Bit	Field	Description
31:10	Reserved	Reserved, must be kept at reset value.
9	EWI	Early wake-up interrupt 1: When the counter value reaches 40h, produce the interrupt. 0: Watchdog interrupt disable This bit can only be cleared by the hardware after reset.
8: 7	WDGTB	Timer base The prescaler time base may be changed as follows: 00: CK timer clock (PCLK1 divided by 4096) divided by 1 01: CK timer clock (PCLK1 divided by 4096) divided by 2 10: CK timer clock (PCLK1 divided by 4096) divided by 4 11: CK timer clock (PCLK1 divided by 4096) divided by 8
6: 0	W[6:0]	7 bit window value Upper window value of the window watchdog.

13.6.3 Status register (WWDG_SR)

Offset address:0x08

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															EWIF
															w0c

Bit	Field	Description
31: 1	Reserved	Reserved, must be kept at reset value.
0	EWIF	EWIF: Early wake-up interrupt flag When the counter value reaches 40h, this bit is set 1 by hardware. It must be cleared by writing 0 by software. To write 1 is invalid.

14 IWDG Independent watchdog

14.1 Overview

The independent watchdog is designed to detect and resolve malfunctions due to software failure. Its principle can be briefly described as follows: a system reset signal is generated when the independent watchdog (IWDG) counter decrements to a given value, thus triggering a system reset and improving the overall safety level of the system.

The independent watchdog is best suited to applications which require the watchdog to run as a totally independent process outside the main application, but have lower timing accuracy constraints.

The independent watchdog is clocked by the internal low-speed clock (LSI) and thus stays active even if the main clock fails.

14.2 Functional block diagram

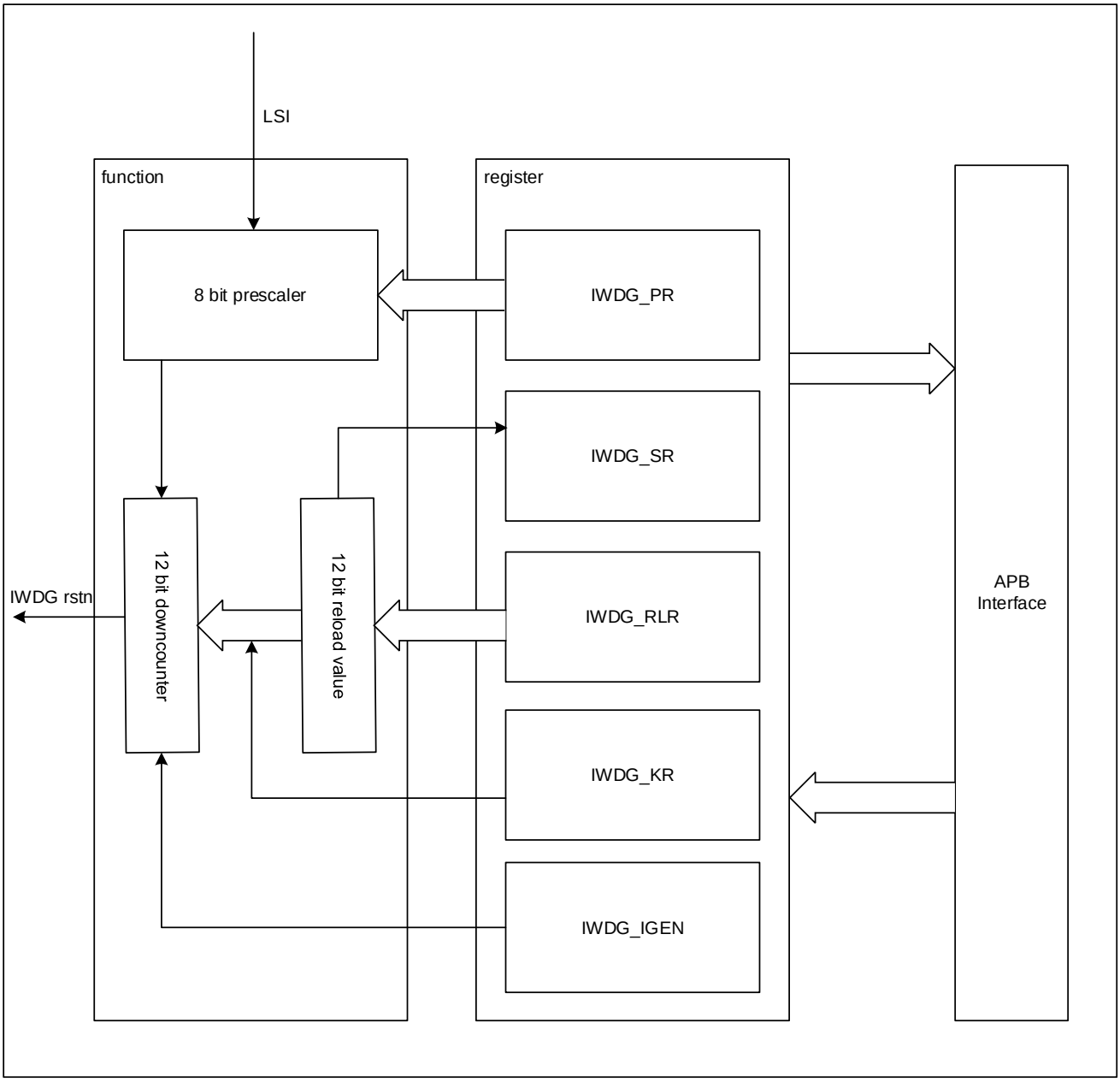


Figure 14-1 Functional block diagram

14.3 Main characteristics

- The software watchdog is the default mode in the chip.
- The hardware watchdog can be enabled by programming the WDG_SW bit of the reset option byte register in the Flash. If enabled, the hardware watchdog operates automatically after it is powered on by a system reset and the internal counter starts decrementing.
- The LSI keeps working in Stop and Standby modes.

- There is a 12-bit free-running downcounter inside the watchdog. A system reset or an interrupt signal is generated when the downcounter value reaches 0x0000.

14.4 Functional description

- The independent watchdog is started by writing the value 0XCCCC in the Key register (IWDG_KR). Meanwhile, the counter starts counting down from the reset value of 0Xfff. Either a system reset signal is generated when the counter reaches 0x0000, or an interrupt is generated when the counter reaches the value of IWDG_IGEN. The IRQ_SEL configuration decides which result will occur.
- Whenever 0XAAAA is written to IWDG_KR, the value in the reload register (IWDG_RLR) will be reloaded into the counter (commonly known as “feeding the dog”) to avoid the generation of a reset signal or interrupt signal.
- A reset signal or interrupt signal is generated if the compare/output program is abnormal and the dog can not be fed properly. Then a system reset or system interrupt is generated.
- Access to the IWDG_PR, IWDG_RLR, and IWDG_IGEN registers is protected. These registers can be modified only by writing he code 0x5555 in the Key register (IWDG_KR). A write access to this register with a different value will break the sequence and register access will still be protected. It is the same case of the reload operation.
- Since the clock of the independent watchdog is provided by LSI, the watchdog can work in the low-power modes such as the Stop and Standby modes.
- The independent watchdog counts normally in the low-power modes. The system will exit the Standby mode if the watchdog is reset.
- In low-power mode, the soft watchdog can be disabled by configuring the RCC register to select whether to turn off the LSI clock upon entering Stop mode.

14.4.1 Flow block diagram

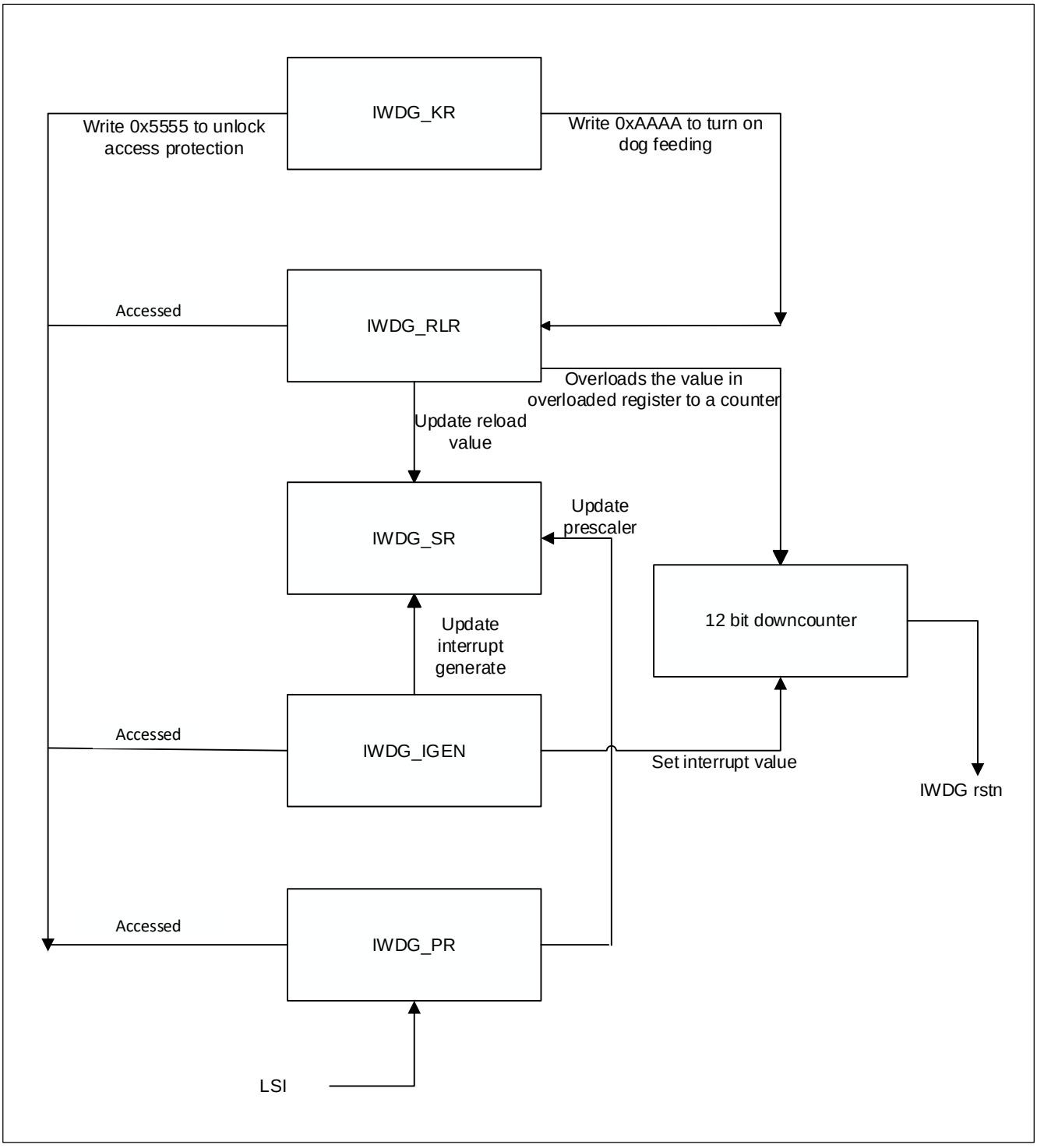


Figure 14-2 Flow block diagram

14.4.2 Independent watchdog timeout

Table 14-1 IWDG timeout (40kHz input clock (LSI))

Prescale Coefficient	PR[2:0] Bits	Min time (ms) RL[11:0]=0x000	Max time (ms)
/4	0	0.1	409.6
/8	1	0.2	819.2
/16	2	0.4	1638.4
/32	3	0.8	3276.8
/64	4	1.6	6553.6
/128	5	3.2	13107.2
/256	(6 or 7)	6.4	26214.4

Timeout (overflow) is calculated as below:

Tout=((4×2^PR)×RLR)/40

where: the unit of Tout is millisecond.

The clock frequency LSI = 40K; a watchdog clock period is the minimum timeout.

Maximum timeout = (maximum value of IWDG_RLR register) x watchdog clock period

14.4.3 Interrupt

- If the CR.IRQ_SEL is set, the IWDG generates an interrupt when the counter overflows.

14.5 Register

14.5.1 Overview of registers

Table 14-2 Overview of IWDG registers

Offset	Acronym	Register Name	Reset
0x00	IWDG_KR	Key register	0x00000000
0x04	IWDG_PR	Prescaler register	0x00000000
0x08	IWDG_RLR	Reload register	0x00000FFF
0x0C	IWDG_SR	Status register	0x00000000
0x10	IWDG_CR	Control register	0x00000000
0x14	IWDG_IGEN	Interrupt generate register	0x00000FFF
0x18	IWDG_CNT	Counter register	0x00000001

14.5.2 WDG_KR Key Register

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY															
w															

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value
15:0	KEY	Key value (write-only register) These bits must be written by software at regular intervals with 0XAAAA to feed the dog, otherwise a reset signal is generated to reset the system when the counter decrements to 0x0000. Writing 0x5555 by software may disable the protection and enable access to other configuration registers (IWDG_PR, IWDG_RLR, IWDG_CR (bit0), IWDG_IGEN). Writing 0xCCCC by software starts the watchdog.

14.5.3 IWDG_PR Prescaler Register

Address offset: 0x04

Reset value: 0x0000 0000 (reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.													PR		
													rw		

Bit	Field	Description
31:3	Reserved	always read as 0
2:0	PR	Prescaler divider These bits are set to select the prescaler dividers of the LSI clock. To modify prescaler dividers, the protection should be disabled at first (write 0x5555 to IWDG_KE). After the prescaler dividers are updated, The PUV bit in the register becomes 0. At this time, the value read from this register is valid. 000: prescaler divider = 4 100: prescaler divider = 64 001: prescaler divider = 8 101: prescaler divider = 128 010: prescaler divider = 16 110: prescaler divider = 256 011: prescaler divider = 32 111: prescaler divider = 256

14.5.4 IWDG_RLR Reload Register

Address offset: 0x08

Reset value: 0x0000 0FFF (reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				RL											
				rw											

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11:0	RL	Watchdog counter reload value These bits serve to define the reload value of the watchdog counter. This value will update to the counter each time the dog is fed (write 0xAAAA to IWDG_KR register). Subsequently, the watchdog counter counts down from this value. To modify the reload values, the protection should be disabled at first (write 0x5555 to IWDG_KR). After the reload values are updated, the RUV bit in the register becomes 0. At this time, the value read from this register is valid. The watchdog timeout period is a function of this reload value and the prescaler.

14.5.5 IWDG_SR Status Register

Address offset: 0x0C

Reset value: 0x0000 0000 (not reset by Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												UPDATE	IVU	RVU	PVU
												r	r	r	r

Bit	Field	Description
31:4	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
3	UPDATE	Watchdog reload value update flag The UPDATE is set when 0XAAAA is written into the IWDG_KR register. It is cleared automatically when the watchdog counter is updated and the reload value is written into the counter.
2	IVU	Watchdog Interrupt Generate value update This bit is set by hardware to indicate that an update of the interrupt generate value is ongoing. It is cleared by hardware when the interrupt generate value update operation is completed in the VDD voltage domain (takes up to 5 oscillator cycles at 40KHz). The interrupt generate value can be updated only when IVU bit is cleared.
1	RVU	Watchdog counter reload value update This bit is set when an update of the reload value is ongoing. It is cleared when the reload value update operation is completed (takes up to 5 oscillator cycles at 40KHz). The reload value can be updated only when RVU bit is cleared.
0	PVU	Watchdog prescaler value update This bit is set when an update of the prescaler value is ongoing. It is cleared when the prescaler value update operation is completed (takes up to 5 oscillator cycles at 40KHz). The prescaler value can be updated only when PVU bit is cleared.

Note: If several reload values, prescaler values or interrupt generate values are used by application, it is necessary to disable the register protection (write 0x5555 to IWDG_KR) at first. Then configure the IWDG_PR, IWDG_RLR and IWDG_IGEN registers. Wait until the corresponding bits (PVU, RVU, IVU) in the status register are cleared, which indicates that the reload value, prescaler value, and interrupt generate values are configured. Next, feed the dog, or wait for a reset or interrupt signal which is generated from the autodecrement of the counter.

14.5.6 IWDG_CR Control Register

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														IRQ_CLR	IRQ_SEL
														rw	rw

Bit	Field	Description
31:2	Reserved	Reserved, must be kept at reset value
1	IRQ_CLR	IWDG interrupt clear 1: Interrupt cleared 0: No effect, interrupt flag still pending Note: It is not necessary to disable the KEY protection before writing to this bit.
0	IRQ_SEL	IWDG overflow operation select 1: interrupt generation enabled after overflow 0: reset generation enabled after overflow

14.5.7 IWDG_IGEN Interrupt Generate Register

Address offset: 0x14

Reset value: 0x0000 0FFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				IGEN											
				rw											

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11:0	IGEN	IWDG Interrupt Generate value These bits serve to define the interrupt generate value of the watchdog. An interrupt is generated when the counter counts down until it reaches this value. The protection should be disabled before changing it. After the value update operation is completed, the value read from this register is valid only when the IVU bit in the IWDG_SR register is cleared.

14.5.8 WDG_CNT Counter Register

Address offset: 0x18

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.													IWDG_CNT		
													r		

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IWDG_CNT								IWDG_PS							
r								r							

Bit	Field	Description
31:19	Reserved	Reserved, must be kept at reset value
18:8	IWDG_CNT	IWDG counter value
7: 0	IWDG_PS	Value of the prescaler counter of the IWDG clock

15 Universal Asynchronous Receiver Transmitter (UART)

15.1 Overview

The universal asynchronous receiver transmitter (UART) can flexibly conduct full duplex data exchange. By the fraction baud rate generator, UART can select a wide range of baud rates. Asynchronous single-way communication and semi-duplex single-line communication, IrDA infrared function can also be supported. Besides, UART also supports communication among multiple processors.

15.2 Function block diagram

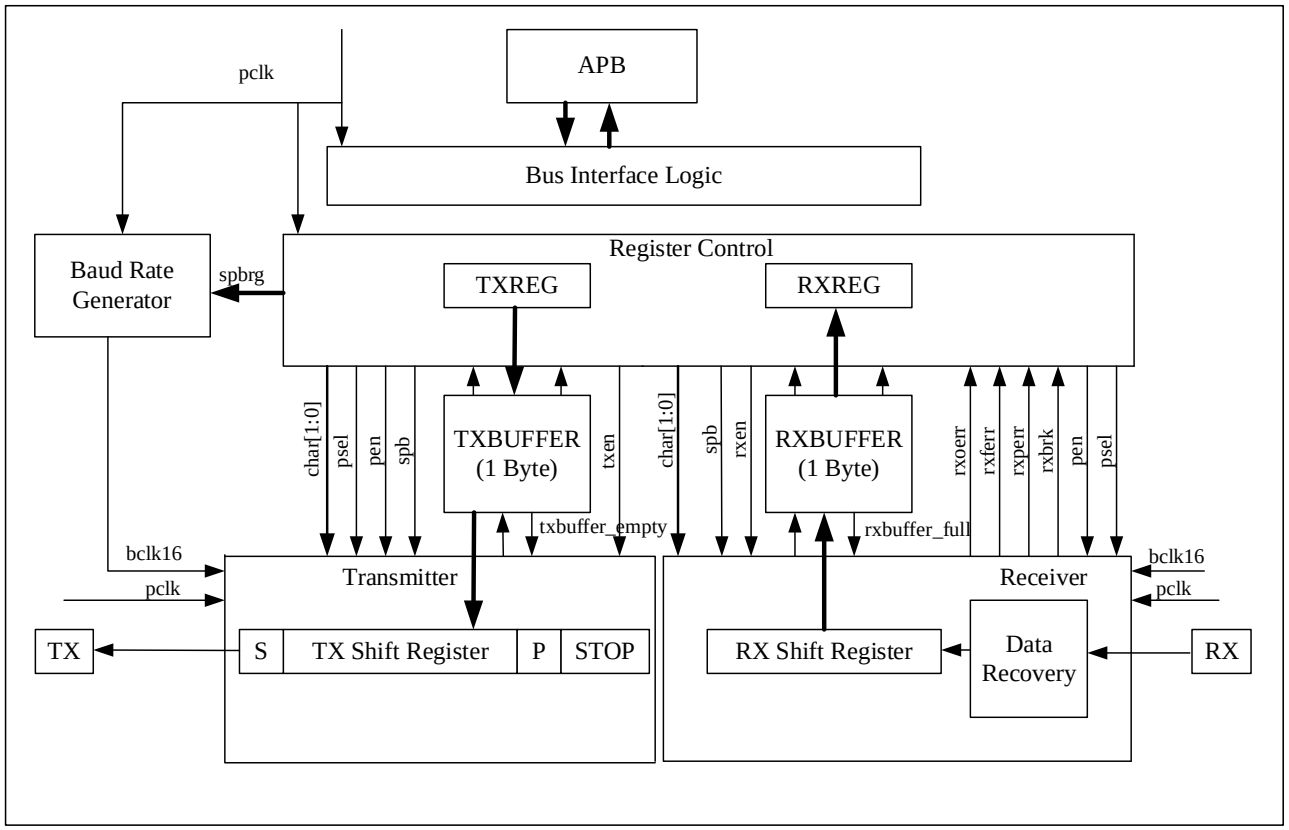


Figure 15-1 UART function block diagram

15.3 Main characteristics

- Support RS-232S protocol under the asynchronous mode, compliance with the asynchronous serial communication 16550 chip standard
- Full duplex asynchronous operation
- Fraction baud rate generator

- Programmable baud rate, for use by the transmitter and receiver, min division factor 1
- Independent transmission and receipt buffer register
- Built in 1 byte transmission and 1 byte receipt buffer
- The transmission and receipt data low bit begins from the previous start bit, followed by the data bit. The data length can be 5~8 bit, followed by the stop bit. Select whether to have the parity check bit, after the data bit and before the stop bit.
- The 9th bit may serve as the synchronous frame configuration
- Support hardware odd/even check generation and detection
- Break frame generation and detection
- Idle frame generation and detection
- Support transmission and receipt of Break frame under the LIN protocol.
- Support signal receipt and transmission interchange, receipt and transmission inversion
- Support baud rate adaptive function
- Support IrDA SIR ENDEC infrared function

15.4 Function overview

Under the full duplex condition, at least assign two pins to UART: receipt data transmission (RX) and transmission data output (TX).

RX: External serial data is transmitted to UART receiver via the pin. For the transmission noise, differentiate and remove it from the data by the sampling technique and obtain the primary data.

TX: The internal serial data inside the UART transmitter is transmitted via the pin. When the transmitter is enabled and there is no data transmission, TX pin is at high level. When the transmitter is enabled, TX pin is restored to its I/O port configuration status.

The idle status is the initial status of the bus before the transmission or receipt begins.

The start bit is one bit, expressed by 0.

One data (5,6,7 or 8 bit), and the transmission and receipt sequence is from the lowest bit to the highest bit.

The stop bit uses ‘1’ to express the completion of one frame. The bit can be configured as 0.5, 1, 1.5 and 2.

Fraction baud rate generator -Configure 16 bit integer and 4 bit decimal, thus improving the baud rate accuracy.

Enable UART_GCR register SWAP bit, and exchange the receipt and transmission terminal signal.

Enable UART_GCR register RXTOG/ TXTOG bit, and invert the receipt/transmission terminal signal (start bit and stop bit are also inverted prior to input/output).

15.4.1 UART characteristics description

Configure UART_CCR.CHAR bit, and the character bit length can be programmed as 5~8 bit. The transmitter will lower TX pin at the transmission start bit, and raise TX pin in the transmission stop bit.

It includes the stop bit, a full data frame composed of '1', defined as an idle frame. The start bit of the next data frame follows the idle frame.

It includes the stop bit, a full data frame composed of '0', defined as a Break frame. After the Break frame, the transmission terminal will again send one '1' stop bit so that the start bit of the next frame can be identified (the produced falling edge is detected by the receipt terminal).

After control by the enable position bit of the transmitter and receiver, the clock produced by the baud rate generator is used for transmission or receipt.

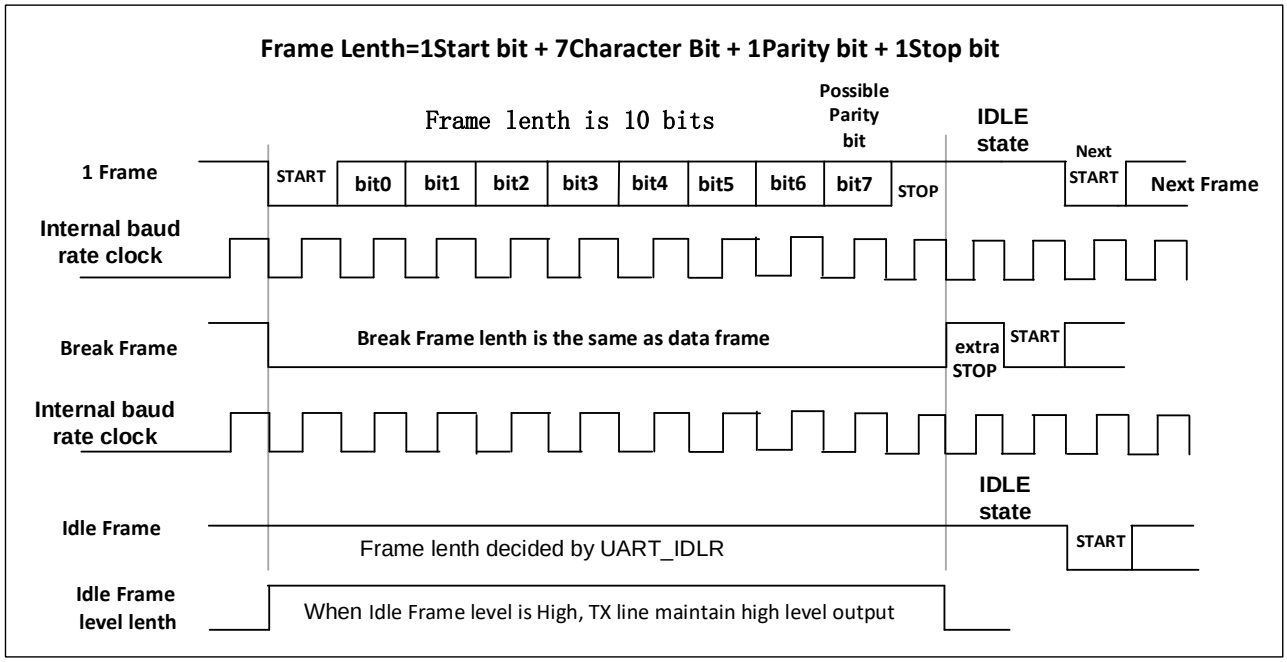


Figure 15-2 UART sequence

15.4.2 Fraction baud rate generator

Set BRR and FRA register, set the corresponding baud rate. Refer to the formula below:

$$f_{baudrate} = \frac{f_{PCLK}}{16 \times UARTDIV}$$

$$UARTDIV = BRR + \frac{FRA}{16}$$

$$f_{baudrate} = \frac{f_{PCLK}}{16 \times BRR + FRA}$$

BRR register's min value is 1. FRA's value is configured within the range of 0~15.

15.4.3 Sampling

UART has a test circuit to test the beginning of one frame of data, and RX pin is sampled. UART uses the 16 times baud rate clock to sample RX pin data. Take the sampling value of the 7th, 8th and 9th falling edge.

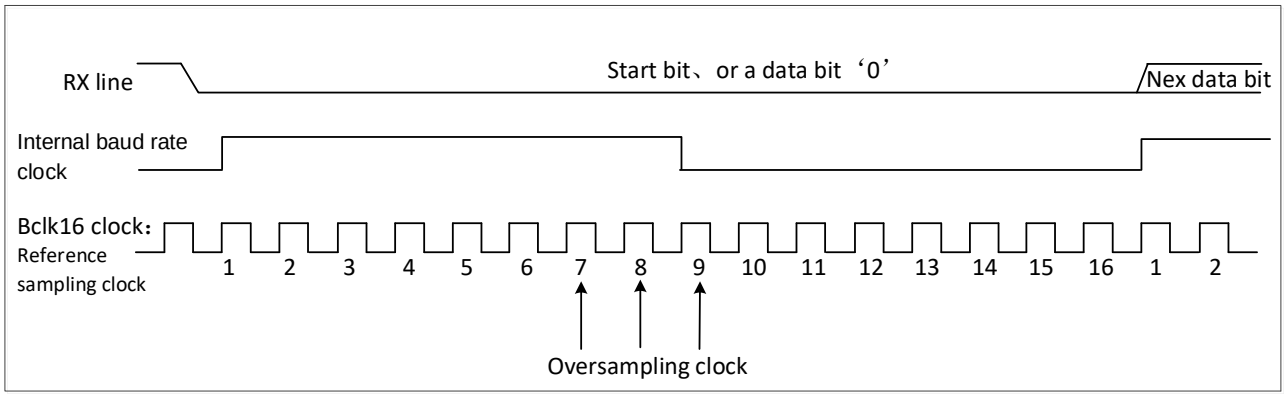


Figure 15-3 RX pin sampling scheme

15.4.4 Tolerance

The factor such as the clock deviation causes the transmission and receipt baud rate error. In order to improve the receipt terminal's tolerance for the change of baud rate, in STOP status, when three data samplings finish, the receipt terminal begins to prepare receipt of the next frame data. When the receipt and transmission baud rate deviation is within ±3%, it can still guarantee normal data transmission.

15.4.5 Parity check control

The parity check (generate a parity bit during transmission, conduct parity check during receipt) can be activated by setting UART_CCR.PEN bit as '1'. In case of the parity check error, the invalid data will continued to be transmitted from the shift register to UART_RDR register. Set UART_ICR RXPERR bit as '1' to clear the UART_ISR.RXPERR_INTF flag.

When the smart card auto response (UART_SCR.SCAEN='1') enables, detect the parity check error in STOP status. It produces a 1 bit low level signal (RXNACK), indicating frame error.

Even check: 1 number of the check bit plus the data is even.

Odd check: 1 number of the check bit plus the data is odd.

The parity check bit output or receipt judgement is auto completed by the hardware, without software configuration.

For example: Data= 8'b1000_0110, there are 3 '1'.

Even check: (UART_CCR PSEL = 1), the check bit is '1'.

Odd check: (UART_CCR PSEL = 0)，the check bit is '0'.

When the parity check enables, the parity check bit transmission will follow the MSB bit. If the parity check fails, the hardware auto sets the UART_ISR.RXPERR_INTF flag bit. In case of interrupt generation, configure interrupt enable register in advance.

When the smart auto response SCAEN enables, detect the parity check error in STOP. It produces a 1 bit low level signal (RXNACK). The smart card output end will lower 1 baud clock period, indicating frame error.

Note: When 9bit function is enabled, the parity check is invalid.

15.4.6 Transmitter

UART_CCR.CHAR bit configuration determines the bit of transmission data (5~8 bit). Configure the UART_CCR.TXEN bit as '1', enable the transmitter, and the data will be serially transmitted to the TX pin.

15.4.6.1 Character transmission

During UART transmission, the data is written from UART_TDR register. After one byte buffer, ultimately transmit the shift register, and export in the TX pin from the min byte to max byte.

Transmission data sequence: 1 bit start bit, character, 1 bit parity check bit (yes or no), stop bit.

The start bit of a low level precedes the character transmission. After the character transmission, transmit the high level stop bit.

Configure UART_CCR.CHAR[1:0] bit to configure the character length; configure UART_CCR.SPB[1:0] bit to configure the stop bit number.

Before the current data transmission finishes, don't clear UART_GCR.TXEN bit. Or else, the baud rate generator will stop the clock generation. The clock will miss a part of data.

15.4.6.2 Configuration steps

1. Configure UART_GCR.UARTEN bit as '1' and enable UART.
2. Configure UART_CCR.CHAR bit and define the character length.
3. Configure UART_CCR.SPB[1:0] bit, and set the stop bit number.
4. Configure UART_BRR and UART_FRA and produce the required baud rate clock.

5. Configure UART_GCR .TXEN bit as ‘1’, and enable the transmitter.

6. Write the transmission data into UART_TDR register (auto clear ‘0’UART_ISR.TX_INTF bit by hardware). The buffer is one byte. It only writes one piece of data at a time. Write the transmission data to the UART_TDR register only when the buffer is empty.

15.4.6.3 Communication process

Write ‘1’ and clear ‘0’ by hardware for the UART_ISR.TX_INTF. When this bit is set ‘1’ by hardware, indicating:

1. Data transmission begins. Data begins to transmit from the UART_TDR register to the shift register

2. Write the next transmission data to the UART_TDR register (without affecting the current data transmission)

If UART transmitter is in the transmission status, after the software writes the UART_TDR register, the hardware will auto write the data into the shift register for serial output, and set UART_ISR.TX_INTF flag and UART_CSR.TXEPT flag. If UART_IER.TX_IEN bit is set as ‘1’, it produces the interrupt.

Write the next transmission data into UART_TDR register via TX_INTF interrupt for continuous transmission.

When the stop bit transmission finishes, and there are no follow up data to be transmitted, set UART_ISR.TXC_INTF flag and UART_CSR.TXC flag. It means all transmissions have finished; if UART_IER.TXC_IEN bit is set as ‘1’, and the interrupt will also produce.

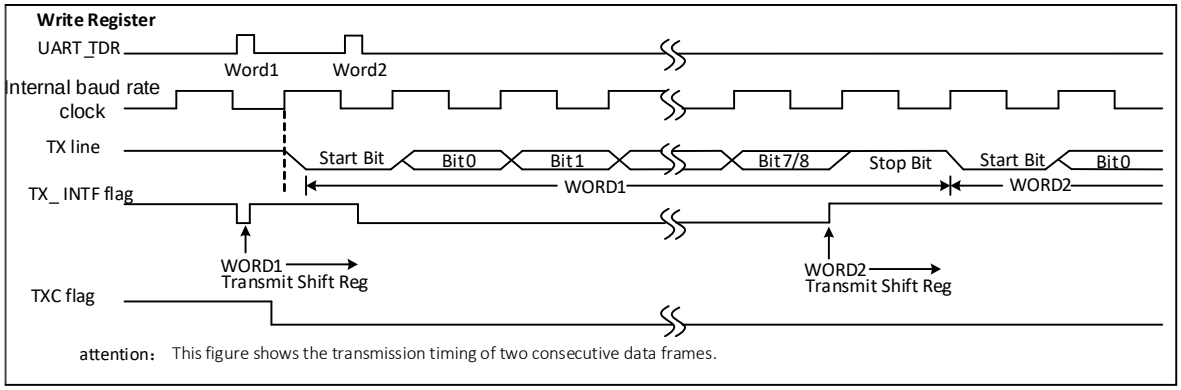


Figure 15-4 Status bit change during transmission

15.4.6.4 Break frame

The Break frame transmission is only allowed when UART_CSR.TXC is 1. Configure UART_CCR.BRK bit as ‘1’ and transmit the Break frame. After the Break frame transmission, the hardware auto clears BRK bit to ‘0’, and transmits a bit of high level for testing the start bit of the next frame of data.

The total length of Break frame is composed of ‘start bit + data bit + (check bit) + stop bit’ and the length varies with the setting of data bit, check bit and stop bit. When the Break frame transmission finishes, set the UART_ISR.TXBRK_INTF flag.

15.4.6.5 Idle frame

The idle frame length is determined by the UART_IDLR register value. When UART_IDLR is not 0, after enabling TXEN, immediately transmit one idle frame before the transmission of valid data.

15.4.7 Receiver

15.4.7.1 Character receipt

For UART receiver and RX serial transmission data, the receipt sequence goes from the min valid bit of the data to the max valid bit. Build in a receipt shift register and a receipt data register for double buffer structure, thereby achieving continuous receipt of data.

Configuration steps:

1. Set ‘1’ UARTEN bit of the UART_GCR register to enable the UART.
2. Configure UART_CCR.CHAR bit and set the character length.
3. Configure UART_CCR.SPB bit and set the stop bit number.
4. Configure UART_BRR and UART_FRA, and produce the required baud rate clock.
5. Set UART_GCR.RXEN bit as 1. Enable the receiver and wait for the detection start bit.
6. Upon receipt of a complete frame, the receipt shift register data is transmitted to the RDR register. The hardware sets the UART_ISR.RX_INTF flag and UART_CSR.RXAVL flag, indicating that the data is received and may be read. If the hardware detects the data receipt error, the corresponding error flag is set.
7. Configure UART_IER.RX_IEN bit as ‘1’ and enable receipt interrupt.
8. When detecting the frame error during receipt, or the overflow error or parity check error, the hardware set the corresponding error flag.
9. The software reads UART_RDR register or configure UART_ICR.RX_ICLR bit as ‘0’ to clear ‘0’ UART_ISR.RX_INTF bit. In order to prevent the overflow error, RX_INTF bit must be cleared before the next character receipt finishes.

Upon receipt of data, don’t clear RXEN bit by the software. In case of clearing RXEN bit during receipt, RX pin data won’t be received.

15.4.7.2 Break frame

When UART receiver identifies a Break frame, set RXBRK_INTF interrupt, and the data is not written into the receipt buffer. The Break frame is composed of ‘start bit +data bit +(check bit) +stop bit’. The length varies with the different setting of the data bit, check bit and stop bit.

15.4.7.3 Idle frame

The idle frame length is set by the UART_IDLR register (default value 0x0C), set UART_IDLR not as 0. In case of receipt of an idle frame, UART will assert RXIDLE_INTF interrupt, and the data won’t be written into the receipt buffer.

15.4.7.4 Overflow error

When UART_RDR receives a character before read, produce the overflow error. In case of overflow error:

1. The hardware will set the UART_ISR.RXOERR_INTF flag.
2. The unread UART_RDR content won’t miss. Read UART_RDR register will get the previous data.
3. The later received character will miss.

If UART_IER.RXOERR_IEN bit is configured as ‘1’, it produces the interrupt.

15.4.7.5 Frame error

When the stop bit doesn’t receiver and identify the detected frame error during the expected time. When the frame error is detected:

1. UART_ISR.RXFERR_INTF bit is set by hardware.
2. UART_RDR register won’t refresh.

If UART_IER.RXFERR_IEN bit is set as ‘1’, produce the interrupt.

15.4.8 Auto baud rate detection

UART can auto detect and set UART_BRR register according to the received signal. The auto baud rate configuration:

1. Configure UART_ABRCR register Former_edge and Latter_edge, and select the edge (rising or falling) before and after the auto baud rate.
2. Configure UART_ABRCR.Abr_bitcnt and select the length of the auto baud rate detection bit.

3. Configure UART_ABRCR.Abren as ‘1’ and enable auto baud rate detection function.

4. Configure UART_IER.ABREND_IEN bit as ‘1’, and produce the interrupt after auto baud rate finishes.

5. Configure UART_IER.ABRERR_IEN bit as ‘1’, and produce the error interrupt after the auto baud rate detection.

- Example 1: Set Former_edge=1(rising edge), Latter_edge=0 (falling), Abr_bitcnt=2 (4 bit length).
- In case of receipt of 0x0F, conduct baud rate detection of the lowest four bit 0xF of RX online data (start bit ->MSB bit: 0_1111_0000), and then reset UART_BRR register. Set UART_ISR.ABREND_INTF flag.
- Example 2: Set Former_edge=0 (falling edge), Latter_edge=1 (rising edge), Abr_bitcnt=2 (4 bit length).
- In case of receipt of 0x08, conduct baud rate detection of the ‘start bit +lowest three bit data’ of RX online data (start bit ->MSB bit: 0_0001_0000), and then reset UART_BRR register. Set UART_ISR.ABREND_INTF flag.

Note: The configuration of detection edge and data length should match with the receipt data. For example, in case of Former_edge=1, and the lowest bit of receipt data is not ‘1’, the hardware will assert the UART_ISR.ABRERR_INTF flag.

Besides, when the auto baud rate detection begins, the difference of baud rate between the software initial configuration and the transmission side is too large, could leading to failure of auto baud rate function.

15.4.9 9 bit data communication

Conduct relevant configuration of UART_CCR register, and achieve 9 bit data communication. Specific configuration:

When B8EN is ‘1’, enable 9 bit data transmission and receipt.

When B8EN is enabled, the parity check enable bit PEN is invalid.

During data transmission, first set B8TXD; B8TXD, as transmission data MSB, is transmitted together with UART_TDR value. When B8TXD and B8POL are the same, it indicates that the data serves as the address frame or sync frame. When B8TOG is ‘1’, B8TXD auto rolls when the current frame transmission ends. It’s not required to configure B8TXD for the next data frame.

During data receipt, the 9th bit of receipt data can be read from B8RXD. When B8RXD and B8POL are same, the hardware asserts the UART_ISR.RXB8_INT flag.

15.4.10 Multiprocessor communication

Communication among multiple processors can be achieved by connecting several UARTs into one network. As the main equipment UART, its TX transmission is connected to the RX input of other auxiliary device. TX logic of other auxiliary device is connected to the main device RX input.

The host seeks the target auxiliary unit by address, and other non-target auxiliary units should keep a lower power mode to prevent resource waste.

When the auxiliary device is not the target auxiliary unit, configure the mute mode to reduce power consumption. In the mute mode:

1. All receipt status flag bits won't be asserted by hardware.
2. All receipt interrupts won't produce.

- Both software and hardware can rewrite UART_CCR.RWU bit. Set '1' UART_CCR.RWU bit. It means the receiver enters the mute mode. When setting the address flag wake-up, if the receipt buffer is not empty (receipt address doesn't match the auxiliary unit), the software can't configure RWU bit as '0' to exit the mute mode.
- Configure UART_CCR.WAKE bit, and select the test mode to exit the mute mode:

1. WAKE=0: Conduct idle bus detection.
2. WAKE=1: Conduct address flag detection.

15.4.10.1 Idle bus detection

In the mute mode, UART_CCR.WAKE=0, the slave device detects the idle frame. When the idle frame is detected on the RX line, the hardware auto clears '0'UART_CCR.RWU bit, and exits the mute mode. Without normal frame input, the hardware won't assert the UART_ISR.RX_INTF flag.

15.4.10.2 Address mark detection

In the mute mode, UART_CCR.WAKE=1, the slave device conducts the address flag detection.

The highest significant bit of valid bit of receipt byte acts as the identify flag of address frame or data frame. Therefore, the highest significant bit B8POL means that this byte is the address frame; otherwise, it's the data frame.

The slave receiver takes the configuration value of UART_RXADDR register as the slave unit address, and makes comparison with the address frame. It should be noted that UART_RXMASK register has the address mask function. Only when its certain bit is 1, the corresponding address bit will participate in the comparison; in case of 0, it directly deems that this bit match succeeds.

If the address frame received by the slave unit fails to match after address mask, the hardware

asserts the UART_CCR.RWU bit and enters the mute mode. Because the slave unit enters the mute mode, the received address frame won't lead to the relevant hardware response (such as trigger flag bit, interrupt request)

If the address frame received by the slave unit matches successfully after address mask, the hardware clears the UART_CCR.RWU bit and exits the mute mode. Because the slave unit exits the mute mode, the address will trigger the UART_ISR.RX_INTF flag, and then the receiver works normally.

15.4.10.3 Configuration steps

- The multiprocessor communication is achieved based on enabling the 9 bit data communication. Steps below are required:
- For host transmission, refer to the 1.6.6.2 configuration steps (transmitter). Precautions below:

1. Before enabling TXEN, configure B8EN bit as '1' and configure B8TXD and B8POL bit. When configuring B8TXD=B8POL=1, the slave unit detects the high level of the highest significant bit and deems it as the address frame. When configuring B8TXD=B8POL=0, the slave unit detects the low level of the highest significant bit and deems it as the address frame. When configuring B8TOG bit as '1', B8TXD configuration in the step 3 is not required.

2. After enabling TXEN, rewrite TDR data as the address value, transmitted to the TX line together with B8TXD as the address frame.

3. And then configure B8TXD as the reverse value of B8POL, and then write TDR, and then transmit the data frame.

4. According to the transmission data type (address frame or data frame), operate based on step 3. Rewrite B8TXD and TDR to achieve the continuous transmission of the host (address frame+data frame+...).

For the slave unit receipt, please refer to 1.6.7.1 character receipt (receiver). Precaution:

1. Before enabling TXEN, configure B8EN bit as '1' and configure B8POL bit (as the flag to judge whether the current character is address frame or data frame; compare with B8RXD bit, and deem it as the address when same; otherwise, it's data frame).

2. Before enabling TXEN, also configure WAKE bit to select the detection mode for the slave unit to exit the silence mode. When configuring WAKE=1, it's also required to configure the slave address (configure RXADDR and RXMASK register).

3. After enabling RXEN, in case of WAKE=0, write RWU=1 and the slave unit enters the mute mode. And then the slave receiver will detect the idle frame and auto exit the mute mode (the hardware will clear the RWU bit).

4. After enabling RXEN, in case of WAKE=0, configure RWU bit as '1' by software and the slave unit enters the mute mode. And then, the salve receiver will conduct the address flag detection. Besides,

the hardware will also assert RWU and enter the mute mode when the address match fails.

15.4.11 Single wire half duplex communication

Configure UART_SCR.HDSEL bit as ‘1’, and select the single wire half duplex mode. The chip’s internal logic will be connected with TX and RX. It should be noted that the UART_SCR.SCEN bit should be configured as ‘0’.

During single wire half duplex:

1. RX pin is suspended, and doesn’t participate in transmission. UART TX is directly connected to another UART TX.
2. During data transmission, TX is always occupied till the stop bit transmission finishes.
3. Without data transmission, TX is in the released status. Therefore, it’s a standard I/O port in the idle status or receipt status. Namely, TX corresponds to I/O, and it’s not driven by UART. It’s mandatory to configure the suspension input (or open drain output high)

Except for the single line pin configuration, the other configurations are consistent with normal transmission.

Before the data transmission, both UART UART_GCR.RXEN are enabled and in the wait receipt status. When communication is required, the two UARTS need to agree who will send data : RXEN of the sender is closed and TXEN is enabled. If both UARTs attempt to transmit data, the transmission will be conflict (hardware doesn’t block UART transmission; when the transmission enable bit TXEN opens, as long as UART_TDR is written, TX will transmit data).

15.4.12 Smart card

Configure UART_SCR.SCEN bit as ‘1’ and enter the smart card mode, and the hardware will be auto in the half duplex mode (no need to set the UART_SCR.HDSEL bit).

In this mode, UART conforms to ISO7816-3 standard, and steps below are required before enabling TXEN or RXEN:

1. 8 bit data bit plus check bit: UART_CCR register configures CHAR=11, PEN=1
2. Both transmission and receipt are recommended as 1.5 stop bits; UART_CCR register configures SPB1=1, SPB0=1.
3. If it’s required to enable NACK signal; UART_SCR register configures SCAEN=1.
4. Smart card protection time configuration: UART_SCR register configures SCFCNT.

Let’s give an example. TX line signal has difference in case of two scenarios of check error and no check error:

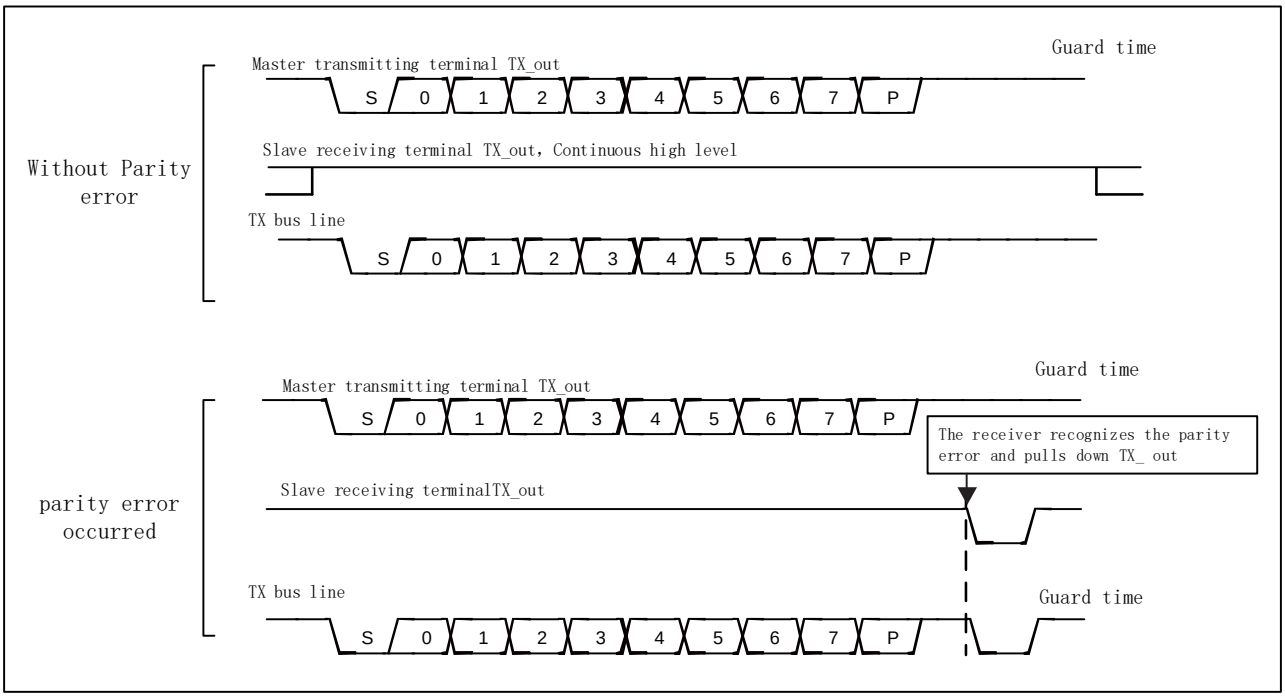


Figure 15-5 UART parity check block diagram

As the host, UART enables TXEN to identify the check error signal from the slave unit (1 bit low level: NACK signal). After the stop bit transmission, disable TXEN. Meanwhile, TX is pulled to the high level (TX should be configured as the open drain mode).

As the host, UART needs the TX single line to drive the two way line of the smart card end. TX and RX in essence use the same IO port.

Smart card communication protocol:

1. The slave unit data is transmitted to the shift register, and serially exported to the TX pin. And it will be at least 1/2 baud clock period compared with the transmission time under the normal mode. The time for data appearing on the TX pin is the next baud clock edge of the full time of transmission shift register. And the time of data appearing on TX pin in the smart card mode will be 1/2 baud clock slower in the normal mode.

2. During the data frame stop bit (setting as 1.5 preferred) receipt period, the slave unit receiver detects the parity check error during data receipt, and will lower one baud clock period of the data line (NACK signal). Because the data line is pulled down, it means the host transmits a frame error (stop bit will be pulled down as ‘0’). When the host receives the feedback, the software can use the feedback signal and resends the transmission data according to the protocol. It should be noted the NACK signal may only be produced by configuring the UART_SCR.SCAEN bit as ‘1’.

3. When smart card mode there is guard time that configure UART_SCR.SCFCNT[7:0] to extend the UART_CSR.TXC flag pending time; after the parity check, the corresponding protection time counter will enable increment. When the counter value ramps up to the value of the configured protection time register, TXC flag will be pending. Under the normal transmission mode(non smart card), the conditions for the hardware to assert TXC is that the transmission buffer is empty and there is no followup data to

write into the TDR.

4. The smart card mode won't affect that TXC flag is auto cleared by the hardware because of hardware writing TDR.

5. If the host receives NACK signal from the slave unit, the host receiver won't take NACK as the detection start bit.

6. After the slave unit detects the check error, NACK signal fed back to the host won't be deemed as the start bit.

Note:

a. Smart card mode doesn't have the definition for the Break frame. One full '0' data frame including the stop bit will be deemed as the data frame error.

b. Smart card mode doesn't have the definition for the Idle frame. During the changeover of TXEN, the smart card doesn't deem to receive the idle frame.

The figure below explains in detail UART's sampling logic for NACK signal. Enable UART receipt function block, and check data integrity and NACK signal (stop bit as 1.5 bit, where 1 NACK bit is in the later phase of stop bit).

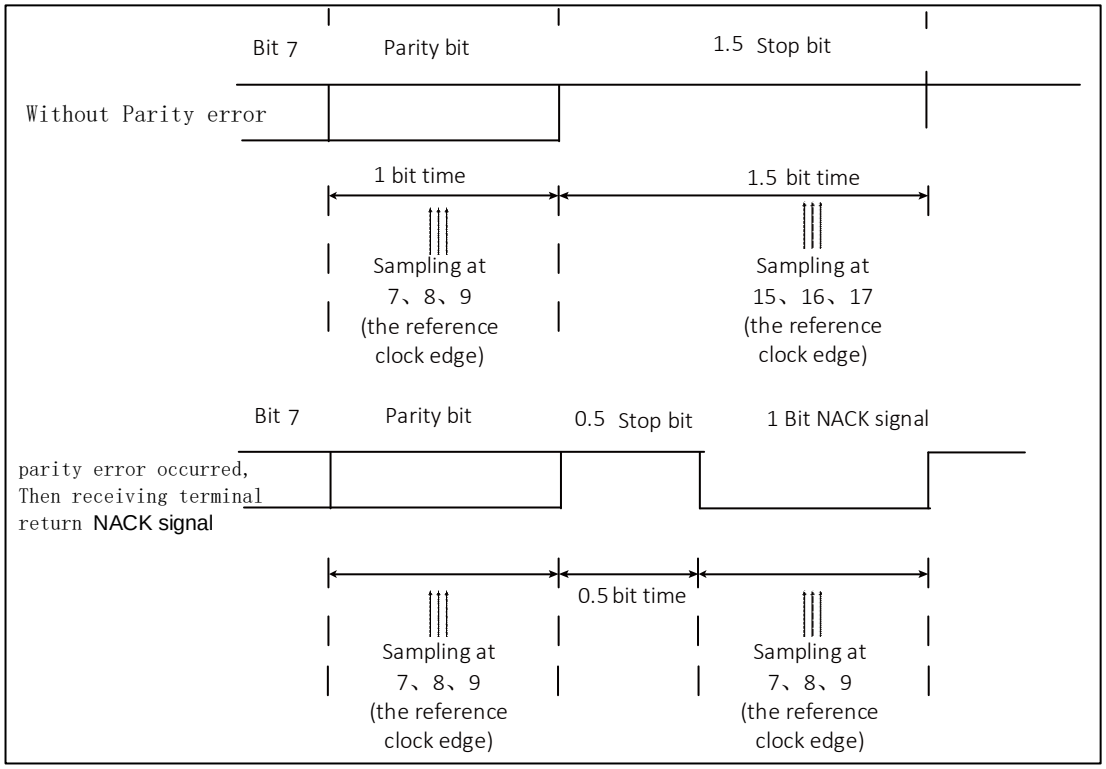


Figure 15-6 UART sampling NACK signal block diagram

15.4.13 Infrared IrDA function

UART module supports IrDA (infrared data association) SIR ENDEC specification; IrDA SIR physical layer specifies the use of reverse phase clearing regulation scheme (RZI).

The scheme uses an infrared light pulse to represent the logic “0”. In the normal mode, “0” pulse width is 3/16 baud16 baud rate. In the low power mode, the frequency range (1.42 MHz <sirlp_clk<2.12 MHz), and the pulse width sirlp_clk*3，sirlp_clk. Configure UART_IRDA.PSC_REG[7: 0] and divide the system clock to reach the value.

SIR transmission encoder and SIR receipt decoder achieve UART data flow and infrared pulse flow conversion.

IrDA is a half duplex communication protocol, which cannot go together with the encode and decoder; in the IrDR mode, UART_CCRL register STOP bit must be configured as 1 stop bit.

Steps below are required before TXEN or RXEN enable:

1. Enable infrared function: UART_IRDA register is configured as Siren=1.
2. During low power mode: Configure in advance UART_IRDA register PSC_REG and Sirlp=1.

The figure below illustrates the sample of data format conversion:

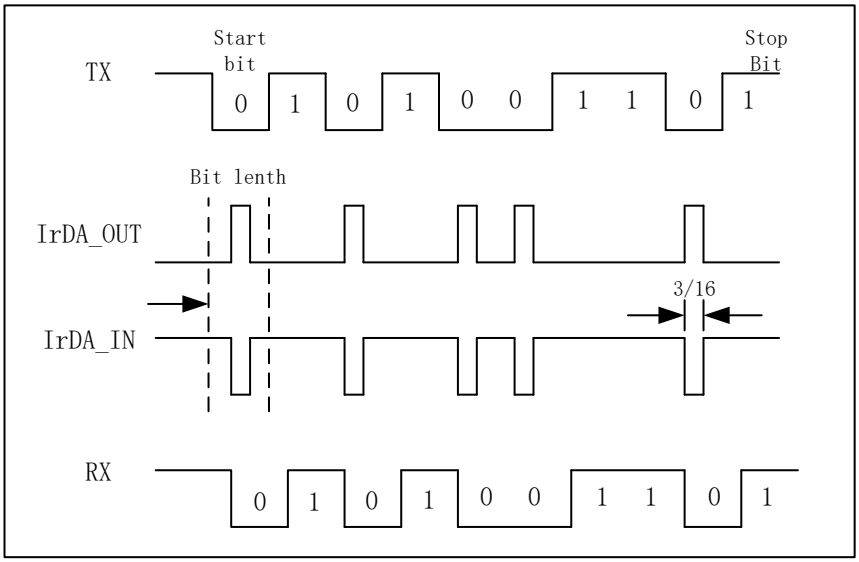


Figure 15-7 IrDA receipt and transmission in the ordinary mode

15.5 Interrupt

Support the interrupt source below:

- Transmission BUFFER empty
- Receipt data valid
- Receipt overflow error
- Frame error
- Parity check error
- Receipt Break frame
- Transmission shift register finish

- Transmission Break frame finish
- Receipt sync frame
- Idle frame detected
- Auto baud rate finish
- Auto baud rate error

Table 15-1 UART interrupt request

Interrupt event	Interrupt status	Enable bit
Transmission buffer empty	TX_INTF	TXIEN
Receive valid data	RX_INTF	RXIEN
Transmission shift register finish	TXC_INTF	TXC_EN
Receipt overflow error	RXOERR_INTF	RXOERREN
Parity check error	RXPERR_INTF	RXPERREN
Frame error	RXFERR_INTF	RXFERREN
Receipt Break frame	RXBRK_INTF	RXBRKEN
Transmission Break frame	TXBRK_INTF	TXBRK_EN
Receipt sync frame	RXB8_INTF	RXB8_EN
Receipt idle frame	RXIDLE_INTF	RXIDLEN
Auto baud rate finish	ABREND_INTF	ABRENDIEN
Auto baud rate error	ABRERR_INTF	ABRERRIEN

When setting the corresponding interrupt enable control bit, these settings can produce the corresponding interrupt。

15.6 Register description

15.6.1 Overview of registers

Table 15-2 UART register overview

Offset	Acronym	Register Name	Reset
0x00	UART_TDR	UART Transmission Data Register	0x00000000
0x04	UART_RDR	UART Receipt Data Register	0x00000000
0x08	UART_CSR	UART Current Status Register	0x00000009
0x0C	UART_ISR	UART Interrupt Status Register	0x00000000
0x10	UART_IER	UART Interrupt Enable Register	0x00000000
0x14	UART_ICR	UART Interrupt Clear Register	0x00000000
0x18	UART_GCR	UART Global Control Register	0x00000000
0x1C	UART_CCR	UART Universal Control Register	0x00000000
0x20	UART_BRR	UART Baud Rate Register	0x00000001
0x24	UART_FRA	UART Fraction Baud Rate Register	0x00000000
0x28	UART_RXADDR	UART Receipt Address Register	0x00000000
0x2C	UART_RXMASK	UART Receipt Mask Register	0x000000FF

Offset	Acronym	Register Name	Reset
0x30	UART_SCR	UART SCR Register	0x00000000
0x34	UART_IDLR	UART Idle Data Length Register	0x0000000C
0x38	UART_ABRCR	UART Auto Baud Rate Control Register	0x00000000
0x3C	UART_IRDA	UART Infrared Function Control Register	0x00000100

15.6.2 UART_TDR Transmission Data Register

Address offset:0x00

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.							TXREG								
							rw								

Bit	Field	Description
31: 9	Reserved	Reserved, always read as 0
8: 0	TXREG	Transmission data register

15.6.3 UART_RDR Receipt Data Register

Address offset:0x04

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.							RXREG								
							r								

Bit	Field	Description
31: 9	Reserved	Reserved, always read as 0
8: 0	RXREG	Receipt data register This register is only readable

15.6.4 UART_CSR Current Status Register

Address offset:0x08

Reset value:0x0000 0009

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												TXEPT	TXFULL	RXAVL	TXC
												r	r	r	r

Bit	Field	Description
31: 4	Reserved	Reserved, always read as 0
3	TXEPT	Transmission buffer empty flag bit 0:Transmission buffer is not empty 1:Transmission buffer is empty
2	TXFULL	Transmission buffer full flag bit 0:Transmission buffer is not full 1:Transmission buffer is full
1	RXAVL	Receipt valid data flag bit When the receipt buffer receives a complete byte data, the bit is asserted. 0:Receipt buffer is empty 1:The receipt buffer receives a complete and valid byte data
0	TXC	Transmission complete flag bit 0: Transmission is not complete 1:Transmission is complete Note: This flag bit is relevant with the guard time (UART_SCR.SCFCNT configuration value) when smart card mode (ISO7816).

15.6.5 UART_ISR Interrupt Status Register

Address offset:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				ABRERR_INTF	ABRENDINTF	RXIDLE_INTF	RXB8_INTF	TXBRK_INTF	RXBRK_INTF	RXFERR_INTF	RXPERR_INTF	RXOERR_INTF	TXC_INTF	RX_INTF	TX_INTF
				r	r	r	r	r	r	r	r	r	r	r	r

Bit	Field	Description
31: 12	Reserved	Reserved, always read as 0

Bit	Field	Description
11	ABRERR_INTF	UART auto baud rate error interrupt flag bit 0:No auto baud rate error 1:Auto baud rate error
10	ABREND_INTF	UART auto baud rate end interrupt flag bit 0:No end of auto baud rate 1:Auto baud rate end
9	RXIDLE_INTF	UART receipt frame idle interrupt flag bit After the stop bit, RX pin receives several (UART_IDLR) high levels after a while. 0: No idle frame 1: Detection idle frame
8	RXB8_INTF	UART sync frame interrupt flag bit In the 9 bit communication mode, when the 9th bit data received is sames as the register CCR.B8POL, RXB8_INT is asserted. The bit can serve as the interrupt request signal 0: No receipt sync frame 1: Receipt sync frame
7	TXBRK_INTF	UART Break frame transmission end interrupt flag bit 0:Shift register is empty or transmitting shift transmission device 1: Shift register Break frame data transmission end Note: No continuous transmission of Break frame.
6	RXBRK_INTF	UART receipt frame break interrupt flag bit After the anomaly stop bit, RX pin receives 10 or more than 10 bit low levels after a while. 0:No Break frame 1:Detection Break frame
5	RXFERR_INTF	Frame error interrupt flag bit Frame error occurs in the detection of anomaly stop bit. 0:No frame error 1:Detection one frame error
4	RXPERR_INTF	Parity check error interrupt flag bit 0:No parity check error 1:Detection parity check error
3	RXOERR_INTF	Receipt overflow error interrupt flag bit 0: No overflow error 1: Receipt overflow error
2	TXC_INTF	UART transmission shift register end interrupt flag bit 0: Shift register is not empty or data transmission is on going 1: Shift register data transmission is complete Note: This interrupt flag bit is relevant with the guard time (UART_SCR.SCFCNT configuration value) when smart card mode (ISO7816).

Bit	Field	Description
1	RX_INTF	Receipt valid data interrupt flag bit The bit is asserted when the receipt buffer receives a complete byte data. 0:Receipt buffer is empty 1:Receipt buffer has valid byte data
0	TX_INTF	Transmission buffer empty interrupt flag bit 0:Transmission buffer is not empty 1: Transmission buffer is empty

15.6.6 UART_IER Interrupt Enable Register

Address offset:0x10

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				ABRERR_IEN	ABREND_IEN	RXIDLE_IEN	RXB8_IEN	TXBRK_IEN	RXBRK_IEN	RXFERR_IEN	RXPERR_IEN	RXOERR_IEN	TXC_IEN	RX_IEN	TX_IEN
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31: 12	Reserved	Reserved, always read as 0
11	ABRERR_IEN	Auto baud rate error enable bit 0:Interrupt disable 1:Interrupt enable
10	ABREND_IEN	Auto baud rate end enable bit 0:Interrupt disable 1:Interrupt enable
9	RXIDLE_IEN	Receipt frame idle interrupt enable bit 0:Interrupt disable 1:Interrupt enable
8	RXB8_IEN	UART sync frame interrupt enable bit 0:Disable receipt sync frame interrupt 1:Enable receipt sync frame interrupt
7	TXBRK_IEN	UART Break frame transmission end interrupt enable bit 0:Disable transmission Break frame end interrupt 1:Enable transmission Break frame end interrupt
6	RXBRK_IEN	UART receipt Break frame interrupt enable bit 0: Interrupt disable 1: Interrupt enable

Bit	Field	Description
5	RXFERR_IEN	Frame error interrupt enable bit 0: Interrupt disable 1: Interrupt enable
4	RXPERR_IEN	Parity error interrupt enable bit 0: Interrupt disable 1: Interrupt enable
3	RXOERR_IEN	Receipt overflow error interrupt enable bit 0: Interrupt disable 1: Interrupt enable
2	TXC_IEN	UART transmission shift register end interrupt enable control bit 0: Interrupt disable 1: Interrupt enable
1	RX_IEN	Receipt buffer interrupt enable bit 0: Interrupt disable 1: Interrupt enable
0	TX_IEN	Transmission buffer empty interrupt enable bit 0: Interrupt disable 1: Interrupt enable

15.6.7 UART_ICR Interrupt Clear Register

Address offset:0x14

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				ABRERR_ICLR	ABRENDD_ICLR	RXIDLE_ICLR	RXB8_ICLR	TXBRK_ICLR	RXBRK_ICLR	RXFERR_ICLR	RXPERR_ICLR	RXOERR_ICLR	TXC_ICLR	RX_ICLR	TX_ICLR
				w	w	w	w	w	w	w	w	w	w	w	w

Bit	Field	Description
31: 12	Reserved	Reserved, always read as 0
11	ABRERR_ICLR	UART auto baud rate error clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
10	ABREND_ICLR	UART auto baud rate end clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt

Bit	Field	Description
9	RXIDLE_ICLR	UART receipt frame idle interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
8	RXB8_ICLR	UART sync frame interrupt flag clear control bit 0: Write 0 meaningless 1: Write 1 to clear receipt sync frame interrupt
7	TXBRK_ICLR	UART Break frame transmission end interrupt flag clear control bit. 0: Write 0 meaningless 1: Write 1 to clear Break frame transmission end interrupt
6	RXBRK_ICLR	UART receipt Break frame interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
5	RXFERR_ICLR	Frame error interrupt enable bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
4	RXPERR_ICLR	Parity error interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
3	RXOERR_ICLR	Receipt overflow error interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
2	TXC_ICLR	Transmission end interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
1	RX_ICLR	Receipt interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
0	TX_ICLR	Transmission buffer empty interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt

15.6.8 UART_GCR Global Control Register

Address offset:0x18

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.					TXTOG	RXTOG	SWAP	SELB8	Res.		TXEN	RXEN	Res.		UARTEN

	rw	rw	rw	rw		rw	rw		rw

Bit	Field	Description
31: 11	Reserved	Reserved, always read as 0
10	TXTOG	Transmission inversion bit 0:Invalid 1:Transmission inversion
9	RXTOG	Receipt inversion 0:Invalid 1:Receipt inversion
8	SWAP	Input and output exchange 0:Invalid 1:Input and output exchange Note: After SWAP, GPIOx_CRL register MODE should be changed, e.g. import mode turning into export mode
7	SELB8	Selection B8 data receipt or transmission valid or not 0:Invalid 1:B8 data receipt and transmission are valid When UART_CCR bit B8EN is valid, transmit 9 bit data; receipt register bit CHAR+1 data length.
6: 5	Reserved	Reserved, always read as 0
4	TXEN	Transmission enable bit 0: Transmission disable. Clear TX BUFFER 1: Transmission enable
3	RXEN	Receipt enable bit 0:Receipt disable. Clear RX BUFFER. 1:Receipt enable
2: 1	Reserved	Reserved, must be kept at reset value..
0	UARTEN	UART module selection bit 0:UART module disable 1:UART module enable

15.6.9 UART_CCR Universal Control Register

Address offset:0x1C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	LIN	WAKE	RWU	B8EN	B8TOG	B8POL	B8TXD	B8RXD	SPB1	CHAR		BRK	SPB0	PSEL	PEN
	rw	rw	rw	rw	rw	rw	rw	r	rw	rw		rw	rw	rw	rw

Bit	Field	Description
31: 15	Reserved	Reserved, always read as 0
14	LIN	UART LIN protocol receipt and transmission Break frame 0:LIN protocol invalid 1:LIN protocol is valid
13	WAKE	Wake-up method. This bit determines UART wake-up method. 0: Idle bus to wake-up from mute mode 1: Address Mark to wake-up from mute mode
12	RWU	Receiver in mute mode This bit is used to determine whether to put UART in the mute mode. This bit can be set or cleared by the software. When the wake-up sequence is available, the hardware will also clear it. 0: Receiver in normal work mode 1: Receiver in the mute mode When setting the address flag wake-up, if the receipt buffer is not empty, the software change is disabled
11	B8EN	UART sync frame 9th bit enable control bit. When the bit is enabled, the check enable bit PEN doesn't work. 0:Disable sync frame 9th bit transmission 1:Enable sync frame 9th bit transmission
10	B8TOG	UART sync frame transmission 9th bit auto roll control bit 0:Disable 9th bit value auto reverse 1:Enable 9th bit value auto reverse Note: When B8TXD and B8POL values are same, after configuring the data register, the 2nd transmission the 9th bit data begins to reverse, and the 1st data is the address bit by default.
9	B8POL	UART sync frame 9th bit polarity control bit 0:Sync frame 9th bit low level valid 1:Sync frame 9th bit high level valid
8	B8TXD	UART sync frame transmission data 9th bit 0:Transmission sync frame 9th bit as low level 1:Transmission sync frame 9th bit as high level
7	B8RXD	UART sync frame receipt data 9th bit Only readable 0:Receipt sync frame 9th bit is low level 1:Receipt sync frame 9th bit is high level
6	SPB1	Stop selection bit, plus SPB0, set the stop bit.
5: 4	CHAR	UART data width bit 00 :5 bit； 01 :6 bit 10 :7 bit； 11 :8 bit

Bit	Field	Description
3	BRK	UART transmission Break frame 0: No Break frame is transmitted 1: Break frame (logic 0 as Break frame) will be transmitted
2	SPB0	Stop bit selection Set transmission stop bit. SPB1, SPB0 00:1 stop bit SPB1, SPB0 01:2 stop bit SPB1, SPB0 10:0.5 stop bit SPB1, SPB0 11:1.5 stop bit
1	PSEL	Check selection bit After the parity check enable, this bit is used to select whether to use even check or odd check. 0:Odd check 1:Even check
0	PEN	Parity enable bit 0:Disable parity check 1: Parity check enable

15.6.10 UART_BRR Baud Rate Register

Address offset:0x20

Reset value:0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa															
rw															

Bit	Field	Description
31: 16	Reserved	Reserved, always read as 0
15: 0	DIV_Mantissa	Integer These 16 bits define the integer of UART divider factor (UARTDIV) DIV_Mantissa min value 1

15.6.11 UART_FRA Fraction Baud Rate Register

Address offset:0x24

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.												DIV_Fraction			
												rw			

Bit	Field	Description
31: 4	Reserved	Reserved, always read as 0
3: 0	DIV_Fraction	UARTDIV Fraction setting The low 4 bits define the decimal of UART divider factor (UARTDIV)

15.6.12 UART_RXADDR Receipt Address Register

Address offset:0x28

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								RXADDR							
								rw							

Bit	Field	Description
31: 8	Reserved	Reserved, always read as 0
7: 0	RXADDR	UART sync frame data host match address In case of RXMASK =0xFF, the received sync frame data is same as the host match address, produce RXB8_INTF. The address 0 is broadcast address, and will respond after receipt.

15.6.13 UART_RXMASK Receipt Mask Register

Address offset:0x2C

Reset value:0x0000 00FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								RXMASK							
								rw							

Bit	Field	Description
31: 8	Reserved	Reserved, always read as 0
7: 0	RXMASK	When the data bits are all“0”, any receipt data will produce sync frame interrupt request (RXB8_INTF). If the data bits whichone is “1”, then the corresponding bit in RDR and RXADDR has to be the same, only that can produce the sync frame interrupt request.

15.6.14 UART_SCR SCR Register

Address offset:0x30

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			HDSEL	SCFCNT								Res.	NACK	SCAEN	SCEN
			rw	rw									r	rw	rw

Bit	Field	Description
31: 13	Reserved	Reserved, always read as 0
12	HDSEL	Single wire half duplex mode selection 0:Disable half duplex mode 1:Enable half duplex mode
11: 4	SCFCNT	ISO7816 protection counter. During data transmission, the time to delay TXC flag bit assertion when the current data transmission by the register finished. Before reaching the count period, the transmitter will disable the start bit for the next data. 1: This setting value is forbidden. 2~255: The protection time is 2~255 baud rate time (in baud rate time) Note: When setting the counter as 0x0 or 0xFF, it’s 255 baud rate count time.
3	Reserved	Reserved, always read as 0
2	NACK	Active receipt frame response bit Receipt low level during protection period,this bit will be set to 1 by hardware.
1	SCAEN	ISO7816 check auto response bit 0:Disable auto response 1:Enable auto response
0	SCEN	ISO7816 enable control bit 0:Disable ISO7816 function 1:Enable ISO7816 function (Smartcard mode)

15.6.15 UART_IDLR Idle Data Length Register

Address offset:0x34

Reset value:0x0000 000C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDLR															
rw															

Bit	Field	Description
31: 16	Reserved	Reserved, always read as 0
15: 0	IDLR	UART idle data length register Data length is not allowed to set as 0.

15.6.16 UART_ABRCR Auto Baud Rate Control Register

Address offset:0x38

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.											Latter_edge	Former_edge	Abr_bitcnt		Abren
											rw	rw	rw		rw

Bit	Field	Description
31: 5	Reserved	Reserved, always read as 0
4	Latter_edge	One edge selection after the auto baud rate 0:Falling edge 1:Rising edge
3	Former_edge	One edge selection before the auto baud rate 0:Falling edge 1:Rising edge

Bit	Field	Description
2: 1	Abr_bitcnt	Auto baud rate detection length Detect the bit length between the previous edge and the following edge. 00:1 bit 01:2 bit 10:4 bit 11:8 bit
0	Abren	Auto baud rate enable Enable auto baud rate only when UART is idle. After enabling, detect the edge of the receipt signal. After auto detection of auto baud rate, the hardware auto sets UART_BRR and UART_FRA register. 0:Auto baud rate disable 1:Auto baud rate enable

15.6.17 UART_IRDA Infrared Function Control Register

Address offset:0x3C

Reset value:0x0000 0100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC_REG								Res.						Sirlp	Siren
rw														rw	rw

Bit	Field	Description
31: 16	Reserved	Reserved, always read as 0
15: 8	PSC_REG	Prescaler value Under the infrared low power mode, get low power mode frequency sirlp_clk for UART source clock (pclk) division: 0000 0000 : Reserved，cannot write the value 0000 0001 : Source clock 1 division 0000 0010 : Source clock 2 division 0000 0011: Source clock 3 division ..... Note: Guarantee clock frequency after division is between 1.42 MHz <sirlp_clk <2.12 MHz.
7: 2	Reserved	Reserved, always read as 0
1	Sirlp	IrDA low power mode 0 : Ordinary mode 1 : Low power mode

Bit	Field	Description
0	Siren	IrDA mode enable 0 : Disable infrared mode 1 : Enable infrared mode

16 SPI_I2S Serial Peripheral Interface

16.1 SPI_I2S function block diagram

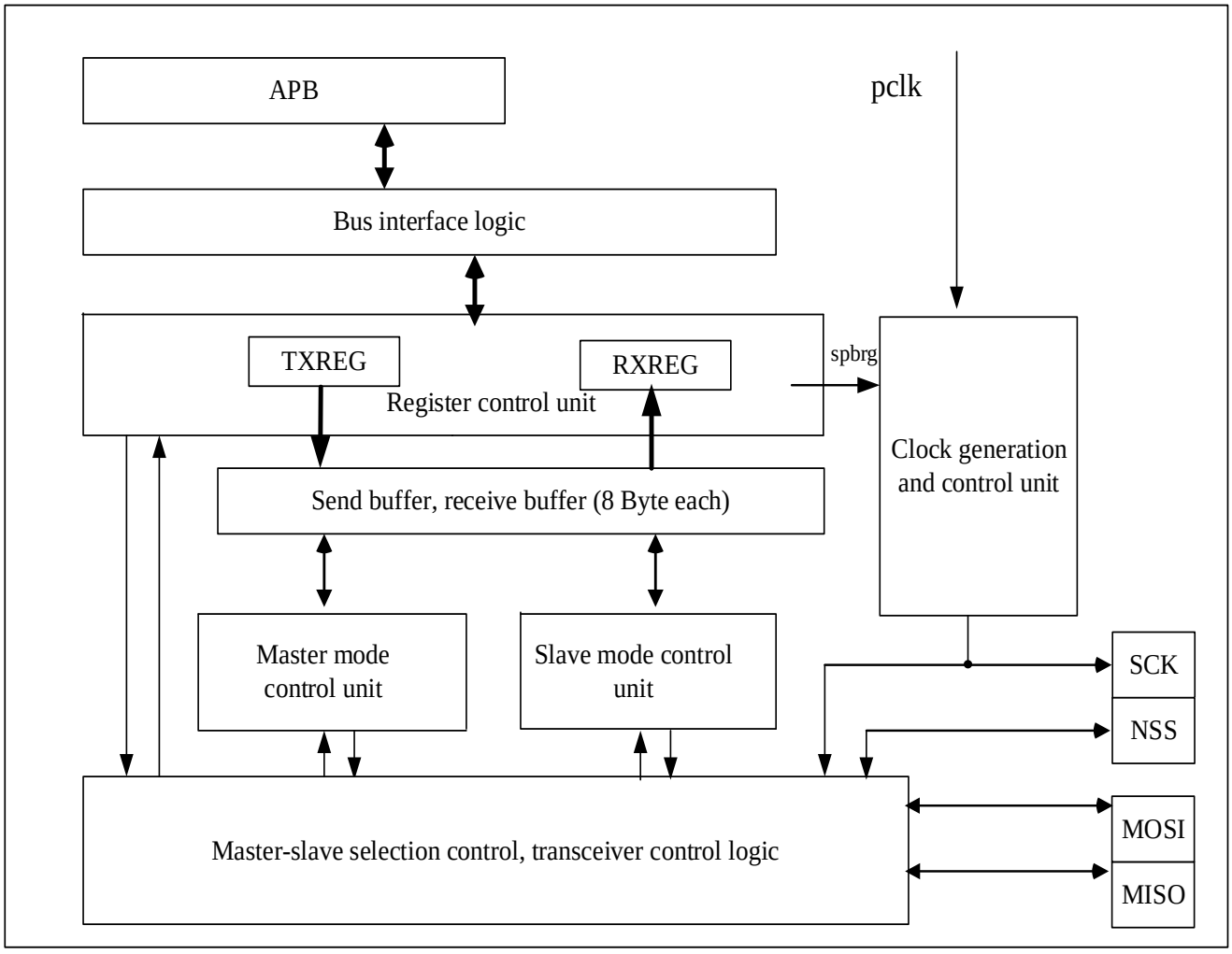


Figure 16-1 SPI_I2S function block diagram

16.2 SPI_I2S overview

The SPI (Serial Peripheral Interface) interface is widely used on board level communication between different devices, such as extended serial Flash and ADC. Many IC manufacturers produce devices that support SPI interfaces.

SPI allows the MCU to communicate with external devices in full duplex, synchronous and serial mode. The application software can communicate by the query status or SPI interrupt.

I2S (Inter-IC Sound) bus, also called the integrated circuit built-in audio bus, is a bus standard formulated by Philips for the audio data transmission among the digital audio device. It uses an

independent wire to respectively transmit the clock and data signals, and prevents the distortion caused by the time difference. Therefore, it's widely used in various multimedia systems for data transmission among audio devices.

16.3 SPI functional description

16.3.1 Overview

SPI supports the simultaneous receipt and transmission of 1~32 bit data. SPI can be configured as the slave mode or the master mode under the host environment. The software configures the CPOL bit and the CPHA bit of the universal control register (CCTL) and select four possible transmission sequences among the clock and data. Configure LSBFE bit to select whether MSB is in front or LSB is in front for data transmission.

SPI transmits data in the rising edge or falling edge of SCK, and receives data in the effective edge of the opposite clock.

SPI is used to exchange data. Data must be read after the transmission, even if the data is not valid data. Besides, the clock phase bit and polarity of the host and other communication slave unit must be same.

SPI is connected with the external device via 4 pins:

- MISO: Master equipment input, slave device output pin. The transmission direction is from the slave equipment to the master equipment.
- MOSI: Master equipment output, slave device input pin. The transmission direction is from the master equipment to the slave equipment
- SCK: SCK:Serial port clock, generated by master equipment and transmitted to the slave equipment.
- NSS: Slave equipment selection. This pin is the selectable function under SPI protocol for master equipment to select the slave equipment to communicate with it (when there are multiple slave equipment) to achieve the one-to-one communication between the master and slave equipment and avoid the conflict between equipment on the data line. When the NSS pin function is activated, configure SPI_I2S_GCTL.MODE as '1' and SPI works in the master mode. Configure SPI_I2S_NSSR.NSS as '0' and NSS pin exports the low level, thereby enabling data communication between the slave equipment connected with the master equipment (configure SPI_I2S_GCTL.MODE as '0') and it.

The figure below shows one-to-one interconnected communication between the master equipment and slave equipment:

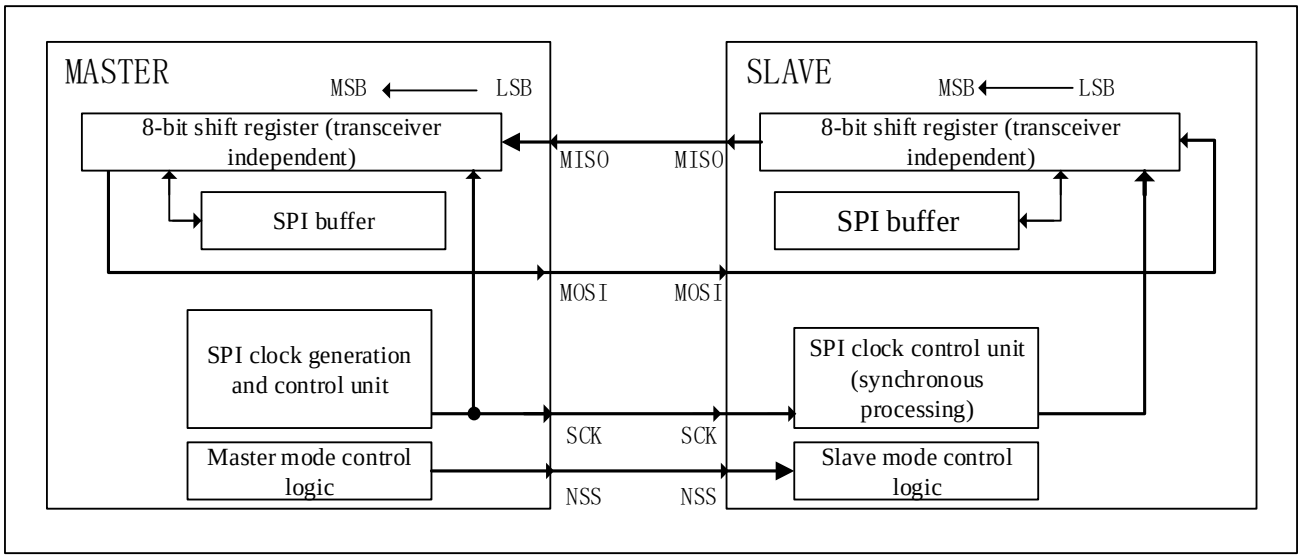


Figure 16-2 Single master and slave application

Same pins of SPI master and slave equipment are connected. The data transmission in the figure is serial communication from the highest valid bit to the lowest valid bit.

The master equipment is responsible for initiating the communication request and the slave equipment is responsible for response. The slave equipment gets the clock signal provided by master equipment via SCK pin, thereby enabling master and slave equipment to use the same clock for synchronous full duplex communication.

For the slave equipment, MOSI pin inputs the transmission data from the master equipment, and MISO pin exports the response data to the master equipment.

16.3.1.1 Phase bit and polarity of clock signal

CPOL and CPHA bit of SPI_I2S_CCTL register respectively control the clock polarity and phase bit of the clock. By combination,4 sequence relations of clock/data can be made available.

Clock polarity refers to the level of SCK clock under the clock idle status: When configuring CPOL bit as ‘0’, SCK clock keeps the low level under the idle status; otherwise, SCK clock keeps the high level under the idle status. Both master and slave equipment will be affected by CPOL control bit.

The clock phase bit determines the input data sampling sequence: When configuring CPHA bit as ‘0’, the first data bit will be sampled in the second clock edge of SCK; otherwise, the first data bit will be sampled in the first clock edge of SCK.

Besides, CPHASEL bit is ‘0’ after system power on and reset. When the software adjusts this bit as ‘1’, change the data sampling sequence and CPHA bit function may change. For example during CPHASEL=1, CPHA=0: The first data bit will be sampled in the first clock edge of SCK (CPOL bit ‘0’ rising edge; ‘1’ falling edge).

Therefore, the expected use of clock/data sequence relations should be based on the configuration

of CPOL, CPHA and CPHASEL bits.

Precautions for the time sequence configuration:

- SPI cannot change CPOL/CPHA during work (When change is needed, first close the SPI enable bit SPIEN).
- The master/slave equipment communicate in step. Therefore, the time sequence configuration should be consistent.
- In the idle status, SCK level must be consistent with the corresponding polarity of CPOL configuration.

The figure below illustrates 4 scenarios of different CPHA and CPOL bit combinations during SPI transmission (configure CPHASEL bit as '1'), as well as the sequence of master/slave equipment SCK/MISO/MOSI/NSS pins.

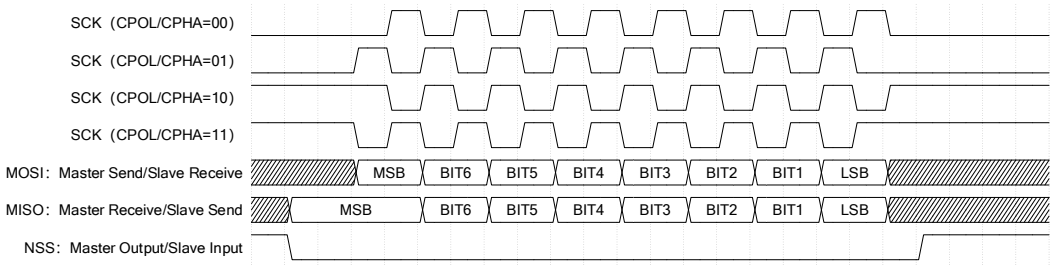


Figure 16-3 Data clock sequence

It should be noted, configure SPI_I2S_GCTL.NSS=1, namely when NSS pin function is auto controlled by hardware. After data communication, the hardware will auto pull up this pin (as illustrated in the figure above). When configuring SPI_I2S_GCTL.NSS=0, NSS output status will be controlled by the NSS bit of register SPI_I2S_NSSR from the chip (changeover of NSS pin output status by software).

16.3.1.2 High speed transmission

With response to the board level sensitivity in the high speed transmission mode, configure SPI_I2S_CCTL register TXEDGE/RXEDGE bit to adjust the time point of data transmission/receipt sampling. During high speed transmission, the baud rate generator for the master equipment is configured at SPBRG<=4 and output SCK clock is quicker (>=10MHz); during low speed transmission, the baud rate generator for the master equipment is configured at SPBRG>4, and output SCK clock is slower (<10MHz).

- In the slave mode, when TXEDGE bit is '1', once the software configures TXREG to write data, rather than waiting for SCK clock input edge, make transmission to the MISO pin line; when TXEDGE bit is '0', the slave equipment will always wait for a valid clock edge before transmitting data to MISO pin line.
- In the master mode, when TXEDGE bit is '1', move back the sampling data time point, and sample data in the tail clock edge of transmission data bit; when TXEDGE bit is '0', sample data

in the middle point of transmission data bit.

16.3.1.3 Data frame format

Configure LSBFE bit of SPI_I2S_CCTL register, and determine the data bit output sequence, from the lowest valid bit to the highest valid bit when the LSBFE bit is ‘1’or from the highest valid bit to the lowest valid bit in case of ‘0’ (default).

Configure SPILEN bit of SPI_I2S_CCTL register, and determine the data frame length. When SPILEN bit is ‘1’ (default), the data frame length is 8 bit. In case of ‘0’, the data frame length is 7 bits. SPI transmission and receipt is controlled by the data frame format configuration.

Besides, configure SPI_I2S_GCT.DW8_32=1 and register SPI_I2S_EXTCTL to achieve the data format of any frame length (frame length range within 1~32 bit); any frame length configuration also supports LSBFE bit function (LSB or MSB preferred).

16.3.2 Main characteristics of SPI

- Fully compatible with Motorola SPI specification
- Support full duplex sync transmission in 3 lines
- 16 bit programmable baud rate generator
- Support master and slave mode
- Support communication between one host and multiple slave units
- When the SPI is in the master/slave mode, the clock can be up to PCLK/2、PCLK/4 (PCLK as APB clock)
- Programmable clock polarity and phase bit
- Programmable data frame length (fixed 8 bit or 7 bit frame length, any frame length of 1~32 bits)
- Programmable data sequence, MSB is in front or LSB is in front (support transmission and receipt of data with any frame length of 1~32 or fixed frame length)
- 8 byte receipt/ transmission buffer, the following interrupt event or status should be available for software configuration and use:
  - Transmission buffer is empty
  - Transmission buffer and transmission shift register are simultaneously empty
  - Transmission end underflow
  - Receive valid byte
  - Receipt buffer overflow
  - Receipt buffer full

- Receive specified byte in the master mode

16.3.3 SPI slave mode

When SPI serves as the slave equipment, SCK pin put comes from the serial clock of the master equipment. Therefore, when the baud rate generator is not used while the slave equipment works, it's not necessary to configure the register SPI_I2S_SPBRG (invalid under the slave equipment).

16.3.3.1 Configuration steps

1. Set SPI_I2S_GCTL.SPILEN to define the data frame format as 7 bit or 8 bit.
2. Configure the CPOL, CPHA/CPHASEL bit of the register SPI_I2S_CCTL to determine the sequence mode.
3. Configure SPI_I2S_CCTL.LSBFE, and determine the data frame receipt and transmission sequence (LSB or MSB bit preferred).
4. Configure MODE bit of register SPI_I2S_GCTL as '0' (slave mode), SPIEN bit as '1' (SPI function enable), and configure GPIO function pins required for SPI operation.
5. Configure TXEN and RXEN bit of register SPI_I2S_GCTL as '1', and open transmission and receipt permission (during transmission, it's required to write data into the register SPI_I2S_TXREG). SPI will receive MOSI pin data under the slave mode, and export data from MISO pin.

Note: It's mandatory to keep consistent the sequence mode and data frame receipt and transmission of master and slave equipment to guarantee normal transmission of data.

16.3.3.2 Data transmission

Write data to the transmission data register SPI_I2S_TXREG, and the whole data is transmitted to the transmission buffer.

When the slave equipment receives the SCK clock signal as well as the first data bit of MOSI pin. The slave equipment uses SCK change edge and transmit the data to the MISO pin bit by bit. The transmission data process conforms to the relevant sequence of data/clock (determined by the CPOL, CPHA/CPHASEL bit).

However, during high speed transmission (configure SPI_I2S_CCTL.TXEDGE=1), the data won't change according to the input SCK clock edge, but instead transmit data to MISO pin in advance along the internal PCLK clock edge (it won't be earlier than the SCK clock edge of the previous bit of data receipt sampling).

When the first bit of data is transmitted, the hardware will assert SPI_I2S_INTSTAT.TX_INTF flag, and the software will use this flag to write TXREG for continuous data transmission (configure

SPI_I2S_INTEN.TX_IEN bit as ‘1’ to produce CPU interrupt).

Note: The slave unit clock signal is provided by the host. Therefore, the precondition of continuous transmission must be that the host can provide the continuous clock.

16.3.3.3 Data receipt

When the slave equipment receives a complete data from the MOSI pin:

- This data along the shift register will be transmitted to the receipt buffer in the final sampling clock edge. The hardware will simultaneously assert the SPI_I2S_INTSTAT.RX_INTF flag. And then the software will read the SPI_I2S_RXREG and acquire the data from the receipt buffer.
- The software configures SPI_I2S_INTEN.RX_IEN bit as ‘1’ to open the interrupt enable, and use CPU interrupt to acquire the receipt data.

16.3.4 SPI master mode

When SPI serves as the master equipment, export the serial clock to the SCK pin for use by the slave equipment.

16.3.4.1 Configuration steps

1. Set SPI_I2S_SPBREG to define the serial clock baud rate.
2. Configure SPI_I2S_CCTL register CPOL, CPHA/CPHASEL, and determine the time sequence mode.
3. Configure SPI_I2S_CCTL.SPILEN to define 8 or 7 bit data frame format. Configure SPI_I2S_GCTL.DW8_32 as ‘1’, and configure SPI_I2S_EXTCTL register to define any frame format (SPILEN should be fixed as ‘1’).
4. Configure SPI_I2S_CCTL.LSBFE to determine the data receipt and transmission sequence (LSB or MSB bit preferred).
5. In case of only receipt of data rather than transmission, configure SPI_I2S_RXDNR register to define the received byte (upon receipt of the specified number of byte, SCK clock output will end and keep CPOL bit configuration status).
6. Configure MODE bit of the register SPI_I2S_GCTL as ‘1’ (master mode), and SPIEN bit as ‘1’ (SPI function enable), and configure GPIO function pin required for the SPI work.
7. Configure TXEN and RXEN bit of register SPI_I2S_GCTL as ‘1’, and open transmission and receipt permission (during transmission, it’s required to write data into the register SPI_I2S_TXREG after opening TXEN). SPI will export clock SCK and sync data MOSI to the pin in the master mode, and sample input data from MISO pin. NSS is the optional output function of master equipment.

Note: It's mandatory to keep consistent the sequence mode and data frame receipt and transmission of master and slave equipment to guarantee normal transmission of data.

16.3.4.2 Data transmission

Configure TXEn bit as '1', and write data to the transmission data register TXREG. The master data will be transmitted to the transmission buffer, and the master equipment begins transmission. The master equipment serially exports SCK clock and MOSI data to the pin according to the preset baud rate, and this process conforms to the relevant sequence of data/clock (as determined by CPOL, CPHA/CPHASEL). Besides, LSBFE bit determines the data serial transmission sequence.

When the 1st bit data is transmitted, the hardware will assert the SPI_I2S_INTSTAT.TX_INTF flag. The software uses this flag to write TXREG and achieve continuous data transmission (configuring SPI_I2S_INTEN.TX_IEN bit as '1' to produce CPU interrupt).

16.3.4.3 Data receipt

When the slave equipment receives a complete data from the MOSI pin:

- This data along the shift register will be transmitted to the receipt buffer in the final sampling clock edge. The hardware will simultaneously assert the SPI_I2S_INTSTAT.RX_INTF flag. And then the software will read the SPI_I2S_RXREG and acquire the data from the receipt buffer.
- The software configures SPI_I2S_INTEN.RX_IEN bit as '1' to open the interrupt enable, and use CPU interrupt to acquire the receipt data.
- In case of only receipt, and after receipt of the number of bytes as defined by RXDNR, the hardware will assert the SPI_I2S_INTEN.RXMATCH_INTF flag. Meanwhile, the master equipment won't transmit the clock signal, and SCK output will be kept in the CPOL bit configuration status (fixed high or low level).

16.3.5 Baud rate setting

SCK pin output clock frequency conforms to the baud rate configuration, which is obtained by the division of internal clock PCK according to the configuration value of SPI_I2S_SPBRG register. The register SPBREG controls the count period of one 16-bit counter.

According to the expected baud rate and Fpclk (APB module PCLK clock frequency), and use the formula of the table below to calculate the configuration value to the register SPBRG (X in the table below), X within the range of 2~65535.

Table 16-1 Baud rate formula

Mode	Formula
SPI mode	Baud rate = Fpclk/X

16.3.6 Interrupt

16.3.6.1 Status flag

To facilitate software operation, the application may use 4 current status flags and 7 interrupt flags to monitor SPI bus status.

The current status flag is only readable, and auto set and cleared by the hardware.

The interrupt status flag bit is set during event occurrence, and produces CPU interrupt during interrupt enable, cleared by software.

SPI has a 8 byte transmission buffer and receipt buffer. According to SPI_I2S_GCTL register DW8_32 bit setting, CPU may read and write 1 or 4 bytes each time. According to DW8_32 setting, the transmission and receipt buffer respectively have one byte or one valid data status flag.

Note: Configure SPI_I2S_GCTL.DW8_32=1, and receipt and transmission buffer have two valid data at most. When the frame length is configured as 8bit and below, one valid data is 1 byte. It's 2 bytes when the frame length is configured as a valid data within the range of 9~16bit. It's 3 bytes when the frame length is configured as a valid data within the range of 17~24bit. It's 4 bytes when the frame length is configured as a valid data within the range of 25~32bit.

Table 16-2 SPI status

Class	Status flag	Buffer and signal status
Interrupt status	TX_INTF	Transmission buffer is empty, according to DW8_32 setting to complete the write operation of transmission data register TXREG
	RX_INTF	According to DW8_32 setting, there is at least one valid data space, and complete the read operation of one receipt data register RXREG
	UNDERRUN_INTF	Transmission buffer is empty and repeat transmission
	RXOERR_INTF	Receipt buffer is not empty and covered
	RXMATCH_INTF	Non empty, the final one data is transmitted to the receipt buffer (valid in the master mode)
	RXFULL_INTF	Receipt buffer is full, and cannot receive new data
	TXEPT_INTF	Transmission buffer and shift register both are empty
Current status	RXAVL_4BYTE	Receipt buffer has the valid data over 4 bytes
	TXFULL	Transmission buffer is full
	TXEPT	Transmission buffer and shift register both are empty
	RXAVL	Receipt buffer is not empty

16.4 I2S functional description

16.4.1 Main characteristics of I2S

- Half duplex communication (only transmitter or receiver)
- Master/slave operation
- 9 bit programmable linear prescaler for precise audio sampling frequency (8KHz~192KHz)
- Data format may be 16 bit, 24 bit or 32 bit
- Data packet frame is fixed as 16 bit (16 bit valid data) or 32 bit (16 bit, 24 bit, 32 bit valid data)
- Programmable clock polarity (steady status)
- Underflow flag in the transmission mode (only the slave unit), overflow flag in the receipt mode (master/slave unit) and frame error flag (only the slave unit) in the receipt/transmission mode
- The transmission and receipt 32 bit register is two audios for time division multiplex access
- The data direction is always MSB preferred
- Support I2S protocol:
- -Philips standard
- -MSB alignment standard (MSB bit left alignment)
- -LSB alignment standard (MSB bit right alignment)
- -PCM standard (short frame and long frame sync mode)
- Configure MCLK clock output to drive the external audio component, baud rate fixed at 256×Fs (FS audio sampling frequency)

16.4.2 I2S bus interface

I2S and SPI share four common pins:

- SD: Serial data (mapped on MOSI pin) for transmitting or receiving two multiplex data channel (only in half duplex mode).
- WS: Audio selection (mapped on NSS pin), master data control signal output mode and auxiliary mode input.
- CK: Serial clock (mapped on SCK pin), serial clock output in the master mode and serial clock input in the slave mode.
- MCK: Optional drive clock (mapped on MISO pin) for driving the external audio component (used only when the external audio equipment needs clock input, provided by the master mode).

16.4.3 Data format

The three-line bus processing audio data must go through the two audios of the time division multiplex access: right audio and left audio. However, there is only one 32 bit register for transmission or receipt. Therefore, the software configures in sequence the value of each audio of the register TXREG or read the register RXREG audio data.Transmit the left audio before the right audio according to the I2S protocol.

Note: SPI_I2S_CSTAT.CHSIDE status flag is invalid under PCM protocol.

Data can be transmitted in the following format:

- 16 bit data packed in the 16 bit frame
- 16 bit valid data packed in the 32 bit data packet frame (software configuration invalid bit as '0')
- 24 bit data packed in the 32 bit data packet frame (software configuration invalid bit as '0')
- 32 bit data packed in the 32 bit data packet frame

When sending the 16 bit valid data or 24 bit valid data along the 32 bit data packet frame, the software should configure invalid data bit as '0' (e.g. for data 0x12345678, only send 16 bit valid data, write TXREG 为 0x00005678; send 24 bit valid data and write 0x00345678).

16.4.4 Communication standard

I2S interface supports four audio standards, and configure the register SPI_I2S_I2SCFGR I2SSTD[1:0], PCMSYNC bit for changeover. The data format will be selected by configuring DATLEN[1:0], CHLEN.

For all communication standards and data formats, always send the highest bit (MSB preferred).

16.4.4.1 Philips standard

For Philips standard, the WS signal is used to indicate the ongoing transmission channel. The transmitter locks data in the CK falling edge, and the receiver reads data in the CK rising edge. For this standard format signal, no matter how many bits of valid data, the highest data bit is always in the 2nd CK pulse of WS change.

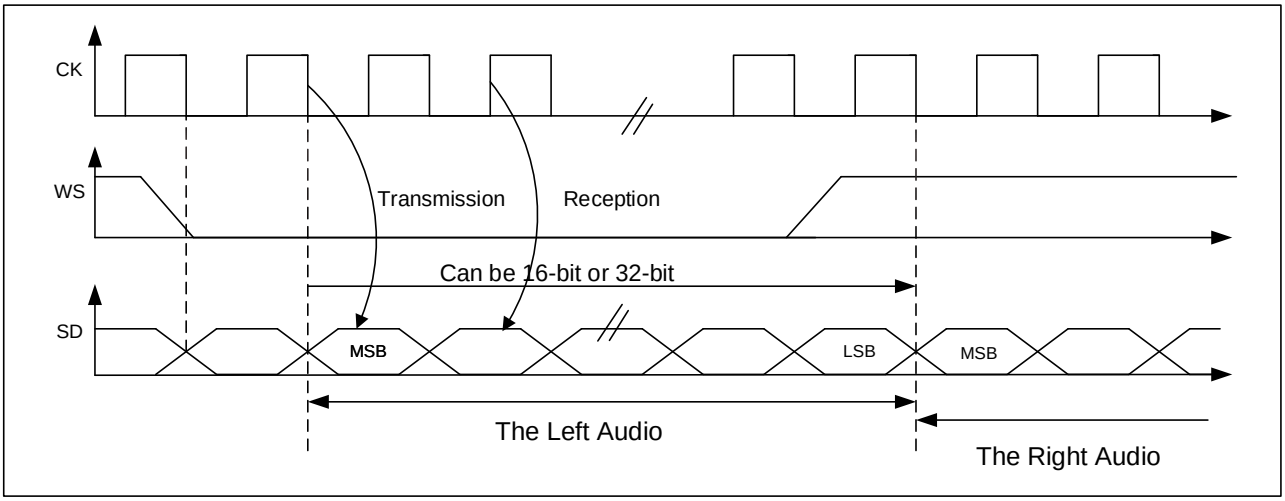


Figure 16-4 Philips standard

16.4.4.2 MSB alignment standard

For MSB alignment standard, the highest valid bit (MSB bit) of each data packet frame always appears in the 1st CK pulse period of WS change.

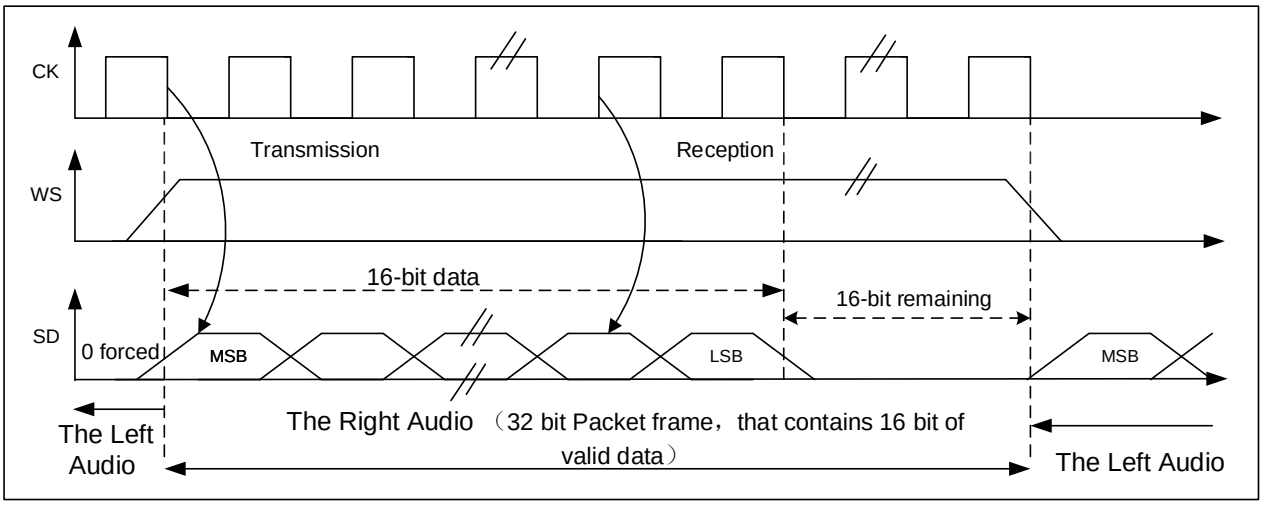


Figure 16-5 MSB alignment standard

16.4.4.3 LSB alignment standard

For LSB alignment standard, the lowest valid bit (LSB bit) of each data packet frame always appears in the 1st CK pulse period of WS change.

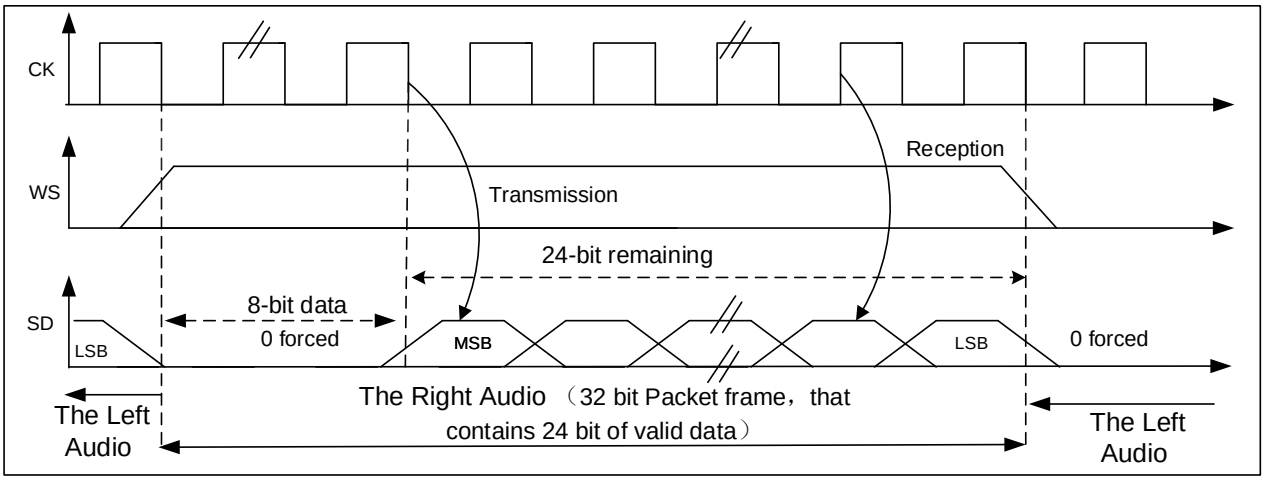


Figure 16-6 LSB alignment standard

16.4.4.4 PCM standard

For PCM standard, WS pin should not use as the audio information (CHSIDE flag bit invalid).

PCM standard has two modes: short frame mode and long frame mode. Configure SPI_I2S_I2SCFGR.PCMSYNC bit for changeover. In the PCM mode, the output signal (WS, SD) is locked in the rising edge of CK clock. The input signal (WS, SD) is captured in the CK falling edge. In the master mode, CK clock and WS pin are configured as output.

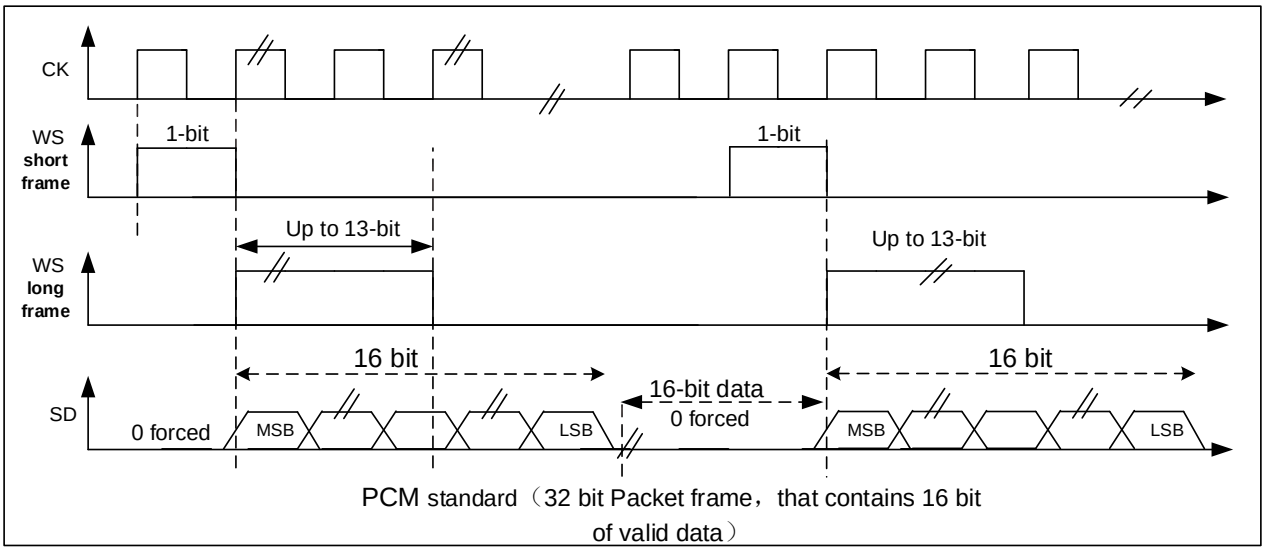


Figure 16-7 PCM standard

16.4.5 Slave mode

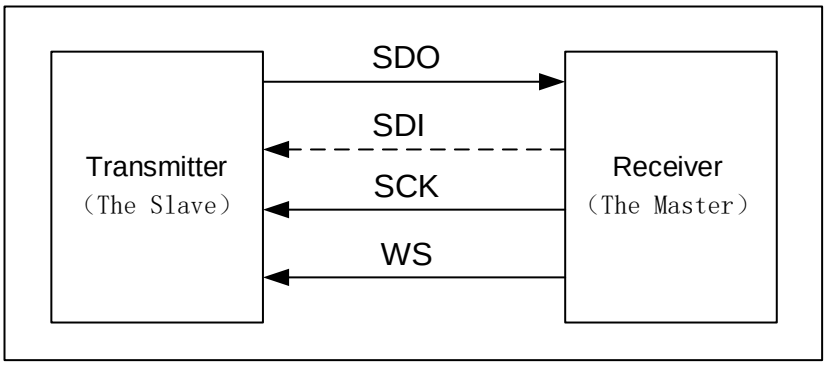


Figure 16-8 I2S auxiliary mode (SD is configured as SDO transmitter, and SDI receiver)

The slave mode supports transmission or receipt. Because I2S only support half duplex, and SD pin is configured as SDO output or SDI input. The receipt and transmission status and achieving method are similar to the SPI slave mode. Configuration steps:

1. Configure SPI_I2S_GCTL.SPIEN bit as '1' and enable the module;
2. Configure SPI_I2S_GCTL.MODE bit as '0', and set the module function as slave mode;
3. Configure SPI_I2S_CCTL.CPOL bit, and select the input level of clock idle status;
4. Configure the register SPI_I2S_I2SCFGR I2SDIV[8:0], DATLEN and CHLEN bit to conform to the desirable audio sampling frequency and data packet frame format. For the calculation method, please refer to chapter 16.4.7 Clock prescaler;
5. Configure the SPI_I2S_I2SCFGR.SPI_I2S bit as '1', and enable I2S transmission;
6. Configure the register SPI_I2S_I2SCFGR I2SSTD[1:0], PCMSYNC bit, and select the communication standard governing I2S transmission;
7. Open the half duplex transmission permission, namely configure the register SPI_I2S_GCTL TXEN or RXEN as '1' (TXEN, RXEN cannot be simultaneously configured as '1').

Note: During the slave mode transmission, prior to the detection of WS edge, it's required to conduct one data write to the register SPI_I2S_TXREG.

Besides, during the slave mode transmission, prior to the configuration of RXEN bit as '1', it's needed to keep WS input signal at the high level.

16.4.6 Master mode

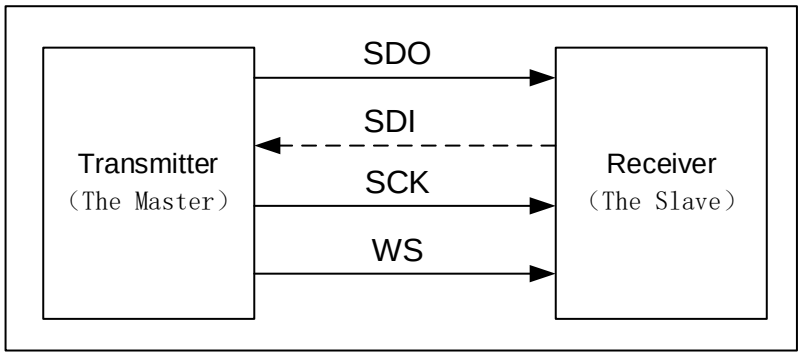


Figure 16-9 I2S master mode (SD is configured as SDO transmitter, and SDI receiver)

The master mode also supports transmission or receipt. Besides, under the half duplex mechanism, SD pin is only configured as SDO output or SDI input. The receipt and transmission status and achieving method are similar to the SPI slave mode. Configuration steps:

1. Configure SPI_I2S_GCTL.SPIEN bit as '1' and enable the module;
2. Configure SPI_I2S_GCTL.MODE bit as '1', and set the module function as master mode;
3. Configure SPI_I2S_CCTL.CPOL bit, and select the input level of clock idle status;
4. Configure SPI_I2S_I2SCFGR.MCKOE bit, and select whether to provide the MCK drive clock for the external element;
5. Configure the register SPI_I2S_I2SCFGR I2SDIV[8:0], DATLEN and CHLEN bit to conform to the desirable audio sampling frequency and data packet frame format. For the calculation method, please refer to chapter 16.4.7 Clock prescaler;
6. Configure the SPI_I2S_I2SCFGR.SPI_I2S bit as '1', and enable I2S transmission;
7. Configure the register SPI_I2S_I2SCFGR I2SSTD[1:0], PCMSYNC bit, and select the communication standard governing I2S transmission;
8. Open the half duplex transmission permission, namely configure the register SPI_I2S_GCTL TXEN or RXEN as '1' (TXEN, RXEN cannot be simultaneously configured as '1').

Note: During the master mode receipt, when the received byte number reaches the register SPI_I2S_RXDNR configuration value, stop transmission immediately (CK clock transmission end, and fixed as CPOL level level status).

16.4.7 Clock prescaler

I2SCLK clock is provided by the system APB clock (SPI modules are respectively integrated in APB1 or APB2). The figure illustrates the prescaler circuit structure of I2S module:

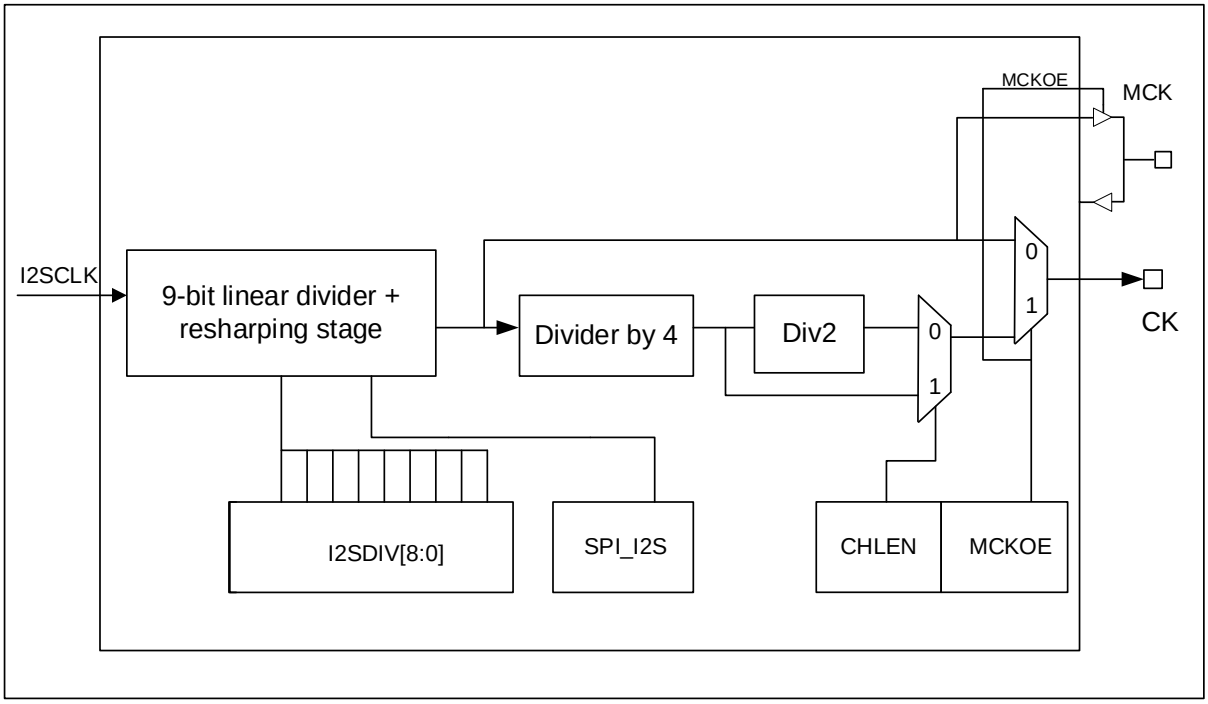


Figure 16-10 I2S clock prescaler

According to the figure above, when MCKOE bit is '0', the chip doesn't need to export MCK clock, and the prescaler directly divides I2SCLK to CK; when MCKOE bit is '1', the chip will export MCK clock. The prescaler will divide I2SCLK and obtain MCK, and then get Ck after division (division times can be selected as 4 or 8 by CHLEN).

The audio sampling rate can be 192KHz, 96 KHz, 48 KHz, 44.1 KHz, 32 KHz, 22.05 KHz, 16 KHz, 11.025 KHz and 8 KHz. Therefore, according to the circuit function of I2S clock prescaler, configure the register I2SCFGR I2SDIV[8:0], MCKOE and CHLEN bit to get the expected audio sampling rate.

During I2S transmission data, the bit rate calculation formula is expressed as the table below (CK output's one clock period corresponding to the transmission 1 bit data, therefore bit rate =CK frequency F_{CK}).

Table 16-3 Bit rate calculation formula

MCKOE	CHLEN	Bit rate calculation formula
0	0	$F_{CK} = F_{I2SCLK}$ / I2SDIV[8:0]
0	1	$F_{CK} = F_{I2SCLK}$ / I2SDIV[8:0]
1	0	$F_{CK} = F_{I2SCLK}$ / (I2SDIV[8:0]×4×2)
1	1	$F_{CK} = F_{I2SCLK}$ / (I2SDIV[8:0]×4)

Audio sampling rate (Fs) and I2S bit rate correlation is defined in the formula below:

Fs = I2S bit rate/(channel length×channel number) = F_{CK} /(channel length×channel number)

Note: Channel length, namely data packet frame length, configured as 16 bit or 32 bit; channel number is left and right audio, value fixed as 2.

Above all, according to the I2SDIV[8:0], MCKOE and CHLEN bit configuration, the correlation of the obtained audio sampling rate and F_{I2SCLK} (APB clock frequency):

Table 16-4 Audio sampling rate Fs and F_{I2SCLK} correlation

MCKOE	CHLEN	Audio sampling rate Fs and F_{I2SCLK} correlation	Precaution
0	0	F_{I2SCLK} = I2SDIV[8:0]×32×Fs	I2SDIV[8:0] should be configured within the range of 2~511
0	1	F_{I2SCLK} = I2SDIV[8:0]×64×Fs	I2SDIV[8:0] should be configured within the range of 2~511
1	0	F_{I2SCLK} = I2SDIV[8:0]×256×Fs	I2SDIV[8:0] should be configured within the range of 1~511
1	1	F_{I2SCLK} = I2SDIV[8:0]×256×Fs	I2SDIV[8:0] should be configured within the range of 1~511

When MCKOE bit is '1', the chip will export MCK drive clock, and comply with $F_{MCLK} = 256×F_s$; assume F_s = 192KHz, I2SDIV[8:0] = 1, it's required that APB clock (F_{I2SCLK})= 1×256×192KHz = 49.152MHz.

For the commonly used audio sampling rate, when calculating according to F_{I2SCLK}=48MHz, the error range is shown in the table below:

Table 16-5 I2S audio sampling rate error

		MCKOE=1				MCKOE=0			
F_s	Audio bit width	I2SDIV ideal value	I2SDIV configuration value	Real F_s	Error (%)	I2SDIV ideal value	I2SDIV configuration value	Real F_s	Error (%)
192000	32	0.976563	1	187500	-2.34375	3.90625	4	187500	-2.34375
192000	16	0.976563	1	187500	-2.34375	7.8125	8	187500	-2.34375
96000	32	1.953125	2	93750	-2.34375	7.8125	8	93750	-2.34375
96000	16	1.953125	2	93750	-2.34375	15.625	16	93750	-2.34375
48000	32	3.90625	4	46875	-2.34375	15.625	16	46875	-2.34375
48000	16	3.90625	4	46875	-2.34375	31.25	31	48387.1	0.806452
44100	32	4.251701	4	46875	6.292517	17.0068	17	44117.65	0.040016
44100	16	4.251701	4	46875	6.292517	34.01361	34	44117.65	0.040016
32000	32	5.859375	6	31250	-2.34375	23.4375	23	32608.7	1.902174
32000	16	5.859375	6	31250	-2.34375	46.875	47	31914.89	-0.26596

		MCKOE=1				MCKOE=0			
22050	32	8.503401	9	20833.3333	-5.51776	34.01361	34	22058.82	0.040016
22050	16	8.503401	9	20833.3333	-5.51776	68.02721	68	22058.82	0.040016
16000	32	11.71875	12	15625	-2.34375	46.875	47	15957.45	-0.26596
16000	16	11.71875	12	15625	-2.34375	93.75	94	15957.45	-0.26596
11025	32	17.0068	17	11029.41176	0.040016	68.02721	68	11029.41	0.040016
11025	16	17.0068	17	11029.41176	0.040016	136.0544	136	11029.41	0.040016
8000	32	23.4375	23	8152.173913	1.902174	93.75	94	7978.723	-0.26596
8000	16	23.4375	23	8152.173913	1.902174	187.5	188	7978.723	-0.26596

16.4.8 Interrupt

16.4.8.1 Status flag

Share the same interrupt vector with SPI

Table 16-6 I2S interrupt status

Interrupt event	Interrupt register flag
TX BUFFER empty	TX_INTF
RX BUFFER non empty	RX_INTF
TX BUFFER underflow	UNDERRUN_INTF
RX BUFFER overflow	RXOERR_INTF
Frame transmission error	FRE_INTF

TX BUFFER empty (TX_INTF): Indicate that the current BUFFER data is read.

RX BUFFER non empty (RX_INTF): Indicate that the current BUFFER data is valid, and APB bus is readable.

TX BUFFER underflow (UNDERRUN_INTF): Indicate that the slave mode sends BUFFER data, which has been transmitted twice in the I2S bus.

RX BUFFER overflow (RXOERR_INTF): Indicate that the last receipt data is not read, and covered.

Frame transmission error (FRE_INTF): Indicate that I2S transmission protocol doesn't match (only valid in the slave mode).

Besides, in the I2S mode, there are 2 special status flags:

Busy flag (BSY): Indicate that I2S is transmitting.

Audio flag (CHSIDE): Indicate the ongoing transmission channel is left or right.

16.5 Register description

16.5.1 Overview of registers

Table 16-7 SPI_I2S register overview

Offset	Acronym	Register Name	Reset
0x00	SPI_I2S_TXREG	Transmission Data Register	0x00000000
0x04	SPI_I2S_RXREG	Receipt Data Register	0x00000000
0x08	SPI_I2S_CSTAT	Current Status Register	0x00002001
0x0C	SPI_I2S_INTSTAT	Interrupt Status Register	0x00000000
0x10	SPI_I2S_INTEN	Interrupt Enable Register	0x00000000
0x14	SPI_I2S_INTCLR	Interrupt Clear Register	0x00000000
0x18	SPI_I2S_GCTL	Global Control Register	0x00000004
0x1C	SPI_I2S_CCTL	Universal Control Register	0x00000008
0x20	SPI_I2S_SPBRG	Baud Rate Generator	0x00000002
0x24	SPI_I2S_RXDNR	Receipt Data Number Register	0x00000001
0x28	SPI_I2S_NSSR	Slave Chip Selection Register	0x000000FF
0x2C	SPI_I2S_EXTCTL	Data Length Control Register	0x00000008
0x30	SPI_I2S_I2SCFGR	I2S Configuration Register	0x00010000

16.5.2 SPI_I2S_TXREG Transmission Data Register

Address offset:0x00

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TXREG															
rw															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXREG															
rw															

Bit	Field	Description
31: 0	TXREG	Transmission data register Valid data bit is controlled by DW8_32: DW8_32=0, only valid in low 8 bit DW8_32=1, TXREG[31: 0] valid

16.5.3 SPI_I2S_RXREG Receipt Data Register

Address offset:0x04

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RXREG															
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXREG															
r															

Bit	Field	Description
31: 0	RXREG	Receipt data register Valid data bit is controlled by DW8_32. DW8_32=0:Only valid in low 8 bit DW8_32=1:RXREG[31:0] valid Note: The register is readable, but not writable.

16.5.4 SPI_I2S_CSTAT Current Status Register

Address offset:0x08

Reset value:0x0000 2001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.		CHSIDE	BUSY	RXFADDR				TXFADDR				RXAVL_4BYTE	TXFULL	RXAVL	TXEPT
		r	r	r				r				r	r	r	r

Bit	Field	Description
31: 14	Reserved	Always read as 0
13	CHSIDE	Audio flag bit 0: Ongoing transmission audio is left 1: Ongoing transmission audio is right Note: SPI mode is not usable, and I2S PCM is meaningless in the standard mode.
12	BUSY	Busy flag bit Indicate I2S or SPI transmission is ongoing
11: 8	RXFADDR	Number of valid byte in the current buffer
7: 4	TXFADDR	Number of valid byte in the transmission buffer

Bit	Field	Description
3	RXAVL_4BYTE	Valid data in the receipt buffer reaches 4 byte flag bit 0: Data in receipt buffer is less than 4 byte 1: Data in receipt buffer exceeds 4 byte Note: In the I2S mode, after receipt of one audio data, the bit is set. (for example CHLEN =0, the bit is set after receipt of 16bit).
2	TXFULL	Transmission buffer full flag bit 0: Transmission buffer is not full 1: Transmission buffer is full
1	RXAVL	Receipt valid byte data message bit When the receipt buffer receives a complete byte data, the bit is set. 0: Receipt buffer is empty 1: Receipt buffer is not empty Note: This bit is only readable, auto set and cleared by the hardware.
0	TXEPT	Transmission empty bit 0: Transmission buffer or shift register is not empty 1: Transmission buffer and shift register both are empty Note: This bit is only readable, auto set and cleared by the hardware.

16.5.5 SPI_I2S_INTSTAT Interrupt Status Register

Address offset:0x0C

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								FRE_INTF	TXEPT_INTF	RXFULL_INTF	RXMATCH_INTF	RXOERR_INTF	UNDERRUN_INTF	RX_INTF	TX_INTF
								r	r	r	r	r	r	r	r

Bit	Field	Description
31: 8	Reserved	Reserved, always read as 0
7	FRE_INTF	Frame transmission error interrupt flag bit Indicate I2S transmission protocol doesn’t match (only valid in the slave mode)
6	TXEPT_INTF	Transmission empty interrupt flag bit Auto set by the hardware, clear by writing the INTCLR. TXEPT_ICLR bit as ‘1’. 0: Transmission buffer or shift register is not empty 1: Transmission buffer and shift register both are empty Note:This bit is the interrupt status signal while TXEPT is the status signal.

Bit	Field	Description
5	RXFULL_INTF	Receipt buffer full interrupt flag bit Auto set by the hardware, clear by writing INTCLR. RXFULL_ICLR bit as ‘1’. 0: RX buffer is not full 1: RX buffer is full
4	RXMATCH_INTF	Receipt specified byte interrupt flag bit Auto set by the hardware, clear by writing INTCLR.RXMATCH_ICLR bit as ‘1’. 0: Still has not received specified byte in RXDNR register 1: Receive specified byte in RXDNR register
3	RXOERR_INTF	Receipt overflow error interrupt flag bit Auto set by the hardware, clear by writing INTCLR. RXOERR_ICLR bit as ‘1’. 0: No overflow error 1: Overflow error
2	UNDERRUN_INTF	SPI auxiliary mode underflow interrupt flag bit Auto set by the hardware, clear by writing INTCLR.UNDERRUN_ICLR bit as ‘1’. 0: No underrun error 1: Underrun error
1	RX_INTF	Receipt data valid interrupt flag bit Auto set by the hardware, clear by writing INTCLR. RX_ICLR bit as ‘1’. When the receipt buffer receives a complete valid data. 0: Receipt buffer is empty 1: Receipt buffer has received a complete valid data Note： a complete valid data’s byte numbers, refer to chapter 16.3.6.1 Status flags.
0	TX_INTF	Transmission buffer empty interrupt flag bit (buffer is empty, can write TXREG) Auto set by the hardware, auto clear when the transmission buffer is not empty. 0: Transmission buffer is not empty 1: Transmission buffer is empty

16.5.6 SPI_I2S_INTEN Interrupt Enable Register

Address offset:0x10

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								FRE_IEN	TXEPT_IEN	RXFULL_IEN	RXMATCH_IEN	RXOERR_IEN	UNDERRUN_IEN	RX_IEN	TX_IEN
								rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31: 8	Reserved	Reserved, always read as 0
7	FRE_IEN	Frame transmission error interrupt enable (FRE_IEN): 0: Interrupt disable 1: Interrupt enable
6	TXEPT_IEN	Transmission empty interrupt enable bit 0: Interrupt disable 1: Interrupt enable
5	RXFULL_IEN	Receipt buffer full interrupt enable bit 0: Interrupt disable 1: Interrupt enable
4	RXMATCH_IEN	Receipt specified byte interrupt enable bit 0: Interrupt disable 1: Interrupt enable
3	RXOERR_IEN	Receipt overflow error interrupt enable bit 0: Interrupt disable 1: Interrupt enable
2	UNDERRUN_IEN	SPI auxiliary mode underflow interrupt enable bit 0: Interrupt disable 1: Interrupt enable
1	RX_IEN	Receipt data interrupt enable bit 0: Interrupt disable 1: Interrupt enable
0	TX_IEN	Transmission buffer empty interrupt enable bit 0: Interrupt disable 1: Interrupt enable

16.5.7 SPI_I2S_INTCLR Interrupt Clear Register

Address offset:0x14

Reset value:0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								FRE_ICLR	TXEPT_ICLR	RXFULL_ICLR	RXMATCH_ICLR	RXOERR_ICLR	UNDERRUN_ICLR	RX_ICLR	TX_ICLR
								w	w	w	w	w	w	w	w

Bit	Field	Description
31: 8	Reserved	Reserved, always read as 0
7	FRE_ICLR	Frame transmission error interrupt clear: 0: Write 0 meaningless 1: Write 1 to clear interrupt
6	TXEPT_ICLR	Transmission empty interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
5	RXFULL_ICLR	Receipt buffer full interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
4	RXMATCH_ ICLR	Receipt specified byte interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
3	RXOERR_ ICLR	Receipt overflow error interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
2	UNDERRUN_ ICLR	SPI auxiliary mode underflow interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
1	RX_ICLR	Receipt interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt
0	TX_ICLR	Transmission buffer empty interrupt clear bit 0: Write 0 meaningless 1: Write 1 to clear interrupt

16.5.8 SPI_I2S_GCTL Global Control Register

Address offset:0x18

Reset value:0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.														PAD_SEL	
														rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAD_SEL			Res.	DW8_32	NSS	Res.	Res.				RXEN	TXEN	MODE	INTEN	SPIEN
rw				rw	rw						rw	rw	rw	rw	rw

Bit	Field	Description
31: 18	Reserved	Reserved, always read as 0

<table>
<tr><th>Bit</th><th>Field</th><th>Description</th></tr>
<tr><td>17: 13</td><td>PAD_SEL</td><td>PAD0，PAD1，PAD2，PAD3 respectively correspond to SCK, MOSI, NSS and MISO pins of pins. Mapping change of signal SCL, MOSI, NSS, MISO and PAD:
<table>
<tr><th>PAD_SEL</th><th>SCL</th><th>MOSI</th><th>NSS</th><th>MISO</th></tr>
<tr><td>0</td><td>PAD0</td><td>PAD1</td><td>PAD2</td><td>PAD3</td></tr>
<tr><td>1</td><td>PAD0</td><td>PAD1</td><td>PAD3</td><td>PAD2</td></tr>
<tr><td>2</td><td>PAD0</td><td>PAD2</td><td>PAD1</td><td>PAD3</td></tr>
<tr><td>3</td><td>PAD0</td><td>PAD2</td><td>PAD3</td><td>PAD1</td></tr>
<tr><td>4</td><td>PAD0</td><td>PAD3</td><td>PAD1</td><td>PAD2</td></tr>
<tr><td>5</td><td>PAD0</td><td>PAD3</td><td>PAD2</td><td>PAD1</td></tr>
<tr><td>6</td><td>PAD1</td><td>PAD0</td><td>PAD2</td><td>PAD3</td></tr>
<tr><td>7</td><td>PAD1</td><td>PAD0</td><td>PAD3</td><td>PAD2</td></tr>
<tr><td>8</td><td>PAD1</td><td>PAD2</td><td>PAD0</td><td>PAD3</td></tr>
<tr><td>9</td><td>PAD1</td><td>PAD2</td><td>PAD3</td><td>PAD0</td></tr>
<tr><td>10</td><td>PAD1</td><td>PAD3</td><td>PAD0</td><td>PAD2</td></tr>
<tr><td>11</td><td>PAD1</td><td>PAD3</td><td>PAD2</td><td>PAD0</td></tr>
<tr><td>12</td><td>PAD2</td><td>PAD0</td><td>PAD1</td><td>PAD3</td></tr>
<tr><td>13</td><td>PAD2</td><td>PAD0</td><td>PAD3</td><td>PAD1</td></tr>
<tr><td>14</td><td>PAD2</td><td>PAD1</td><td>PAD0</td><td>PAD3</td></tr>
<tr><td>15</td><td>PAD2</td><td>PAD1</td><td>PAD3</td><td>PAD0</td></tr>
<tr><td>16</td><td>PAD2</td><td>PAD3</td><td>PAD0</td><td>PAD1</td></tr>
<tr><td>17</td><td>PAD2</td><td>PAD3</td><td>PAD1</td><td>PAD0</td></tr>
<tr><td>18</td><td>PAD3</td><td>PAD0</td><td>PAD1</td><td>PAD2</td></tr>
<tr><td>19</td><td>PAD3</td><td>PAD0</td><td>PAD2</td><td>PAD1</td></tr>
<tr><td>20</td><td>PAD3</td><td>PAD1</td><td>PAD0</td><td>PAD2</td></tr>
<tr><td>21</td><td>PAD3</td><td>PAD1</td><td>PAD2</td><td>PAD0</td></tr>
<tr><td>22</td><td>PAD3</td><td>PAD2</td><td>PAD0</td><td>PAD1</td></tr>
<tr><td>23</td><td>PAD3</td><td>PAD2</td><td>PAD1</td><td>PAD0</td></tr>
</table></td></tr>
<tr><td>12</td><td>Reserved</td><td>Reserved, must be kept at reset value..</td></tr>
<tr><td>11</td><td>DW8_32</td><td>Transmission and receipt data register valid data selection
0:Only valid in low 8 bit
1:Valid in 32 bit data
Note: Fixed as 1 in the i2S mode</td></tr>
<tr><td>10</td><td>NSS</td><td>NSS output in the hardware or software control master mode
0:Control by the NSSR register value
1:Auto control by the hardware during data transmission
Note: Fixed as 0 in the i2S mode</td></tr>
<tr><td>9: 5</td><td>Reserved</td><td>Reserved, must be kept at reset value..</td></tr>
<tr><td>4</td><td>RXEN</td><td>Receipt enable bit
0: Receipt disable. Clear RX buffer at the same time
1: Receipt enable
Note: When the SPI only works in the host receipt mode, TXEN must be set as 0.</td></tr>
</table>

Bit	Field	Description
3	TXEN	Transmission enable bit 0:Transmission disable. Clear TX buffer at the same time 1: Transmission enable Note: In the host mode, the transmission and receipt occur simultaneously
2	MODE	Host mode bit 0: Slave mode (serial clock from external host) 1: Host mode (serial clock produced by the internal BRG)
1	INTEN	SPI/I2S interrupt enable bit 0: Disable SPI/I2S interrupt 1: Enable SPI/I2S interrupt
0	SPIEN	SPI/I2S selection bit 0 : SPI/I2S disable (reset status) 1 : SPI/I2S enable

16.5.9 SPI_I2S_CCTL Universal Control Register

Address offset:0x1C

Reset value:0x0000 0008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									CPHASEL	TXEDGE	RXEDGE	SPILEN	LSBFE	CPOL	CPHA
									rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31: 7	Reserved	Reserved, always read as 0
6	CPHASEL	CPHA polarity inversion selection 0 : CPHA setting remains unchanged. 1: Invert CPHA setting value. In case of CPHA as 1, the first data bit sampling begins from the second clock edge. In case of CPHA as 0, the first data bit sampling begins from the first clock edge. Note: Fixed as 1 in the I2S mode

Bit	Field	Description
5	TXEDGE	Transmission data phase bit adjustment (slave mode) 0: Transmit data to the bus after one valid clock edge. To be used in the low speed mode (slave mode input clock slower). 1: Immediately transmit data to the data bus. To be used in the high speed mode (slave mode input clock quicker, exceeding 10MHz). Note: It's recommended to configure the bit as 1 to prevent incompliance with AC characteristics because of quicker communication speed, or the data transmission in the slave mode failing to conform to the master sampling sequence requirement.
4	RXEDGE	Receipt data sampling clock edge selection bit (master mode) 0: Sample data in the middle of transmission data bit. 1: Sample clock in the tail clock edge of the transmission data (for high speed mode, move back the sampling time point to respond to the board level wiring and slave mode transmission data latency). Note: It's recommended to configure the bit as 1 to prevent incompliance with AC characteristics because of quicker communication speed, or the incorrect data receipt in the master mode
3	SPILEN	SPI data width bit This bit works after DW8_32 setting (DW8_32 =0); this bit should be kept as 1 during DW8_32=1. 0: 7 bit data 1: 8 bit data (default) Note: Fixed as 1 in the I2S mode
2	LSBFE	LSBFE:LSB front enable bit 0: Data transmission or receipt highest bit in front 1: Data transmission or receipt lowest bit in front Note: Fixed as 0 in the I2S mode
1	CPOL	Clock polarity flag bit 0: Clock in low level in idle status (between two transmissions) 1: Clock in high level in idle status (between two transmissions)
0	CPHA	Clock phase selection bit 0: The first data bit sampling begins from the second clock edge 1: The first data bit sampling begins from the first clock edge Note: Fixed as 0 in the I2S mode

16.5.10 SPI_I2S_SPBRG Baud Rate Generator

Address offset:0x20

Reset value:0x0000 0002

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPBRG															
rw															

Bit	Field	Description
31: 16	Reserved	Reserved, always read as 0
15: 0	SPBRG	SPI baud rate control register for baud rate Baud rate formula: Baud rate = Fpclk/SPBRG (Fpclk is APB clock frequency) Note: Don't write 0 and 1 in the register.

16.5.11 SPI_I2S_RXDNR Receipt Data Number Register

Address offset:0x24

Reset value:0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXDNR															
rw															

Bit	Field	Description
31: 16	Reserved	Reserved, always read as 0
15: 0	RXDNR	The register is used to hold a count of to be received in the next receipt process Note: The register value is valid when SPI is in the host receipt mode. The default value is 1. The register value changes by MCU write value, and don't write the '0' value in the register.

16.5.12 SPI_I2S_NSSR Slave Chip Selection Register

Address offset:0x28

Reset value:0x0000 00FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Res.	NSS
	rw

Bit	Field	Description
31: 1	Reserved	Reserved, must be kept at reset value..
0	NSS	The chip selection output signal in the master mode. Low valid, the bit is invalid in the slave mode 0:Slave device is not selected (allow communication between the slave device and master mode) 1: Slave device is selected

16.5.13 SPI_I2S_EXTCTL Data Length Control Register

Address offset:0x2C

Reset value:0x0000 0008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.											EXTLEN				
											rw				

Bit	Field	Description
31: 5	Reserved	Reserved, always read as 0
4: 0	EXTLEN	Control SPI data length 0 0000: 32 bit 0 0001: 1 bit 0 0010: 2 bit 0 0011: 3 bit …… 1 1100: 28 bit 1 1101: 29 bit 1 1110: 30 bit 1 1111: 31 bit Note: It's valid only when the SPI_I2S_GCTL.DW8_32 bit is '1'; in case of DW8_32=0, it's mandatory to keep the initial value 5'h8). Configuration invalid in the I2S mode: In the I2S mode, in case of CHLEN=1, EXTLEN value is fixed as 5'b00000 (32 bit); In the I2S mode, in case of CHLEN=0, EXTLEN value is fixed as 5'b10000 (16 bit)

16.5.14 SPI_I2S_I2SCFGR I2S Configuration Register

Address offset:0x30

Reset value:0x00010000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.							I2SDIV								
							rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				MCKOE	SPI_I2S	Res.			PCMSYN	I2SSTD		Res.	DATLEN		CHLEN
				rw	rw				rw	rw			rw		rw

Bit	Field	Description
31: 25	Reserved	Reserved, always read as 0
24: 16	I2SDIV	I2S prescaler division factor Note: Only valid in the master mode, and the bit value cannot be configured as 0; don’t export MCK drive clock in the master mode, namely in case of MCKOE =1, this bit is not configured as 1 (don't support 1 division)
15: 12	Reserved	Reserved, always read as 0
11	MCKOE	MCK clock output enable 0: Master mode drive clock MCK output disable 1: Master mode drive MCK clock output enable
10	SPI_I2S	Module function selection 0: Enable SPI function; 1: Enable I2S function. Note: When configuring this bit as 1, namely in the I2S mode, the following register functions will be fixed by the hardware, and software configuration is not required. GCTL register DW8_32, NSS bit; CCTL register CPHASEL, SPILEN, LSBFE, CPHA bit; EXTCTL register EXTLEN bit.
9: 7	Reserved	Reserved, always read as 0
6	PCMSYNC	PCM standard frame sync mode 0: Short frame sync 1: Long frame sync
5: 4	I2SSTD	I2S standard 00: PHILIPS standard 01: MSB alignment standard 10: LSB alignment standard 11: PCM standard
3	Reserved	Reserved, always read as 0

Bit	Field	Description
2: 1	DATLEN	Data length 00:16 bit width 01:24 bit width 10:32 bit width 11: Setting disable Note: In case of CHLEN = 0, DATLEN value is fixed as 2'b00 by the hardware.
0	CHLEN	Audio length (data bit number contained in each audio) 0 : 16 bit width 1: 32 bit width

17 I2C Inter-Integrated Circuit Interface

17.1 Overview

I2C (inter-integrated circuit) bus interface provides support for serial interconnection between chips in the microcontroller. It provides multi-master capability to control all I2C bus-specific sequencing, protocol, arbitration and timing.

The I2C bus is a two-wire serial interface, where the serial data line (SDA) and the serial clock line (SCL) pass information between the devices connected to the bus. Each device is addressable by a unique address and can operate as a transmitter or receiver. Besides, the device can also be considered as a master or slave when it conducts data transfer. A master is a device that initiates the data transfer on the bus and generates clock signals that allow the transfer. At this point, any device addressed is considered as a slave.

I2C has two optional speed modes: Standard mode (data transfer rate up to 100Kbps) and Fast mode (data transfer rate up to 400Kbps).

17.2 Main characteristics

- I2C bus protocol converter/parallel-bus
- Half-duplex synchronous operation
- Master and Slave modes supported
- 7-bit/10-bit address format supported
- Standard mode (up to 100Kbps) and Fast mode (up to 400Kbps) supported
- START, STOP, RESTART, and ACK generation and detection supported
- Respective 2-byte Tx and Rx FIFOs
- Spike-free circuit added to SCLI and SDAI
- Interrupt or polled-mode operation supported
- Multiple slave addresses supported (see the I2C_SLVMASK register description for details)

17.3 Functional description

17.3.1 Functional block diagram

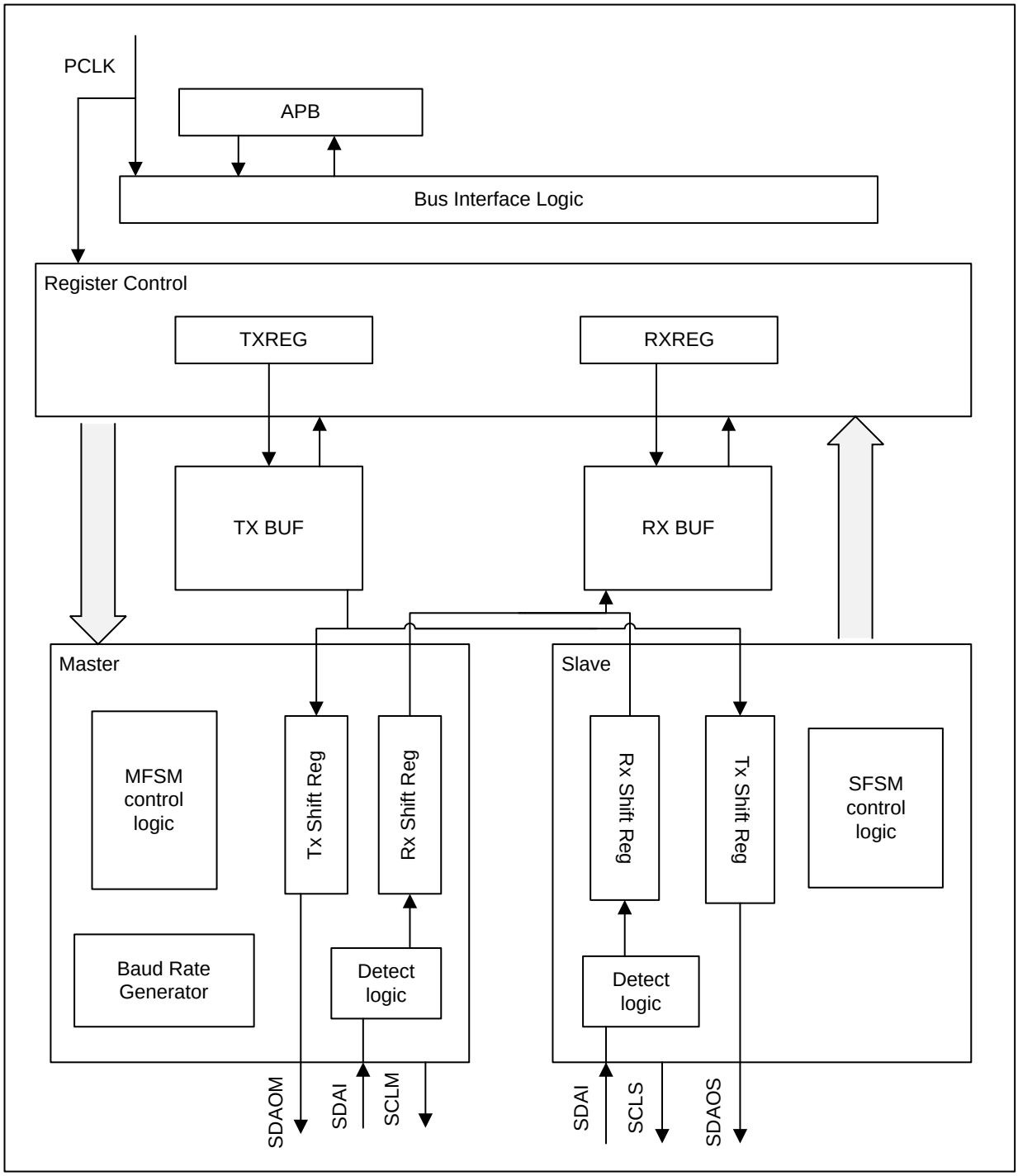


Figure 17-1 I2C functional block diagram

17.3.2 Pin definitions

Table 17-1 Pin definitions

Pin Name	Attribute	Description
I2C_SCL	I/O	I2C clock
I2C_SDA	I/O	I2C data

Note: When used, the I2C_SCL and I2C_SDA should be in open-drain mode by configuration. Please refer to the GPIO section for configuration method.

17.3.3 I2C protocol

17.3.3.1 Start and STOP conditions

When the bus is idle, SCL and SDA are pulled up by external pull-up resistors. The data transfer initiated in the Master begins with a START condition. A high to low transition on the SDA line while SCL is high defines a START condition. The data transfer in the Master is terminated by the Master issuing a STOP condition. A low to high transition on the SDA line while SCL is high defines a STOP condition. A timing diagram of Start and STOP conditions is shown as below. SDA must maintain steady during the data transfer when SCL is 1.

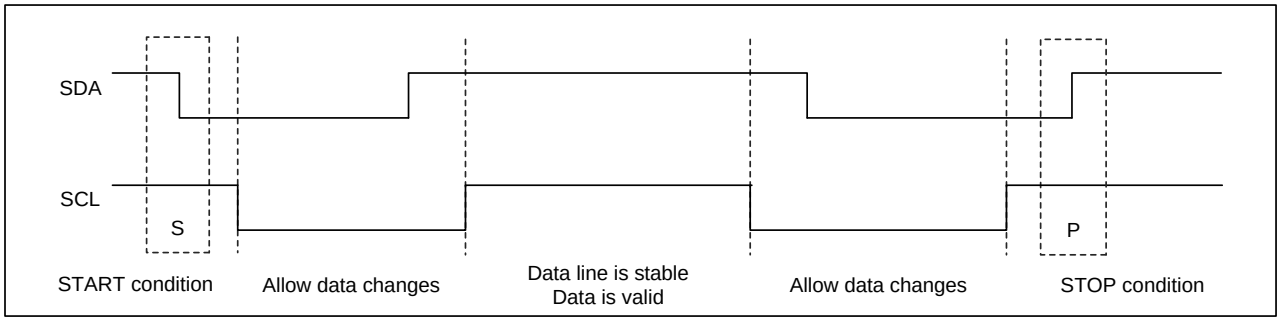


Figure 17-2 Start and STOP conditions

17.3.3.2 Addressing protocol

I2C has two address formats: 7-bit address format and 10-bit address format.

- 7-bit address format

The slave address is the first 7 bits (bit 7:1) of the first byte sent following a START condition (S). The least significant bit (bit 0) determines the data transfer direction. If bit 0 is 0, then the Master writes data to the Slave; if bit 0 is 1, then the Master reads data from the Slave.

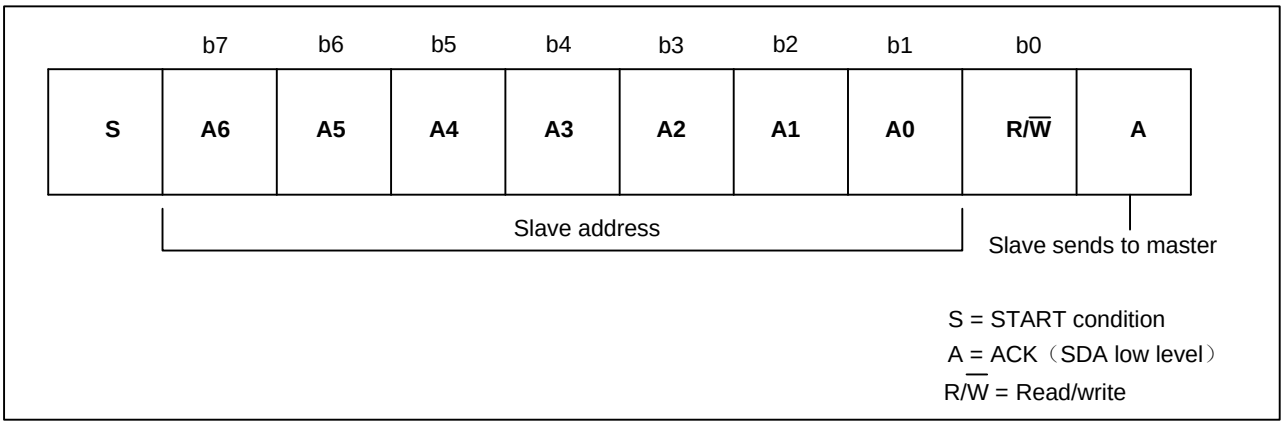


Figure 17-3 7-bit address format

- 10-bit address format

During 10-bit addressing, two bytes should be transferred to set the 10-bit address. The first byte contains the following bit definition. The first five bits (bits 7:3) notify the slaves that this is a 10-bit transfer followed by the next two bits (bits 2:1), which set the slaves address bits 9:8, and the LSB bit (bit 0) determines the data transfer direction (R/W). The second byte sets lower eight bits of the 10-bit address.

For details, see the diagram below:

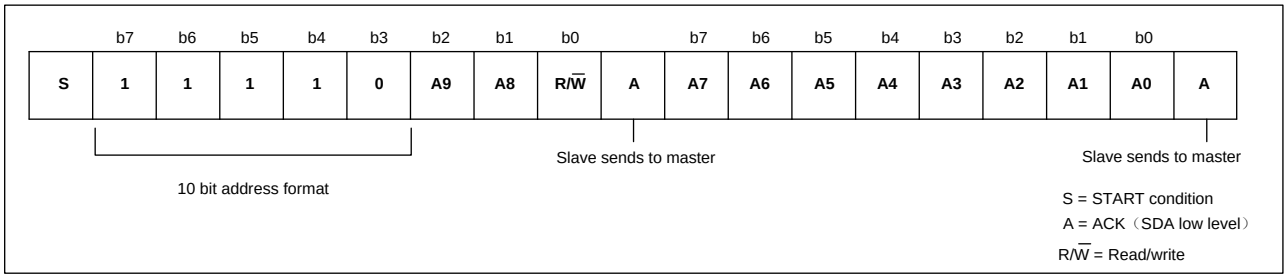


Figure 17-4 10-bit address format

The following table defines the special purpose and reserved addresses of the first byte for I2C:

Table 17-2 First byte for I2C

Slave Address	R/W Bit	Description
0000 000	0	General Call address. I2C places the data in the receive buffer and issues a General Call interrupt.
0000 000	1	Start byte
0000 001	x	CBUS address. I2C ignores these accesses.
0000 010	x	Reserved
0000 011	x	Reserved
0000 1xx	x	Reserved
1111 1xx	x	Reserved
1111 0xx	x	10-bit slave addressing

17.3.3.3 Transmitting and receiving protocol

The master can initiate data transmission and reception to or from the bus, acting as either a master-transmitter or master-receiver. A slave responds to requests from the master, acting as either a slave-transmitter or slave-receiver.

- Master-transmitter and Slave-receiver

All data is transmitted in byte format, with no limit on the number of bytes per data transfer. After the master sends the address and R/W bit or the master transmits a byte of data to the slave, the slave-receiver must respond with the acknowledge signal (ACK). When a slave-receiver does not respond with an ACK pulse, the master aborts the transfer by issuing a STOP condition. If a slave is not able to respond, the slave must leave the SDA line high so that the master can abort the transfer.

If the master-transmitter is transmitting data as shown in the following figure, then the slave-receiver responds to the master-transmitter with an ACK pulse after every byte of data is received.

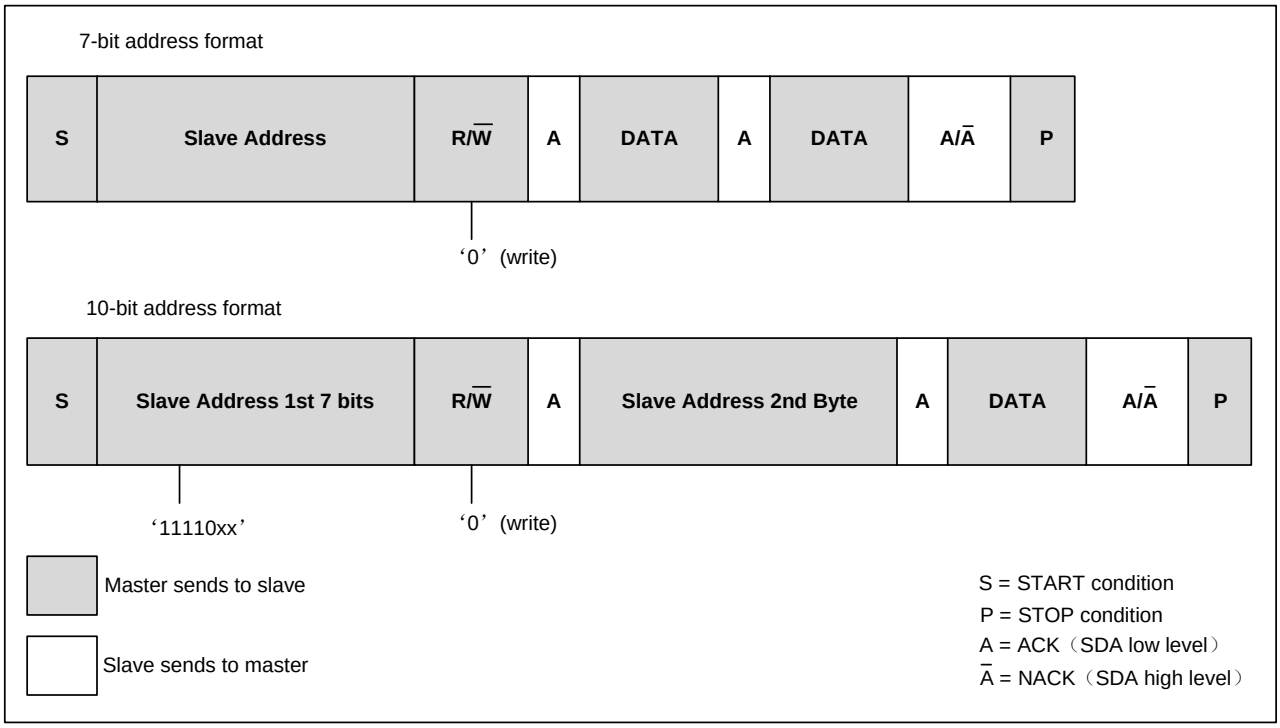


Figure 17-5 Master-transmitter protocol

- Master-receiver and Slave-transmitter

If the master is receiving data as shown in the following figure, then the master responds to the slave-transmitter with an ACK pulse after a byte of data has been received, except for the last byte. This is the way the master-receiver notifies the slave-transmitter that this is the last byte. The slave-transmitter relinquishes the SDA line after detecting the No Acknowledge (NACK) so that the master can issue a STOP condition.

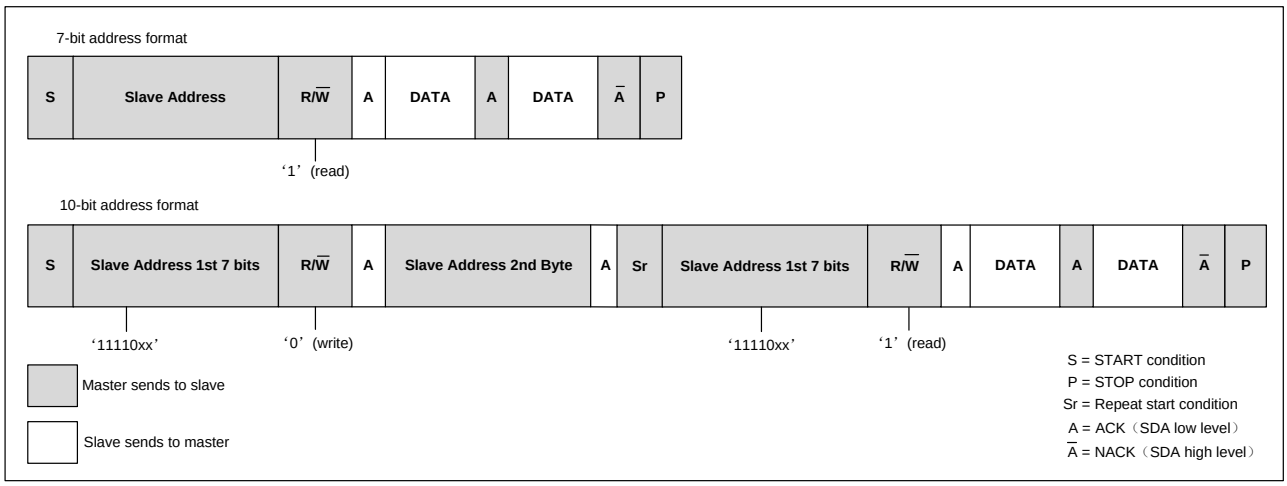


Figure 17-6 Master-receiver protocol

When a master does not want to relinquish the bus with a STOP condition, the master can issue a RESTART condition. This is identical to a START condition except it occurs after the ACK. Operating in Master mode, the I2C interface can then communicate with the same slave using a transfer of a different direction.

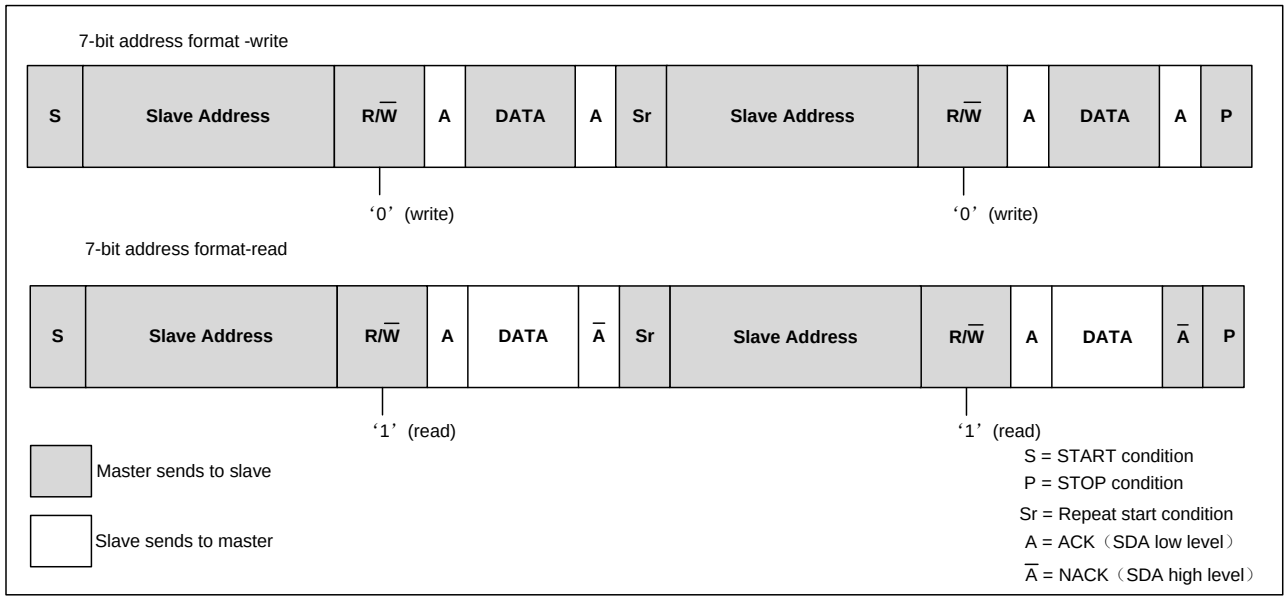


Figure 17-7 Master-transmitter and receiver protocol with a RESTART (SR)

- Start Byte transfer protocol

The Start Byte transfer protocol is set up for systems that do not have a dedicated I2C hardware module. When I2C is set as a master, it supports the generation of Start Byte transfers at the beginning of every transfer in case a slave device requires it.

This protocol consists of seven zeros followed by a 1, as illustrated in the following figure. This allows the processor that is polling the bus to under-sample the address phase. Once the processor detects a 0, it switches from the under sampling rate to the correct rate of the master.

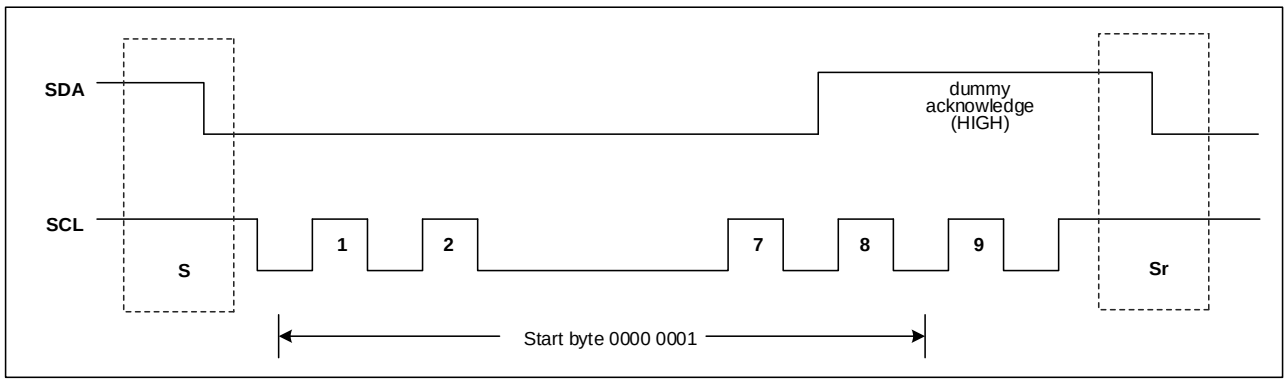


Figure 17-8 Start Byte transfer

The Start Byte procedure:

1. Master generates a START condition
2. Master transmits the Start byte (0000 0001)
3. Master transmits the ACK clock pulse (to conform with the byte handling format used on the bus)
4. No slave responds with an ACK pulse
5. Master generates a RESTART condition

A hardware I2C receiver does not respond to the Start Byte because it is a reserved address and resets after the RESTART condition is generated.

17.3.3.4 Tx FIFO management and Start, Stop, and Restart generation

I2C doesn't generate a STOP condition if the Tx FIFO becomes empty. Instead, it holds the SCL low and stalls the bus until Tx FIFO receives new data. The STOP condition is generated only when I2C_CR.STOP (bit 9) or I2C_ENR.ABORT (bit 1) is set to '1'.

The figure below shows bits in the I2C_DR register.

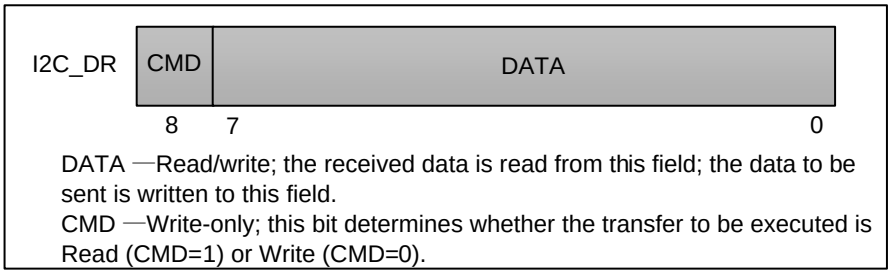


Figure 17-9 I2C_DR register

The following timing diagram illustrates the behavior of I2C operating in the master transmit mode when TX FIFO becomes empty.

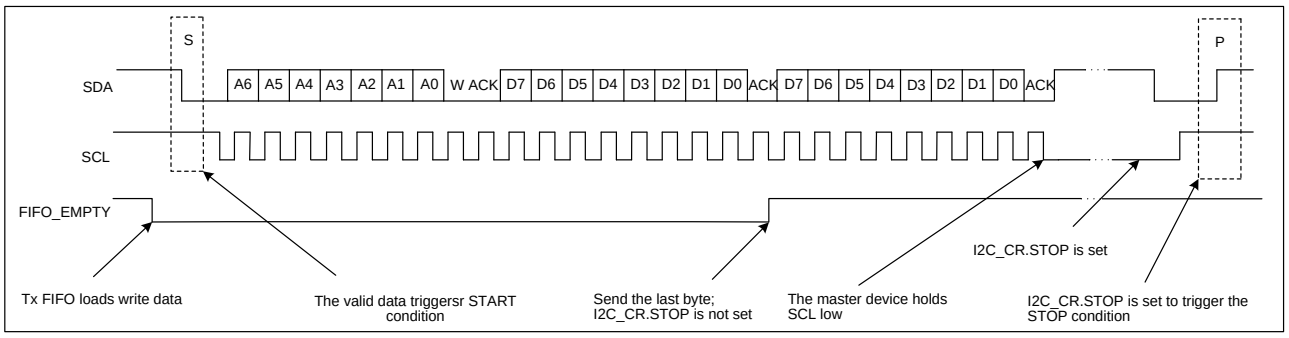


Figure 17-10 Master-transmitter, TX FIFO empties

The following timing diagram illustrates the behavior of I2C operating in the master receive mode when TX FIFO becomes empty.

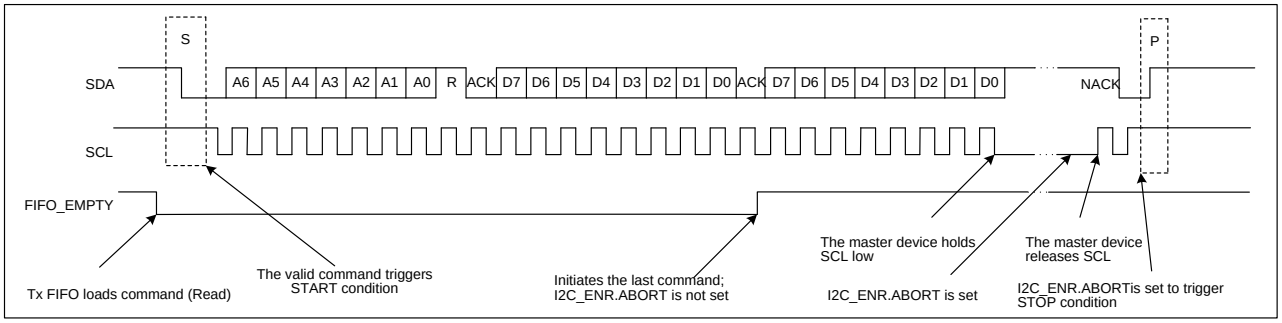


Figure 17-11 Master-receiver, TX FIFO empties

The following timing diagram illustrates the behavior of I2C operating in the master transmit mode when RESTART (I2C_CR.REPEN=1) is generated. If I2C_CR.REPEN (bit 5) is ‘0’, then a STOP followed by a START is issued, in order to replace the RESTART.

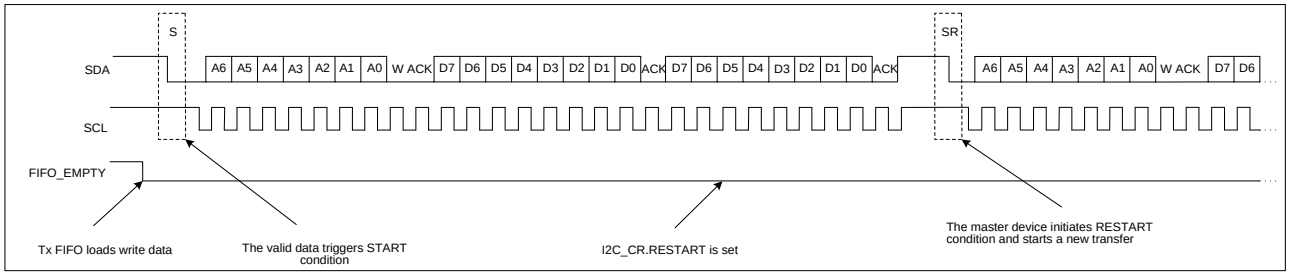


Figure 17-12 Master-transmitter, with RESTART

The following timing diagram illustrates the behavior of I2C operating in the master receive mode when RESTART is generated.

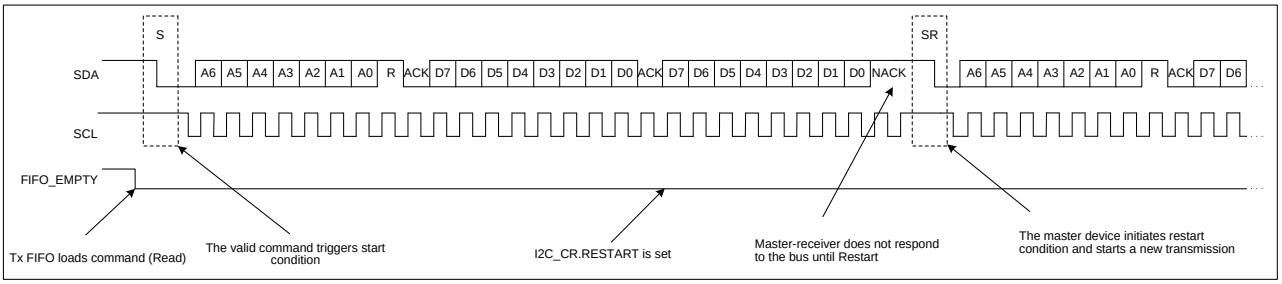


Figure 17-13 Master-receiver, with RESTART

17.3.3.5 Arbitration

If there are several masters on the same bus, there is an arbitration procedure if they try to take control of the bus by generating the START condition at the same time (if several masters try to take control of the bus at the same time, only one master can take control of the bus with intact messages). Once a master has control of the bus, no other master can take control until the first master sends a STOP condition and places the bus in an idle state.

Arbitration takes place on the SDA line, while the SCL line is high. If there are two or more masters trying to send messages to the bus, the master, which transmits a 1 while the other master transmits 0, loses arbitration. The master that lost arbitration can continue to generate clock pulses until the end of the byte transfer. If all masters are addressing the same device, the arbitration could go into the data phase.

Upon detecting that it has lost arbitration, the I2C interface stops generating SCL.

The following figure illustrates the timing of two masters arbitrating on the bus.

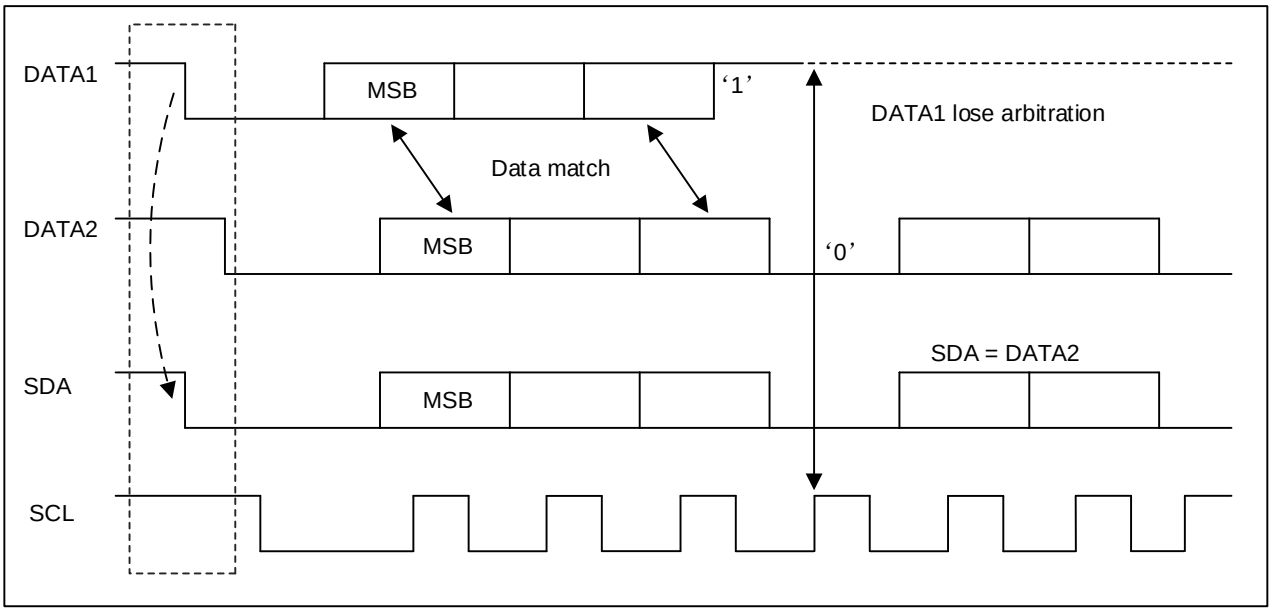


Figure 17-14 Dual master arbitration

17.3.3.6 Clock synchronization

When two or more masters try to transfer information on the bus at the same time, they must arbitrate and synchronize the SCL clock. All masters generate their own clock to transfer messages. Data is only valid during the high period of clock. Clock synchronization is performed using the wired-AND connection to the SCL signal. When the master transitions the SCL clock to 0, the master starts counting the low time of the SCL and transitions the SCL to 1 at the beginning of the next clock period. However, if another

master is holding the SCL line to 0, then the master goes into a wait state until the SCL transitions to 1.

All masters then count off their high time, and the master with the shortest high time transitions the SCL to 0. The masters then counts out their low time and the one with the longest low time forces the other masters into a wait state. Therefore, a synchronized SCL clock is generated, which is illustrated in the following figure.

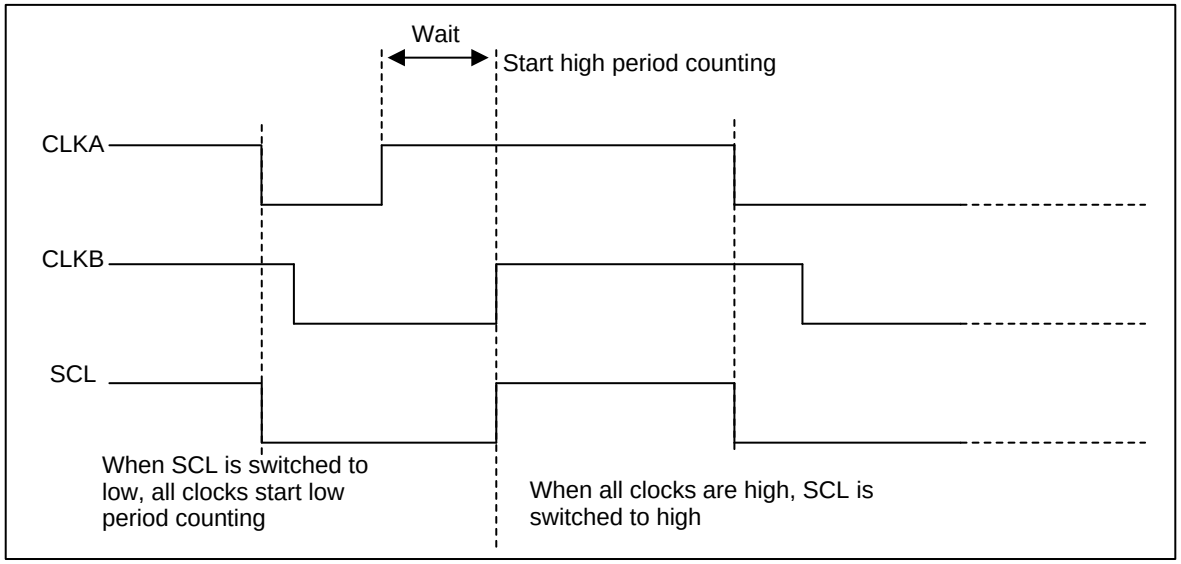


Figure 17-15 Clock synchronization (schematic diagram)

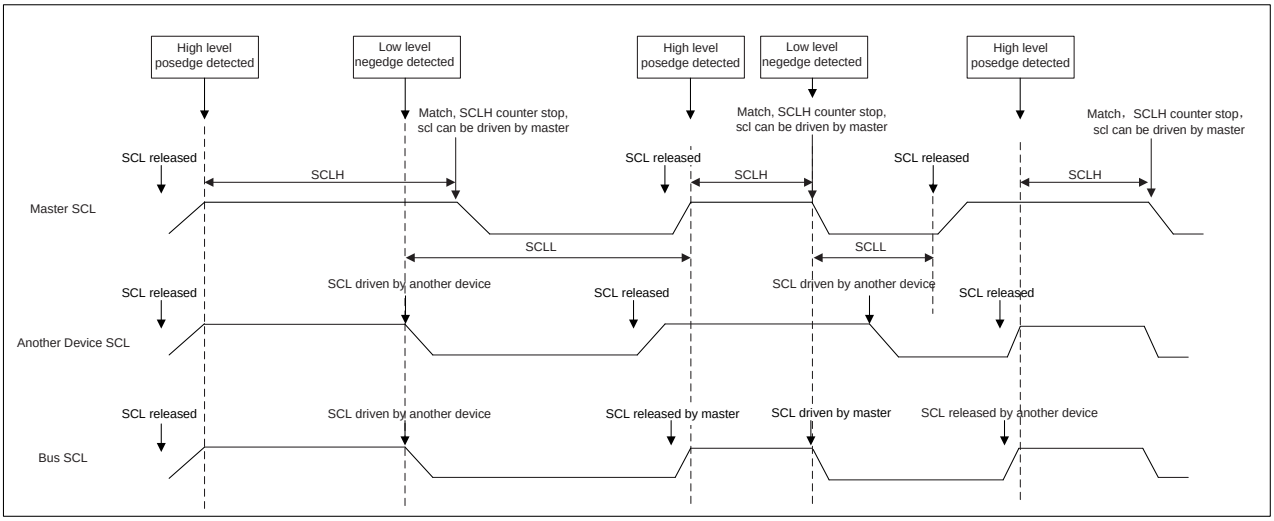


Figure 17-16 Clock synchronization (timing diagram)

17.3.3.7 SCL configuration

For SCL configuration of I2C, refer to the following:

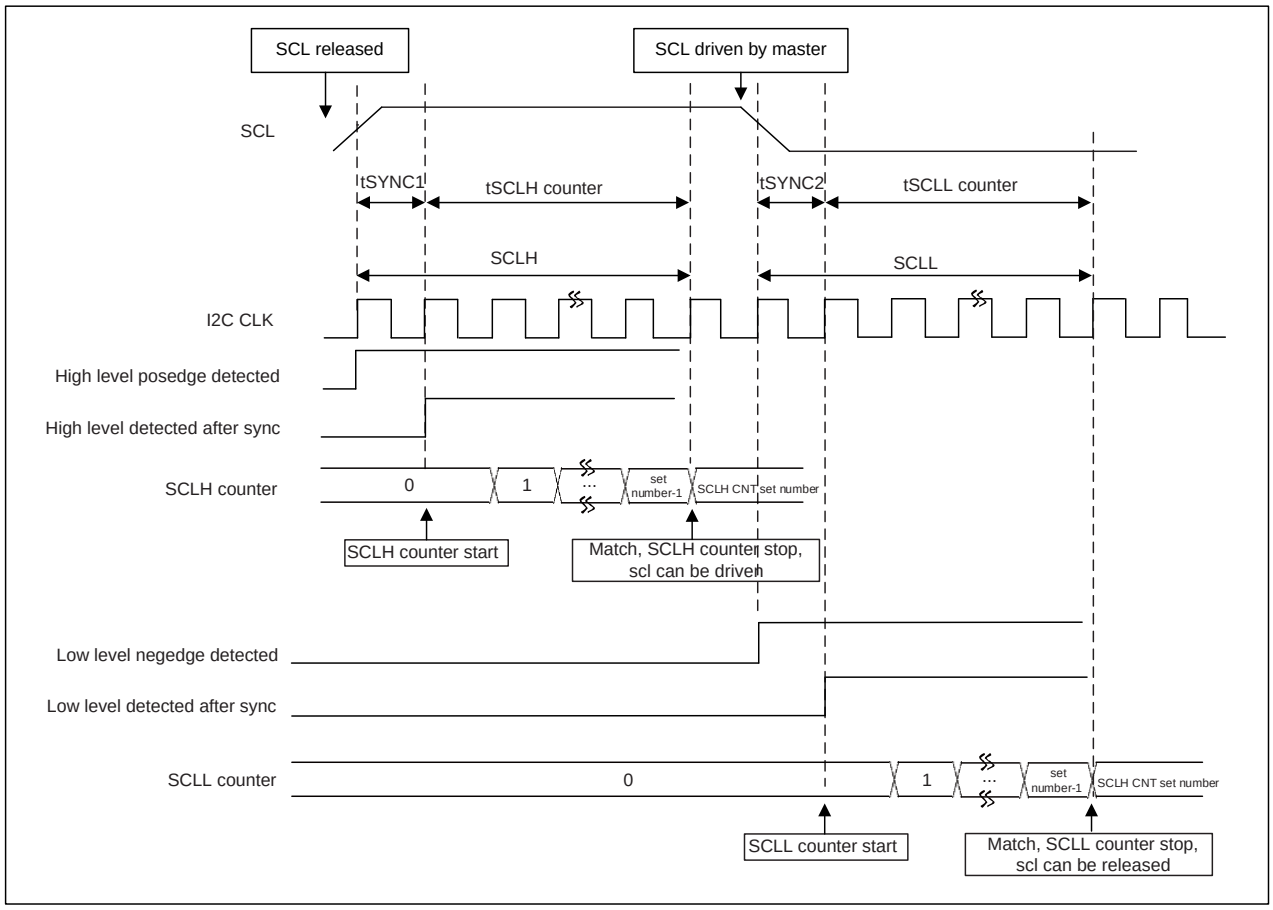


Figure 17-17 SCL generation timing

- Standard mode:

SCLH = (SSHR + 12) x I2C CLK + tSYNC1

SCLL = (SSLR + 1) x I2C CLK + tSYNC2

Note: tSYNC1 is equal to 0 ~ 1 I2C CLK and tSYNC2 is equal to 0 ~ 1 I2C CLK

- Fast mode:

SCLH = (FSHR + 12) x I2C CLK + tSYNC1

SCLL = (FSLR + 1) x I2C CLK + tSYNC2

Note: tSYNC1 is equal to 2 ~ 3 I2C CLK and tSYNC2 is equal to 2 ~ 3 I2C CLK

17.3.4 Operating mode

I2C interface can operate in one of the four following modes:

- Slave transmitter mode
- Slave receiver mode
- Master transmitter mode
- Master receiver mode

Note: I2C can only operate in the Master or Slave mode but it cannot operate in both modes at the same time. Therefore, make sure that the register I2C_CR.DISSLAVE (bit 6) and I2C_CR.MASTER (bit 0) are not set to 0 and 1 (or 1 and 0 respectively).

17.3.4.1 Slave mode

- Initial configuration

1. Disable I2C by writing a 0 to I2C_ENR.ENABLE (bit 0).

2. Configure the I2C_SAR register to set the slave address (this is the address to which the I2C responds).

3. Configure the I2C_CR.SLAVE10 (bit 3) to specify which type of addressing is supported (7- or 10-bit address); write 0 to I2C_CR.DISSLAVE (bit 6) and I2C_CR.MASTER (bit 0).

4. Enable the I2C by setting the I2C_ENR.ENABLE (bit 0) to 1.

- Slave-transmitter operation for a single byte

When another I2C master device addresses the I2C interface and requests data, the I2C interface acts as a slave-transmitter and the following steps occur:

1. The other I2C master device initiates an I2C transfer with an address that matches the slave address in the I2C_SAR register.

2. The I2C interface acknowledges the sent address and recognizes the direction of the transfer to indicate that it is acting as a slave-transmitter.

3. The I2C interface asserts the RD_REQ interrupt (bit 5 of the I2C_RAWISR) and holds the SCL line low. The bus stays in the wait state until the software responds. If the RD_REQ interrupt has been masked (I2C_IMR bit 5 = 0), then it is recommended that CPU does periodic query of the I2C_RAWISR register:

a. Reads that indicate I2C_RAWISR.RD_REQ (bit 5) being set to 1 must be treated as the equivalent of the RD_REQ interrupt being asserted.

b. Software must then act to satisfy the I2C transfer requirements.

c. The timing interval is generally 10 times the SCL clock period (for example, for 400kbps, the timing interval is 25us).

4. If there is any data remaining in the TX FIFO before receiving the read request, the I2C interface asserts a TX_ABRT interrupt (I2C_RAWISR bit 6) to flush the old data from the TX FIFO (when I2C_CR. SLV_TX_ ABRT_DIS = 0).

Therefore, it is necessary for software to clear the TX_ABRT interrupt by reading the I2C_TX_ABRT register before attempting to write into the TX FIFO. If the TX_ABRT interrupt has been masked (I2C_IMR bit 6 = 0), then it is recommended that CPU does periodic query of the I2C_RAWISR register: Reads that indicate I2C_RAWIS.TX_ABORT (bit 6) being set to 1 must be treated as the equivalent of the TX_ABRT

interrupt being asserted.

5. Software writes data to the I2C_DR register and writes a 0 in bit 8.

6. Software must clear the RD_REQ (I2C_RAWISR bit 5) and TX_ABRT (I2C_RAWISR bit 6) interrupts.

7. The I2C interface releases SCL and transmits the byte.

8. The master controls the bus by issuing a RESTART condition or releases the bus by issuing a STOP condition.

- Slave-receiver operation for a single byte

When another master device addresses the I2C interface and sends data, the I2C interface acts as a slave-receiver and the following steps occur:

1. Another I2C master device initiates an I2C transfer with a sending address that matches the slave address in the I2C_SAR register.

2. The I2C interface acknowledges the sent address and recognizes the direction of the transfer to indicate that it is acting as a slave-receiver.

3. I2C interface receives the transmitted data from master and places it in the receive buffer.

4. I2C interface asserts the RX_FULL interrupt (I2C_RAWISR bit 2).

If the RX_FULL interrupt has been masked (I2C_IMR bit 2 = 0), then it is recommended that CPU does periodic query of the I2C_SR register. Reads that indicate I2C_SR.RFNE (bit 3) being set to 1 can be treated as the equivalent of the RX_FULL interrupt being asserted.

5. Software may read the I2C_DR register (bits 7:0) to get the data that was received.

6. The master controls the bus by issuing a RESTART condition or releases the bus by issuing a STOP condition.

- Program flow chart

The following chart shows a programming model when I2C interface is set as a slave:

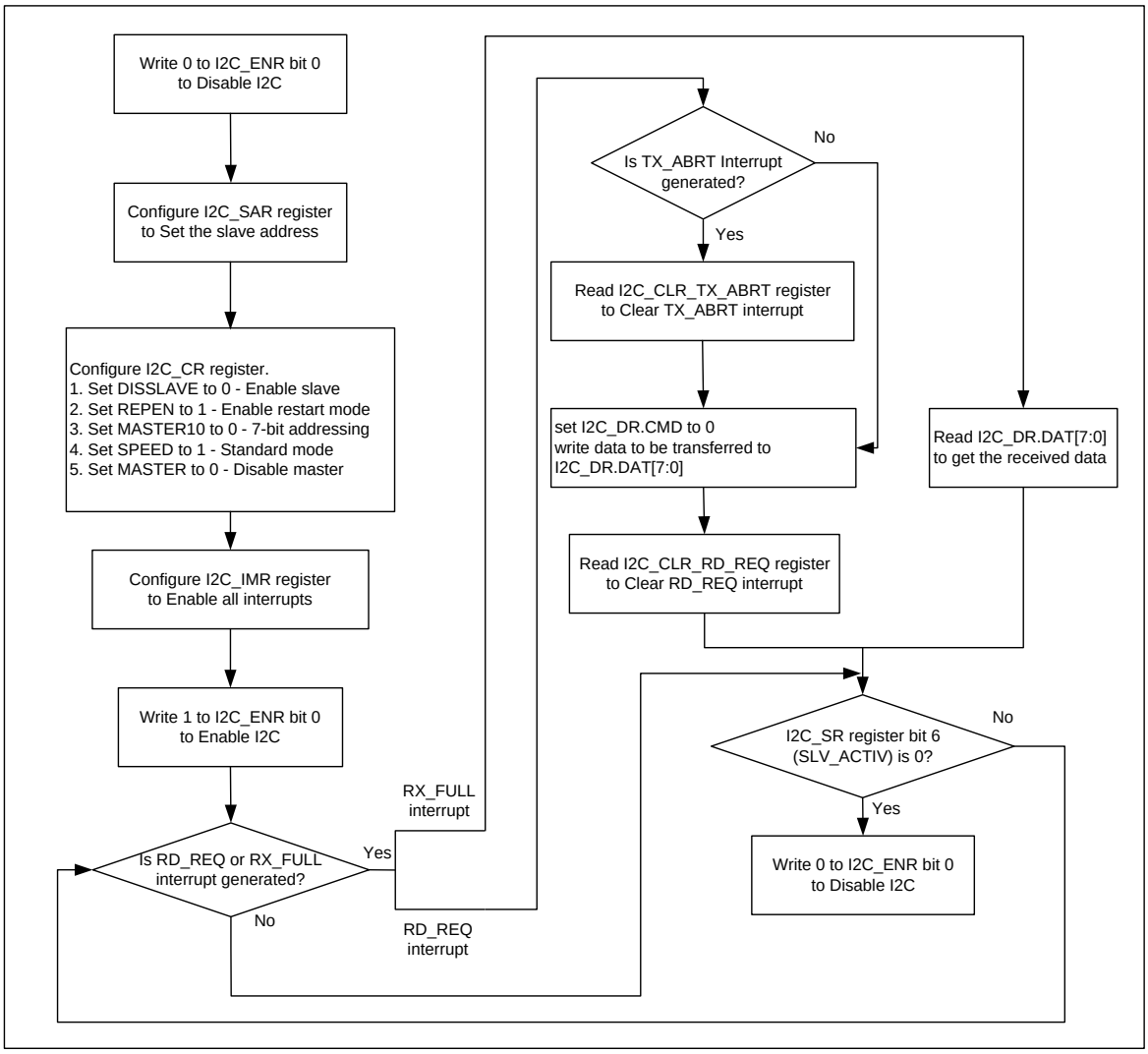


Figure 17-18 Flow chart (I2C interface as a slave)

- Slave-transfer operation for bulk transfers

In the standard I2C protocol, all transactions are single byte transactions and the program responds to a master read request by writing one byte into the slave's TX FIFO. When a slave (slave-transmitter) receives a read request (RD_REQ) from the master (master-receiver), there should be at least one entry placed into the slave-transmitter's TX FIFO. The I2C interface is designed to handle more data in the TX FIFO so that subsequent read requests can receive that data without raising an interrupt. Ultimately, this eliminates latencies significantly being incurred by the interrupt for data each time.

This mode is only applicable when I2C interface is acting as a slave-transmitter. If the master acknowledges the data sent by the slave-transmitter and there is no data in the slave's TX FIFO, the I2C interface holds the SCL line low while it raises the read request interrupt (RD_REQ); moreover, it will not release the SCL line unless the data in the TX FIFO is prepared.

If the RX_REQ interrupt has been masked (I2C_ISR bit 5 = 0), then it is recommended that the software does periodic query of the I2C_RAWISR register. Reads that indicate I2C_RAWISR.RX_REQ (bit 5) being set to 1 can be treated as the equivalent of the RX_REQ interrupt being asserted.

The RD_REQ interrupt is raised upon a read request, and like interrupts, must be cleared when exiting the interrupt service handling routine (ISR). The ISR allows you to write 1 byte or more than 1 byte into the TX FIFO. During the transmission of these bytes to the master, if the master acknowledges the last byte, then the slave must raise the RD_REQ interrupt request again because the master is requesting for more data.

If the master is to receive n bytes from the I2C interface but the program wrote a number of bytes larger than n to the TX FIFO, then when the slave finishes sending the requested n bytes, it clears the TX FIFO and ignores any excess bytes.

17.3.4.2 Master mode

- Initial configuration

1. Disable I2C by writing a 0 to I2C_ENR.ENABLE (bit 0).
2. Set the I2C operation speed mode (Stand mode or Fast mode) by configuring I2C_CR.SPEED (bit 2:1). Meanwhile, make sure I2C_CR.DISSLAVE (bit 6) = 1 and I2C_CR.MASTER (bit 0) = 1.
3. Write to the I2C_TAR (this register can also be set to configure a General Call address or a Start Byte command) the address of the I2C device to be addressed.
4. Enable the I2C interface module by setting the I2C_ENR.ENABLE (bit 0) to 1.
5. Now write the transfer direction and data to I2C_DR. Thus, I2C interface will generate a START condition and send the address byte data.

If the I2C_DR register is written before the I2C interface is enabled, the data and commands are lost as the buffers are kept cleared when the I2C interface is not enabled.

- Master transmit and Master receive

The I2C interface supports switching back and forth between reading and writing dynamically. When transmitting data, write the data to be written to the lower byte of I2C_DR. The I2C_DR.CMD (bit 8) should be written to 0 for write operations. Subsequently, a read command does not require the configuration of lower byte of I2C_DR, and a 1 should be written to I2C_DR.CMD. If the TX FIFO is empty, the I2C module holds SCL low until the next command is written into the TX FIFO.

- Program flow chart

The following flow chart shows a programming model when I2C interface is set as a master:

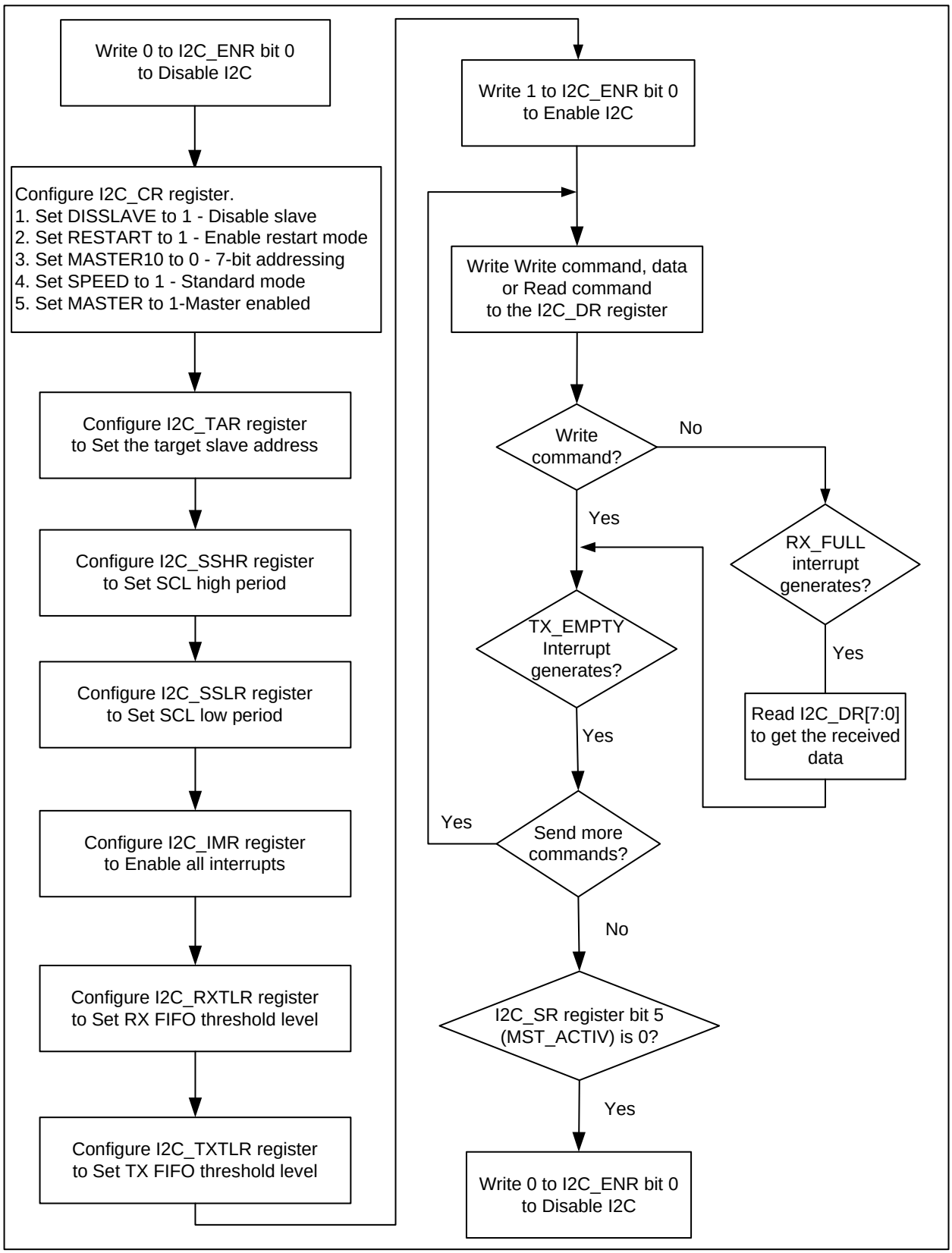


Figure 17-19 Flow chart (I2C interface as a master)

17.3.4.3 Abort transfer

The I2C_ENR.ABORT (bit 1) allows the software to relinquish the I2C bus before completing the issued transfer commands from the TX FIFO. In response to an ABORT request, the I2C module issues the STOP condition over the I2C bus, alongside TX FIFO flush. Aborting the transfer is allowed only in master mode of operation.

- Procedure:

1. Stop filling the TX FIFO (I2C_DR) with new commands
2. Set I2C_ENR.ABORT (bit 1) to 1
3. Wait for the TX_ABRT interrupt

17.3.5 Interrupt

The table below gives the list of I2C interrupt bits, as well as the way to set and clear them. Some bits are set by hardware and cleared by software, others are set and cleared by hardware.

Table 17-3 Setting and clearing the interrupt bits

Interrupt bit	Set by hardware/Cleared by software	Set and cleared by hardware
GEN_CALL	√	x
START_DET	√	x
STOP_DET	√	x
ACTIVITY	√	x
RX_DONE	√	x
TX_ABRT	√	x
RD_REQ	√	x
TX_EMPTY	x	√
TX_OVER	√	x
RX_FULL	x	√
RX_OVER	√	x
RX_UNDER	√	x

The figure below describes the operation of the interrupt registers where the bits are set by hardware and cleared by software.

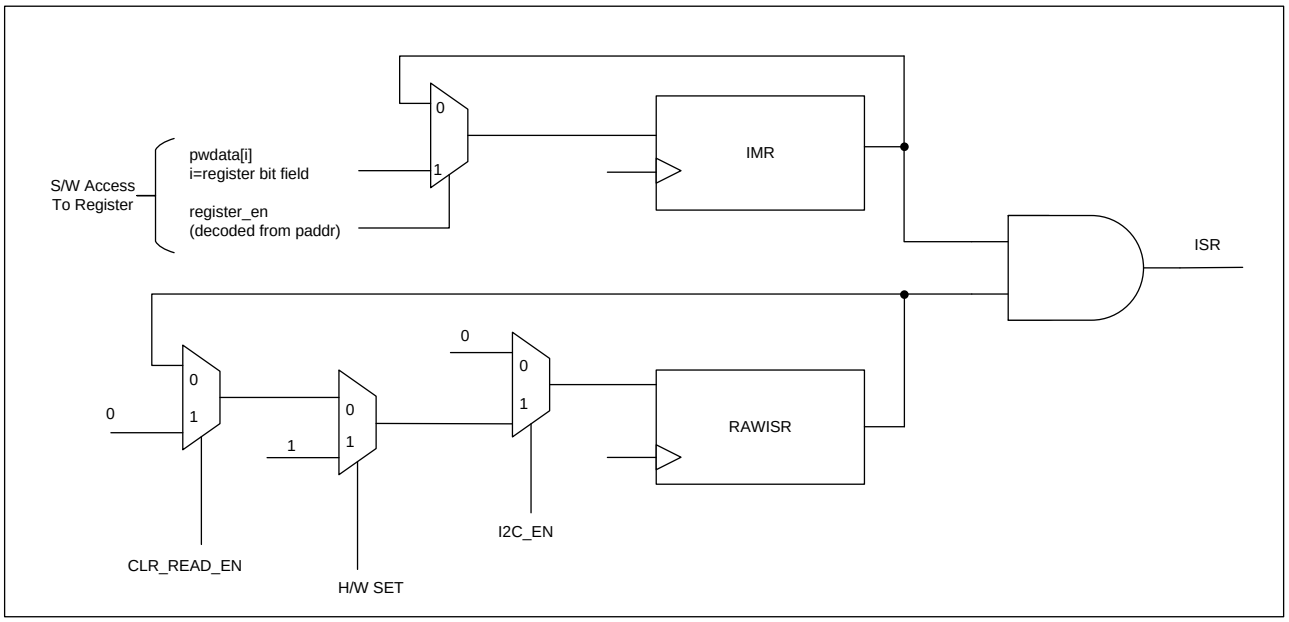


Figure 17-20 I2C interrupt mechanism

17.4 Register description

17.4.1 Overview of registers

Table 17-4 Overview of I2C register

Offset	Acronym	Register Name	Reset
0x00	I2C_CR	Control Register	0x0000007F
0x04	I2C_TAR	Target Address Register	0x00000055
0x08	I2C_SAR	Slave Address Register	0x00000055
0x10	I2C_DR	Data Command Register	0x00000000
0x14	I2C_SSHR	Standard Mode SCL High Count Register	0x00000190
0x18	I2C_SSLR	Standard Mode SCL Low Count Register	0x000001D6
0x1C	I2C_FSHR	Fast Mode SCL High Count Register	0x0000003C
0x20	I2C_FSLR	Fast Mode SCL Low Count Register	0x00000082
0x2C	I2C_ISR	Interrupt Status Register	0x00000000
0x30	I2C_IMR	Interrupt Mask Register	0x000008FF
0x34	I2C_RAWISR	RAW Interrupt Status Register	0x00000000
0x38	I2C_RXTLR	Receive Threshold Register	0x00000000
0x3C	I2C_TXTLR	Transmit Threshold Register	0x00000000
0x40	I2C_ICR	Combined and Independent Interrupt Clear Register	0x00000000
0x44	I2C_RX_UNDER	RX_UNDER Interrupt Clear Register	0x00000000
0x48	I2C_RX_OVER	RX_OVER Interrupt Clear Register	0x00000000
0x4C	I2C_TX_OVER	TX_OVER Interrupt Clear Register	0x00000000
0x50	I2C_RD_REQ	RD_REQ Interrupt Clear Register	0x00000000
0x54	I2C_TX_ABRT	TX_ABRT Interrupt Clear Register	0x00000000

Offset	Acronym	Register Name	Reset
0x58	I2C_RX_DONE	RX_DONE Interrupt Clear Register	0x00000000
0x5C	I2C_ACTIV	ACTIVITY Interrupt Clear Register	0x00000000
0x60	I2C_STOP	STOP_DET Interrupt Clear Register	0x00000000
0x64	I2C_START	START_DET Interrupt Clear Register	0x00000000
0x68	I2C_GC	GEN_CALL Interrupt Clear Register	0x00000000
0x6C	I2C_ENR	Enable Register	0x00000000
0x70	I2C_SR	Status Register	0x00000006
0x74	I2C_TXFLR	Transmit FIFO Level Register	0x00000000
0x78	I2C_RXFLR	Receive FIFO Level Register	0x00000000
0x7C	I2C_HOLD	SDA Hold Time Register	0x00000001
0x94	I2C_SETUP	SDA Setup Time Register	0x00000064
0x98	I2C_GCR	General Call ACK Register	0x00000001
0xB0	I2C_SLVMASK	Slave Address Mask Register	0x000003FF
0xB4	I2C_SLVRCVADDR	Slave Address Receive Register	0x00000000

17.4.2 I2C_CR Control Register

Address offset: 0x00

Reset value: 0x0000 007F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			PAD_SEL	SLV_TX_ABRT_DIS	RESTART	STOP	EMPINT	STOPINT	DISSLAVE	REPEN	MASTER10	SLAVE10	SPEED		MASTER
			rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw

Bit	Field	Description
31:13	Reserved	Reserved, must be kept at reset value
12	PAD_SEL	PAD MUX Swap the PAD corresponding to SCL and SDA 0: PAD0 □ SCL；PAD1 □ SDA 1: PAD0 □ SDA；PAD1 □ SCL
11	SLV_TX_ABRT_DIS	When I2C is acting as a slave-transmitter 0: Clearing TX FIFO after the reception of RD_REQ signal is enabled 1: Clearing TX FIFO after the reception of RD_REQ signal is disabled

Bit	Field	Description
10	RESTART	This bit controls whether a RESTART is issued before transmission or reception. 0: If REPEN is 1, a RESTART is issued only if the transfer direction is changing from the previous command. If REPEN is '0', a STOP followed by a START is issued instead. 1: If REPEN is '1', a RESTART is issued before the data is sent/received (according to the value of CMD), regardless of whether or not the transfer direction is changing from the previous command. If REPEN is '0', a STOP followed by a START is issued instead.
9	STOP	This bit controls whether a STOP is issued after transmission or reception. 0: STOP is not issued after this byte, regardless of whether or not the TX FIFO is empty. The master continues the current transfer by sending/receiving data bytes according to the value of the CMD bit. If the TX FIFO is empty, the master holds the SCL low and stalls the bus until TX FIFO receives new command. 1: Transmission mode: STOP is issued after this byte, if the TX FIFO is empty. If the TX FIFO is not empty, STOP is issued after remaining bytes is transmitted. Reception mode: STOP is issued after this byte. If the TX FIFO is not empty, the master immediately tries to start a new transfer by issuing a START and arbitrating for the bus. Note: To generate a STOP after the reception of data bytes, use the I2C_ENR.ABORT bit. Please refer to the I2C_ENR register.
8	EMPINT	This bit controls whether a TX_EMPTY interrupt is generated. For details, refer to the I2C_RAWISR register.
7	STOPINT	This bit controls whether a STOP interrupt is generated in the Slave mode. 0: STOP interrupt is generated regardless of whether it's addressed or not. 1: STOP interrupt is generated only when it's addressed. Note: During a General Call address, if this bit is set, the slave doesn't generate a STOP interrupt. The STOP interrupt is generated only when the transmitted address is matched with the slave address (I2C_SAR).
6	DISSLAVE	This bit controls whether I2C has its slave disabled 0: slave enabled 1: slave disabled

Bit	Field	Description
5	REPEN	This bit determines whether RESTART conditions may be sent when I2C is acting as a master. 0: disabled 1: enabled RESTART condition can be replaced with a Stop and a subsequent START condition. When RESTART is disabled, I2C acting as a master is incapable of performing the following functions: Send a Start Byte Change transfer direction in the combined format Read operation with a 10-bit address If the above operations are performed, it will result in setting the I2C_RAWISR.TX_ABRT (bit 6) to 1.
4	MASTER10	Addressing mode when I2C is acting as a master 0: 7-bit addressing 1: 10-bit addressing
3	SLAVE10	When I2C is acting as a slave, this bit controls whether it responds to 7- or 10-bit addresses. 0: 7-bit addressing. I2C ignores transactions that involve 10-bit addressing; for 7-bit addressing, only the lower 7 bits of the I2C_SAR register are compared. 1: 10-bit addressing. I2C responds to only 10-bit addressing transfers that match the full 10 bits of the I2C_SAR register.
2:1	SPEED	These bits control at which speed the I2C operates in the Master mode. 1: Standard mode (up to 100Kbps) 2: Fast mode (up to 400Kbps)
0	MASTER	This bit controls whether the I2C master is enabled. 0: master disabled 1: master enabled

I2C_CR.DISSLAVE (bit 6) and I2C_CR.MASTER (bit 0) are set according to the table below:

Table 17-5 DISSLAVE and MASTER settings

DISSLAVE (I2C_CR[6])	MASTER (I2C_CR[0])	Status
0	0	Slave device
0	1	Configuration disabled
1	0	Configuration disabled
1	1	Master device

17.4.3 I2C_TAR Target Address Register

Address offset: 0x04

Reset value: 0x0000 0055

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				SPECIAL	GC	ADDR									
				rw	rw	rw									

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11	SPECIAL	This bit indicates whether software performs a special command (General Call or Start Byte command). 0: Ignore GC (bit 10) and use ADDR (bit 9:0) normally 1: Perform special I2C command as specified in GC bit
10	GC	If bit 11 (SPECIAL) is set to 1, then this bit indicates whether a General Call or START byte command is to be performed by the I2C. 0: General Call After issuing a General Call, only writes may be performed. I2C remains working in General Call mode until the SPECIAL (bit 11) is cleared. 1: START BYTE command
9:0	ADDR	This is the target address for any master transaction. When transmitting a General Call, these bits are ignored. To generate a START BYTE command, the CPU needs to write only once into these bits.

17.4.4 I2C_SAR Slave Address Register

Address offset: 0x08

Reset value: 0x0000 0055

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						ADDR									
						rw									

Bit	Field	Description
31:10	Reserved	Reserved, must be kept at reset value
9:0	ADDR	The slave address of I2C. For 7-bit addressing, only ADDR [6:0] is valid.

17.4.5 I2C_DR Data Command Register

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.							CMD	DAT							
							w	rw							

Bit	Field	Description
31:9	Reserved	Reserved, must be kept at reset value
8	CMD	This bit controls whether a read or a write is performed in the Master mode. 0: Write 1: Read When a command is entered in the TX FIFO, this bit distinguishes the write and read commands. In slave-receiver mode, write operation to this bit is ignored. In slave-transmitter mode, a “0” indicates that data in I2C_DR register is to be transmitted.
7:0	DAT	This register contains the data to be transmitted or received on the I2C bus.

17.4.6 I2C_SSHR Standard Mode SCL High Count Register

Address offset: 0x14

Reset value: 0x0000 0190

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value
15:0	CNT	This register sets the SCL clock high-period count (min valid value is 6) for Standard mode. Note: This register is programmed to a value between 6 and 65525, because I2C uses a 16-bit counter to flag an I2C bus idle condition when this counter reaches a value of CNT+10.

17.4.7 I2C_SSLR Standard Mode SCL Low Count Register

Address offset: 0x18

Reset value: 0x0000 01D6

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value
15:0	CNT	This register sets the SCL clock low-period count (min valid value is 8) for Standard mode.

17.4.8 I2C_FSHR Fast Mode SCL High Count Register

Address offset: 0x1C

Reset value: 0x0000 003C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT															
rw															

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value
15:0	CNT	This register sets the SCL clock high-period count (min valid value is 6) for Fast mode.

17.4.9 I2C_FSLR Fast Mode SCL Low Count Register

Address offset: 0x20

Reset value: 0x0000 0082

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CNT
rw

Bit	Field	Description
31:16	Reserved	Reserved, must be kept at reset value
15:0	CNT	This register sets the SCL clock low-period count (min valid value is 8) for Fast mode.

17.4.10 I2C_ISR Interrupt Status Register

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				GC	START	STOP	ACTIV	RX_DONE	TX_ABRT	RD_REQ	TX_EMPTY	TX_OVER	RX_FULL	RX_OVER	RX_UNDER
				r	r	r	r	r	r	r	r	r	r	r	r

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11:0	ISR	Refer to I2C_RAWISR register for detailed descriptions of these bits.

17.4.11 I2C_IMR Interrupt Mask Register

Address offset: 0x30

Reset value: 0x0000 08FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				GC	START	STOP	ACTIV	RX_DONE	TX_ABRT	RD_REQ	TX_EMPTY	TX_OVER	RX_FULL	RX_OVER	RX_UNDER
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11:0	IMR	These bits mask their corresponding interrupt status bits in I2C_ISR.

17.4.12 I2C_RAWISR RAW Interrupt Status Register

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.				GC	START	STOP	ACTIV	RX_DONE	TX_ABRT	RD_REQ	TX_EMPTY	TX_OVER	RX_FULL	RX_OVER	RX_UNDER
				r	r	r	r	r	r	r	r	r	r	r	r

Bit	Field	Description
31:12	Reserved	Reserved, must be kept at reset value
11	GC	General Call Set when a General Call address is received and acknowledged. It is cleared either by disabling I2C or when the CPU reads the I2C_GC.GC (bit 0). I2C stores the received data in the Rx buffer.
10	START	START condition detection This bit is set to 1 once a START or RESTART condition has occurred on the I2C interface regardless of whether I2C is operating in Master or Slave mode.
9	STOP	STOP condition detection The status of this bit differs based on the status of I2C_CR.STOPINT (bit 7): STOPINT = 0: This bit is set to 1 once a STOP condition has occurred on the I2C interface regardless of whether I2C is operating in Master or Slave mode. In Slave mode, a STOP interrupt is generated regardless of whether the slave device is addressed. STOPINT = 1: In Master mode (MASTER = 1), this bit indicates whether a STOP condition has occurred on the I2C interface; in Slave mode (MASTER = 0), a STOP interrupt is generated only if the slave device is addressed.
8	ACTIV	This bit captures I2C module activity. Once this bit is set, it stays set unless one of the four methods is used to clear it: Disabling the I2C Reading the I2C_ACTIV register Reading the I2C_ICR register System reset Even if the I2C is idle, this bit remains set until cleared.

Bit	Field	Description
7	RX_DONE	Slave transmit done When the I2C is acting as a slave-transmitter, this bit is set to 1 if the master does not acknowledge a transmitted byte. This occurs on the last byte of the transmission, indicating that the transmission is done.
6	TX_ABRT	Transmit abort This bit is set to 1 when I2C, as an I2C transmitter, is unable to complete the data transfer. Note: The I2C flushes the TX FIFO and RX FIFO when there is a transmit abort. The TX FIFO remains in this flushed state until the I2C_TX_ABRT register is read. Once this read is performed, the TX FIFO is then ready to accept new data from the APB interface.
5	RD_REQ	Read request This bit is set to 1 when I2C is acting as a slave and another I2C master is attempting to read data from the I2C interface. The I2C interface holds the bus in a wait state (SCL = 0) until this interrupt is serviced, which means that the I2C interface has been addressed by another master that is asking for data to be transferred. The processor must respond to this interrupt and then write the requested data to the I2C_DR register. This bit is set to 0 just after the processor reads the I2C_RD_REQ register.
4	TX_EMPTY	Transmit buffer empty The status of this bit differs based on the status of I2C_CR.EMPINT (bit 8): If EMPINT = 0, this bit is set to 1 when the transmit buffer is at or below the threshold value; If EMPINT = 1, this bit is set to 1 when the transmit buffer is at or below the threshold value and the most recent address/data transmission from the internal shift register is completed. It is automatically cleared by hardware when the transmit buffer level is above the threshold value.
3	TX_OVER	Transmit buffer over When the transmit buffer is full, the processor writes new data and causes overflow, which sets the bit to 1.
2	RX_FULL	Receive buffer full Set when the receive buffer is above the threshold. It is cleared by hardware when the receive buffer level is at or below the threshold.
1	RX_OVER	Receive buffer over Set if the receive buffer is full and an additional byte is received. The I2C interface acknowledges this, but new data will be lost.
0	RX_UNDER	Receive buffer over Set if the processor attempts to read the I2C_DR register when RX FIFO is empty.

17.4.13 I2C_RXTLR Receive Threshold Register

Address offset: 0x38

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								TL							
								rw							

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value
7:0	TL	Receive FIFO threshold level Controls the level of entries that triggers the RX_FULL interrupt.

17.4.14 I2C_TXTLR Transmit Threshold Register

Address offset: 0x3C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								TL							
								rw							

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value
7:0	TL	Transmit FIFO threshold level Controls the level of entries that triggers the TX_EMPTY interrupt.

17.4.15 I2C_ICR Combined and Independent Interrupt Clear Register

Address offset: 0x40

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															ICR
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	ICR	Reading this register clears all combined and independent interrupts. This bit only clears software clearable interrupts, instead of hardware clearable interrupts.

17.4.16 I2C_RX_UNDER RX_UNDER Interrupt Clear Register

Address offset: 0x44

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															RX_UNDER
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	RX_UNDER	Reading this register clears the RX_UNDER interrupt (I2C_RAWISR[0]).

17.4.17 I2C_RX_OVER RX_OVER Interrupt Clear Register

Address offset: 0x48

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															RX_OVER
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
0	RX_OVER	Reading this register clears the RX_OVER interrupt (I2C_RAWISR[1]).

17.4.18 I2C_TX_OVER TX_OVER Interrupt Clear Register

Address offset: 0x4C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															TX_OVER
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	TX_OVER	Reading this register clears the TX_OVER interrupt (I2C_RAWISR[3]).

17.4.19 I2C_RD_REQ RD_REQ Interrupt Clear Register

Address offset: 0x50

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															RD_REQ
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	RD_REQ	Reading this register clears the RD_REQ interrupt (I2C_RAWISR[5]).

17.4.20 I2C_TX_ABRT TX_ABRT Interrupt Clear Register

Address offset: 0x54

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															TX_ABRT
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	TX_ABRT	Reading this register clears the TX_ABRT interrupt (I2C_RAWISR[6]). This also releases the TX FIFO from the flushed/reset state, allowing more writes to the TX FIFO.

17.4.21 I2C_RX_DONE RX_DONE Interrupt Clear Register

Address offset: 0x58

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															RX_DONE
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	RX_DONE	Reading this register clears the RX_DONE interrupt (I2C_RAWISR[7]).

17.4.22 I2C_ACTIV ACTIVITY Interrupt Clear Register

Address offset: 0x5C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															ACTIV
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	ACTIV	Reading this register clears the ACTIV interrupt (I2C_RAWISR[8]) if the I2C is not active anymore. If the I2C is still active, the ACTIV interrupt bit continues to be set. It is cleared by hardware if the I2C module is disabled and if there is no further activity on the I2C bus. By reading this register, you get status of the I2C_RAWISR.ACTIV (bit 8).

17.4.23 I2C_STOP STOP_DET Interrupt Clear Register

Address offset: 0x60

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															STOP
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	STOP	Reading this register clears the STOP interrupt (I2C_RAWISR[9]).

17.4.24 I2C_START START_DET Interrupt Clear Register

Address offset: 0x64

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															START
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	START	Reading this register clears the START interrupt (I2C_RAWISR[10]).

17.4.25 I2C_GC GEN_CALL Interrupt Clear Register

Address offset: 0x68

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															GC
															r

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value
0	GC	Reading this register clears the GC interrupt (I2C_RAWISR[11]).

17.4.26 I2C_ENR Enable Register

Address offset: 0x6C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														ABORT	ENABLE
														rw	rw

Bit	Field	Description
31:2	Reserved	Reserved, must be kept at reset value
1	ABORT	I2C transfer abort 0: ABORT not initiated or ABORT done 1: ABORT operation in progress The software can abort the I2C transfer by setting this bit when I2C acts as a master. The software cannot clear the ABORT bit once set. In response to an ABORT, the I2C controller issues a STOP and flushes the TX FIFO after completing the current transfer, then sets the TX_ABRT interrupt after the abort operation. The ABORT bit is cleared automatically after the abort operation.

Bit	Field	Description
0	ENABLE	I2C module enable 0: I2C module disabled (TX and RX FIFOs are held in an erased state) 1: I2C module enabled

17.4.27 I2C_SR Status Register

Address offset: 0x70

Reset value: 0x0000 0006

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.									SLV_ACTIV	MST_ACTIV	RFF	RFNE	TFE	TFNF	ACTIV
									r	r	r	r	r	r	r

Bit	Field	Description
31:7	Reserved	Reserved, must be kept at reset value
6	SLV_ACTIV	Slave FSM activity status 0: Slave FSM is in IDLE state so the Slave part of I2C is not active 1: Slave FSM is not in IDLE state so the Slave part of I2C is active
5	MST_ACTIV	Master FSM activity status 0: Master FSM is in IDLE state so the Master part of I2C is not active 1: Master FSM is not in IDLE state so the Master part of I2C is active
4	RFF	Receive FIFO completely full 0: Receive FIFO is not full 1: Receive FIFO is full
3	RFNE	Receive FIFO not empty 0: Receive FIFO is empty 1: Receive FIFO is not empty
2	TFE	Transmit FIFO completely empty 0: Transmit FIFO is not empty 1: Transmit FIFO is empty
1	TFNF	Transmit FIFO not full 0: Transmit FIFO is full 1: Transmit FIFO is not full
0	ACTIV	I2C activity status This bit is an ORed result of MST_ACTIV bit and SLV_ACTIV bit.

17.4.28 I2C_TXFLR Transmit FIFO Level Register

Address offset: 0x74

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														CNT	
														r	

Bit	Field	Description
31:2	Reserved	Reserved, must be kept at reset value
1:0	CNT	Contains the number of valid data entries in the transmit FIFO (0~2)

17.4.29 I2C_RXFLR Receive FIFO Level Register

Address offset: 0x78

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.														CNT	
														r	

Bit	Field	Description
31:2	Reserved	Reserved, must be kept at reset value
1:0	CNT	Contains the number of valid data entries in the receive FIFO (0~2)

17.4.30 I2C_HOLD SDA Hold Time Register

Address offset: 0x7C

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.								RX_HOLD							
								rw							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TX_HOLD															

rw

Bit	Field	Description
31:24	Reserved	Reserved, must be kept at reset value
23:16	RX_HOLD	Sets the internal SDA hold time (valid while SCL is high) in units of APB clock period, when I2C device acts as a receiver.
15:0	TX_HOLD	Sets the SDA hold time (after SCL goes from high to low) in units of APB clock period, when I2C device acts as a transmitter.

17.4.31 I2C_SETUP SDA Setup Time Register

Address offset: 0x94

Reset value: 0x0000 0064

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.								CNT							
								rw							

Bit	Field	Description
31:8	Reserved	Reserved, must be kept at reset value
7:0	CNT	SDA setup time (min value is 2) It is recommended that if the required delay is 1000ns, then for an APB clock frequency of 10MHz, this register should be programmed to a value of 11.

17.4.32 I2C_GCR General Call ACK Register

Address offset: 0x98

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.															GC
															rw

Bit	Field	Description
31:1	Reserved	Reserved, must be kept at reset value

Bit	Field	Description
0	GC	ACK General Call 0: I2C neither responds with an ACK nor generates an interrupt, when it receives a General Call. 1: I2C responds with an ACK when it receives a General Call.

17.4.33 I2C_SLVMASK Slave Address Mask Register

Address offset: 0xB0

Reset value: 0x0000 03FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						MASK									
						rw									

Bit	Field	Description
31:10	Reserved	Reserved, must be kept at reset value
9:0	MASK	Slave address mask 0: a corresponding bit in the I2C_SAR register is masked, not requiring comparison 1: a corresponding bit in the I2C_SAR register requires comparison

17.4.34 I2C_SLVRCVADDR Slave Address Receive Register

Address offset: 0xB4

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.						ADDR									
						r									

Bit	Field	Description
31:10	Reserved	Reserved, must be kept at reset value
9:0	ADDR	Slave address actually received by I2C

18 DBG Debug Support

18.1 Overview

The MCU core includes the debug module mainly used for the function commissioning. When the core gets address (address breakpoint) or accesses data (data breakpoint), the hardware debug module may control the core to stop. The user can enquire the core internal status and system status. After enquiry, the core may continue to execute the current program.

When the chip and debugger connect and begin debugging, the debugger auto calls the core debug module for debug operation.

18.2 Functional description

18.2.1 Function block diagram

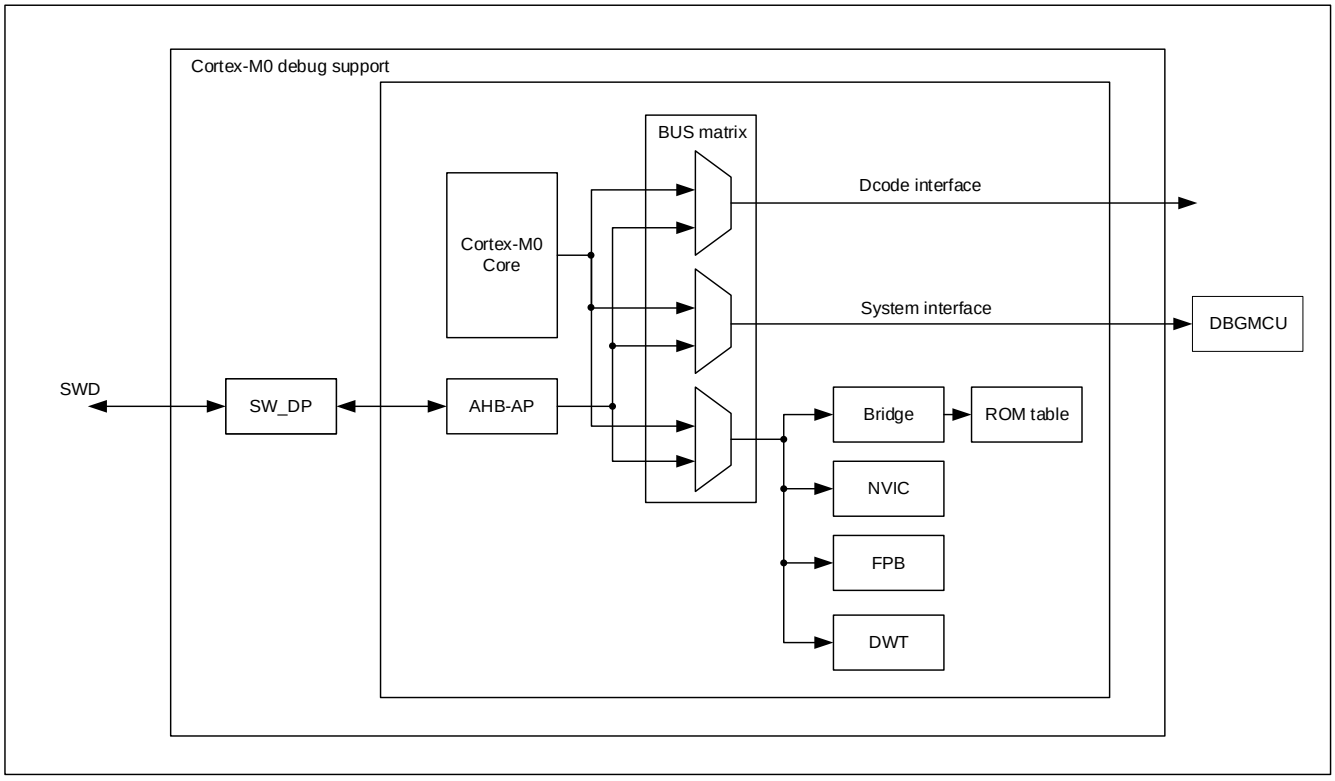


Figure 18-1 Debug function block diagram

Cortex®-M0 core includes the debug unit, and it's composed of:

SWDP: SW debug port

BPU: Breakpoint debug unit

DWT: Data watchpoint and trace

18.2.2 SWD internal pull up and down

SWD pin input directly controls the debug mode, and cannot be floated. To guarantee that I/O level is controllable, SWD pin has the built in pull up and down resistance.

- SWDIO: Internal pull up
- SWCLK: Internal pull down

The software can use these I/O ports as the ordinary I/O port. At this time, the pull up/down function closes by default. Refer to the chapter of General Purpose Input/Output GPIO.

18.2.3 SWJ debug port

The chip's two ordinary I/O ports are used as SWD-DP interface pin. These pins in different packages all support SWD debug port.

Table 18-1 SWD debug port pin

SWJ-DP port pin name	SW debug interface		Pin allocation
	Type	Debug function	
SWDIO	Input/output	Serial data input/output	PA13
SWCLK	Input	Serial clock	PA14

18.3 ID code and lock mechanism

The chip has many ID codes, as seen below:

Table 18-2 ID code

ID Name	Chip
DEV_ID	0x4C50F800
CPU TAP SW ID	0x0BB11477

18.3.1 Microcontroller device ID code

The microcontroller contains the device ID code. This ID defines MCU chip version, and is mapped on the external APB bus. This code can be acquired by the user code and debug interface.

18.3.2 Cortex JEDEC-106 ID code

The microcontroller has a JEDEC-106ID code. It’s located in the 4KB ROM table mapped to the

internal PBB bus address 0xE00FF000_0xE00FFFFF.

18.4 SW debug port

18.4.1 SW protocol introduction

This sync serial protocol uses 2 pins: Clock signal (SWCLK) and two-way data signal (SWDIO) from host to target.

As the two-way data line, SWDIO must be connected to the pull up resistance (ARM's recommended value 100K). SWDIO pin has the built in pull up resistor, and the extra external resistor is not required.

Data transmission begins from the low bit, and allow reading and writing the register DPACC and APACC.

According to the protocol, when SWDIO changes direction, insert one conversion time (one Bit time by default, adjustment by SWCLK). In this period, any device cannot drive the signal line.

18.4.2 SW protocol series

One series contains three phases:

- The host sends 8 bit request packet
- The target sends 3 bit acknowledgement response
- According to the configuration direction, the host or target sends 33 bit (including one check bit) data

Table 18-3 8bit request packet

Bit	Name	Description
0	Beginning	Must be 1
1	APnDP	0: Access DP 1: Access AP
2	RnW	0: Write request 1: Read request
4: 3	A[3 : 2]	DP or AP register address
5	Parity	Previous bit check bit
6	Stop	0
7	Park	Cannot be driven by host. Because of pull up, the target is always read as 1

Note: Each request packet is followed by 1 bit conversion time. For more information on DAPCC and APACC register, refer to the relevant CPU technology data sheet of ARM.

Table 18-4 3bit response package

Bit	Name	Description
0 .. 2	ACK	001: Fail 010: Wait 100: Success

Note: When the response (ACK) signal is one of scenarios above, the response bit is followed by one conversion time.

Table 18-5 33bit data packet

Bit	Name	Description
0 .. 31	WDATA/RDATA	Read or write data
32	Parity	32 bit data parity check bit

Note: Wait for one conversion time after read data bit.

18.4.3 SW-DP status unit (Reset, Idle states, ID code)

SW-DP status unit uses the internal ID code to identify SW_DP. Observe JEP-106 standard. For specific information, please refer to the relevant ARM manual.

SW-DP status unit doesn't work until before the debugger reads ID.

- In case of power on reset, or DP switches from JTAG to SWD, or exceeds high level of 50 periods, SW-DP status unit is in the reset status;
- If the low level of at least 2 periods appear after RESET status, the status unit will switch to the IDLE status;
- When the status unit is in the reset status, first switch to the IDLE status, and read DP-SW ID register. Otherwise, the debugger cannot conduct other normal transmission, and ACK Fault will appear;

18.4.4 DP and AP read/write access

- DP read operation has no delay: The debugger directly acquires data (in case of ACK return success status) or waits (in case of ACK return wait status);
- AP read operation has latency. It means that the previous read operation result can only be acquired during the next operation. If the next operation is not access to AP, it's mandatory to read DP-RDBUFF register to acquire the previous read operation result;
- DP-CTRL/STAT register READOK flag bit is refreshed after AP read operation and RDBUFF read operation to inform the debugger whether AP read operation succeeds;
- SW-DP has the write buffer (DP and AP have the write buffer) so that the write operation can be accepted when other transmissions are ongoing. If the write buffer is full, the debugger will

acquire one wait ACK response. Read IDCODE register, read CTRL/STAT register and write ABORT register operation remain accepted when the write buffer is full;

- Because SWCLK and HCLK are not in step, insert two extra SWCLK periods after the write operation (after parity check bit) to ensure that the internal write operation is correctly completed. These two extra clock periods should be inserted when the line is in IDLE status. This operation step is particularly important during writing CTRL/STAT register to propose one power on request. Otherwise, the next operation (valid operation after core power on) will execute immediately. It will lead to failure;

18.4.5 SW-DP register

In case of APnDP=0, access to these registers below.

Table 18-6 SW-DP register

A[3:2]	Read/write	SELECT register's CTRLSEL bit	Register	Description
00	Write		IDCODE	Fixed as 0x0BB1 1477 (for identifying SW-DP)
00	Write		ABORT	
01	Read/Write	0	DP-CTRL/STAT	Request one system or debug power on operation; configure AP access operation mode; Control compare, check operation; Read some status bits (overflow, power on response).
01	Read/Write	1	WIRE CONTROL	Configure the serial communication physical layer protocol (such as conversion time length, etc.)
10	Read		READ RESEND	Allow to restore data from one error debug transmission rather than repeat the initial AP transmission.
10	Write		SELECT	Select current access port and valid 4 word register window.
11	Read/Write		READ BUFFER	This register will capture the data result of the previous read operation from AP. Therefore, the data can be acquired without again enabling the new AP transmission

18.4.6 SW-AP register

In case of APnDP=1, access and AP register access address is composed of two parts:

- A[3: 2] value
- DP SELECT register's current value

18.5 MCU debug module (DBGMCU)

MCU debug module provides the debugger assist function below:

- Support low power mode
- Breakpoint timer and watchdog clock control

18.5.1 Debug support at low power mode

MCU has multiple low power modes. It can close CPU clock, reduce CPU power consumption, and enter the low power mode by executing WFE or WFI command. FCLK and AHB bus clock HCLK are mandatory for debug operation, and cannot be closed. MCU is equipped with some registers to change the low power mode characteristics, thus supporting the debug code in the low power mode. Specific configurations:

- Enter the sleep mode. In order to make HCLK and FCLK have the same clock, the debugger must set the DBGMUC_CR register DBG_SLEEP bit.
- Enter the stop mode. It's mandatory to configure DBG_STOP bit. The operation will activate the internal oscillator HSI, thus providing the clock for FCLK and HCLK.
- Enter the standby mode. It's mandatory to configure DBG_STANDBY bit. The digital circuit under this configuration won't power off, and will activate HSI to provide the clock for FCLK and HCLK.

18.5.2 Support timer, watchdog

In case of the breakpoint, select the timer's work mode according to the application of timer and watchdog;

- Counter count continues. The application is in the PWM wave control motor
- Counter count stops. The application is in the watchdog counter.

18.6 Register

18.6.1 Register overview

Table 18-7 DBG register overview

Offset	Acronym	Register Name	Reset
0x00	DBG_IDCODE	DBG ID encode register	0x4C50F800
0x04	DBG_CR	DBG control register	0x00000000

18.6.2 DBG_IDCODE ID Encode Register

Address offset:0x40013400 (only support 32-bit access, read only)

Reset value: 0x4C50F800

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DEV_ID															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DEV_ID															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bit	Field	Description
31:0	DEV_ID	Device Identifier Read only register, always read as reset value

18.6.3 DBG_CR Control Register

Address offset: 0x40013404 (only support 32 bit access)

Reset value: 0x0000 0000(POR reset only, not reset by system reset)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.													DBG_TIM14_STOP	Res.	
													rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.			DBG_TIM3_STOP	Res.	DBG_TIM1_STOP	DBG_WWDG_STOP	DBG_IWDG_STOP	Res.				DBG_STOP_FOR_LDO	DBG_STANDBY	DBG_STOP	DBG_SLEEP
			rw		rw	rw	rw					rw	rw	rw	rw

Bit	Field	Description
31:19	Reserved	Reserved, must be kept at reset value.
18	DBG_TIM14_STOP	The counter stops work when the core enters the debug mode 0: The counter still works normally when the timer is selected 1: The counter stops when the timer is selected
17:13	Reserved	Reserved, must be kept at reset value.
12	DBG_TIM3_STOP	The counter stops work when the core enters the debug mode 0: The counter still works normally when the timer is selected 1: The counter stops when the timer is selected
11	Reserved	Reserved, must be kept at reset value.

Bit	Field	Description
10	DBG_TIM1_STOP	The counter stops work when the core enters the debug mode 0: The counter still works normally when the timer is selected 1: The counter stops when the timer is selected
9	DBG_WWDG_STOP	The debug window watchdog stops work when the core enters the debug mode 0: The window watchdog counter still works normally 1: The window watchdog counter stops work
8	DBG_IWDG_STOP	Independent watchdog stops work This bit is not related to whether the core enters the debug status 0: The watchdog counter still works normally 1: The watchdog counter stops work
7:4	Reserved	Reserved, must be kept at reset value.
3	DBG_STOP_FOR_LDO	Debug stop mode LDO status 0: LDO enters the low power mode, and normally enters the STOP mode 1: LDO doesn't enter the low power mode, PLL sustains power on, and cannot truly enter STOP mode. CPU enters DEEPSLEEP, and HCLK closes
2	DBG_STANDBY	Debug standby mode 0: Digital circuit power down, HCLK and FCLK close. Except for the flag bit to indicate exiting the standby mode, STANDBY mode exit and reset are same 1: Digital circuit doesn't power off. FCLK and HCLK clocks are opened and provided HSI. STANDBY mode exit and reset are same.
1	DBG_STOP	Debug stop mode 0:In the stop mode, the clock controller disables all clocks (including HCLK and FCLK). When exiting from STOP mode, the clock configuration is consistent with the configuration after reset. If it's required to again enable PLL, HSE clock, the software must reconfigure the clock control system. 1: In the stop mode, FCLK and HCLK open, and HSI provides the clock. When exiting from STOP mode, make reconfiguration by software if it's required to again enable PLL, HSE clock
0	DBG_SLEEP	Debug sleep mode 0: In the sleep mode, clock FCLK opens, FCLK keeps the configured system clock by default, and HCLK closes. The sleep mode won't reset the configured clock system. When exiting the sleep mode, the software doesn't need to reconfigure the system clock 1: In the sleep mode, FCLK and HCLK clock is provided by the formally configured system.

19 Device Electronic Signature

19.1 Overview

Device electronic signature is an identifier (96bits) that is stored in the Flash memory system storage area to solely identify a microcontroller. Under no circumstances can the user modify the device electronic signature.

Device electronic signature can be read by software and used to achieve the following functions:

- Used as a password to improve the security of code in flash memory by using device electronic signatures in combination with software encryption and decryption algorithms when programming flash memory
- Used as serial number in terminal applications
- Activating the bootloader with safety mechanism

19.2 Register description

Base address: 0x1FFF F7E8

Table 19-1 Register capacity register overview

Offset	Acronym	Register Name	Reset
0x00	UID1	Unique identifier	0xXXXXXXXX
0x04	UID2	Unique identifier	0xXXXXXXXX
0x08	UID3	Unique identifier	0xXXXXXXXX

19.2.1 UID1 Unique Identifier

Address offset: 0x00

Reset value: This value is written by default.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID (31: 16)															
R															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID (15: 0)															
r															

Bit	Field	Description
31: 0	U_ID (31: 0)	U_ID: 31: 0 unique ID bits

19.2.2 UID2 Unique Identifier

Address offset: 0x04

Reset value: This value is written by default.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID (63: 48)															
R															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID (47: 32)															
R															

Bit	Field	Description
31: 0	U_ID (63: 32)	U_ID: 63: 32 unique ID bits

19.2.3 UID3 Unique Identifier

Address offset: 0x08

Reset value: This value is written by default.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID (95: 80)															
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID (79: 64)															
r															

Bit	Field	Description
31: 0	U_ID (95: 64)	U_ID: 95: 64 unique ID bits

20 History Records

Table 20-1 History records

Date	Version	Contents
2022/10/25	V1.01	Modify Timers
2022/08/11	V1.0	Modify Timers, etc.
2022/03/15	V0.51	Overall upgrade in content and format
2022/01/27	V0.5	First public release